



Subscription Order Management

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Subscription Order Management

Subscription Order Management is specifically designed to support the order process of a combination of physical goods, service, and subscriptions to enable you to transition from a product-based model to a service-based one.

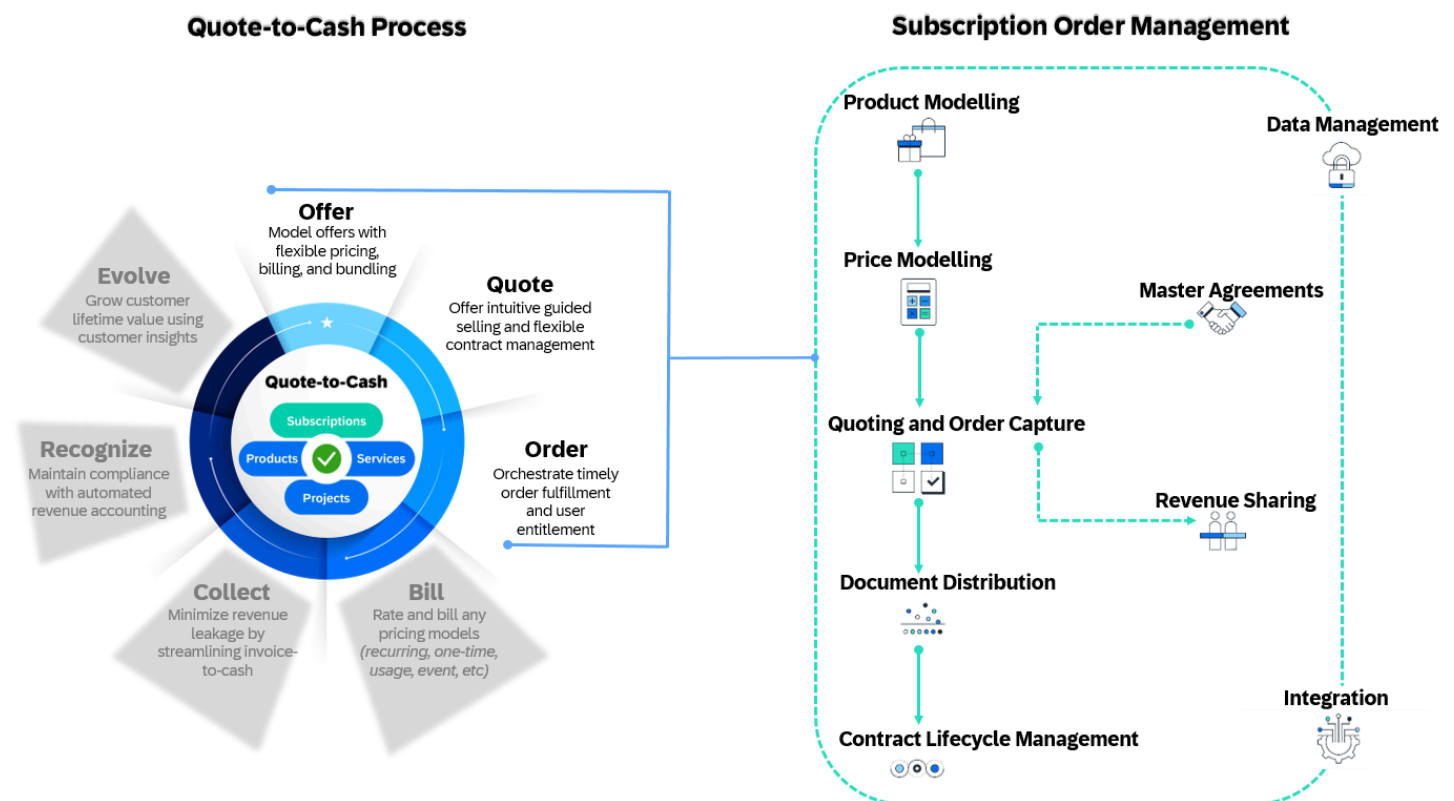
Typical use cases include selling a physical device (such as mobile phones) along with a usage contract, offering packed IT solutions instead of only products, and selling Software as a Service. There are also other key scenarios such as Anything as a Service (XaaS) and Outcome-Based Business Models.

With Subscription Order Management, you can not only manage your contracts flexibly over the whole contract lifecycle (such as adding and modifying elements of contracts and tracking the service usage), you can also easily model new subscription offerings with the right mix of products, recurring and one-time services, usage charges, revenue sharing models, and third-party content.

The following image further explains the functionality of Subscription Order Management.

i Note

The images are interactive and contain links to documentation in which you might be most interested. You can hover over each image for a description and click on them for more information.



Please note that image maps are not interactive in PDF outputs.

Subscription Order Management is a part of the Billing and Revenue Innovation Management landscape. For more information, refer to [Billing and Revenue Innovation Management](#).

Business Roles in Subscription Order Management

Subscription Order Management delivers its functionality via the SAP Fiori Launchpad and the WebClient UI. The following business roles are available:

Business Role	System	Purpose
SAP_S4C_UIU_SOM_REP	Back end server	For the internal sales representative role (S4C_SOM_REP) in the WebClient UI.
SAP_S4C_UIU_SOM_PROD	Back end server	For the product modeler role (S4C_SOM_PROD) in the WebClient UI.
SAP_BR_PROD_CONF_MODEL_R_SOM	Front end server	Controls the Fiori Launchpad for the product modeler.
SAP_BR_INTRNAL_SALESREP_SOM	Front end server	Controls the Fiori Launchpad for order and contract management.

For more information, refer to the Security Guide for SAP S/4HANA.

i Note

The technical profile DEFAULT is assigned to the business role S4C_SOM_REP . The runtime framework profile setting for DEFAULT must have the **Disp.Mode** indicator set in the Customizing ► **Service** ► **UI Framework** ► **UI Framework Definition** ► **Maintain Runtime Framework Profile** ►

Subscription Order Management on the SAP Fiori Launchpad

To find out about the relevant BRIM content for the SAP Fiori launchpad, go to the [SAP Fiori Apps Reference Library](#) and select ► **All apps for SAP S/4HANA** ► **by Application Component** ► Search for CRM-S4-SOM to display SAP Fiori apps for Subscription Order Management or *FI-CA* to display SAP Fiori apps for Convergent Invoicing and Contract Accounts Receivable and Payable (Contract Accounting).

For more information on SAP Fiori, got to [SAP Fiori Overview](#).

OData Services

On the Fiori server, you must activate following additional OData services and assign a system alias (transaction /IWFND/MAINT_SERVICE):

- MD_C_PRODUCT_MAINTAIN_SRV (Manage Product Master Data)

Depending on this service, maintain the following additional services:

- CV_ATTACHMENT_SRV (Gateway: ODATA model for CV - attachments)
- LO_VCHCLF (Variant Configuration and Classification Service)
- SD_PRICING_CONDITIONRECORD_SRV (SD Pricing Condition Record)

For more information, see SAP Note [2405092](#) 📄.

- CRMS4_SOM_PROD_SPEC_SRV (Manage Subscription Product-Specific Data)

For general information on how to set up the Fiori Launchpad and activating services, see the documentation for the Fiori Launchpad.

Master Agreement

A master agreement is a contract between a provider and another company that you use to manage B2B scenarios.

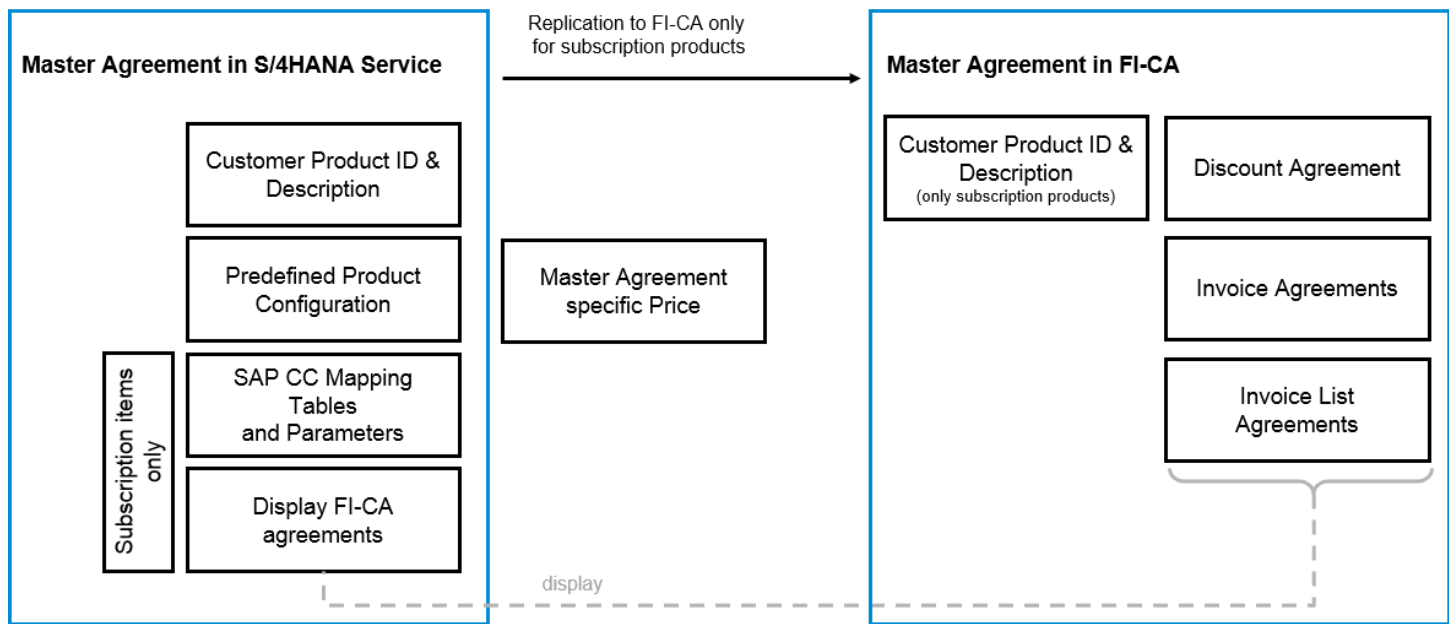
Definition

Here, you define the following individual sales agreements for a specific validity period:

- Authorized groups
- Individual products (define specific product attributes)
- Individual prices and discounts

The terms defined in the master agreement are used as templates for referenced solution quotations and subscription orders. If an authorized company (business partner) refers to an existing master agreement in their order or quotation, individual products and agreements from this master agreement can be accessed here. Agreed prices and products are automatically copied to the subsequent call-off document.

You can use master agreements to establish long-term relationships with your business partners, using individual prices and customer-specific products. The master agreement is first created in Subscription Order Management, where additional attributes such as item details are maintained. Master agreement items referencing subscription products are then replicated to Contract Accounts Receivable and Payable (FI-CA in SAP S/4HANA) and **SAP Convergent Charging** (SAP CC). You can make further settings for subscription items that are replicated to FI-CA master agreement, such as assigning invoice agreements, invoice list agreements, or discount agreements to dedicated products, product groups, customers, or customer groups.



Master Agreement Features Overview

Features

The master agreement is an object used to adapt product attributes such as product ID, product description, price, SAP CC tables and parameters, and product configuration, to fit your customer's requirements. You can then use this as a template to create follow-up objects (call-offs).

Searching for Master Agreements

You can search for master agreements in the search space, using the master agreement ID and other attributes.

Searching for Subscription Orders and Solution Quotations

If you have created a subscription order or solution quotation as a call-off to a master agreement, you can use the master agreement ID to search for the related subscription orders and solution quotations in the respective search space.

Master Agreement Creation and Maintenance

You can adapt individual master agreement items as per customer requirements. The following objects can be adjusted for specific customers:

- Product configuration

You can change the product characteristics in a configurable product using Advanced Variant Configuration. The changes you make here are used as a template for the subsequent subscription order or solution quotation, where they can be changed if the configuration has not been fixed in the master agreement.

- Custom ID

You can define a customer-specific product ID for each item in a master agreement.

- Custom texts

You can edit a customer-specific text for the individual product in various languages.

- Convergent charging tables

You can change the values of the interval or mapping tables that are assigned to the charge plans for the product by redefining them. This creates a new table in SAP CC with the modified values. These values are transferred to the subscription order where they can no longer be changed.

- Convergent charging parameters

You can change the values for the SAP CC parameters that are assigned to the charge plans for the product here. These values are transferred to the subscription order in SAP CC.

- Product list

The product list displays all admissible sub-products. By redefining a sub-product, you can also perform all of the adjustments described above for this sub-product. A redefinition is only required if you want to make changes.

- Sharing group ID

If the used product Cross-Catalog Mapping version supports shared allowances, you must maintain a sharing group ID here. This sharing group ID is used for all subsequent subscription orders or solution quotations, where they cannot be changed.

Maintaining Master Agreement During Contract Changes

You can change or delete the existing master agreement during the Product Change and Change Sold-To Party change processes. After you have selected a new product, the master agreement is validated against it in the change order. In case of an error, you must either remove the existing master agreement or manually assign a relevant one in the [Subscription-Specific Data](#) assignment block in item details.

Sharing Objects

If the used product can be shared with a sharing product, you can create or assign sharing contracts that can be used in call-offs for this master agreement. For more information about this, refer to **Shared Products**.

You can only create a sharing contract with reference to a master agreement once the master agreement item is released. The transaction type you can use for creating a sharing order must be maintained in Customizing for [Service](#) under  [Transactions](#) .

[Settings for Master Agreement](#) > [Define Target Document Type](#) . As a payer, you can select any authorized partner maintained for this master agreement item.

An assignment of a sharing contract is already possible even if the status is **In Process**. You can only assign sharing contracts that belong to an authorized partner of this master agreement item. An assigned sharing contract can be unassigned at any time.

Price Agreement

You can maintain specific prices for a master agreement using one of the following options:

- Directly in the master agreement, using the **Price Agreement** assignment block at item level
- Using transactions VK11 and VK12

i Note

You can also maintain these prices directly from the **Price** agreement tab for condition types. The document category for master agreement is **CMMA**.

You can enable condition types, such as **PSMB** and **PSPB**, from the Customizing for [Sales and Distribution](#) > [Basic Functions](#) > [Pricing](#) > [Pricing Maintenance](#) > [Define Price Maintenance Scenarios](#) . The scenarios must be mapped to the transaction type and/or item categories in the Customizing under [Service](#) > [Transactions](#) > [Basic Functions](#) > [Define Transaction Types and Define Item Categories](#) .

You can transfer these prices from the master agreement to the resultant subscription order or solution quotation by creating an access sequence for the condition type that supports master agreement access. By default, SAP delivers the access sequence PSMB for recurring fees and PSPB for bundle prices as samples.

Releasing Items for Call-Off

To create a solution quotation or a subscription order from a master agreement, you must release the relevant items within the master agreement. Once released, you can use these items to copy over the master agreement details into the new order or quote.

i Note

You can only change the item attributes if the item status is not set to **Released**.

More Information

For more information, see Customizing for [Service](#) under [Transactions](#) > [Settings for Master Agreement](#) .

Related Information

[Shared Products](#)

[Master Agreement: Create/Assign Sharing Contract](#)

[Change Product](#)

[Change Sold-To Party](#)

Creating and Editing Master Agreements

Prerequisites

- You make general settings for master agreements in Customizing for [Service](#) under [Transactions](#) [Settings for Master Agreement](#).
- While creating a price agreement in the master agreement, you can only select the UoM that is assigned to the product in the material master/product master data.

Activities

In the WebClient UI, you can use the role S4C_SOM_REP to create and edit a master agreement. From the business role selection menu, choose the role S4C_SOM_REP to access the overview page for Subscription Order Management.

Go to [Home](#) [Subscriptions](#) [Search](#) and choose [Master Agreements](#) to search for a specific master agreement. You can also enter the corresponding data for a new master agreement under [Create](#). Save your entries.

In the Fiori Launchpad, you can use the [Manage Master Agreements](#) app to search for existing master agreements as well as to create new ones. To use this application, you must have the role SAP_BR_INTRNAL_SALESREP_SOM assigned to your user.

Assigning Business Partners

Use

In Service, all business partners who may be relevant are assigned in the master agreement. A business partner is assigned to the master agreement directly, where the business partner is valid for the entire validity period of the item.

Features

You can manage various business partners in the master agreement.

- Master Agreement Partner

The master agreement partner is the partner with whom you have negotiated the master agreement. A master agreement only has one master agreement partner. The [sold-to party](#) category is assigned to the contract partner.

- Authorized Partners

Authorized partners can create call-offs for the master agreement or can receive goods and services.

These partners are also differentiated according to their roles in the subscription contract.

- Authorized Contract Partners

Authorized contract partners can create call-offs and act as the contract partner (payer), sold-to party, and service recipient in the subscription contract as a default setting.

- Authorized Sold-to Parties

Authorized sold-to parties can create call-offs and act as the sold-to party and service recipient in the subscription contract as a default setting. They cannot, however, act as the contract partner in the subscription contract. The default contract partner is used here.

- Authorized Service Recipients

Authorized service recipients cannot create call-offs or act as the contract partner in the subscription contract. They can be entered as an alternative service recipient in the subscription order.

- Default Contract Partners

The default contract partner is always assigned as the contract partner (payer) in the subscription order if an authorized sold-to party creates a call-off, since this sold-to party cannot act as the contract partner. Each item can only have one default contract partner. This applies to all authorized sold-to parties. The default contract partner cannot perform releases or act as the service recipient.

You can replace the default contract partner and service recipient proposed with another partner that is defined in this role in the master agreement. You cannot use partners who are not assigned in the master agreement as a contract partner or service recipient in the subscription contract.

- Employee Responsible

The individual who has negotiated the terms of the master agreement on behalf of the provider.

When the master agreement is transferred to SAP FI-CA, the system only transfers partners who can act as the contract partner (payer) in the FI-CA provider contract. You can use these partners in SAP FI-CA to create agreements (invoice agreements, invoice list agreements, and discount agreements) in the SAP FI-CA master agreement.

Partner Check Against Rule

The business partner roles described are found in the standard system response. However, you can use a rule, which you record in the master agreement item, to find authorized partners. You do not have to record all business partners in the master agreement to do so. Example: All ADAC members can perform releases.

You can use this rule with the Business Add-In (BAI) **BAI: Check Authorized Partners for Provider Master Agreement** (CRM_PRVMA_PARTNER_CHECK) to record your own checks or to overwrite the checks in the standard system by creating an implementation for the filter value **Initial**.

i Note

While searching for a master agreement in the release transaction, the system only finds master agreement items for which the business partner is maintained directly in the item and therefore exists in the database. If the partner has not been maintained in the master agreement, the release can only be performed directly from the master agreement and not by using the release transaction.

If a master agreement that contains such rules is transferred, mapping to the SAP FI-CA partners (contract partner and service recipient) must be performed in the **BAI: Data Exchange with CI for Master Agreement** (CRMS4_MAGR_DATA_EXCH), if these partners are relevant for creating specific partner groups in SAP FI-CA.

You can find these BAIs in Customizing for **Service** under **▸ Transactions ▸ Settings for Master Agreement ▾**.

Assigning Products

Use

The master agreement is used as a product catalog where you can adapt individual products for specific customer requirements. While executing a call-off for the master agreement, the selected products are copied into the resultant subscription order or solution quotation. Once this is done, it is then no longer possible to make changes to a subscription order or quotation when it is created with reference to a master agreement. However, you can make changes to the product configuration if you have made the corresponding settings in the master agreement.

While you can assign a sales or subscription product to a master agreement, if the product purpose of the material type is **Master Agreement**, then a call-off document (subscription order or solution quotation) can only be created with a master agreement reference. The product type is mapped to the product purpose in the Customizing under **▸ Service ▸ Master Data ▸ Settings for Subscription Master Data ▸ Subscription Product-Specific Data ▸ Assign Product Purpose to Product Types ▾**

Features

You can then define a customer-specific product version at item level and assign a **Custom ID**. This is a customer-specific product ID that describes a unique product version in the master agreement, including all redefined subitems. It can only be edited at main item level since this is the level from which you subsequently perform the release. This means that all main items collectively represent the product catalog specified for the master agreement.

The customer-specific product ID can be modified until you release the item. If you want to change a customer-specific product ID after an item is released, you must create a new item in Subscription Order Management, where the additional items represent time changes. However, these changes are not always relevant for SAP FI-CA. The SAP FI-CA master agreement does not, therefore, contain any items and the system only saves the customer-specific product ID. You can reference this customer-specific product ID in FI-CA to create product groups or agreements (invoice agreements, discount agreements, and invoice list agreements). You can therefore also edit language-dependent texts in the master agreement, which are replicated to SAP FI-CA.

You can enter customer-specific product IDs in SAP FI-CA product groups and then assign SAP FI-CA agreements. You can, for example, record information used to determine the billing cycle or invoice forms for printing the invoice or alternative payers in an invoice agreement. You can record information in a discount agreement about discount determination, which is performed when the billable items are processed. The customer-specific product ID always remains the same. If you make changes to the product, the system defines a new product, and the previous product expires.

Product Configuration

If your product is marked as configurable in the material master data, you must set the values for the given characteristics within the variant configuration. You can also create master agreements using configurable products where multi-level configuration using Advanced Variant Configuration along with a BOM has been enabled. For more information about this, refer to **Multi-Level Configuration of Product Bundles in Master Agreements**.

For multi-level configurable products, the following must be considered:

- You can use the configuration date to influence the date which is used to access the related BOM hierarchy. This configuration date can only be set on main item and is read-only for subitems.
- If you have a plant-specific BOM, only the plant assigned as a delivery plant in the product master is considered in the master agreement. When creating a call-off, no new plant determination is executed for on the used business partner.
- Master agreement does not support quantities. If your BOM explosion depends on the quantity, this is not supported. But you still can change the quantity in the call-off document as long as the BOM hierarchy is not changed
- You cannot add manual subitems. Subitems are only added/deleted by changing the configuration of the main item.
- When creating a master agreement with a multi-level configurable product bundle via the **Master Agreement** OData API, the same product can occur multiple times as a subitem. You can identify the different instances of such a product using the BOM identifier attribute, `BillOfMaterialItemNodeNumber`.

i Note

You will be notified to configure the product and cannot release the item until you do so.

If you set the **Configuration** to **Fixed**, you can no longer change these characteristics in the resultant subscription order/solution quotation. To be able to do so, you must set the **Configuration** to **Not Fixed**. If the configuration is not fixed and you are using a multi-level configurable product, you can only change those characteristics in the call-off document which do not change the exploded BOM hierarchy from the master agreement. In case you use a multi-level configurable bundle product, you can fix the configuration on configurable subscription subitems, because a bundle product will be split into separate follow-up documents, such as a subscription contract and a sales order.

Product/Item Structure in Master Agreement

A product can consist of several sub products or components. However, the main item in the master agreement always contains the product to be sold. The sub-products of a product bundle are not automatically expanded into subitems, as is the case in the subscription order.

The **Product List** assignment block shows the maximum expansion of a product bundle with all the possible sub-products. This list only contains each product once even if you have used it several times in the product definition. You can generate subitems for products from this list, which you want to redefine, for example if a different price is to apply or you only want to allow a specific configuration.

For other kinds of hierarchy, for example, AVC BOM hierarchy or a manually created item hierarchy, the product list is not displayed and you see the item hierarchy only in the item list. You can access the subitems directly from the item list.

Status Management

A main item can have the following statuses in the master agreement:

- In Process

A new item is created, and all changes can be made. Follow-up documents can exist.

- Released

The item has been created completely and is legally valid. You can start subsequent processes (releases). You can only reset the status to **In Process** if there are no follow-up documents or there are only canceled follow-up documents. The following can no longer be modified:

- Customer-specific product ID
- Fixed configuration
- Configuration
- Parameters
- Cross-catalog mapping version
- Billing cycle
- Sharing group ID
- Configuration date

You can still make changes to the following objects:

- Mapping tables

You can change all records in a redefined mapping table since they have their own time dependency. However, you cannot reset a mapping table that has been redefined to the default version or redefine a default table.

- Date fields

These must not overlap in the master agreement.

- Texts
- Partners

Changes do not affect existing subscription contracts and only apply to new releases.

- Sharing objects

You can create new sharing contracts or assign existing ones. These are available for new call-offs from the master agreement.

- Completed

The item is completed meaning that you cannot perform any new actions (releases). The item is no longer selected during the release. You can only reset the status to **Open** if there are no follow-up documents or there are only canceled follow-up documents.

Displaying FI-CA Agreements

You can determine the FI-CA agreements available in a master agreement.

Use

You can use the **Agreements** assignment block and the date search in the SAP FI-CA system to find the following agreement types for a specific key date:

- Invoice agreement
- Invoice list agreement
- Discount agreement

All agreements in the master agreement are displayed in the document header. You can select an item and restrict the search using a customer-specific ID.

i Note

You can use additional filters to further restrict the validity of an agreement in the SAP FI-CA system. These filters cannot be evaluated during display in the master agreement in Subscription Order Management. The system therefore may display agreements that are not valid for this item.

Activities

To display the master agreement, use the transaction launcher to navigate to the SAP FI-CA system.

Creating Call-Offs with Reference to Master Agreements

Use

During the creation of subscription orders/solution quotations, the system transfers the agreed upon terms defined in the master agreement to this subscription order/solution quotation, for example, authorized partners, individual products, prices, and discounts.

Integration

Relationship Between Master Agreement and Call-Off Objects

When a subscription order/solution quotation is created, the following objects are transferred from the master agreement:

- Product ID

The product ID is used to individually select characteristic values, which are linked to the associated product model. A configurable product is then assigned to an item accordingly and configured according to the customer requirement. This customer-specific product contains its own product ID in the master agreement.

- Product bundle

The system only creates subitems for a product in the master agreement if a higher-level product exists. The system first checks in the call-off document whether these higher-level products exist in the master agreement. If they exist, the configured data is used and if not, the default values from the product are used. If a subitem is redefined in the master agreement, the data from this subitem is copied to the call-off document. If a subitem is not redefined but is created in the call-off document from the bundle explosion, it is created with its default values.

- Partners

The system checks whether the various roles in the call-off document are used in the corresponding roles in the master agreement. If various items are reported, the system performs the following checks for the partner roles:

- Sold-to Party: Is checked against the "authorized contract partner" and "authorized sold-to party".
- Payer: Is checked against the "authorized contract partner" and "default contract partner".
- Ship-to Party: Is checked against the "authorized contract partner", "authorized sold-to party", and "authorized ship-to party".

- Subitems

While creating a call-off document from a master agreement, all subitems in the master agreement are copied over to the call-off document. In the call-off document, you can remove the subitems that are either optional in a product bundle or have been added manually in the master agreement. You can choose to add any of the removed subitems while executing a change process on the subscription contract. When adding a subitem in a change order, only those subitems that have a reference to the master agreement are available in the product list on clicking the [Add Subitem](#) button.

The following scenarios are available for subitems:

- The subitems that are already part of a product bundle can be directly selected or deselected in the call-off document and need not to be copied over from the master agreement to the call-off document. Only the manually added subitems are copied over.
- In case of a multi-level configurable product, you cannot add manual subitems. Thus, the [Add Subitem](#) button is not available in the call-off document or the change order.
- In case of a manual hierarchy, all the subitems are added manually and can be copied over into the call-off document.

The [Add Subitem](#) button only displays the direct subitems of the selected item. If such a subitem has its own subitems, they will be automatically copied into the order when the item is selected.

i Note

The [Add Subitem](#) button is only available in initial and change orders but not in solution quotations. Thus, you must enter a product ID and a higher-level item number to add subitems in solution quotations.

- Product configuration

The product configuration is transferred from the master agreement to the call-off document.

In case of multilevel configuration, the complete item hierarchy is copied from the master agreement to the call-off document. If configuration is not fixed in the master agreement, only those configuration changes that do not change the item hierarchy are allowed.

- Price agreement

Modified price agreements are transferred from the master agreement to the call-off document, and cannot be changed.

- Sales organization

The sales organization is transferred from the master agreement to the call-off document.

- Contract date

The start date for the subscription order item or subscription item in solution quotation must be within the master agreement validity period.

- Mapping tables

There is only a reference to the SAP CC mapping tables. In the call-off document, you can only redefine the tables that are not already redefined in the master agreement.

- Cross catalog mapping versions

A cross catalog mapping version is transferred to the subscription contract as a reference.

- Parameters

There is only a reference to the SAP CC parameters. In the call-off document, you can only redefine the parameters that are not already redefined in the master agreement.

- Billing cycle

The billing cycle is copied from the master agreement to the call-off document. The corresponding field is available in the item enhancement. If this field is not edited in the master agreement, the system transfers the default value for the product. It is then not possible to make any changes to the subscription contract.

- Sharing Group ID

The sharing group ID is transferred from the master agreement to the call-off document.

- Account Assignment Object

You can manually maintain WBS elements in the **Account Assignment** field in the master agreement at the header and item level. These are copied over to the call-off document (either a subscription order or a solution quotation).

- Configuration Date

When a master agreement is created for a configurable product, the configuration date is copied from the master agreement into the call-off document (subscription order or solution quotation). This date cannot be changed as long as the reference to the master agreement exists.

Subitems should inherit the Main item target quantity only if subitem is manually added. Quantity changes of AVC multilevel sub items or bundle sub items are not allowed.

Features

Apart from creating a subscription order or a solution quotation directly from the master agreement, you can also establish a reference using the automatic master agreement check when entering a new product in the subscription order or solution quotation. In this case, the order item can be created from the selected master agreement item.

i Note

You can use a master agreement item if the payer of the subscription order and the authorized contract partner are the same partners. If you create the first subscription order main item with reference to the master agreement item and the master

agreement item has several authorized contract partners, you can replace the payer of the subscription order with another authorized contract partner from the master agreement. Having done so, the subscription order payer cannot be changed and only the master agreement items with this payer as authorized contract partner can be used as reference for subscription order items.

New order items can be created directly from master agreements that have already been referenced using the toolbar in the item tree.

You can also perform a mass creation of call-off documents from a master agreement.

Target and Released Quantity in a Master Agreement

The target quantity and released quantity fields in a master agreement enable you to implement partial consumption of a predetermined quantity of a specific product. The target quantity of this product is defined in the master agreement and the quantity entered while creating call-off documents is updated as the released quantity in the master agreement.

❖ Example

Consider that you need a 100 printers for your organization that need to be delivered in different quantities across multiple locations. In this case, you would define the target quantity as 100 in the master agreement, and then use call-off documents to order smaller quantities for specific locations, such as 50 printers for City A and 20 for City B. Once the call-off documents are created, which means you save a subscription order or a solution quotation, the released quantity is updated from 0 to 70.

If a subscription order or solution quotation is rejected, the released quantity is updated in the master agreement accordingly. The reject option is only available until the subscription contract is either not yet created or is still inactive. Once a subscription contract is active, you can't cancel or withdraw to return the ordered quantity.

❖ Example

Let's consider the previous example. 20 printers have been ordered for City B and the order is now rejected. In this case, the value of the released quantity is updated from 70 to 50.

You can update the target quantity at any time during the validity period of the master agreement. However, you can only increase the value of the target quantity to ensure there are no inconsistencies in the existing call-off documents.

In case your master agreement consists of product bundles or configurable product bundles, only the main item will have the target quantity. The subitems, whether bundle items or manually added subitems, will inherit their target quantity from the main item.

If the ordered quantity in the call-off documents is modified using change processes such as [Change Quantity](#) and [Create Follow-Up Order](#), the released quantity is updated in the master agreement accordingly. In case the resultant released quantity exceeds the target quantity, you will be blocked by an error until you modify the ordered quantity to fit the target quantity or you increase the target quantity in the master agreement to accommodate the higher demand.

The following measures are put in place to ensure consistency between target and released quantities and the quantity in the call-off documents:

- The target quantity in a master agreement can only be updated by one user at a time. In case multiple users try to edit the master agreement at the same time, they will get an error.
- A call-off document can only be saved if the master agreement is not locked.
- The master agreement is locked when the released quantity is updated by call-off documents.

The Master Agreement OData API uses the fields **Master Agreement Target Quantity** (SrvcTransMstrAgrmtItmTargetQty), **Master Agreement Released Quantity** (SrvcTransMstrAgrmtItmRelQty), and **Master Agreement Quantity ISO Unit** (SrvcTransMstrAgrmtQtyISOUnit) at the master agreement item level to support target quantity in external applications as well.

i Note

You can't implement the partial consumption scenario for existing call of documents because the target quantity and released quantity are only available for newly created master agreements.

Activities

You can create a subscription order or a solution quotation with reference to a master agreement using the role S4C_SOM_REP in WebClient UI or the role SAP_BR_INTRNAL_SALESREP_SOM on SAP Fiori.

You can create a subscription order/solution quotation using the following ways:

- Using an existing master agreement and creating a call-off
- By creating a new master agreement and creating a call-off
- Directly creating a subscription order/solution quotation for which automatic master agreement determination applies

Related Information

[Mass Creation of Call-Off Documents from Master Agreement](#)

WBS Elements as Account Assignment Objects in Master Agreements

You can manually maintain WBS elements in the **Account Assignment** field in master agreements on the UI and Master Agreement OData API, at the header and item level. This data is transferred to the call-off document (either a subscription order or a solution quotation) at a header and item level, and then distributed to the FI-CA provider contract. You can manually change the WBS element in the order or quotation to a different one than the one referred to in the master agreement.

For product bundles, the WBS element is copied over from the main item to the subitem, and can be changed manually. Once the main item is released, you can no longer make any changes to the WBS element at the subitem level.

i Note

If you have maintained the WBS element at the item level, it takes precedence over the WBS element maintained at the header level (if any). If a WBS element is not maintained at the item level, then the one at header level (if any) is mapped to the item, and then distributed to the FI-CA provider contract.

Customer Hierarchies in Master Agreement

Use customer hierarchies to leverage parent-child relationships between customers with respect to their sales areas.

While creating a master agreement, you can assign a customer hierarchy either as an **Authorized Contract Partner** or an **Authorized Sold-To Party**. You do so by choosing **Parties Involved** > **Select Partner Number Type** > **Customer Hierarchy Node** >

A customer hierarchy node is assigned to a master agreement using the customer hierarchy key (**Node_ID**). This **Node_ID** uses the number range object **SD_CUSHIER**.

i Note

You must first configure the number range and generate the customer hierarchy node ID before the master agreement can use the customer hierarchy node. The node IDs for existing customer hierarchy nodes are generated using the report **KNVH_GEN_ALT_KEY**. For more information about this, refer to the Customizing ► **Sales and Distribution** ► **Master Data** ► **Business Partners** ► **Customers** ► **Customer Hierarchy** ► **Generate Alternate Key for Customer Hierarchy** ►.

Any changes to the customer hierarchy, such as moving a node (business partner) to another hierarchy or removing a node (business partner) entirely, may impact any existing subscription contracts associated with the master agreement. In this case, you may manually change the existing master agreement reference either by assigning another relevant master agreement or by removing the reference altogether. You may do so using the **Create Follow-Up Order** change process. If no associated master agreement is found for the changed customer hierarchy, the existing master agreement reference is removed from the subscription contract. For more information about this, refer to SAP Note [3396116](#) 📄 (Allow master agreement change when BP is moved to a different customer hierarchy or removed).

Related Information

[Customer Hierarchies](#)

Archiving Master Agreements

Use

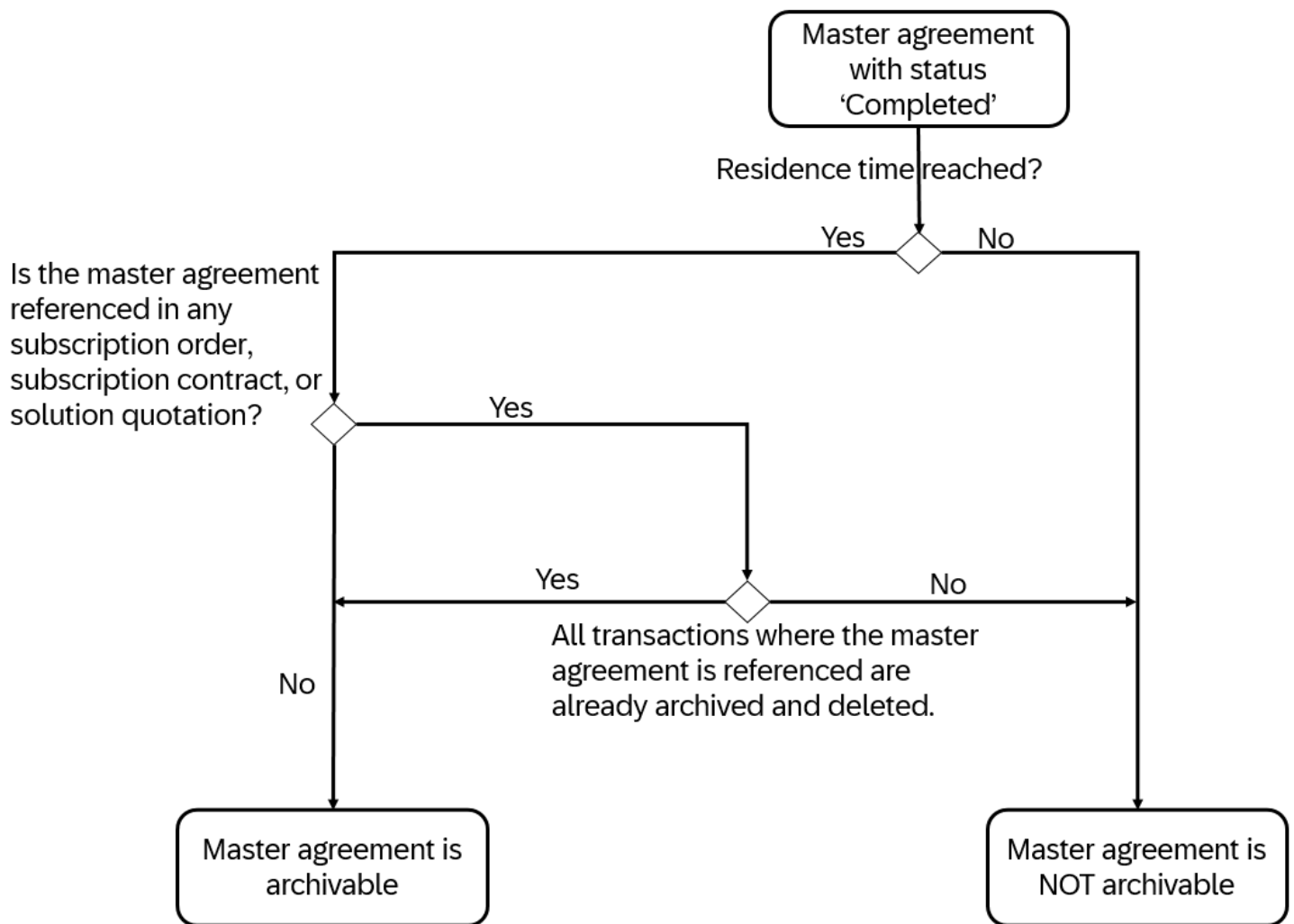
If a business relationship with a contract partner ends, the master agreement associated with this business relationship can be archived. The archiving object CRM_PRVMA is available for doing so.

Prerequisites

If you want to archive a master agreement, the following prerequisites must be met:

- The header status must be **Completed**
- The residence time (number of days after the last change to the master agreement) must have elapsed.
- All subscription orders, solution quotations, and subscription contracts created with a reference to the master agreement must be archived .

If you have set up the ILM integration for the CRM_PRVMA ILM object, your ILM rules must also allow the archiving. For more information, see **ILM Objects in Subscription Order Management**.



Archiving Master Agreements

Related Information

[ILM Objects in Subscription Order Management](#)

Enabling Credit Check on Subscription Items

Prerequisites

- You must activate SAP Credit Management.

i Note

It can be deactivated with an entry in the view V_UKM_IS_ACTIVE or the BAdI implementation for BAdI UKM_R3_ACTIVATE.

- The SAP_BR_CREDIT_CONTROLLER business role must be assigned to your user so that you can access the [Manage Documented Credit Decisions](#) and [Display Credit Exposure](#) Fiori apps.
- You must assign the [UKM – SAP Credit Management](#) role to the business partner and maintain a credit limit and credit segment as well.
- You must mark the transaction type as relevant for credit check in Customizing under [Service > Transactions > Basic Settings > Credit Management > Activate Credit Management in Service Transactions](#).

The credit check is executed only for subscription orders and solution quotations that are maintained in the Customizing for subscription-specific transactions and are marked with either the [Standard Subscription Order](#) or [Solution Quotation](#) document purpose.

- You must mark the item category as relevant for credit check in Customizing under [Service > Transactions > Basic Settings > Credit Management > Activate Credit Management on Item Types](#).

i Note

Even if you mark a service or sales item as relevant in this Customizing, it will not be considered in a solution quotation/subscription order while creating a real credit exposure. Such items are only considered in a simulation mode.

Overview

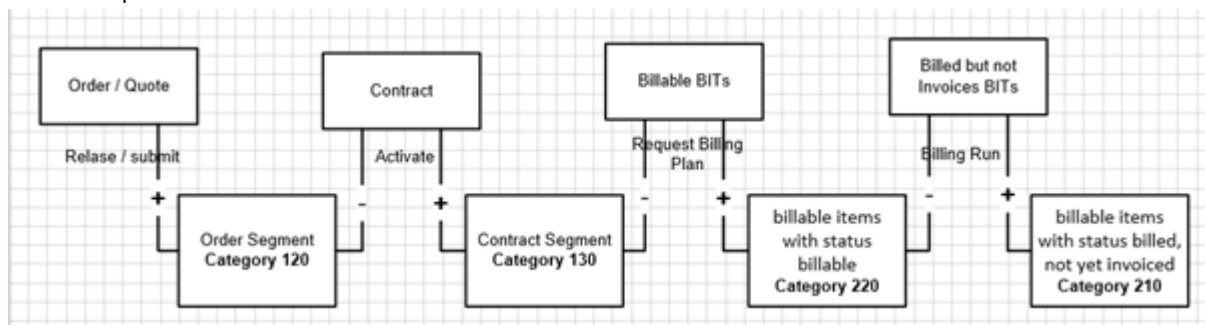
SAP Credit Management (FIN - FSCM - CR) determines the risk of losses on receivables from your business partners early on and helps in making credit decisions efficiently and, in some cases, automatically.

When a customer signs a subscription with a service provider, the service provider must ensure the creditworthiness of this customer. To do so, a credit check is executed during the order (or quotation) capture process. In a subscription order or solution quotation, subscription items are enabled to run credit management functionality, such as credit check, credit exposure, or Documented Credit Decision. If an item is relevant for integration, by default, the item gross value is used to execute the credit check and create the credit exposure. The gross value is calculated using the contract end date or the end of contract term (if the contract end date is unavailable). This value can be influenced by the CRMS4_SOM_CREDIT_CHECK_VALUE BAdI in the Customizing, [Service > Transactions > Settings for Subscription Transactions > Integration > Integration with Credit Check > BAdI: Credit Check Value](#).

i Note

Only items with recurring fees or one-off fees are relevant because items with usage fee do not have a gross value.

The credit check is always executed on a payer level. If there are different payers in a document, the amount per payer is calculated across all relevant items and sent as a single amount to SAP Credit Management. If a credit check fails, a Documented Credit Decision (DCD) is created in SAP Credit Management for the corresponding payer and the subscription order or solution quotation is blocked for further processing. The subscription order or solution quotation can be submitted/accepted only if all items pass the credit check successfully, and only then will the subscription contract be created. A credit manager can check such a DCD and either release it manually or do a recheck. Items that are sent to credit check are updated with system status [Credit Check OK](#) or [Credit Check not OK](#), depending on the result of the credit check. A cumulative status across all items is displayed at the header level. Completed items are not considered in this header status.



End-to-End Flow for Integration with Credit Check Management

If the credit check is successful, the credit exposure is created in the subscription order or solution quotation with exposure category 120 (Subscription Order & Solution Quotation). On contract activation, it is copied over to the contract with exposure category 130 (Subscription Contract). During billing of billable items, the exposure is transferred to one of the billing categories, 220 (Unbilled BIT), 210 (Billed BIT), or 200 (Open Item), depending on the BIT status when data is uploaded to SAP Credit Management. When the open item is paid, the credit exposure is finally cleared.

The subscription specific credit exposure is created on an item/contract level with the following object keys in credit management:

Credit Exposure	Key Format
In the subscription order/solution quotation	Document ID-Item Number
Inactive subscription contracts	Subscription Contract Document ID-Item Number
Active subscription contracts	Subscription Contract Document ID-Contract Number

For more information, refer to [SAP Credit Management \(FIN-FSCM-CR\)](#).

Constraints

- SAP Credit Management must run in the same SAP S/4HANA system as Subscription Order Management. A distributed landscape is not supported.
- In a subscription order or solution quotation, only subscription items are enabled to create a real credit exposure or to block the creation of follow-up documents by creating a DCD. Service and sales items are only supported in simulation mode but the real credit exposure is only created in the follow-up documents.
- Integration with billing is only supported for Convergent Invoicing/FI-CA on a BIT level.
- You can release or a trigger a recheck on a Documented Credit Decision only from the [Manage Documented Credit Decisions](#) Fiori app.
- You can change neither the document currency nor the payer once a credit exposure is triggered for an item.
- The pricing procedure must be in sync between the subscription order or solution quotation and subscription contract. That is, the price calculation must always return the same value for all objects, otherwise there will be inconsistencies in credit exposure and during delta calculation in a change order.
- There are also certain constraints when using APIs:
 - No credit check is executed in a change quotation. It is executed during change processing in the change order.
 - If a solution quotation is created and accepted in a batch process and the credit check fails, only an error is returned. No DCD can be created because no solution quotation is created in this case.

Related Information

[Executing Credit Check in Subscription Orders and Solution Quotations](#)

[Billing Integration](#)

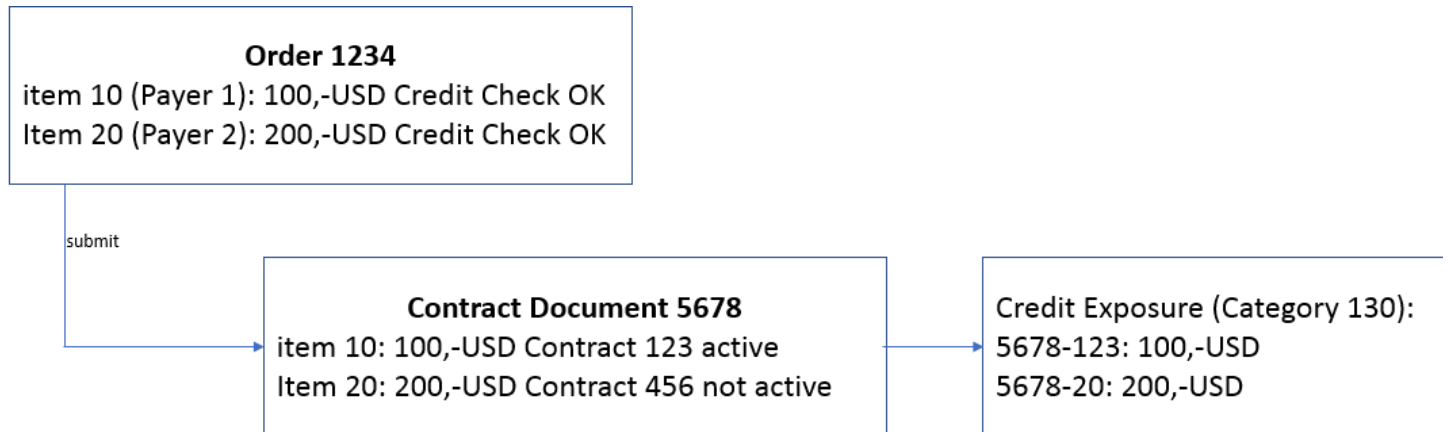
Executing Credit Check in Subscription Orders and Solution Quotations

Credit Check in Subscription Orders

An initial check is executed in a subscription order, either if the item is released manually and the order is saved or when the order is submitted. After a credit check was executed the first time, it will always be triggered when the order is saved. If the credit check is successful, a corresponding credit exposure is created on exposure category 120. If the credit check fails, a Documented Credit Decision (DCD) is created for the payer and an error is raised in the order. An order cannot be submitted if even a single item has a failed credit check.

If the item price is changed, a new credit check is executed for the delta amount. If successful, the exposure is adjusted accordingly. However, if the check fails, the DCD is created for the delta amount and the existing exposure for corresponding items will be deleted. If item price is reduced or item is rejected/deleted, an existing exposure is used to reduce the value for credit check for other items for same payer. On item rejection/deletion, an existing DCD is closed automatically, and an existing credit exposure is cleared if applicable.

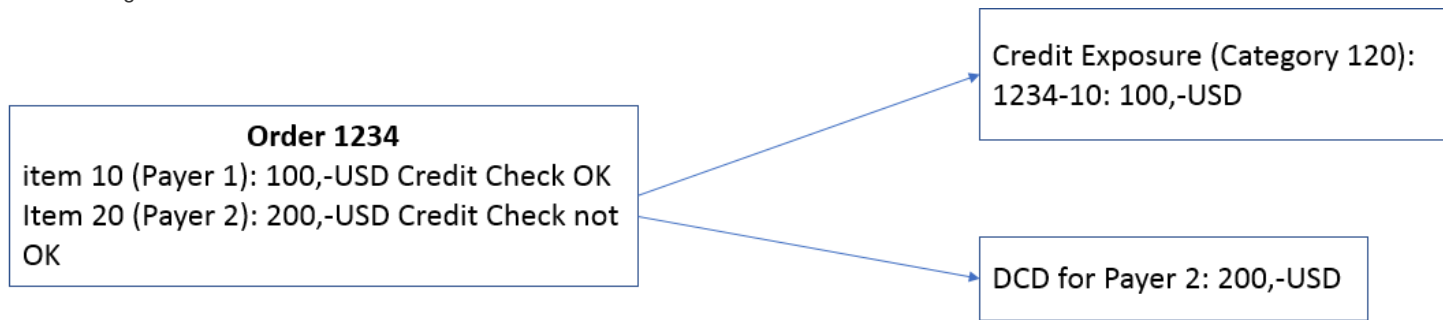
On submitting the credit exposure, items relevant for subscription contract are moved from category 120 to category 130. Exposure for one-off items remain on category 120.



1. Credit check successful for Payer 1 (item 10) and Payer 2 (item 20).
2. On submitting the order, data is copied to the contract.
3. Contract 123 for Payer 1 is active – Credit exposure for the active contract is transferred to category 130 with object key 5678-123 (contract level).
4. Contract 456 for Payer 2 is not active – Credit exposure for the inactive contract is transferred to category 130 with object key 5678-20 (item level).

Credit Management - Successful Check in Order

If the credit check is executed during the Submit operation and fails for one item, no data is copied to the contract and the credit exposure remains in category 120.



1. Credit check successful for Payer 1 (item 10).
2. Credit check failed for Payer 2 (item 20).
3. On submitting the order, no data is copied to the contract, not even for Payer 1.
4. Credit exposure for successful item 10 is created with the order category 120, with object key 1234-10.
5. A DCD is created for Payer 2.

If the contract start date is in the future, the order item is still changeable after submitting. If item price is changed after submitting, the exposure that was already moved to the contract item will be considered and the relevant amount will be reduced for credit check in the order. If the item price is decreased, this will be reflected in the contract exposure when the order is submitted again. This decreased exposure is not visible until the order is submitted (no negative exposure for the order item).

Credit Check in Solution Quotations

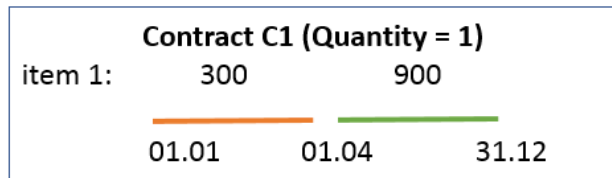
When a solution quotation is accepted, the credit check is automatically executed. When the quotation item is released automatically, a credit simulation is triggered and a warning is displayed if the credit limit has been exceeded. If the credit check is successful, a corresponding credit exposure is created with exposure category 120. If it fails, a Documented Credit Decision is created for the payer and an error is raised in the quotation. A quotation cannot be accepted if even a single item has a failed credit check.

If the item price is changed, a new credit check is executed for the delta amount. If successful, the exposure is adjusted accordingly. However, if the check fails, the DCD is created for the delta amount and the existing exposure for corresponding items will be deleted. If item price is reduced or item is rejected/deleted, an existing exposure is used to reduce the value for credit check for other items for same payer. On item rejection/deletion, an existing DCD is closed automatically, and an existing credit exposure is cleared if applicable.

On accepting the solution quotation, the credit exposure that has been created for items that are relevant for the subscription contract are moved from category 120 to category 130. The exposure for one-off items remain on category 120.

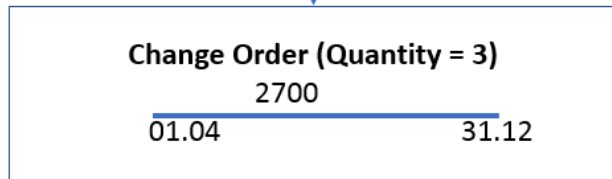
Credit Check in Change Orders

In a change order, only the delta amount must be sent for credit check, and not the complete gross value of the order item. After the change order is submitted, the delta amount is added to the existing contract exposure.



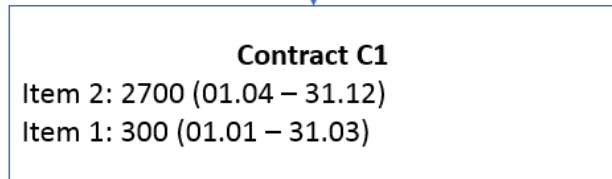
Contract Exposure:
1200,-

Change Quantity



Value for Credit Check (only delta)
 $2700 - 900 = 1800$

Submit change and activate contract

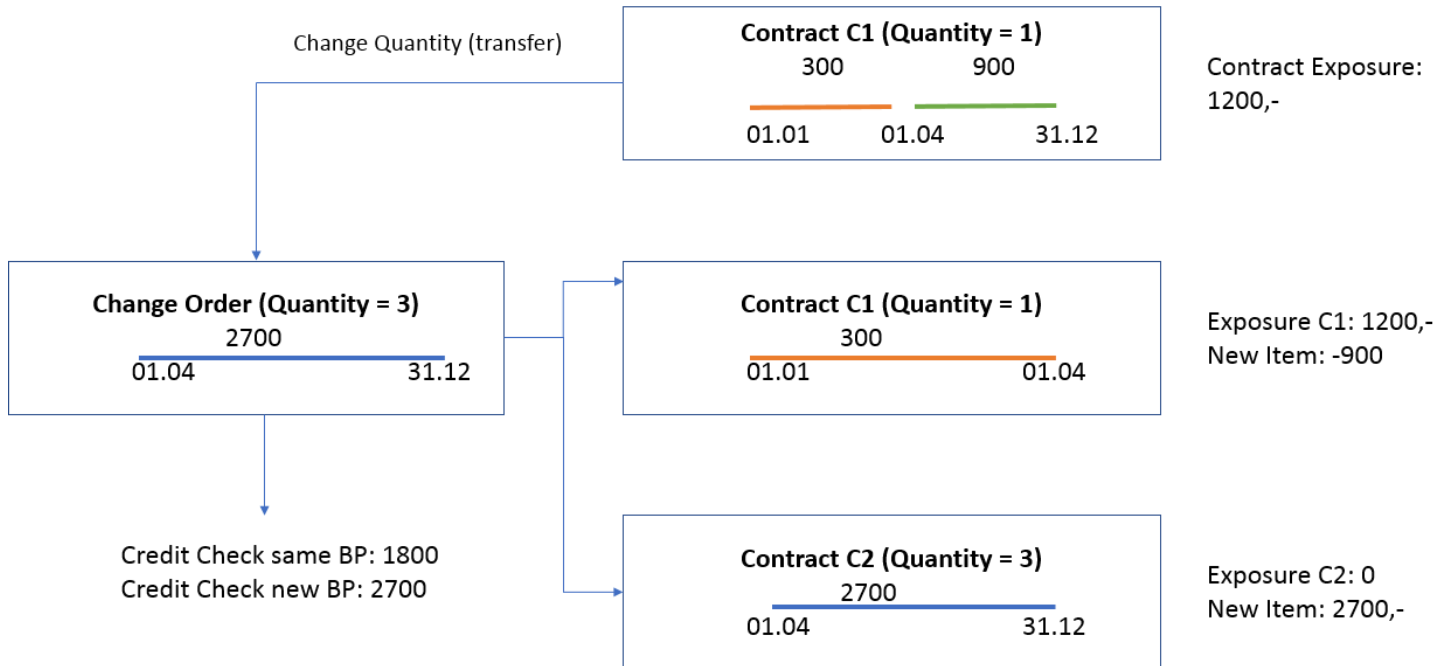


Contract Exposure after
activation:
3000,-

Credit Management - Change Order

This delta amount is calculated based on the old gross value of the active contract item. So, it is only an estimated value, because no new pricing is executed to get this delta amount.

In case of a contract transfer, including a change of Sold-to Party, the scenario is as below:



Credit Management - Contract Transfer

In case a change order is created from a change quotation and the credit check fails, a DCD is created for the change order. If you release the DCD, the error in the change order is removed but the order is only submitted when the corresponding change quotation is processed again.

If the payer is changed using a change order, the complete gross value of the order item is sent to credit check. Otherwise, the delta amount is sent, as for any other change process. When the change order is submitted, the old contract exposure is reduced, and a new contract exposure is created on the new contract. In the following contract change situations, no credit check is executed:

- In change processes where no change order is created, no credit check is executed, and the contract exposure is not adjusted.
- In case of a cancellation change request, no credit check is executed for the item. When the cancellation is submitted and the contract becomes active, the contract exposure, if applicable, is reduced.

Copy Exposure to Subscription Contract

When the contract item is created, the exposure is moved from category 120 to category 130. As long as the contract is not active, the exposure is created with the object key "Document ID-Contract Item Number" in SAP Credit Management. When the contract is activated, this exposure is added to the final contract object key "Document ID-Contract Number".

When the contract is activated, all open exposures or existing DCDs for a change order that is not yet submitted, will be cleared. Once a contract is set to completed, the related contract exposure will also be cleared.

Simulate Credit Check

An automatic credit check is only triggered when you accept a solution quotation or save a subscription order with items that have been released. If you want to know whether the current value of the document is within the credit limit before saving any items in released status, you can simulate a credit check from the solution quotation/subscription order header. By doing so, you can avoid the creation of unwanted DCD cases. You can only execute the credit check simulation when the document is error free and in edit mode.

In a solution quotation you can simulate a credit check using one of the following two options:

- Simulate all items where items with status open, in process, or released are included in the simulation.
- Simulate the released items where only the items with status released are included in the simulation.

In a subscription order, items with status open, in process, or released are always included in the simulation. When the simulated credit check result shows OK, all items in the simulation can pass the credit check. If the simulated credit check result is not OK, the system prompts you with detailed information, such as which partner has violated the credit limit and by how much the credit limit was exceeded. You leverage this information and decide how to proceed further. For example, you may delay the release of subscription items with higher values and only release those items with lower values to pass the credit check.

i Note

Since the simulation runs in real time, its result only represents the current, temporary state. The real credit check result may differ from the result of the simulation. It is also important to note that the simulation contains all the document items whereas the real credit check contains only the subscription items.

Billing Integration

Prerequisites

- The following Customizing must be configured for the billing integration:
 - You must set the communication type to **Local (Same System)** and the **Billing Active** indicator in the Customizing **►Financial Accounting ► Contract Accounts Receivable and Payable ► Integration ► Financial Supply Chain Management ► Credit Management ► Activate Credit Management ►**. This enables the credit exposure to automatically move from contract category to a billing relevant category.

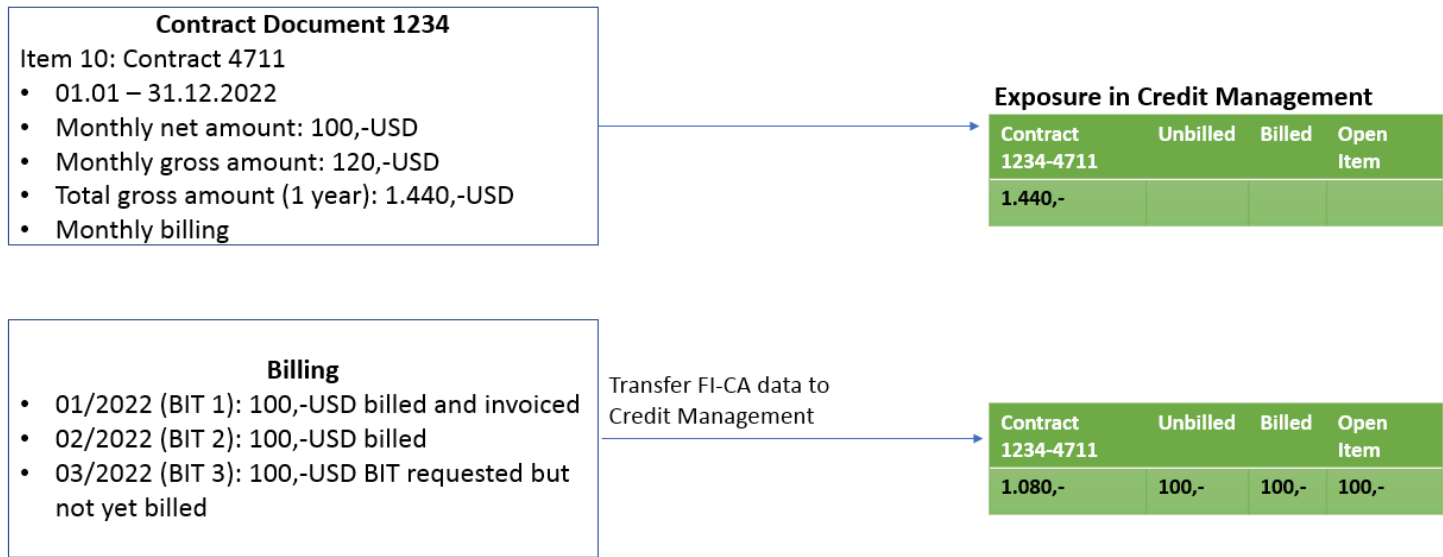
- o You must define the billable item classes that should be considered for integration with SAP Credit Management. The BIT class must contain the custom field CM_OBJECT_KEY that can be assigned in the Customizing, [Financial Accounting > Contract Accounts Receivable and Payable > Convergent Invoicing > Integration > Subscription Order Management > Credit Management > Billable Item Classes Relevant for Subscription Transactions](#).

i Note

In case you use one BIT class for usage and recurring BITs, you can use certain attributes to mark the BITs that are not relevant for the credit check. You do so in the Customizing [Financial Accounting > Contract Accounts Receivable and Payable > Convergent Invoicing > Integration > Subscription Order Management > Credit Management > Integration SOM with Credit Management - Billable Items](#).

- You must implement SAP Note [3222198](#) (FI-CA: FPCM1 does not fill Account Document Reference) to fix a known issue when uploading different billing categories in one message.

If a BIT that is relevant for integration with SAP Credit Management is created (one-off BIT, BIT from a billing plan, or recurring BIT from SAP CC), the field CM_OBJECT_KEY is filled with an indicator so that the corresponding contract exposure can be reduced when billable items are loaded to SAP Credit Management (Either via the transaction FPCM1 or via the SAP menu [Accounting > Financial Accounting > Contract Accounts Receivable and Payable > Periodic Processing > Forward Data to SAP Credit Management > Transfer Credit Data](#)). It is assumed that a credit check is executed for all relevant BITs, if a credit exposure exists for the corresponding BIT. For example, if you have a billing plan for the contract main and subitem, it is assumed that both items are sent for a credit check. If this is not the case and for some reasons only main item was credit checked, you must implement a corresponding check in the BADI BADI_FKK_SOM_CM_BIT, so that BITs from the subitem are not considered as relevant when reducing contract exposure.



- When FI-CA data is sent to SAP Credit Management, the contract exposure is reduced on the gross amount:
 $3 \times 120 = 360,-\text{USD}$
- Each BIT is transferred as the net/gross amount (Depending on whether the tax is already included in the BIT amount), along with its state, using either the “Unbilled”, “Billed”, or “Open Item” as exposure category.

Credit Management - Billing Integration

If a BIT is reversed/excepted, this is also reflected on the contract exposure. If a BIT that already has reduced the contract exposure is reversed/excepted, the contract exposure is increased again.

i Note

Contract exposure is reduced based on a gross amount because it is also created using a gross amount. However, the BIT can also be a net amount. In this case, the gross amount is calculated based on the tax information from the corresponding contract item. If BIT amount is a net amount, there will be a gap in credit exposure after data is moved from the contract to BIT

exposure. To update SAP Credit Management with the gross value for BITs, you must implement FI-CA event 4515 and calculate the corresponding gross amount for the BIT.

For more information about the general integration of FICA into credit management, refer to:

- [Integration with Accounts Receivable Accounting \(FI-AR\)](#)
- [Calculating and Transferring Credit Exposure](#)

Credit Check Reports

The following reports are delivered for rebuilding credit management data and rechecking credit for subscription transactions.

Rebuild Credit Exposure for Subscription Items

Credit Management offers the possibility to rebuild data relevant for credit management to ensure consistency and accuracy of the credit exposure. For more information about this, refer to [Rebuild Credit Management Data](#). You use the program UKM_COMMITMENT_REBUILD to rebuild data for subscription items (exposure category 120 for subscription orders or solution quotations and 130 for subscription contract), typically in the following scenarios:

- If you activate credit management in your system, where subscription transactions (subscription orders, solution quotation with subscription items, or subscription contracts) already exist, credit checks are only performed for the new transactions and not the previously existing transactions. When a credit check is performed for the new transactions and no changes are made to the old transactions, you need to rebuild your credit management data to ensure that the credit exposure is updated with the correct value for both old and new subscription transactions. After the credit exposure is updated for the old transactions, the credit status of the old transactions is not changed.
- You want to change the current credit segment assignment after you have assigned your sales area to a credit segment and created transactions using this configuration. You need to rebuild your credit management data after you change the current credit segment assignment.
- You change the Customizing for relevant transaction types and item categories so that either some transactions are no longer relevant, or some transactions become relevant.

In such scenarios, you can run this report to delete existing data in credit management and rebuild the credit exposure based on the transactions in the system. No credit check is executed during this rebuild. If the existing credit exposure does not need to be deleted, you can also run the program CRMS4_SOM_CREDIT_CMTMT_REBUILD, which only creates a credit exposure for existing subscription transactions.

→ Recommendation

You should run this report out of business hours to ensure that the rebuild data is as accurate as possible. Once the rebuild has been performed, some old transactions/items that have not gone through a credit check may also be updated to the Credit Management credit exposure. However, their credit status will remain unchanged.

The following scenarios are specific for subscription transactions and must be kept in mind:

- The rebuild is only executed for the selected business partners. In case the **Change Sold-to** change process has been executed for an existing subscription contract and you do not select the old and new business partners, the credit exposure will be created only for the selected business partner.
- Credit exposure for subscription contracts and one-off items are reduced by billable items when they are uploaded to credit management. In this rebuild scenario, the credit exposure is directly reduced if a billable item exists for a relevant

subscription item. Only those billable items that are already in the system are considered for reducing the exposure, that is, archived billable items are not considered.

For more information about this, refer to SAP Note [3245076](#)  (Integration Credit Management - Rebuild Report for Subscriptions).

Subscription Item Credit Recheck

You use the report CRMS4_SOM_CREDIT_RECHECK to perform an automatic credit check for subscription items (in a solution quotation or subscription order) for which the credit check has failed. This enables you to reduce the workload of a credit controller by scheduling an automatic credit check.

- You must specify either the credit information, such as the credit account (used as the payer in subscription item), or the credit segment that must be considered for the recheck and to restrict the results.
- The processing order of the selected items is based on the contract start date. For items without a contract start, such as a one-off item, the creation date is used. This means an item with an earlier start/creation date is processed first.
- As a result, you receive a list of all payers (business partners) and the related subscription items for which the credit check has been executed. If the credit check is successful for all items of a payer within a transaction, the system status **Credit Check Ok** is set.

i Note

If the subscription item is part of a solution quotation, only the credit status is changed. No automatic acceptance of the quotation is triggered.

For more information about this, refer to SAP Note [3253694](#)  (Credit Check - Recheck Report for Subscription Items).

Quoting and Order Capture

Solution quotations and subscription orders are used to offer customers a combination of products, product bundles, and services. Once released, they automatically trigger the creation of subscription contracts, and other follow-up documents like sales orders, or service contracts.

For more information about this, refer to [Subscription Orders](#) and [Solution Quotation in Subscription Order Management](#).

Solution Quotation in Subscription Order Management

Solution quotations are legally binding offers with fixed conditions.

With a solution quotation, you can offer your customers solutions that combine different types of products (physical goods, services, subscriptions). These combinations are modeled as product bundles in your master data. When you enter product bundles as items in a quotation, the bundle components are displayed automatically as subitems. You can also create a solution quotation as a call-off from a master agreement or you can determine a master agreement during creation of a solution quotation. The end values for prices in solution quotation items for recurring fees are calculated dependent on quantity and contract runtime. Once a quotation is released and accepted, follow-up transactions are generated automatically and subscription items in the quotation are copied into a subscription contract. When you accept a solution quotation, the subscription items for one-offs will trigger an ODI step to create a one-off BIT. Releasing the items and accepting the solution quotation will also trigger a credit check for the business partner. For more information about this, refer to **Integration with SAP Credit Management**.

You can perform these and multiple other operations on solution quotations using the **Manage Solution Quotations** Fiori app or the WebClient UI (business role S4C_SOM_REP - Internal Sales Rep. (Subscr.)). For more information about solution quotations and the related Customizing, refer to **Solution Quotation Management**.

Features

In addition to the settings for the service solution, the solution quotation also supports subscription items. The business transaction for this solution quotation is type SRVP. The secondary BUS2000265 subscription order must be assigned to the transaction type to process subscription-specific items in the solution quotation. The following subscription-specific item categories are available in addition to those available for solution quotations:

- PRQP – Subscription Item
- PRQR – Subscription Item - Recurring
- PRQF – Subscription One Off Item
- PRQM - Subscription Item in Quote for multi AVC
- Bundle explosion is supported for BUS2000155 Subscription Order Item

The predelivered pricing procedure has been enhanced with the following condition types:

- PSPB (Bundle Price)
- PSMB (Monthly Subscr Fee)
- VA00 (Variant Price)
- VASE (Variant Price – Duration)

Terms of Payment

You can assign different **Terms of Payment** for subscription and sales items, at item level, when FI-CA is active. This enables you to manage payment due dates specific to each item in a subscription contract. For subscription items, the **Terms of Payment** are first copied to the subscription contract and then distributed to the FI-CA provider contract, where a one-to-one mapping between the terms of payment and FI-CA payment conditions determines the relevant payment condition to be considered during invoicing in Convergent Invoicing. This one-to-one mapping is maintained in the Customizing **►Service ►Transactions ►Settings for Subscription Transactions ►Integration ►Integration with Provider Contract ►Map Terms of Payment to FICA Payment Conditions ►**(table TFK_MAP_ZAHLKOND). Only those terms of payment that are mapped to a payment condition are available for use in a solution quotation. For sales items, the **Terms of Payment** will be considered during invoicing in Convergent Invoicing based on the implementation of FI-CA event 8240. If this event is not implemented, the payment condition defined in the contract account is considered during invoicing in Convergent Invoicing.

❖ Example

If a company sells software (on a subscription basis) and hardware, you can set the **Terms of Payment** for the software to indicate that the payment is due immediately and for the hardware to indicate that the payment is due in one month, to allow some time to set up.

Account Assignment Objects

You can manually maintain WBS elements in the **Account Assignment** field in solution quotations via the UI and API. These can be maintained at item level for subscription items. The account assignment objects are then copied over to the subscription contract, and further distributed to the FI-CA provider contract. You can use any open change order associated with the contract to change the assigned WBS element.

Related Information

[Solution Quotation Management](#)

[Enabling Credit Check on Subscription Items](#)

Subscription Orders

A subscription order is a special sales order that you can use to sell products or product bundles that consist of subscriptions to services (usage-based charges, recurring and one-time charges) and physical goods.

Use

A new subscription contract is created automatically when a subscription order is released, whereby items with a short lifetime (sales items, one-time fees) stay in the order and items with a long lifetime (rate plans, rate plan options) become part of the contract. Whenever contract changes are required, a new subscription order (change order) is created. When the maintenance of this subscription change order is completed, these changes are transferred to the subscription contract. These contract changes can be made using change processes. For more information about change processes, refer to **Maintenance Framework for Business Transactions** and **Contract Change Processes**.

The subscription order can be used as follows:

- Subscription order for selling products and services (such as usage-based charges, recurring, and one-time charges); for the creation of a subscription contract.
- Change order for subscription contract changes.

When you confirm the subscription order, the application automatically creates or updates the relevant subscription contract, according to the status.

Features

Searching for Subscription Orders

You can search for existing subscription orders using different search parameters such as the order ID, the customer ID, the technical ID, the contract account ID, and navigate to the order.

Subscription Order Creation and Maintenance

You create a subscription order using the transaction type PRV0 and maintain the following data at header level:

- **Organizational data**

Select the sales organization, distribution channel, and division. The sales organization can be automatically determined based on the organization data and the organizational data profile for the transaction type. For more information, see **Organizational Management in Service**.

- **Customer data**

Select an existing business partner as the sold-to party, ship-to party, and payer. The sold-to party entered is also used as the ship-to party and payer by default.

Maintain and display the following data at item level:

- **Product or item data**

The products entered are represented as items in a subscription order.

- **Master agreement**

Once you assign a product, the master agreement determination is executed, and you can choose a relevant master agreement item from the pop-up window. Click on **Cancel** to proceed without choosing a master agreement. If you have assigned a master agreement, you can use the **Add from Master Agreement** button to assign additional products to the subscription order.

i Note

Only those products that are linked to the assigned master agreement are available in this action.

- Enter the product or item related data, such as the contract start date, contract duration, technical resources, and contract account.
- Some of the information entered at header level can be redefined at item level.

❖ Example

- The Ship-to Party can be redefined at item level but the Sold-to Party and Payer cannot be redefined at item level.
- The account assignment object can be redefined at an item level. The value at item level will take precedence over that defined at the header level.

❖ Note

The account assignment object is supported at the header and item level for subscription and sales items in subscription orders. However, it is supported only at item level for subscription items in solution quotation. For sales items in solution quotation, the account assignment object is not supported.

- **Plant and Storage Location**

Add and update the plant and storage location when you create a subscription order.

By default, the plant and storage location values are fetched from the existing values in the system. You can modify these values until you submit the subscription order. If you have entered undefined or invalid values, the system alerts you with an error message. When this happens, you can use the **Redetermine Plant/Location** button to replace the incorrect values with valid values from the system.

i Note

The plant and storage location values are optional for subscription items.

- **Technical resources**

Enter additional information that is necessary for the sale and subsequent operation of the service. For example, in the telecommunications industry, technical connection data such as the telephone number or SIM card number is entered. The entry of such data enables you to perform a simple search for provider orders. For more information, see **Technical Resource**.

- **Contract account**

Select an existing or create a new contract account. A contract account contains the information about which bank or card ID is to be used for incoming and outgoing payments. For more information, see **Contract Accounts in Subscription Orders**.

- **SAP Convergent Charging tables**

These tables are mapping or range tables in the SAP Convergent Charging system. They are used in the SAP CC system for pricing and in the charge plan for price determination. Originally, these tables are referenced to a product in Cross-Catalog Mapping. They can be displayed or even redefined for a subscription order item here.

- **Allowances**

Display the allowances that are activated automatically after contract creation. For more information, see **Display Allowances During Order Capturing**.

- **Sharing contract**

If the product qualifies for sharing, you can enter a reference to a sharing contract in the subscription-specific data. Once the order is copied to a contract, this reference cannot be changed in a change process.

- **Sharing group ID**

If the used product Cross-Catalog Mapping version supports shared allowances, you must enter a sharing group ID. Once the order is copied to a contract, this reference cannot be changed in a change process.

Trade Compliance Check

The integration with SAP Global Trade Services (SAP GTS) allows you to perform embargo checks, sanctioned party list screening, and legal controls for subscription orders. For more information about this, refer to [Integration with SAP Global Trade Services](#).

Reject Order

You can **Reject All Items** in a subscription order as long as the respective subscription contract items are not technically active. Furthermore, you can also **Reject** individual items in the item details. If the FI-CA provider contract has already been created with early distribution, it will be deleted, and the subscription order and contract are set to rejected. If the FI-CA provider contract has already been updated with a new item due to early change distribution, it will be reversed, and the end date of the old item will be updated.

Release a Subscription Order for Subscription Contract Creation

When the maintenance of a subscription order and all its items is finished, you can submit the subscription order for contract creation. All subscription order items are then copied to the contract and the contract is distributed.

Maintenance of Change Order

If a contract needs to be changed, a change order is created to track this change request. When the changes maintained in the change order can be released for a contract change, they are copied back to the contract.

Other functions supported by the subscription order also include updating and controlling contract changes.

More Information

The following transaction types are available for subscription orders and subscription contracts in the standard system:

- Standard Document Purpose
 - PRV0 - Subscription Order
 - PRVC - Subscription Contract Document
- Special Document Purpose
 - PRPA - Partner Agreement

- PRPO - Partner Agreement Order Item
- Document Purpose Sharing Order
 - PRCO - Sharing Order
 - PRCC - Sharing Contract

You make general settings for subscription transaction types in the Customizing for **Service** under ► **Transactions** ► **Basic Settings** ► **Define Transaction Types** ► and special settings for subscription transaction types in the Customizing for **Service** under ► **Transactions** ► **Settings for Subscription Transactions** ► **Define Settings for Transaction Types** ►.

The following item categories are available for subscription orders and contract items in the standard system:

- PROP - Subscription Order Item
- PRSP - Subscription Sales Package
- PROF - Subscription One Off Item
- PROS - Subscription Order Sales Item
- PRCP - Subscription Contract

For special purposes in the standard system, the following item categories are available:

- PRPO - Partner Agreement Order Item
- PRPC - Partner Agreement Item
- PROM - Subscription Order Item for multi AVC
- PRCM - Subscription Contract Item for multi AVC

You make general settings for subscription item categories in the Customizing for **Service** under ► **Transactions** ► **Basic Settings** ► **Define Item Categories** ►.

You make special settings for subscription item categories in the Customizing for **Service** under ► **Transactions** ► **Settings for Subscription Transactions** ► **Define Settings for Item Categories** ►.

Related Information

[Maintenance Framework for Business Transactions](#)

[Contract Change Processes](#)

[Organizational Management in Service](#)

[Technical Resource](#)

[Contract Accounts in Subscription Orders](#)

[Display Allowances During Order Capture](#)

Item Quantity in Subscription Orders

Use

The Subscription Item Quantity feature allows you to modify the quantity of a subscription item. This quantity then applies to the recurring fees and reflects it to the subscription order, subscription contract, CI billing plan, and revenue accounting.

i Note

The quantity at an item level is used for subscription products with recurring and one-time charges. It is not intended to be used for subscription products with only usage-based charges. You can control this using item categories.

Features

Enable/Disable Item Quantity based on Item Category

The item category group 'PRSR', and item categories 'PROR' and 'PROM' in subscription order and 'PRQR' and 'PRQM' in solution quotation are delivered as SAP standard to support the item quantity functionality.

The following descriptions give you an overview of the recommended Customizing for this function:

- For products with recurring charges, you must do the following to enable the item quantity functionality:
 1. In the product master data, assign the item category group PRSR to the relevant product.
 2. Setup item category determination based on group PRSR (see item category determination as discussed above).
 3. Select the **Enable Quantity** flag in the Customizing for the respective item category (see settings for PROR, PROM, PRQR, or PRQM under **Service > Transactions > Settings for Subscription Transactions > Define Settings for Item Categories**).
- For products with usage-only charges, it is not recommended that you enable the quantity field for input.
 1. In the product master data, assign the item category group PRRP to the appropriate product.
 2. Setup item category determination based on group PRRP (see item category determination as discussed above).
 3. Disable the **Enable Quantity** flag in the Customizing for the respective item category (see settings for PROP or PRQP under **Service > Transactions > Settings for Subscription Transactions > Define Settings for Item Categories**).

Special Handling for Product Bundles

For pre-configured product bundles, you can change the quantity for the main item only. All the subitems are automatically updated with the multiple of the quantities as configured in the bundle. You cannot change the quantity at a subitem level.

When a subitem is added manually in the order, the behavior of the quantity functionality at the subitem level is independent of the main item.

i Note

When a product with usage-only charges is part of a product bundle, it is recommended that you set the **Quantity Value** to 1 and select the **Fixed Quantity** option during the setup of the product bundle.

Calculate a Quantity-Based Recurring Price

The system calculates the quantity-based recurring fees (quantity * price per unit) based on the calculation type in the condition type. The following calculation types are allowed for quantity-based recurring fees:

- M Quantity - Monthly Price
- N Quantity - Year Price
- O Quantity - Daily Price

- P Quantity - Weekly Price

The relevant settings for the calculation type in the condition type should be configured in the Customizing under [Sales and Distribution](#) [Basic Functions](#) [Pricing](#) [Pricing Control](#) [Define Condition Types](#).

Support for Scaled Prices

The system uses the scale basis condition type to calculate the scaled price on the basis of quantity. The scale basis in the condition type must be set to 'C Quantity Scale' in the Customizing under [Sales and Distribution](#) [Basic Functions](#) [Pricing](#) [Pricing Control](#) [Define Condition Types](#).

Transfer Quantity-Based Price to FI-CA Provider Contract

The quantity-based price calculated in a subscription order or solution quotation is copied to the subscription contract during contract creation. Upon distribution of the subscription contract, the calculated price is copied over to the FI-CA provider contract during its creation.

Transfer Quantity-Based Price with Billing Cycle to CI Billing Plan

During distribution of the subscription contract during the creation of the CI Billing Plan along with FI-CA provider contract, the calculated price is copied over to the Billing Plan item.

Change Process: Change Quantity

You can use the process [Change Quantity](#) process to update the quantity of items contained within a subscription contract.

Related Information

[Change Quantity](#)

Subitems with Start of Business in the Future

Use

You can create a subscription contract where the subitem starts in the future while the main item starts immediately. You do so by defining a different **BUSINESSSTRT** (Start of Business) date for the subitem than that of the main item, while the **CONTSTART** for the main item and subitem remains the same. If the **CONTBILS** (Start of Billing) for the subitem is the same as **BUSINESSSTRT**, it effectively means that the subitem is only billable in the future, even if its technical validity begins immediately. For example,

	Contract Start Date	Start of Business	Start of Billing	Contract End Date
Main item	01.01.2023 08:00:00	01.01.2023 08:00:00	01.01.2023 08:00:00	01.01.2024 08:00:00
Subitem	01.01.2023 08:00:00	01.02.2023 08:00:00	01.02.2023 08:00:00	01.01.2024 08:00:00

Changing a Subscription Contract with Future Dated Subitems

When executing a change process on a subitem with start of billing and business in the future, the following scenarios are possible:

- If the business start date for the main item is in the past, the activation date of the change process defaults to the current date.

- If the business start date for the main item is in the future, this activation date defaults to the contract start date.

Purchase Order Numbers in Subscription Items

You can manually maintain purchase order numbers in the **External Reference** field in a subscription order or solution quotation, on the header and subscription item level. These external references are then copied over to the subscription contract, and further distributed to the FI-CA provider contract as a **Customer Reference**.

i Note

As long as you have maintained the purchase order number at the item level (Ship-to PO Number), it is distributed to the FI-CA provider contract by default. If the item level number is not provided, the header level (Ship-to PO Number), is distributed to the FI-CA provider contract.

You can change the existing external references in a change order, via the **Create Follow-Up Order** change process.

Related Information

[Create Follow-Up Order](#)

WBS Elements as Account Assignment Objects in Subscription Transactions

You can manually maintain WBS elements in the **Account Assignment** field in subscription orders on the UI and in the API scenario. These can be maintained on the header and item level for subscription and sales items. The account assignment objects are then copied over to the subscription contract, and further distributed to the FI-CA provider contract.

i Note

If you have maintained the WBS element at the item level, it takes precedence over the WBS element maintained at the header level (if any). If a WBS element is not maintained at the item level, then the one at header level (if any) is distributed to the FI-CA provider contract.

You can update the assigned WBS element using any open change order associated with the subscription contract or by directly executing the **Change WBS Element** change process.

Related Information

[Change WBS Element](#)

Terms of Payment in Subscription Items

Terms of Payment are considered for calculating the due date for incoming payments when the invoice document is created.

You can assign **Terms of Payment** for subscription and sales items, separate from those defined in the assigned contract account, when FI-CA is active. This enables you to manage payment due dates specific to each item in a subscription contract. For subscription items, the **Terms of Payment** are first copied to the subscription contract and then distributed to the FI-CA provider contract, where a one-to-one mapping between the terms of payment and FI-CA payment conditions determines the relevant payment condition to be considered during invoicing in Convergent Invoicing. This one-to-one mapping is maintained in the

Customizing ► [Service](#) ► [Transactions](#) ► [Settings for Subscription Transactions](#) ► [Integration](#) ► [Integration with Provider Contract](#) ► [Map Terms of Payment to FICA Payment Conditions](#) ► (table TFK_MAP_ZAHLKOND). Only those terms of payment that are mapped to a payment condition are available for use in a subscription order. For sales items, the [Terms of Payment](#) will be considered during invoicing in Convergent Invoicing based on the implementation of FI-CA event 8240. If this event is not implemented, the payment condition defined in the contract account is considered during invoicing.

❖ Example

If a company sells software (on a subscription basis) and hardware, you can set the [Terms of Payment](#) for the software to indicate that the payment is due immediately and for the hardware to indicate that the payment is due in one month, to allow some time for device set up.

Technical Resource

Definition

You can use technical resources in subscription contracts. To do this, you define which technical resources must be assigned to a subscription contract for it to be fulfilled at contract product level. During contract entry, you save the technical keys (IDs) of the resources for the contract, which assigns them to the contract. Using Business Add-Ins (BAIs), you can implement communication with an external system to reserve resources or simply check their availability during data entry. During contract distribution, the technical resources are distributed to external systems with the contract and can be used to identify the technical service within the system landscape.

Structure

Product Maintenance

To use the resource management function for a product, you must connect it to an assignment schema in product master maintenance. For more information about this, refer to [Subscription Product-Specific Attributes of Product Master](#).

Assignment Schema and Resource Types

An assignment schema represents all technical resource data assigned to a contract item. The assignment schemas assigned to products are maintained in Customizing for [Service](#) under ► [Transactions](#) ► [Settings for Subscription Transactions](#) ► [Technical Resources](#) ► [Define Assignment Schema](#) ►.

An assignment schema consists of elements representing single fields, resource types, and additional controls. These controls define the sort order and cardinality of these resource types, which is the minimum and maximum number of optional or required entries for a resource type that can be maintained or that need to be maintained.

The sort order **1** defines the leading resource for a subsequent contract. If maintenance of several resources is allowed for this resource type, the first one is the leading resource. IDs of leading resources are displayed whenever a contract is shown in a list or overview that does not allow all technical resources to be displayed at once.

A resource type represents a field within the set of technical data found in an assignment schema. Resource types can have the following categories, which are split into two groups:

- Internal
 - Local

These fields represent extensions of the contract item. This means that they are not stored in separate objects. This is the easiest means of capturing technical resource information for a contract.

- External

These technical resources can be managed in external systems but can also consist of customer objects (tables) in the CRM system. The following external categories exist, depending on how a resource is assigned:

- Resource is assigned automatically
- Resource is assigned using list selection
- Resource is assigned using ID-based value help
- Resource is assigned using advanced, extended value help

The first three of these four external categories are implemented using the following Business Add-Ins (BADIs):

- CRM_TC_TECH_RES_BADI
- CRM_ISX_TR_VALUE_HELP_BADI

You can make the relevant settings in Customizing for [Customer Relationship Management](#) under [Transactions](#) [Settings for Subscription Transactions](#) [Technical Resources](#) [Business Add-Ins \(BADIs\)](#).

These BADIs are also used in other contexts, such as validations, and are explained in more detail below.

i Note

All implementations for these BADIs are filtered by the resource type. This means that a BADI implementation has to be performed for each resource type where BADI functionality is required.

Validating Values for Resource Types

The **Unique** field controls the system checks for whether the resource is unique. Since such a check can only be performed for resources in Subscription Order Management, this indicator can only be maintained for resources that belong to the **local** categories.

The **Allocation Check Time** field controls when the system requires an entry for a specific resource type (during order capture or only before activation).

Syntactic and semantic checks are defined by the customer in the validation BADI CRM_TC_TECH_RES_BADI. These validations are performed as soon as an entry is made. This means that for a resource type whose **Allocation Check Time** is **before activation**, these checks are also performed if a value is already entered for this **resource type** during order capture.

The methods to be implemented in the BADI CRM_TC_TECH_RES_BADI for these checks are defined by the resource category:

- The method TR_ID_CHECK is used for the internal categories.
- The methods TR_BEFORE_ASSIGN and TR_BEFORE_UNASSIGN can be used for the external categories if the customer wants to implement additional checks for the resources provided by the external systems.
- Some final checks can be performed in the method TR_BEFORE_SAVE.

Assigning Values to External Resources

Assigning values to externally managed resources require a project implementation that uses pre-defined spots in the SAP standard process. Different BADI methods are to be used according to the external resource category to which a resource type belongs:

- The method TR_BEFORE_ASSIGN in the BAdI CRM_TC_TECH_RES_BADI has to be implemented for automatically assigned external resources.
- The method TR_GET_LIST in BAdI CRM_ISX_TR_VALUE_HELP_BADI is relevant for resources assigned using list selection.
- The method TR_GET_VALUE_HELP_BY_ID in the BAdI CRM_ISX_TR_VALUE_HELP_BADI is relevant for resources assigned using ID-based value help.
- Resources entered using advanced value help require their own customer implementation of UIs and search help logic.

Depending on the business requirements and the way the external resource management systems work, using these methods and customer development during order capture may not be sufficient in all cases. For example, assigning a value managed in an external system during order capture may only result in this value being blocked or reserved within that system. The actual assignment of the external resource is subject to a successful order capture or an activation process. If order capture is successful, the method TR_BEFORE_SAVE in the BAdI CRM_TC_TECH_RES_BADI can perform the final assignment of the resource. The external systems need to be updated for activation processes.

Integration

Maintenance of Technical Resources in Subscription Orders and Contracts

Maintain or display technical resources in the assignment block **Technical Resources** of the **Subscription Order Item** or **Subscription Contract** detail screen. The list of technical resources is sorted according to the settings made in the resource assignment schema for the corresponding product.

Multiple resources of the same type are distinguished using the column, **Slot**. The resources are numbered sequentially starting from 1 for each resource type. The resource with slot 1 for the resource type with sort number 1, normally the first resource in the table of technical resources, is the 'leading' resource for this contract. The leading resource is the single resource that is shown in search results containing this contract.

While capturing subscription orders, the current status of the technical resources entered is displayed in the **Items** assignment block. The status is **Incomplete** as long as mandatory resources are missing or resources entered fail. The status switches to **Complete** once the resources are complete according to the assignment schema settings.

Example

Mobile phone

A telephone number is mandatory for a mobile phone contract. Optionally, one or two email addresses can be provided to contact the customer if problems occur. In this example, the schema has two lines:

- The first line with the resource type telephone number with a minimum and maximum of 1. In this case, the resource type is mandatory without the option to add further telephone numbers.
- The second line with the resource type email address with a minimum of 0 and a maximum of 2. In this case, the resource type is optional with the option to add up to the two email addresses.

Related Information

[Product Master](#)

Configuring Subscription Items using Variant Configuration

You can use this function to configure products in subscription items. For more information about this, refer to [Advanced Variant Configuration](#).

You can further create and change subscription items with pre-configured cross-plant subscription products by using material variants. For more information about cross-plant material variants, refer to **Creating Cross-Plant Material Variants**.

Related Information

[Advanced Variant Configuration](#)

[Creating Cross-Plant Material Variants](#)

Redefinition and Assignment of SAP CC Tables

Use

Convergent charging tables (mapping or range) are tables in an SAP CC system that are used during the charging and rating process in SAP CC. You can reference the identifier of a table in a charge plan using a parameter, where the parameter value becomes the default value. When you create a provider contract in SAP CC, you can use a different parameter value (different table identifier).

In Subscription Order Management, charge plan parameters can be overwritten in Cross-Catalog Mapping of a subscription product. For parameters of the type **mapping** or **range** table, you can create a new table and assign its identifier to a parameter value. When the subscription product is used in a subscription order, such a table parameter can be overwritten. Finally, during the contract creation process, a specific parameter value (mapping or range table identifier) is determined and sent to the SAP CC system.

You can maintain table parameters using the **Convergent Charging Tables** assignment block in the subscription item. For more information, see **Tables** and **Convergent Charging Tables**.

Prerequisites

You must assign a charge plan that contains table parameters in Cross-Catalog Mapping, for the assigned product.

Features

Subscription Order

The default table that is defined in the Cross-Catalog Mapping of the product is copied as the default table in the subscription order item. If an object-specific table (redefined or assigned) is defined in the product, it is copied as a default table into the subscription order item.

The table parameters of the subscription order item can be modified.

Assigned Table

An assigned table can be a table from the Convergent Charging (CC) standard catalog, an agreement table, or a table from the subscriber account assigned in the subscription order. A table from the subscriber account is available only if the account split functionality is not activated and the subscriber table must be object independent, that is, tables with object-specific usage = **Yes** are not allowed.

Redefined Table

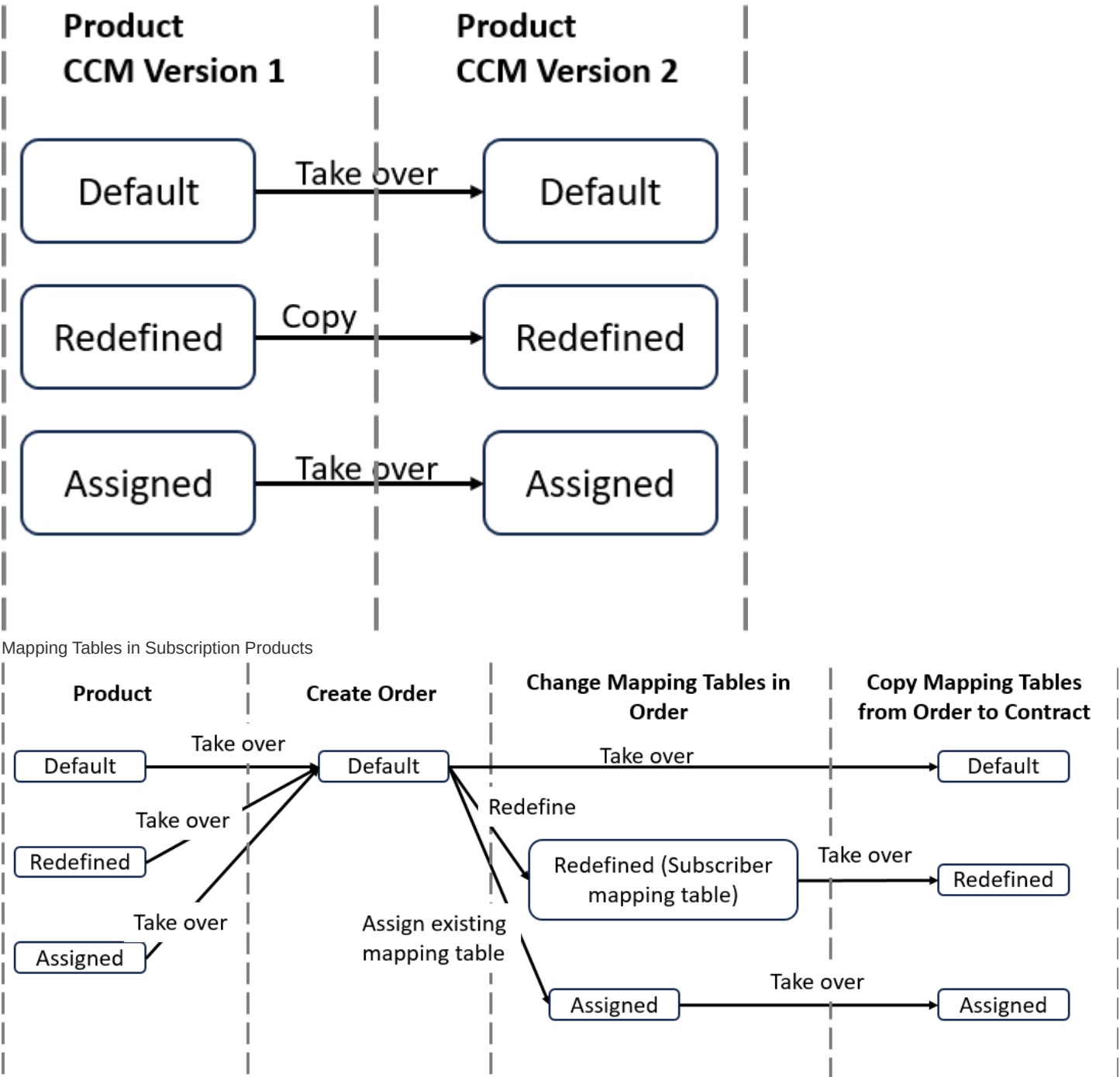
If a redefined table is created in the subscription order, it is always stored in the subscriber account for the SAP CC system. You can change the contents of a redefined table.

A redefined table always has the object-specific usage = Yes, that is, it can only be used in the current context. It cannot be assigned to another object.

Subscription Contract

The table parameters are copied from the subscription order (default and object-specific table) and cannot be modified; however, the content of a redefined table can be changed.

The usage of the convergent charging tables is shown in the following graphics:



Mapping Tables in Subscription Orders and Contracts

Related Information

[Tables](#)

Display Allowances During Order Capture

Use

You can display allowances to be activated after submitting a subscription order during maintenance. As this automatic activation of allowances may depend on complex conditions containing any contract details, it helps to keep track of the current offer and observe any changes to it during order capture.

This is a display function for allowances that are activated automatically if the current subscription order is submitted without further changes. For more information, see Allowances.

Prerequisites

The Customizing for the item category is maintained to show allowances in the Customizing for [Service](#) under [Transactions](#) [Settings for Subscription Transactions](#) [Define Settings for Item Categories](#).

Features

For subscription orders supporting the use of allowances, the [Allowances Included](#) assignment block is available at item level. The list is populated by checking the conditions maintained in the allowance definition group assigned to the current product. The list shows all allowance definitions that are available for the current type of order (sales order or change order) and whose conditions are met with the current order data.

The list consists of the following fields:

- Allowance definition ID
Identification of the allowance
- Product and charge plan ID
The product and the corresponding charge plan the allowance will be created for
- Technical resource type and ID, service ID
Identifies the service for which the allowance is activated

Related Information

[Allowances in Subscription Order Management](#)

Discount Assignment

Use

Billing discounts are defined, calculated, and posted in invoicing documents in Convergent Invoicing. In Subscription Order Management, these discounts can be assigned to subscription contracts. Together with the contract, discount assignments are replicated to SAP CI and then calculated during the billing and invoicing steps for this contract in SAP CI. In SAP CI, discounts can be volume-based, and they can be related to a product, contract, contract account, or discount group.

❖ Example

Example for a volume-based discount: A specific discount is only applied when a specific usage was reached.

Prerequisites

The item category used in the corresponding product has to be defined to be used for discounts in Customizing. Otherwise, the billing discounts cannot be handled during order capturing.

You make the settings for this in Customizing for **Service** under **► Transactions ► Settings for Subscription Transactions ► Define Settings for Item Categories ►**

You have either assigned billing discounts to the subscription product that is used for order capturing or you have implemented the enhancement spot CRM_ISX_BT_X_DISCOUNTS.

Features

Discounts can be assigned to subscription products. For more information, see **Discounts and Charges**. When this product is used in a subscription order or subscription contract, the defined discount is assigned to the item.

Discounts can also be determined in the enhancement spot CRM_ISX_BT_X_DISCOUNTS independently of the discount assignment to a subscription product. For more information, see the **Enhancement Spot** documentation.

Activities

Discount Assignment

Discounts can be found in the assignment block **Discounts** on the item overview page. A discount will have the contract relevance set to either **Always Relevant** or **Optional**. This is defined in product assignment or in the enhancement spot.

A discount that is always relevant is added to the item automatically and cannot be removed. An optional discount can be added to the item by choosing **Add**. It can be removed by choosing **Delete**. If the item category is not defined to be used for discounts, the assignment block does not appear.

Discount Duration

A discount has a **Valid From** and a **Valid To** date. The default valid-from date is the time slice start date of the item. This means that if a discount is applied in the subscription order, it is the contract start date. If it is applied in a change order, it is the activation date. The valid-to date of the discount is unlimited as a default setting (technically it is 31.12.9999 23:59:59). Another duration can be defined in the product assignment or in the enhancement spot.

❖ Example

A discount should only apply for the first three months of a contract.

The following units for discount duration exist:

- Hour: The end of duration is defined by the time stamp. For example, if the discount is valid for 3 hours and the contract starts at 14:50:11, the discount ends at 17:50:11.
- Day: The end-date of the discount is the end of the day plus one second.
- Month: The end-date of the discount is the end of the day.

Recommendation for Discount Determination

Define your discounts either in the subscription product or in the enhancement spot CRM_ISX_BT_X_DISCOUNTS. If you implement the enhancement spot, you must not define discounts in the product.

i Note

These billing discounts are only calculated and posted in SAP CI. In Subscription Order Management, these discounts are not calculated and are not part of the prices displayed. You can only display the discount assignments (discount key and discount description) in a subscription contract.

Related Information

[Discounts and Charges](#)

Account Split

Use

Typically, business partners are mapped to subscriber accounts in SAP CC in a 1:1 relationship. However, you should be able to support B2B customers where one subscriber account is mapped to multiple provider contracts. This leads to performance degradations in SAP CC, since parallelization is not possible within one subscriber account. The functionality of account split successfully counters the effects of this 1: many mapping. When a provider contract is transferred to SAP CC, the business partner is used as subscriber account in SAP CC. With account split, a new alternative Subscriber Account (aSUAC) is created for each provider contract in SAP CC only. This aSUAC is not a real business partner in SAP S/4HANA. Thus, each provider contract has its own subscriber account, and the workload can be better distributed on the SAP CC raters.

If you redefine a mapping or range table in the order a subscriber specific table is created in SAP CC for the business partner since the aSUAC is not yet known. When the subscription contract is transferred to SAP CC your ODI schema must include ODI step STCO (Copy Subscriber Table for aSUAC). In this step the subscriber specific table is copied to the new aSUAC.

If you assign a table in the subscription order, you can either use a catalog table or an agreement table created for this business partner.

i Note

Account split is not available for a sharing order and a subscription order with reference to a sharing contract.

Prerequisites

You must consider the following prerequisites for account split:

- This function is supported only with SAP CC release 5.0 SP04.
- The catalog web service version 3 (namespace: <http://catalog.v3.highdeal.com/>) must be used to connect the SAP S/4HANA system with SAP CC. A logical port for the consumer proxy CO_CRMS4_SOM_CC_V3_CATALOG_SER must be created in the transaction SOAMANAGER.

Determine Account Split

To be able to create multiple provider contracts based on a business partner, there must be a logic to decide when an account split is required. You can implement it using one of the following options:

- Mark an item category as split relevant from the Customizing for [Service](#) under [Transactions](#) > [Settings for Subscription Transactions](#) > [Define Settings for Item Categories](#).

Set a flag in the column [Split](#) if a split must be executed for an item category.

- Implement BAdI CRMS4_SOM_BADI_CC_ACC_SPLIT

The indicator for the account split is only determined while creating a subscription order and is then copied to the subsequent subscription contract. In all change processes, no new determination is executed.

Contract Handling

When a subscription contract is transferred to FI-CA and CC, a new alternative subscriber account is created in case of an active account split. This subscriber account is a concatenation of a business partner number and provider contract number. However, this aSUAC is not a real business partner and is used only in the SAP CC provider contract as a subscriber. The distribution of the new aSUAC to SAP CC is handled by the FI-CA master data distribution, which sends the subscriber and provider contracts to SAP CC.

Related Information

[Convergent Charging Tables](#)

[Agreement Tables](#)

Auto Renewal

Use

During order capture you can setup a pre-agreement, for a subscription contract to continue beyond the current expiration date. It is a mechanism by which the execution of the change process [Contract Extension](#) can be performed without user intervention.

For subscription contracts with an expiration date, that is, [CONTEND](#), auto-renewal would mean the extension of the expiration date based on the selected extension period. Previously, extension of such expiring contracts would have required user intervention.

For subscription contracts without an expiration date, auto-renewal would mean the extension of the [End of Term](#) or [End of Runtime](#), that is, [ISTRUNTMEND](#). Although such subscription contracts would not have ceased or expired and would still be available for use, extension of such contracts is necessary in specific situations (such as when Revenue Accounting and Reporting is involved), and which works with a defined [End of Runtime](#).

For phased subscription contracts, auto renewal should be configured on the last phase.

You can modify the auto renewal start period and extension period using the [Subscription Contract Change Process \(Asynchronous\)](#) SOAP API.

Prerequisites

- You must include the following in the date profile assigned to the item category

Duration [AR_EXTENSION](#) - The duration for which the subscription contract should be extended

Duration [AR_PERIOD](#) - The duration (for example, 5 days or 2 weeks) before the contract end or end of term when the auto-renewal should be processed

Date Type **AR_START** - The date on which the auto-renewal process shall be triggered. Calculated as contract end or end of term – **AR_PERIOD**

You can make this setting under **Service > Basic Functions > Date Management > Define Date Profile**

- You must include the action definition **CONTRACT_AUTO_RENEWAL** in the action profile assigned to the item category(s) used in subscription contracts.

You can make this setting under **Service > Basic Functions > Basic Functions > Actions > Change Actions and Conditions > Define Action Profiles and Actions**

- You must schedule a batch job for the selection report **RSPPFPROCESS** which will trigger the execution of the action **CONTRACT_AUTO_RENEWAL** once the start condition is satisfied
- You must activate the change process **EXTENSION**.

You can make this setting under **Service > Transactions > Settings for Subscription Transactions > Actions > Change Actions and Conditions > Define Action Profiles and Actions > Contract Changes > Settings for the Maintenance Framework (BTMF)**

Subscription with Recurring Delivery of Physical Products

Handle the periodic delivery of physical goods or consumables related to a subscription, based on a predetermined delivery schedule.

Subscription Order Management enables you to handle subscriptions that include recurring delivery of physical goods or consumables using product bundles or Advanced Variant Configuration (AVC) BOMs. The main item is always a subscription product. The subitems within the product bundle can be subscription, one-time sales, and recurring sales items. Consider the following examples:

- Industrial printers are used to put important markings such as barcodes and QR codes on product packaging. Manufacturers and distributors can provide this equipment on a subscription basis to packaging companies. The subscription can include a payment model based on the number of printed sheets and can also include the regular replenishment, or recurring delivery, of raw materials such as high-temperature ink.
- In the medical industry, hospitals use imaging equipment such as MRI and CT machines. Manufacturers and distributors can provide this equipment on a subscription basis to hospitals. The subscription can include a payment model based on the number of scans and can also include regular replenishment of consumables such as contrast dye solutions and imaging film.

Use

When you create a subscription order or a solution quotation using a product bundle with a recurring sales item, the system creates delivery plan lines for the recurring sales item based on the default recurring period. You can view these delivery plan lines on the item details page in the **Delivery Plan** section. The recurring period is derived from the default date rule set in the delivery plan Customizing. You can change this period as needed, such as from monthly to quarterly or from quarterly to yearly. The number of delivery plan lines changes based on the selected recurring period. The system calculates the estimated total quantity as item recurring quantity multiplied by the number of delivery plan lines. You can modify the quantity of the individual delivery plan lines as needed until you submit the subscription order or accept the solution quotation. When you change the quantity per delivery plan line, you must ensure that the planned total quantity remains equal to the estimated total quantity. The **First Delivery Date** is the delivery date of the first delivery plan line. The system calculates the delivery date for every subsequent delivery plan line based on this date. For example: if the first delivery date is 1st January 2025 and the recurring period is monthly, then the delivery date for the second item is 1st February 2025, for the third is 1st March 2025, and so on.

The system initially creates the delivery plan lines with the status **Open**. When you submit the subscription order or accept the solution quotation, the delivery plan lines are copied over to the subscription contract and their status is updated to **Copied to Contract** in the order or quotation. When the subscription contract includes a product bundle with a one-time and a recurring sales item, both these items are visible on the **Manage Subscription Contract Lifecycle** app. You can navigate to the **Manage Subscription Contract Item Details** app using the  icon. There you can see the delivery plan details in the **Delivery Plan** and **Delivery Plan Lines** sections.

As the delivery plan lines in the subscription contract are processed and the sales orders are created, their status is updated from **Open** (the initial status) to **Processed** in the contract. You can create sales orders for recurring delivery sales items eligible on a particular day or during a specific period using the report CRMS4_R_DLVPLAN_PROCESS (**Process Delivery Plan Items**). You can configure this report to run periodically as a background job. Although this report runs in the background, you can also trigger it manually for specific dates, date range, or sales items in Customizing under  **Service**  **Transactions**  **Settings for Subscription Transactions**  **Delivery Plan** .

i Note

You can create subscription orders with recurring sales items using either the **Manage Subscription Orders** app or the **Subscription Order (Version 0002)** OData API. Pass the delivery plan recurring period and the first delivery date at the item level. If you don't provide the recurring period, the system uses the value configured in the Customizing. If you don't provide the first delivery date, the system uses the start of timeslice. You must provide delivery plan items in an 1:n association to the item. If you don't provide the delivery plan items, the system generates them based on the recurring period rule and the first delivery date.

i Note

If you create a subscription order or solution quotation with a product bundle containing a one-time and a recurring sales item, both these items are copied into the subsequent subscription contract. However, if your subscription order or solution quotation contains the system copies this item directly into a sales order. It doesn't appear in the subscription contract.

If you implement a change process on a subscription contract with recurring sales items, you must ensure that the activation date for the change is **after** the last processed delivery date. This is because the previous delivery plan lines aren't impacted by your change.

Configuration

The business object type BUS2000173 (**Recurring Sales Item**) represents physical products related to a subscription that need to be delivered periodically. It's assigned to the item categories PCRS (**Subscription Contr Item for Recrrg Sales**), PORS (**Subscription Order Item for Recrrg Sales**), and SQRS (**Subscription Item in Qtan for Rec Sales**) in the Customizing activity **Define Item Categories** under  **Service**  **Transactions** .

You can make the following settings for the delivery plan in the Customizing activities under  **Service**  **Transactions**  **Settings for Subscription Transactions** .

- Define a delivery plan type in the Customizing activity **Define Delivery Plan Types**. Assign the date rule (**Rule Period**) to determine how the system creates delivery plan lines, for example, monthly, quarterly, or yearly.
- Assign the delivery plan type to a combination of transaction type and item category in the Customizing activity **Assign Delivery Plan Type to Item Category**.

Constraints

The following features are currently not available for recurring delivery of physical products:

- You can't create a solution quotation with a recurring sales item using an API. You can only do so in the user interface.
- The CRMS4_R_DLVPLAN_PROCESS ([Process Delivery Plan Items](#)) report can only process a single delivery plan line in each run.

Contract Accounts in Subscription Orders

Use

To control processes in invoicing, contract accounts receivable and payable, taxation, and correspondence, a contract account must be assigned to a subscription order. You can enter payment-related information in the contract account, such as an alternative payer or payee, or which bank or card ID is to be used for incoming and outgoing payments.

During the billing process items can either be combined in a single invoice or listed separately. This is defined in part by whether items are only assigned to one contract account or several contract accounts.

Prerequisites

The following prerequisites must be considered when using contract accounts:

- While creating a subscription order, you must assign a contract account to each item in a subscription order.

Features

Contract accounts are handled explicitly in subscription orders. That is, from the contract account and payment data views, you can see which contract account is used, including the contract account ID, description, payment data, and other attributes. The system also provides a list of all other subscription contracts that are related to the relevant contract account, in the [Contract Account](#) block. According to customer requirements, you can use this information to decide which contract account should be used for a new subscription order. If a suitable contract account does not exist, you can also create a new contract account. You do so by entering not only the payment data for incoming payments, but also all other information that defines a contract account.

You can assign different contract accounts at order item level. In case only one contract account exists for the payer, it will be assigned by default. An overview of all contracts assigned to the contract account is displayed when you access the contract account.

Payment Card Processing

You can save one or more payment cards in a subscription order.

When you create a subscription order, you can add an existing payment card from the payer (business partner) master data. You can also register a new payment card that isn't registered to the payer. If references to multiple cards is maintained in the system, the subscription items inherit the card with no limit.

You can replace this default card with another card registered for the payer. Use the [Add Payment Card](#) button in the [Payment Method](#) assignment block to do this. You can also register a payment card while creating a subscription order by using the [Register Payment Card](#) button in the [Payment Method](#) assignment block. The newly registered card doesn't need to be linked to the associated business partner. If you register a payment card at the header level, the system copies it to the items. You can also register a payment card specifically at the item level until you save the order. Registering a new card creates a unique token number for the card. The token number differs from the reference number, which is created when you add a payment card. If you register a new payment card, the system uses the token number and leaves the reference number field blank.

i Note

If you add multiple payment cards for a transaction, only one card can have an unrestricted card limit. You must define a card limit for all other listed payment cards.

The payment card information you provide in the subscription order or solution quotation is copied to the follow-up documents – subscription contract and sales order. The system distributes the payment card information on the subscription contract to the FICA provider contract. The reference number or token number is mapped to the incoming payment card of the provider contract. The default payment method maintained in the customizing is mapped to the incoming payment method. The Use Payment Data from Provider Contract flag is also checked. You can update the payment card assignment in a specific contract by using the [Create Follow-Up Order](#) change process.

Prerequisites

You need an active integration with the digital payments add-on. For more information, refer to: [Integration with SAP Digital Payments Add-On](#).

You maintain payment cards for the relevant transaction types in Customizing: [Service](#) > [Transactions](#) > [Basic Settings](#) > [Define Transaction Types](#).

Related Information

[Create Follow-Up Order](#)

Archiving Subscription Orders

Use

If a subscription order is completed, you can archive the associated document. The archiving object CRM_PRVO is available for doing so.

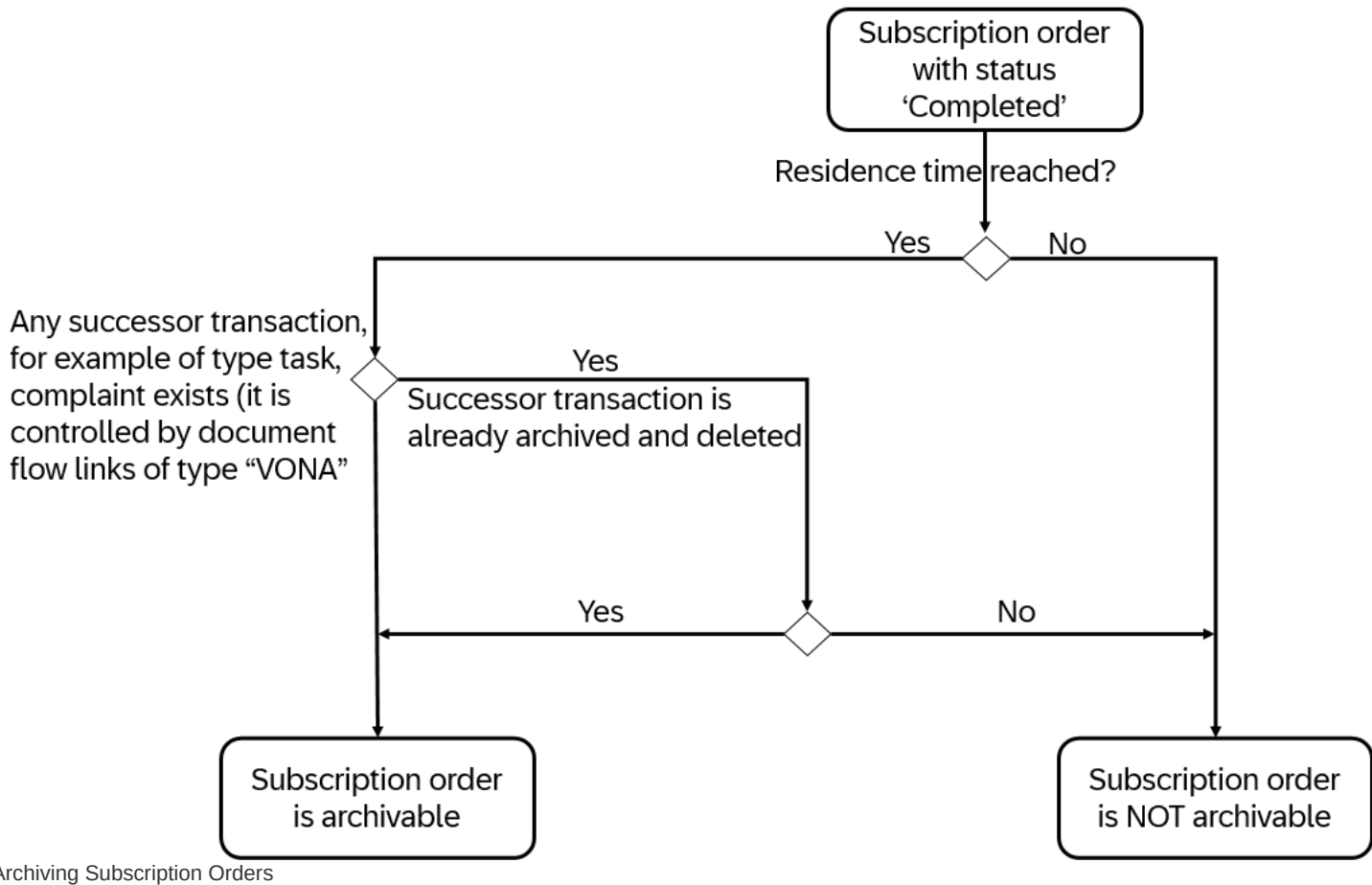
Subscription orders can be archived independently from subscription contracts. This means that the system does not check whether the associated subscription contract document or the associated contract time slice can be archived. It does not check whether the preceding time slice in the subscription contract document can be archived for subscription change orders.

Prerequisites

If you want to archive a subscription order, the following prerequisites must be met:

- The **Completed** status must be set at header level. The residence time (number of days after the last change to the subscription document) must have elapsed.
- Subsequent documents (such as tasks or complaints) must already be archived and deleted from the system.

If you have set up the ILM integration for the CRM_PRVO ILM object, your ILM rules must also allow the archiving. For more information, see [ILM Objects in Subscription Order Management](#).



Archiving Subscription Orders

Related Information

[ILM Objects in Subscription Order Management](#)
[Data Management](#)

Subscription Contracts

A subscription contract manages the long-term agreement for a technical service that the provider company offers to the customer.

Use

It is represented by a main subscription contract item within a subscription contract document. A subscription contract document contains one or more subscription contracts, where the individual subscription contracts are not dependent on each other.

A typical subscription contract contains important information, such as technical resources (technical IDs/references of services), contract history (an overview of all the changes related to the contract and previously created orders), subscription contract options (subitems of a subscription contract, containing their own data), and business partner data (for example, alternative addresses for specific address types).

A subscription contract is closely associated with a subscription order or a solution quotation, since a subscription contract document is automatically created when subscription orders are successfully released, or solution quotations are accepted. Contract-relevant data is copied into the contract document. Subscription order/solution quotation items become subscription contract items in the subscription contract document. You can only make changes to a subscription contract using change processes.

You can find a complete overview of a subscription contract using the [Manage Subscription Contract Lifecycle](#) app and, from there, navigate to the [Manage Subscription Contract Item Details](#) app for further details of a specific contract item.

Features

You can use the following functions:

- Search for subscription contracts
- Create subscription contracts as follow-up documents of subscription orders
- Change subscription contracts

You can use the change processes delivered as standard by SAP and adapt them to your requirements. You can also implement your own change processes. For more information about using the maintenance framework to develop your own change processes, see **Maintenance Framework for Business Transactions**. When you make changes to contracts, change fees, or existing subscriptions can change. You can view these fees on the UI for subscription order, either at item or at header level.

- Early contract distribution

When a subscription contract is created or changed with activation date in the future, it will be distributed to FI-CA with an Inactive status. Once the activation date is reached, the contract will be distributed to FI-CA in active status and to SAP Convergent Charging. The status of the FI-CA contract will remain active in case of early distribution of contract changes. This feature is enabled only when the ODI step with step type **PCER** is determined.

- You can use the [Reject All Items](#) button to reject all items of a subscription contract document. For more information, refer to **Reject Items**.

Transaction Types

The following transaction types are available for subscription orders and subscription contracts in the standard system:

	Transaction Types
Standard Document Purpose	PRVO - Subscription Order PRVC - Subscription Contract Document For more information about this, see Business Transaction .
Special Document Purpose	PRPO - Partner Agreement Order PRPA - Partner Agreement PROM - Subscription Order Item for multi AVC PRCM - Subscription Contract Item for multi AVC
Document Purpose Sharing Order	PRCO - Sharing Order PRCC - Sharing Contract

You can make settings for subscription transaction types in Customizing for [Service](#) under [Transactions](#) [Settings for Subscription Transactions](#) [Define Settings for Transaction Types](#).

You can make settings for subscription contracts, such as contract change processes and pricing for contract changes in Customizing for [Service](#) under [Transactions](#) [Settings for Subscription Transactions](#).

Document Distribution

The document distribution framework provides functions for document distribution to downstream systems, that can be controlled using customer exits and scheduling mechanisms. You can make settings for the document distribution framework in the Customizing for [Service](#) under [Transactions](#) > [Settings for Subscription Transactions](#) > [Document Distribution](#) .

Situation Handling

This app uses Situation Handling, which automatically informs users responsible about matters that require their attention. For more information, see [Situation Handling](#).

Related Information

[Subscription Orders](#)

[Business Transaction](#)

[Maintenance Framework for Business Transactions](#)

[Price Modelling](#)

[Contract Change Processes](#)

[Reject Items](#)

[Subscription Contract Item Due to Expire](#)

[Situation Handling](#)

Replication of Subscription Contracts to FI-CA and SAP CC

Use

Replication of Subscription Contracts to Provider Contracts in FI-CA

The Subscription Order Management system replicates the subscription contract to the FI-CA system using the Order Distribution Infrastructure (ODI).

The ODI calls the Common Object Layer (COL), which ensures that the provider contract is created in the FI-CA system and in the SAP Convergent Charging (SAP CC) system.

Replication of Subscription Contracts to SAP CC

With this function, you can use a set of Customizing settings with the implementation of Order Distribution Infrastructure (ODI) step types for distribution to FI-CA. You can use the ODI to set up a typical scenario during the implementation of the distribution of contracts in a system landscape with an SAP Convergent Charging system for rating and a Convergent Invoicing system for billing/invoicing.

The distribution of contract data to FI-CA is relevant for the following processes:

- Create
- Change
- Cancel

Creating and changing contract data is relevant for the activation of ODI. The cancellation of contract data is relevant for the deactivation of ODI.

Replication of Subscription Contracts for Revenue Accounting and Reporting

This functionality is part of the integration of [SAP Revenue Accounting and Innovation Management](#) (SAP BRIM) into [SAP Revenue Accounting and Reporting](#). The goal of this integration is to enable the accounting and reporting of customer revenues that are managed by SAP BRIM solutions as operational systems in [SAP Revenue Accounting and Reporting](#).

The creation or change of revenue accounting items is triggered in Convergent Invoicing. To enable SAP CI to transfer the correct data to the [SAP Revenue Accounting and Reporting](#), the required information has to be sent from the subscription contract to the FI-CA provider contract. Therefore, the replication of contract data using the [Common Object Layer](#) (COL) is enhanced. The enhancement includes price data, contract term data, service IDs, change reason, and additional information to group all items across different components (SOM, FI-CA, CI, SD, RA).

If you want to change existing subscription contracts, you must perform contract change processes. The definition of change processes contains the CHANGE_REASON field. This field is transferred to FI-CA and the [SAP Revenue Accounting and Reporting](#) to improve understanding when reorganizing the revenue accounting items after a contract change.

More Information

You make the Customizing settings for the replication of subscription contracts to FI-CA and SAP CC under [►Service► Transactions► Settings for Subscription Transactions► Document Distribution► Define Settings for Document Distribution►](#).

The change reason for subscription contract replication for revenue accounting and reporting has to be defined in Customizing for [Service](#) under [►Transactions► Settings for Subscription Transactions► Contract Changes► Settings for the Maintenance Framework \(BTMF\)►](#).

Related Information

[ODI Framework](#)

[Standard Steps](#)

Allowances in Subscription Contracts

Use

You can display active or expired allowances for a contract and check counter values for active allowances.

Prerequisites

To display allowances created in SAP CC either automatically during contract creation and contract changes or manually in the allowance refill, the item category is maintained to show allowances in the Customizing for [Service](#) under [►Transactions► Settings for Subscription Transactions► Define Settings for Item Categories►](#).

Features

For contracts supporting allowances, the [Allowances](#) assignment block is available. The list is populated by requesting the information from SAP Convergent Charging (SAP CC). Initially, the list shows all currently active allowances and allowances to be activated for this contract. Choose [Show All](#) to also display expired allowances. Choose [Hide Expired](#) to switch back.

The [Refresh](#) button triggers an update of the list. This may be necessary to display allowances that have just been activated manually or to observe changes to counters.

The list consists of the following fields:

- Allowance definition ID, Allowance ID, Allowance plan

Identification of the allowance

- Counter description and value

Counter identification and current counter value

Per default, all counters are displayed. If you want to display only a number of counters, you must define a view ID in the Business Add-In (BAI) method in Customizing for [Service](#) under [Transactions > Settings for Subscription Transactions > Integration > Integration with Convergent Charging > Business Add-Ins \(BAIs\) > BAI: Display SAP CC Counters in Subscription Contracts > SET_VIEW_ID](#). In addition, maintain the counter descriptions for the counters you want to display in Customizing for FI-CA.

You can also define the output format of the counter values in the Customizing for FI-CA.

i Note

The counter for inactive and expired allowances can contain incorrect values, for example, counters of an allowance to be activated may still need to be defaulted. The values of shared allowances cannot be displayed.

- Validity dates

- Status

Active – the allowance is currently active

Inactive – the allowance is not yet in use

Expired – the validity period of this allowance has ended

- Parameters

Choose [Display Parameters](#) to display the allowance parameters

- Source Contract

Identifies the item to activate the allowance. This can be useful when multiple services exist in one contract (for example, partner cards)

Related Information

[Allowances in Subscription Order Management](#)

[Refill Allowance](#)

[Display Charging Data for Subscription Contracts](#)

Equipment Integration

In response to changing customer behavior, industries that traditionally sold hardware and/or related products are now evolving their business models to become a part of the subscription-based economy. Models such as DaaS (Device-as-a-Service) have come into the picture, where service providers take a typical hardware device (such as a laptop, desktop, tablet, printer, or mobile phone), bundle it with a variety of services and software, and offer it to customers for a recurring subscription fee

Example

- A company that offers laptop devices, IT support, and software (including 3rd party ones) bundled into a subscription on a fixed recurring price per device
- A company that sells exercise equipment outright but offers related training classes on a subscription basis

To support such models where the subscription is either for the hardware or dependent on the hardware, an integration between the equipment master and the subscription contract has been provided. The ability to bundle different kinds of products (hardware, service, and subscription) was already available and hardware products resulted in the creation of SD sales orders from which downstream processes such as Delivery, Goods Issue, etc. could be carried out. This has been taken further now, where if the hardware results in the creation of an equipment, then such an equipment can be linked back to the related subscription contract. With this integration, some dependent processes can be built. For example, activation of subscription and start of billing will occur only after the hardware is delivered to the customer and installed.

Prerequisites

The relevant item category must be marked as relevant for equipment using the **Obj List** flag in the Customizing **►Service► Transactions►Settings for Subscription Transactions►Define Settings for Item Categories►**.

Features

- Maintaining equipment IDs on subscription contract

Equipment IDs for a subscription item in subscription order or solution quotation can be maintained under the Object List. Additionally, you can search for equipment based on their **Equipment ID**, **Serial Number**, and **Reference Product ID**. The maintained equipment IDs are copied to the subscription contract. If the equipment IDs are unknown during the creation of subscription order or solution quotation, the subscription contract will be created without the equipment IDs and the contract will remain inactive. Equipment IDs that are determined after downstream processes, such as delivery of the hardware, must be maintained on the related subscription contract. The **Maintain Equipment ID** change process is available to support the maintenance of equipment IDs later on in the process.

i Note

This process can be used only if the contract is inactive or the preceding document (solution quotation) is not open, and distribution is not in progress.

- Auto-update of equipment IDs based on serial number assignment

A sample implementation of how equipment IDs can be automatically maintained on a subscription contract based on serial number assignment in the corresponding delivery has been provided. If equipment master is created and activated when serial numbers for a product are created, it is possible to identify the relevant equipment IDs during the delivery. By tracking back from the delivery, it should be possible to identify the relevant sales item in a subscription order or solution quotation. If the products are structured in such a way that the sales item is a subitem of a subscription item or vice versa, then the equipment ID must be maintained on the subscription item. You can maintain equipment IDs on a subscription contract either by using an open change process or by using the **Maintain Equipment IDs** change process. This system-based auto update is achieved by using **Event Type Linkages** (see transaction SWE2). When serial numbers are assigned in a delivery and the document is saved, the event CREATED/CHANGED for the object LIKP is raised. An event receiver SOM_EQUI_DELV is registered against this event, which facilitates the maintenance of the equipment IDs on the subscription contract.

❖ Example

The following examples are useful but it is important to note that each customer might implement these processes as per their requirement:

- Customer A has implemented creation of equipment master data when serial numbers are created for a product. In this case, it is possible to identify the relevant equipment IDs, during delivery creation, based on serial numbers. These are used to trigger the maintenance of the equipment IDs on the inactive subscription contract. The sample implementation mentioned above caters to this.
 - Customer B has implemented creation of equipment master data only when the hardware is delivered to the end user. In this case, it is not possible to identify the relevant equipment during delivery creation. A custom implementation can be built to support this.
- Trigger subscription contract activation on equipment activation

Similar to above, a sample implementation of how subscription contract distribution and activation can be triggered based on a status on the equipment has been provided. When the status of an equipment changes, the event CHANGED for the object EQUI is raised. If the ID of the equipment which is changed is maintained on a subscription contract, then different change process can be triggered depending on the status. In this example, when the status of an equipment is set to 'ECUS(At Customer)', the distribution and activation process of the subscription contract is triggered. An Event Receiver 'SOM_EQUI_ACTV' is registered against this event, which facilitates the trigger.

i Note

The standard sample implementation is based on status ECUS, but this can be changed by implementing the BADI CRMS4_SOM_BADI_EQUI_INTEG.

- Distribution and activation of subscription contract

The subscription contract will not be distributed and activated until the equipment IDs are maintained and the relevant status (For example, At Customer, Ready for Use) is set on the equipment. This is achieved by using the ODI framework. A standard sample schema EQUI has been provided with steps:

- EQU (Equipment data maintained?): The status of this step will be set to executed only if the equipment IDs are maintained on the subscription contract.
 - EQUA (Are all Equipment active?): The status of this step will be set to executed only if the linked equipment has a certain status. Standard implementation checks for 'ECUS(At Customer)'.
 - PCEA (Call FICA Message contract activation): The standard distribution step that triggers the creation of the FICA and CC contracts which will be executed only once the above two steps are successful.
- Billing start date based on equipment activation

Typically with a subscription contract, the date from which the recurring charges are applicable is the same as the contract start date. When equipment is considered, there are scenarios where the recurring charges are applicable only once the relevant equipment is activated. Additionally, it could also be 'once the equipment is activated' or 'X number of days after contract start', whichever is earlier. To support such scenarios, a date type CONTBILS(Start of Billing) has been provided. As a standard, this date type is set to the date on which the step EQUA is executed successfully and can be changed by implementing the BADI CRMS4_SOM_BADI_EQUI_INTEG.

- Equipment ID as technical resource

The ID associated with the hardware (serial number or equipment ID) might be relevant in the calculation of usage-based charges in CC. To facilitate this, equipment IDs maintained in the object list are copied to technical resources and distributed to the CC provider contract. The prerequisite for such a copy is that the **Technical Resource Type** should be defined with the category **Equipment**. The standard implementation copies the equipment IDs to the technical resources. In scenarios where serial number is setup to be different from equipment IDs and serial number is to be used as technical resource or if the technical resource is to be derived based on the equipment ID, this can be done by implementing the BADI CRMS4_SOM_BADI_EQUI_INTEG. In scenarios where the equipment IDs will not be available at the time of subscription order or solution quotation creation, the technical resource should be set up in such a way that the time of allocation check should be set to 'BA-Before Activation' to suppress the check during order capturing.

- Early distribution when equipment activation is delayed

Typically, early distribution of subscription contract is applicable only when the start date is in the future. With equipment, we have the scenario of where the contract start date might be immediate, but the actual activation and distribution should happen only once the related equipment is activated which could be in the future. With Revenue Accounting and Reporting in picture, an early view into such a contract might be required even though it is not yet active. To support this, a feature has been provided that allows for the execution of early distribution even when the start date is immediate, but the actual distribution is to be deferred. This can be configured at item category level at [►Service ► Transactions ► Settings for Subscription Transactions ► Define Settings for Item Categories ►](#)

- Change processes - Replace Equipment

You can replace equipment IDs on an active subscription contract using the [Replace Equipment](#) change process. The object list view is displayed in the change process popup where list of equipment IDs can be maintained.

i Note

This change process only replaces equipment IDs with no further processing behind it. For example, triggering of return process in SD is not supported.

- Change Processes – Scale Up/Down Equipment

You can increase or decrease the quantity on an active subscription contract using the [Scale Up/Scale Down Equipment](#) change process. The quantity and object list views are displayed in the change process popup where the new quantity and list of equipment IDs can be maintained. In case of a scale up, if the product bundle contains a relevant sales product, it will be added with the delta quantity in the change order.

Related Information

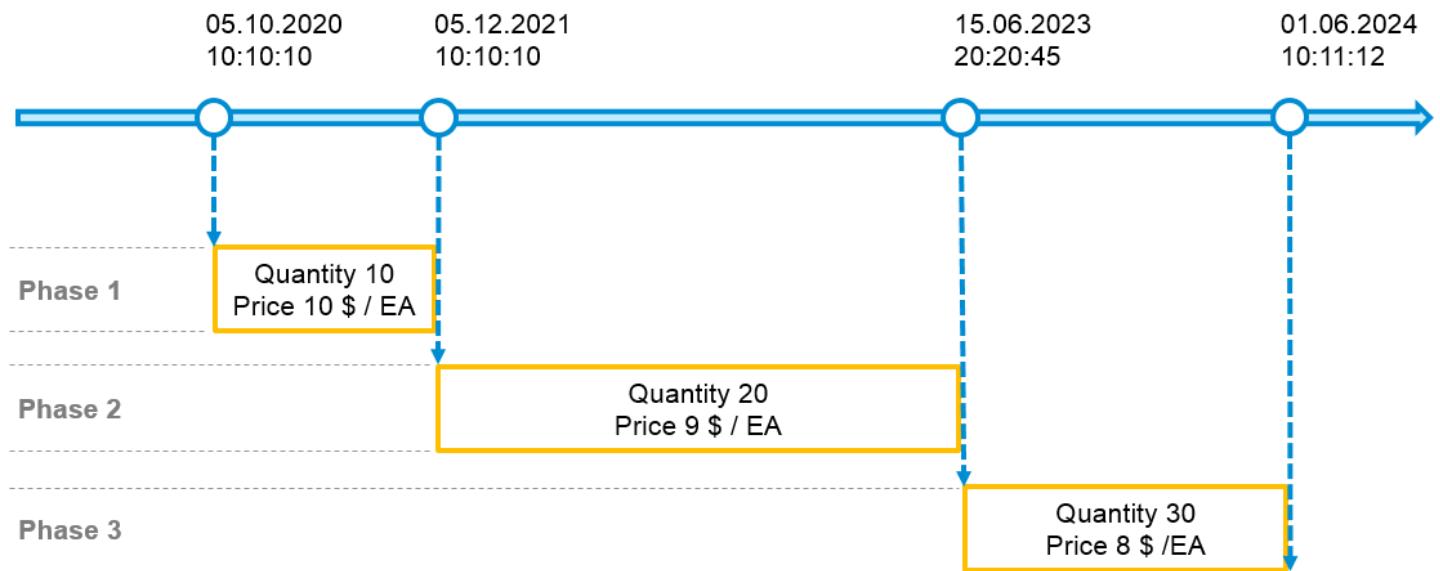
[Replace Equipment](#)

[Scale Up/Down Equipment](#)

Phased Contracts

Use

Phased contracts are subscription contracts where customers choose to split the contract into several phases, each with different requirements. Each phase has a start date and an end date, so that one phase begins right after the end of the previous phase. The phases don't need to be the same length, but they must be continuous. There can't be any gaps or overlaps. Attributes such as quantity and configuration can differ across phases.



Structure of Phased Contracts

Prerequisites

The following are prerequisites for phased contracts in Subscription Order Management:

- The Payer and Sold-to Party must be the same across all phases of a single subscription contract.
- Phases can have different lengths but there must be no gaps or overlaps between these phases.
- The start of timeslice of the first phase and end of timeslice of the last phase must match the start and end dates of the contract, respectively.
- If you create a phased contract using the user interface, you need to provide a contract end date. You can't create an evergreen phased contract.
- The following prerequisites are important if you create or change phased contracts using the [Solution Quotation \(A2X\)](#) OData API:
 - The phase indicator and predecessor item number must be included in the API structure of the subscription item.
 - Information of all phases, such as the FI-CA contract number and item number in the subscription contract, must be provided during a contract change along with the change process information.

Features

Phased contracts have the following features:

- You can create phased contracts using the [Manage Subscription Orders](#) and [Manage Solution Quotations](#) Fiori apps. You can implement changes on the phased contracts using any of the change processes available on the current phase. Alternatively, you can create and change phased subscription contracts using the [Solution Quotation \(A2X\)](#) OData API.

If you create phased contracts using the [Manage Subscription Orders](#) and [Manage Solution Quotations](#) Fiori apps, you can use the [Add Successor Phase](#) button in the item details to add the new phases (new items). The system copies data from the predecessor phase into the successor phase, that is, the main and subitems are copied, and a reference is created between the items. You can provide a start of timeslice for the successor phase on the [Add New Phase](#) pop-up window. The default start of timeslice is set to the current timestamp. The end of timeslice for the predecessor phase is set to one second before the time provided on the pop-up window. After you create the new phase, you can modify details such as the product, the quantity, and prices.

If you create phased contracts using the [Solution Quotation \(A2X\)](#) OData API, the system creates phased subscription contracts with individual products as a follow-up to the solution quotation. To change phased contracts using the [Solution Quotation \(A2X\)](#) OData API, you must also pass the executed change process group for the future phases.

To change the current and future phases of a phased contract, you must provide data for all the phases along with the change data in the change order.

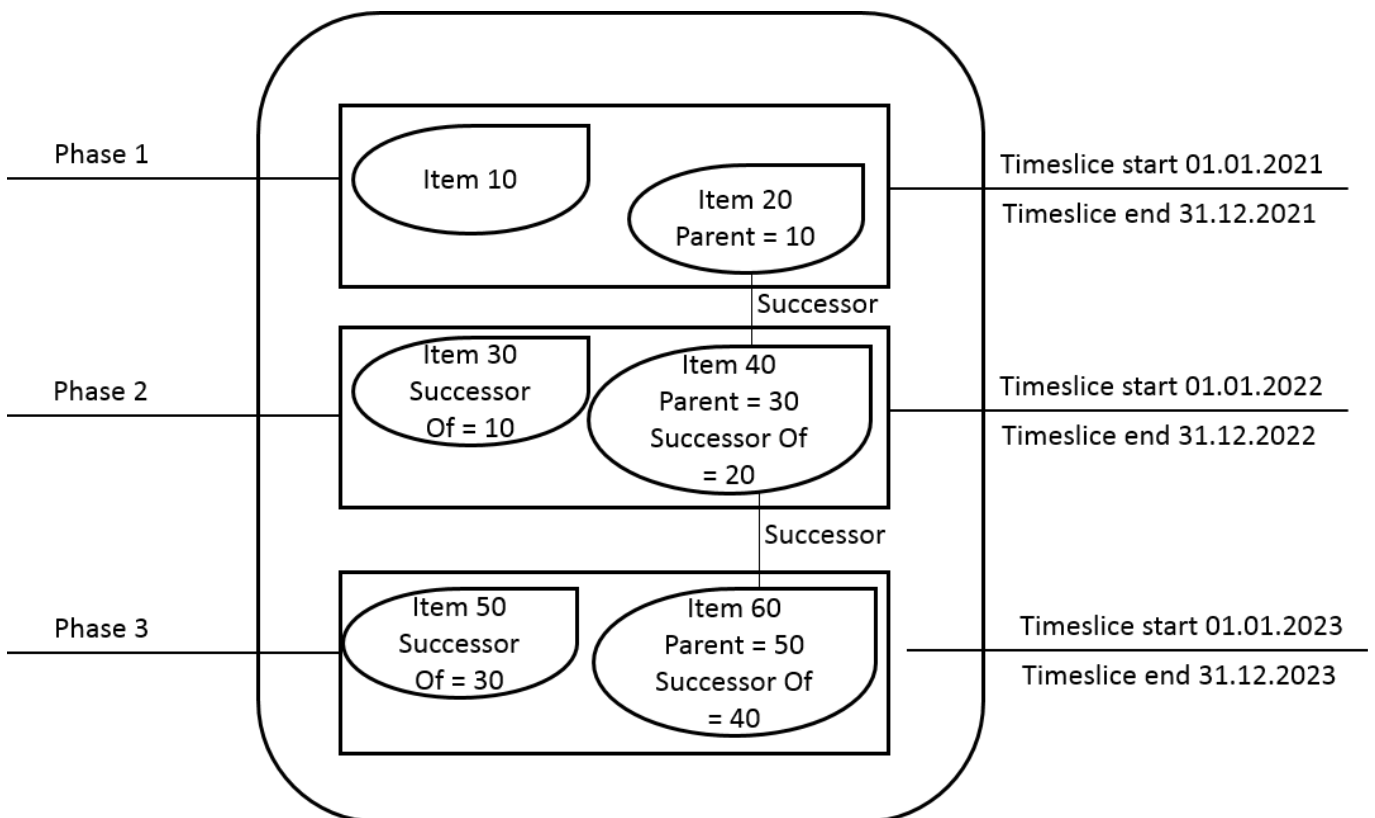
- The following sets of dates are important to phased contracts and are visible under the [Dates](#) assignment block:
 - Contract start and end dates indicate original contract duration
 - The planned contract duration is defined by the term start and end dates
 - Timeslice start and end dates help to define the length of a phase
- You can view the phases on the subscription contract UI and the phase data on the subscription item UI.
- If you create a phased contract with items and subitems using the [Solution Quotation \(A2X\)](#) OData API, you must also define a relationship between the subitems of each phase using the Predecessor Of number.

i Note

It's also possible to have individual subitems for a main phase where the subitems don't have a Predecessor Of relationship.

→ Recommendation

If the phased contract contains preconfigured product bundles, you must avoid marking any subcomponent as mandatory, default, or associated with a BRF+ function. This ensures that the items that are provided using the API are the only ones included in the solution quotation, and no additional items are included due to the bundle explosion.



Predecessor Of (Phased Contract with Subitems)

It's only possible to cancel a subitem in an active phase.

Extension of Phased Contracts

You can extend a phased contract using either the [Extend](#) or the [Add Extension Phase](#) change processes. The following table highlights the features of these scenarios:

Feature	Contract Extension	Add Extension Phase
Restriction	Contract extension can only be implemented on the last phase once it's active.	An extension phase can be added independent of which phase is currently active.
Editable attributes	Only the attribute CONTDURA (extension duration) is considered.	Different attributes, such as quantity, can be provided through the change quotation. The change process uses these values while creating the new phase.
Result	Creates a new phase at the end of the current last phase. Attributes of the current last phase are copied over to this new phase.	Creates a new phase at the end of the current last phase. The details for the new phase must be explicitly provided via the change quotation.

Distribution of Phased Contracts

Active phases of a phased contract are distributed to FI-CA contract, CI billing plan, and SAP Convergent Charging. Future phases are **distributed early** to the SAP FI-CA provider contract and CI billing plan only.

i Note

When the first phase itself is in the future, only the first phase is early distributed. The future phases are distributed only after the first phase is activated.

Constraints

The [Subscription Order \(Version 0002\)](#) OData API does not support phased contracts.

Related Information

[Solution Quotation \(A2X\)](#)

[Extend](#)

[Add Extension Phase](#)

Add Extension Phase

Use

You use this change process to add a new phase after the last phase of a phased contract.

i Note

You can implement any of the standard change processes on phased contracts by using the user interface or the [Solution Quotation \(A2X\)](#) OData API. The [Add Extension Phase](#) change process is available only for phased contracts.

Prerequisites

This change process is available only on the current active phase of the contract.

Activities

Technical Details

Process Name	Process Type	Process Class	Process View
ADD_EXTN_PHASE	ISTA	CL_CRMS4_BTMTF_EXTN_PHASE	PROVDLPR/PhaseExtensionVS

Add the following parameters to the process container when implementing the change process on a phased contract using the [Solution Quotation \(A2X\)](#) OData API (API_BUS_SOLUTION_QUOTATION_SRV):

Process Name	Process Class
ADD_EXTN_PHASE	CL_CRMS4_BTMTF_EXTN_PHASE

i Note

The payload for creating the change quotation and phased contract must contain information for the current active phase and all following phases.

Add the following parameters to the process container when implementing the change process using the [Subscription Contract Change Process \(Asynchronous\)](#) SOAP API (II_CRMS4_PRVC_CHANGEPROC_IN):

Parameter Name	Description
DURATION	Value of the extension period
TIME_UNIT	Unit of the extension period (DAY, MONTH, YEAR)

The START method adds the new phase into a change order which is then copied into the phased contract on submission.

Example

Consider a phased contract with a duration of three years, from 1st January 2025 to 31st December 2027. The contract includes the following phases:

- **Phase 1:** 1st January 2025 to 31st December 2025
- **Phase 2:** 1st January 2026 to 31st December 2026

The second phase is currently active.

- **Phase 3:** 1st January 2027 to 31st December 2027

The [Add Extension Phase](#) change process is available during the active phase of the contract, which is the second phase in this example. When you use this change process to add a new phase, the system automatically adds the new phase after the last existing phase. The contract would then include the following phases:

- **Phase 1:** 1st January 2025 to 31st December 2025
- **Phase 2:** 1st January 2026 to 31st December 2026

- **Phase 3:** 1st January 2027 to 31st December 2027
- **Phase 4:** 1st January 2028 to 31st December 2028

Related Information

[Phased Contracts](#)

[Solution Quotation \(A2X\)](#)

Contract Repair

Use

You can use the contract data repair function when the distribution of subscription contract data fails due to inconsistent data during order capturing. You can correct the data by adjusting the preceding subscription order document and resubmitting it.

In some cases, a preceding subscription order document may not be available to perform the correction. For example, a preceding subscription order document cannot be produced by migrating contract data from a legacy system or when the preceding document is in status **Completed** and is not open for change. For example, Solution Quotation.

In such cases, you can use change processes of type **Contract Repair** (ISTG process type in Business Transaction Maintenance Framework (BTMF)) to correct or change data on inactive subscription contracts. This is done by creating an intermediate order which is treated as a predecessor to the contract. By submitting the order document, corrections are transferred to the existing subscription contract and the activation process is triggered again.

ISTG processes are applicable when the following conditions are met:

- The subscription contract is inactive
- The preceding document of the subscription contract does not exist. For example, in the case of migration scenarios or if the preceding document is completed, for example in the case of contracts created out of a solution quotation
- The distribution of the contract has ended in error or has not begun yet. For example, when the activation date of the contract is in the future.

The delivered Change Processes for contract repair and their prerequisites are listed below:

Create Repair Order

This change process must be used in migration scenarios when the preceding document is not available as part of the migration and the distribution of the contract has resulted in an error.

- The Subscription Contract is not active
- The preceding document does not exist. For example, in migration scenarios when only the contract is migrated and not the preceding order
- The distribution of the contract has resulted in an error

Correct Errors

To be used in scenarios when the preceding document is completed (for example, Solution Quotation) and distribution of the contract has ended in error.

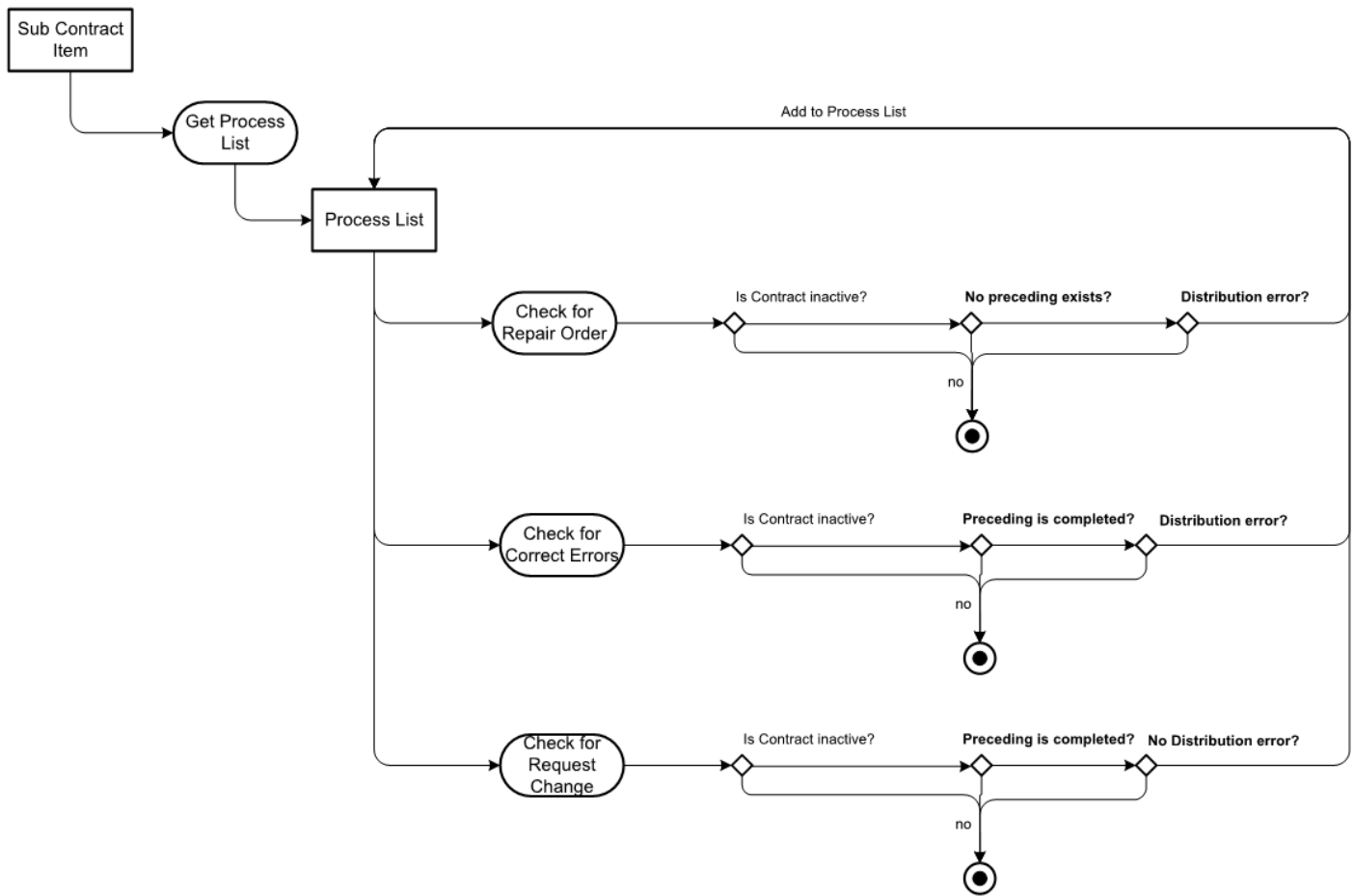
- The Subscription Contract is not active

- The preceding document exists but is completed. For example, Contract created out a Solution Quotation that is completed
- The distribution of the contract has ended in error

Request Change

To be used in scenarios when the preceding document is completed and there are requests for changes before the activation date of the contract.

- The Subscription Contract is not active
- The preceding document exists but is completed
- The distribution of the contract has not started, for example, activation date is in the future



Related Information

[Repair](#)

Contract Change Overview

Displays an overview of the executed change processes.

Use

You can also display an overview of the changed data when a contract change process has been executed. The overview compares the data changes between a change order and the previous contract time slice.

Integration

The **Contract History** assignment block on the contract UI contains the **Change Overview** column. When choosing a link in this column, the system displays a dialog box containing the data changes for the corresponding change order and its previous contract time slice.

Activities

If you want to display your own objects in the change overview, you must execute the following steps:

1. API

Implement the method IF_CRM_ISX_PRV_DELTA~ADJUST_DELTA for the BAdI CRM_ISX_PRV_DELTA. Add your own data to the table CT_DELTA in this method. The table contains the following fields:

- NUMBER_INT: Item number
- DESCRIPTION: Product description
- ORDERED_PROD: Product ID
- OBJECT: Technical object name
- OBJECT_DESCR: Object description
- SUB_OBJECT: Technical sub object name (value grouping)
- SUB_OBJECT_DESCR: Sub object description
- CHANGE_INDICATOR: Change indicator (changed, new, deleted)
- ATTRIBUTE: Technical name of attribute
- ATTRIBUTE_DESCR: Attribute description
- VALUE_OLD: Old value (previous time slice)
- VALUE_OLD_DESCR: Old value description
- VALUE_NEW: New value (from change order)
- VALUE_NEW_DESCR: New value description

i Note

If you do not want to display the assignment block for a specific object, you must delete the corresponding entries in the delta table within the BAdI implementation.

2. Web Client UI

For displaying the object changes in a new assignment block, you must execute the following steps in the UI component ISXDELTA:

- Runtime Repository

Create a component usage for the interface view ISXDELTA/SingleWindow. The naming convention has to be as follows: usage+<Object name>, for example, usageOBJECT

Add the new component usage to the view area the overview page of the view set ISXDELTA/OverView.

- View Configuration

- Create a new view configuration for the view ISXDELTA/Delta with the name of the component usage. In the configuration, specify the columns of the delta table to be displayed in the assignment block for the customer object.
- Add this new view configuration as assignment block to the view configuration of the overview page ISXDELTA/OverView.

Maintenance Framework for Business Transactions

Use

The maintenance framework for business transactions (BTMF) is used to execute defined processes for subscription contracts. A BTMF process is always executed for a specific subscription contract item. Processes can be executed either via the UI or they can be called using an API.

Prerequisites

You must define change processes before using the BTMF framework. Typically, different kinds of change processes and different types of process types are available. For more information, see **Process Types**.

You can make the relevant settings in Customizing of **Service** under **Transactions** > **Settings for Subscription Transactions** > **Contract Changes** > **Settings for the Maintenance Framework (BTMF)**.

Features

While various change processes are already available for implementation, they are based on best practices and may not be sufficient for all requirements. Therefore, SAP provides a change framework for change processes.

Related Information

- [APIs and Events for Subscription Order Management](#)
- [Define Change Processes](#)
- [Process Types](#)
- [Mass Changes of Subscriptions](#)
- [Manage Mass Processing of Subscriptions](#)

Process Types

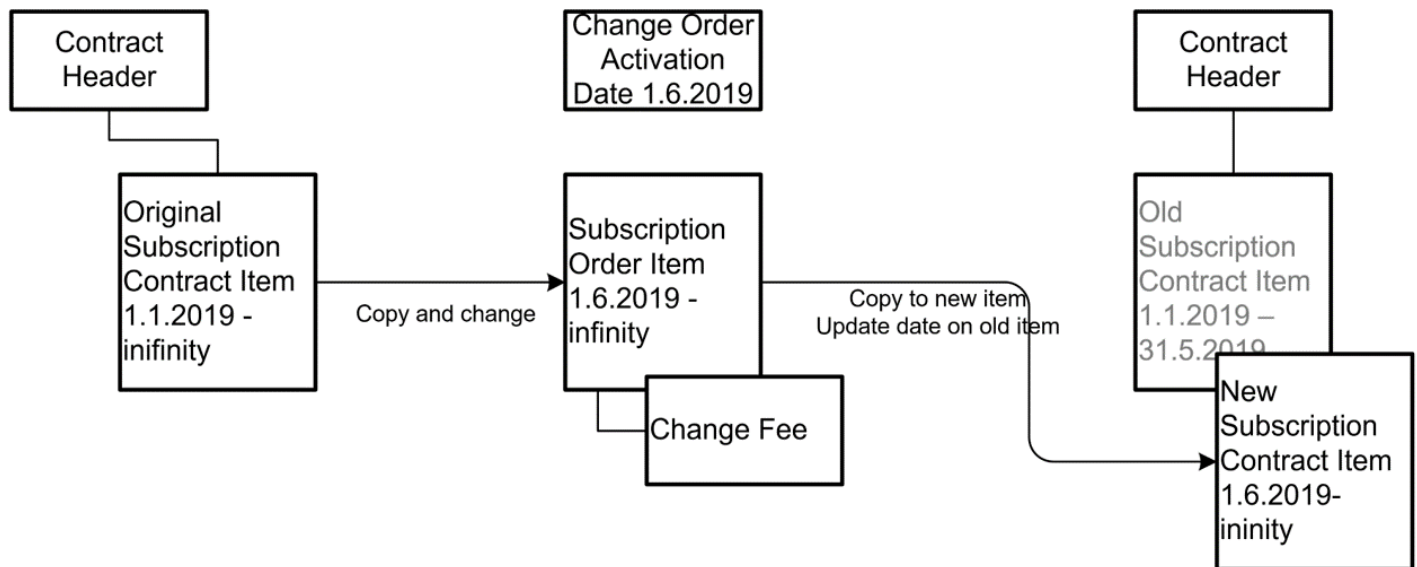
Use

The maintenance framework for business transactions (BTMF) is used to execute defined processes for subscription contracts. There are different types of change processes and therefore, different process types are available:

Process Type	Description
ISTA	Contract change (for example, product change, contract extension, maintain technical resources, or cancellation) For this type of change process, BTMF creates a change order. Initially, the data of the active contract time slice is copied to the

Process Type	Description
	change order. This change order can be modified using the process implementation class. A process can have implementations which are automatically executed and a process-specific UI in which you can maintain changes. A change order has an activation date assigned. It defines when the changes shall become active in the subscription contract. Several ISTA processes can be executed in one change order with the same activation date, so all changes become active at same time. i Note There can be only one change order with a future activation date at the same time, so several planned changes in future are not possible. The change order needs to be released for contract change. If this is done, the system automatically creates a new time slice in subscription contract document and the new contract data is copied from the released change order. In general, change processes are used to update a contract. Contracts are never updated directly, but by means of change orders. Depending on the process type, the update of the contract can differ. From the contract, all relevant data is copied to the change order. The change order changes the validity dates of the existing contract item and copies the new data of the change order to a new contract item with relevant dates to the contract.
ISTF	Change process that does not create a new time slice in the contract but allows to trigger some logic in downstream systems.
ISTG	In some cases, you can create one subscription contract without a referenced subscription order, for example when the contract is migrated from an external system and has no predecessor order. If the activation of the contract failed during distribution, the contract cannot be changed and distributed again. To correct the contract, this process copies the contract item to a new subscription order. The new subscription order is treated as the predecessor to this contract. You can make changes in the subscription order and submit it to update the contract item. This is now the standard subscription order, and you can use the standard process.
ISTH	As described for process type ISTG there can be a subscription contract without a preceding subscription order. While the ISTG process allows a change by creating a new subscription order to perform this change, the ISTH change processes now support changes to existing subscription contracts without creating an intermediate subscription order. Such changes can be applied on contracts that are still inactive.
ISTC	The process type ISTC is used for locking/unlocking subscription contracts.
ISTE	The process type ISTE is used for modifying subscription contracts that are past their end date.

An ISTA change order updates a contract as follows (example):



ISTA change order update of a contract (example)

Processes of different process types are independent and can be triggered in parallel.

More Information

For more information, see Customizing of **Service** under **Transactions** > **Setting for Subscription Transactions** > **Settings for the Maintenance Framework (BTMF)**.

Related Information

[Contract Change Overview](#)

[Contract Repair](#)

Define Change Processes

Use

The maintenance framework for business transactions (BTMF) is used to execute defined processes for subscription and service contracts. A BTMF process is always executed for a specific contract item.

Prerequisites

If a process requires a specific UI to enter data, the process view can also be defined here. To use the BTMF framework, you must define and activate change processes. You can make these settings in Customizing for **Service** under **Transactions** > **Settings for Subscription Transactions** > **Contract Changes** > **Settings for the Maintenance Framework (BTMF)**.

Features

Various change processes are available that customers can use during the implementation. These change processes are based on best practices. In some cases, this might not be sufficient for all requirements. This is why a change framework for change processes is available. The following options are available to adapt change processes:

- Check for process validity

Processes are not valid for all products at all times. BTMF provides modeling rules using the Business Rules Framework plus (BRF+) that checks whether processes are valid according to the current state of the contract. Alternatively, checks can be coded using the standard ABAP code in the process implementation. You are only offered processes that return the status **Valid**. Example: This could be used to define a start time from which a contract can be canceled, for instance three months before the contract end date.

- Define business logic

BTMF implements customer-specific process classes to specify the business logic for each change process, for example to define default values, consistency checks or follow-up actions.

- Define UI

You can specify UI views for each change process that are displayed during the contract change process. Enter the corresponding views in the Customizing activity used to define change processes. This means you can add fields to standard processes or create new processes.

- Enable for Fiori menu

You can use the **Fiori Menu Enabled** checkbox to display/hide a change process from the **Change Contract** quick access menu in the **Manage Subscription Contract Lifecycle** and **Manage Subscription Contract Item Details** apps, and the **Change Contracts** quick access menu in the **Manage Subscription Contract Documents** app.

To use the BTMF framework, you must define and activate change processes. The business logic of a BTMF process is implemented by an ABAP class, whereas the class interface IF_CRM_BTMF defines the required methods for BTMF processes. A default implementation is available for subscription contract processes in the class CL_CRM_BTMF_CONTENT_TELCO that must be used as a super class. This avoids having to redefine all interface methods in each process class. For more information, see BTMF Interface.

Once a change order has been updated and the contract distribution is started, the order cannot be changed. If an error occurs when the contract is distributed, for example during activation, the contract and the change order are defined as containing errors. This opens the change order again for modifications. Once the changes have been made, the modified change order overwrites the latest contract item with the new correct data.

The implementation of a process must be able to work within a certain context, for example, a process triggered from the subscription order API in the background must not rely on input data provided from the WebClient UI.

Persistent Contract History

The contract history is not determined dynamically, it is stored instead. In the standard system, one record is recorded for the contract history in the database for every BTMF process. A reject order is stored in the database, but not displayed in the contract history. A record is only stored for the selected item in the change process. Changes to subitems of the selected items, which are triggered by inheritance, are not recorded.

You can also enhance the functionality using your own logic, since the implementation is made in the BTMF classes for the change processes. Implement the method DETERMINE_CONTRACT_HISTORY in your process class. If you do not implement this method, system uses the method in the superclass.

Activities

You can make the relevant settings in Customizing for **Service** under **► Transactions ► Settings for Subscription Transactions ► Contract Changes ► Settings for the Maintenance Framework (BTMF) ►**.

If a process requires a specific UI to enter data, the process views must be created in the BSP workbench. You can specify the change processes used by the views here. To do so, provide the BSP application, the view name, and a description.

To define a change process, you can maintain the following attributes:

- BTMF process
- Scenario: SERVICE/PROVIDER
- Process type: ISTA/ISTF/ISTG/ISTH/ISTC/ISTE
- BTMF Process class: The process implementation class controls process execution. The class must implement the interface IF_CRM_BTMF. If no process implementation class is specified, the system uses the standard implementation class CL_CRM_BTMF_CONTENT_TELCO to execute the process during at runtime.
- BTMF BRF+ Function: You can use BRF+ functions to control which processes can be selected at runtime. You can for example specify that the contract extension change process can no longer be selected for a contract that has been terminated.
- Change reason
- Text: Text that appears in the process selection assignment block.

i Note

Each process class is comparable to a Business Add-in (BAI). You can use the methods in each class to implement your own coding. Several interaction time points are provided.

In the scenario dependent attributes for PROVIDER, you can maintain:

- Service Active: You can change the technical data of a product in the change process.
- Desc. Link Active: Description of the product that appears as a link that you can use to navigate to the product details.
- Dur. Editable: You can change the duration of a rate plan in a change process.
- Business Event: For change processes of type ISTA, you can define if a specific business event shall be raised when the change process is executed. You can select from the following predefined events:
 - STATECHANGED - Generic change process event
 - CANCELED - Change process for cancellation
 - PRODUCTCHANGED - Change process for product change
 - CONTRQTYCHANGED - Change process for quantity change
 - EXTENDED for EXTENSION - Change process for contract extension

For the scenario PROVIDER, you can also assign one or more views associated with the process. If you do not assign any views to the process, the system executes the process directly.

Example

Corresponding classes for the interface IF_CRM_BTMF are implemented for the change processes that are delivered as standard. You can use these sample implementations to define your own change processes and adapt them to your requirements.

Define Change Process Group

Use

The BTMF change process group is a set of changes that are to be executed together on a contract. The change process group follows an 'all or nothing' principle, where the changes will be committed to the contract only if all the change processes are executed successfully.

Change process groups allow the individual change processes to remain at an atomic level while being added into groups, based on the required combinations. This also enables the plug in of customer implemented processes into groups, with minimal enhancements.

i Note

Change process groups currently support subscription contract changes via the Business Solution Quotation API (API_BUS_SOLUTION_QUOTATION_SRV) only. For more information about this, refer to **Solution Quotation (A2X)**.

Prerequisites

Individual change processes must be first defined in the Customizing for **Service** under **Transactions > Settings for Subscription Transactions > Contract Changes > Settings for the Maintenance Framework (BTMF)**.

Features

Change Process Group Processing

The processing of the change process group along with the Business Solution Quotation API has the following features:

- A change process group is executable only if the CHECK for each individual process in the group is successful.
- The execution of a change process group is considered successful and will be committed to the contract only if the execution of each individual change process is successful.
- Currently only the 'PROVIDER' scenario (that is, subscription contracts) are supported.
- All change processes of the process type ISTA and the Add New Contract (process type ISTG) are supported.
- When executed in combination with a change quotation that references multiple contracts, each contract is treated independently. For change processes of type ISTA, a separate change order is created for each individual contract.
- You can assign a processing class (that implements the interface IF_CRMS4_BTMF_GROUP_CONTENT) to a change process group to enable skipping of the CHECK or EXECUTE of individual processes within the group. With this, the number of groups that are required for the various possible combinations can be limited.

Change Process Group - Read

If required by an external system, you can retrieve a list of change process groups, a list of processes that belong to each group, and a list of container attributes that are required for an individual process, using the API API_CHANGEPROCESSGROUP. For more information about this, refer to **Change Process Group - Read**.

Executable Change Process Group(s) for a Contract - Read

For a subscription contract, you should be able to retrieve the list of executable change process group(s) using the API API_SUBSCRIPTIONCONTRACT. For more information about this, refer to **Subscription Contract - Read**.

Activities

You can make the relevant settings in the Customizing for [Service](#) under [Transactions](#) > [Settings for Subscription Transactions](#) > [Contract Changes](#) > [Settings for the Maintenance Framework \(BTMF\)](#) > .

You can define your own change process group(s) and assign the following attributes:

- Scenario: Currently, only PROVIDER is supported.
- BTMF Process Type: Controls the behavior of how the change group process is executed. Only processes that are of the same BTMF process type can be added to the group.

i Note

Currently, only the BTMF process type ISTA is supported.

- Process Group Class: The behavior of the group is determined by an ABAP class, which should implement the interface IF_CRMS4_BTMTF_GROUP_CONTENT. For example, whether the CHECK of an individual process in the group is to be skipped.

A default implementation is available for subscription contract processes in the class CL_CRMS4_BTMTF_GROUP_CONTENT that must be used as the super class. An additional class CL_CRMS4_BTMTF_GRP_PROD_CHANGE is also available and can be used as the super class when product change is part of the group.

- Define Process List: Here, you can define the list of change processes that are to be part of a change process group.

You must define a sequence for these processes and the process with the lowest sequence number plays a special role (hence, it must be set carefully). For example, to determine the transaction type of the change order.

Related Information

[Solution Quotation \(A2X\)](#)

[Change Process Group - Read](#)

[Subscription Contract - Read](#)

[Change Quotation](#)

Change Quotation

Use

You can execute one or more change processes for a subscription contract using the OData service API_BUS_SOLUTION_QUOTATION_SRV for solution quotations by assigning a change process group to the subscription item. The change process group combines one or more single change processes to be executed together.

A change quotation comes into the picture if the solution quotation contains an item with a change process group. The change quotation item is used as a data container to store the new data. The changes are then executed in an asynchronous step triggered by the PPF action CRMS4_SOM_PROCESS_CHANGE_QUOTE, if the quotation has the status [Change Processing](#) (I8919). This status is set when the quotation is accepted and will be deleted after the items are processed. While the items are processed, a change order is created as a copy of the existing contract and the data relevant for each change process is copied from the quotation to the change order. At the end of processing, the change order is submitted, and the contract is updated by copying data from the change order to the new contract item.

i Note

This functionality is only available using API and is not supported in the UI. You can, however, display the new fields in the UI by changing UI configuration for the subscription item in the solution quotation.

All items where the leading change process of a change process group has the document flow strategy set = 'N' (Create New Contract in New Contract Document) will be created within a single new contract document. The same is valid for the specific header change process group **ADD_NEW_CONTRACT**; if no contract reference is assigned, the new contract will also be added to the same new contract document.

All items without a change process group in the same solution quotation will end up in a separate own new contract document.

Prerequisites

The following are the prerequisites for change quotations:

- Change process groups must be set up in the Customizing for **Service** under **►Transactions►Settings for Subscription Transactions►Contract Changes►Settings for the Maintenance Framework (BTMF)►**.
- Change process groups and contract reference data must be provided in these items: ChangeProcessGroup, CAPProviderContract (external ID of the FI-CA provider contract on the subscription contract item), and SubscriptionContractItem.
- The report **RSPPFPROCESS** must be scheduled on a periodic basis to execute the asynchronous action for change quotation processing.
- Validations
 - The sold-to party of the change quotation must be the same for all contracts used within the quotation.
 - All change processes included in a change process group must be executable for the existing contract, otherwise the change process group cannot be processed for this contract.
 - If you use an item hierarchy with main and subitems, you can use different change process groups either on the main and subitem or only on one of the items. However, the same type of change processes must be used on the items.
 - Phased Contract
 - The change process group must be the same for all phases.
 - You can neither add nor delete a phase.
 - If the end of the time slice for a phase has been reached (that is, the phase is historical), it must not be included in the change quotation. Only active and future phases can be used in a change quotation.
 - When a change process group is maintained, a contract reference must be entered in the payload when triggering the POST operation via OData:
 - CAPProviderContract = Subscription Contract ID
 - SubscriptionContractItem = Item number in the subscription contract document

This must be filled for existing items. For newly added subitems (special change process for the main item), this item reference must be initial.
 - A new change quotation is rejected by the system if another unprocessed change quotation exists for the same contract.

Features

Standard Setup

The **ADD_NEW_CONTRACT** header change process group is delivered in the standard Customizing and cannot be changed. This change process group has a slightly different behavior than a normal change process group:

- Used to add a new contract to an existing contract document or to a newly created contract document.
- No reference to a Subscription Contract ID (**CAProviderContract**) is required, instead, a reference to a subscription contract document (**SubscriptionContractReference**) can be provided.
- If a contract document reference is provided, the new contract will be added to this contract document. Otherwise, a new contract will be created in a new contract document.
- In case the new contract is created in a new contract document, a subscription order is created as an intermediate document. The data from the change quotation is copied into the subscription order. From there the new contract document is created. Thus, you need to maintain copy control in the Customizing from solution quotation to subscription order. In a standard setup, we provide a copy of the transaction type from SRVP to PRVO.

Processing of a Change Quotation

- It is possible to combine changes for several contracts as separate items in the same change quotation. You can even add new items in the change quotation to trigger a new contract.
- Items must be released, and the quotation must be accepted. Only then will all items with a change process group be processed by an asynchronous action. All items without a change process group are executed synchronously as in a normal solution quotation (Initial Items).
- An own change order is created for each contract.
- The changes of a change quotation cannot be processed as long as an open change order exists for the same contract.
- A document flow will be created between the change quotation and the corresponding change order (on header and item level).
- After processing an item, an event is raised to signify whether the item was processed successfully or if an error has occurred:
 - You can register a receiver for this event in transaction **SWE2**
 - Object Category: CL – ABAP Class
 - Object Type: **CL_CRMS4_SOM_CHNGQUOTE_BEH**
 - Event name: **PROCESSED**
 - You can use the generic receiver **CL_BEH_QUEUE_MANAGER** if you want to connect to the EMS (Enterprise Messaging).
- If the item was processed successfully, the item status will be changed to Completed. Otherwise, it remains in status Released so that the corrected data can be updated.
- A log of the complete processing can be found in the quotation for the executed PPF action and is additionally stored separately for each item in business application log (BAL), under object SOM, sub-object QCP, and quotation item GUID as external ID.

Change Configuration for AVC Bundles

- The **Change Configuration** change process is only allowed on the main item. In this case, all the changed configurations are also copied from the subitems which have a contract reference (subscription contract ID). The newly added subitems from the Bill Of Material (BOM) which are not configurable are copied to the change order if they have a contract reference (subscription contract ID).

- The subitems that are added in the change quotation only by BOM explosion and do not have a contract reference (that is, they were not part of the payload) will not be copied to the change order. They will automatically be added in the change order by the BOM explosion. If you want to redefine any attributes on such subitems (such as the price and CC parameters), they must be part of the payload and a contract reference (subscription contract ID) must be provided in the payload.
- The subitems that are sent in the payload need the `BillOfMaterialItemNodeNumber` as an identifier if the subitem already exists in the contract. Additionally, the contract references (subscription contract ID + item number) must be provided
- The subitems that are deleted by the BOM explosion will automatically be deleted in the change order.
- The configuration date must be provided in the payload to ensure that BOM hierarchy provided in the payload fits to a BOM explosion executed in the change quotation. The configuration date is always copied from the change quotation to the change order.
- **→ Recommendation**
It is recommended that the payload contains all subitems of the BOM hierarchy. For newly added subitems, only the subscription contract ID must be provided as a contract reference. For existing contract items, the subscription contract ID + item number must be provided as a contract reference.
- The document flow is only written between the items that are provided in the payload and have a contract reference. The subitems that are added in the change quotation by BOM explosion will not have a document flow to the corresponding items in the change order.
- Characteristics that are inherited by the subitems must not be sent in the payload for the subitem otherwise they will be treated as manually entered values and this would break the inheritance.

Using Bundle Products

i Note

If you use a product bundle where some products depend on a BRF+ function, you must ensure that the function works for a change quotation and a change order. For example, it is not sufficient to check for an ISTA process as this is only set in the change order and not in the change quotation. The explosion executed in the change order will be used, independent of the explosion in the change quotation.

- If you use an external system to create the quotation and the quotation in SAP S/4HANA is only used as an integration object, it is not recommended to use a product bundle that explodes automatically in the backend. In this case it is better to use manual main and subitems.
- If you use a product bundle and if a bundle explosion is not needed in the quotation, you can suppress it by using a specific item category.

Error Handling

- If an item is processed with errors, you can send a patch request to update the existing item.

If the error occurs before the change order is saved, no change order will be created. If the error occurs while a change order is being saved, the order is created but it cannot be edited manually.

- You do not see the error directly on the quotation item. You can only see it in the application log of the action processing or in the item specific log mentioned above.
- Once the quotation is accepted again, the item will be processed again by the asynchronous action.
- The related subscription contract cannot be changed until the change quotation item is not processed successfully.

Related Information

[Change Process Group - Read](#)

[Solution Quotation \(A2X\)](#)

[Define Change Process Group](#)

Contract Change Processes

Use

In a subscription contract, you can trigger change processes leading to various contract changes, for example, **Contract Extension**. Only specific change processes are available at any given time, depending on the state of the contract. For example, a canceled contract cannot be extended anymore and only the **Revoke Cancellation** change process will be available in this case. Besides changing subscription contracts from the UI, you can also use APIs to apply changes to existing subscription contracts. For more information, see **APIs and Events for Subscription Order Management**.

Prerequisites

- You have set up the business processes in the **Business Transaction Maintenance Framework** (BTMF). For more information, see **Maintenance Framework for Business Transactions**.
- Before executing a change process, set the document flow strategy to determine whether the change process will trigger a contract transfer. You can do so in the Customizing for **Service** under **Transactions** > **Settings for Subscription Transactions** > **Contract Changes** > **Settings for the Maintenance Framework (BTMF)**.

i Note

By default, the value is set to ' ' (space), in which case the existing contract is updated with details from the change order.

To trigger a contract transfer, you can choose either strategy U (update new contract in existing contract document) or N (update new contract in a new contract document).

Features

The application displays the confirmed subscription contract with the included products and associated product data, along with other contract details such as activation date and start/end of term. Typically, you cannot reuse these processes out-of-the-box and they should be adapted according to your business requirements during system setup.

Once you save the change order, it can either be rejected or released. Releasing a change order will apply the saved changes to the contract.

Related Information

[APIs and Events for Subscription Order Management](#)

[Subscription Contracts](#)

[Maintenance Framework for Business Transactions](#)

Change Product

You use this change process to replace the currently assigned product in a subscription contract, with other existing products.

Prerequisites

The item in the contract must be active.

Activities

To enable the data transfer, implement the Business-Add-In **BAdI: Recover Data When Changing Product** (CRM_RESCUE_DATA). You can maintain the BAdI in Customizing for **Service** under **Transactions > Settings for Subscription Transactions > Contract Changes > Business Add-Ins (BAdIs) > BAdI: Recover Data When Changing Product**.

To customize the list of products during the product change process, implement the Business-Add-In **BAdI: Filter Product List During Product Change Process** (CRMS4_BADI_PROD_CHNG_FILTER). You can maintain the BAdI in Customizing for **Service** under **Transactions > Settings for Subscription Transactions > Contract Changes > Business Add-Ins (BAdIs) > BAdI: Filter Product List During Product Change Process**.

After you have triggered the change process, you choose a relevant subscription product to replace the one that is currently assigned. You can also edit the activation date and time for this product change. Once you **Apply**, either a new change order is created, or this change is added to an existing change order. The system displays the change order with the new product, the associated product data, and relevant prices. When you submit this change order, the changes are activated according to the specified activation date.

A product change can lead to a change of billing cycle. For products with recurring fees, this change may also impact the related billing plans in Convergent Invoicing and billable items requested for them. For further information about this, see **Recurring Fees**. The standard parameter SAP_BILL_CYCLE_DATE from Subscription Order Management is passed to SAP CC and by default, the external parameters that are passed to SAP CC are unmodifiable. However, we recommend that you don't set the SAP_BILL_CYCLE_DATE parameter to unmodifiable in SAP CC because this parameter is updated every time a new item is created in the FI-CA provider contract. To find out how to change this setting in SAP CC, refer to [New Constraint for External Parameters \(New\)](#).

After you have selected a new product, the master agreement is validated against it in the change order. In case of an error, you must either remove the existing master agreement or manually assign a relevant one in the **Subscription-Specific Data** assignment block in item details.

Technical Details

Process Name	Process Type	Process Class	Process View
PROD_CHANGE	ISTA	CL_CRM_BTMF_TELCO_PRODUCT_CHAN	PROVDLPR/ChangeProductVS

i Note

While executing a product change process, you can also replace the assigned technical resources and contract accounts in the same change order.

If you have enabled contract transfer in the Customizing and there are allowances and counters associated with the initial product, these will be transferred to the new contract only after it is activated (at 00:01 the next day).

To execute this change process using the **Subscription Contract Change Process** SOAP API, you must add the following parameters to the process container:

Parameter Name	Description
PROD_ID	Product ID of the new product

The START method checks whether the product from the process container is maintained as a subscription product. If this is the case, a new item containing the new product is added to the change order and the old one is deleted. If the product is not defined, the process is terminated, and an error message is returned.

Related Information

[APIs and Events for Subscription Order Management](#)

[Recurring Fees](#)

[Contract Transfer](#)

Cancel

Use

You use this change process to cancel an active contract by setting a specific cancellation date. In this case, the entire contract is canceled.

i Note

If a contract with future dated billing schedule lines (BSLs) is canceled, a change order is created with the error that billing date must lie within the validity period of the item. To submit the change order, you must first adjust the BSL date to the cancellation date in the change order.

Prerequisites

The main item in the contract must be active.

Features

The following three types of cancellations exist, based on your entries in the **Requested Cancellation Date** and **Requested Cancellation Time**:

- Immediate: If you enter the current date and time, the contract is canceled immediately. You cannot revoke it.
- Deferred: If you enter a date or time in the future, the cancellation is deferred. You can revoke it if the requested cancellation date/time has not been reached.
- Backdated: If you enter a date or time in the past, the contract is canceled immediately with the entered requested cancellation date. You cannot revoke it.

Once the contract has been canceled, all subitems for the main item are also canceled. The contract end date (appointment type CONTENTD, precisely timed) is defined for the entire contract. The requested cancellation time stamp and the activation time stamp of the cancellation must be different. The cancellation dates and procedures are set only on the main item.

Activities

Enter the following parameters:

- Requested cancellation date and time
- Cancellation reasons

You must also enter the activation date and time for this change process. The default setting is that the current system time stamp is set for cancellation dates and the default cancellation procedure 'ISTC', the cancellation party 'CNPR', and the cancellation reason 'K004' are used. These are defined in the implementation class.

Technical Details

Process Name	Process Type	Process Class	Process View
CANCEL	ISTA	CL_CRM_BTMF_TELCO_CANCEL	PROVDLPR/CancellationVS

To execute using the [Subscription Contract Change Process](#) SOAP API, you must add the following parameters to the process container:

Parameter Name	Description	Value
CANC_REASON	Cancellation reason	from Customizing (default: K004)
CANCPARTY	Cancellation party	from Customizing (default: CNPR)
CANCPROC	Cancellation procedure	from Customizing (default: ISTC)
CANC_REQUEST_DATE	Requested cancellation date	(Default: current system date)
CANC__REQUEST_TIME	Requested cancellation time	(Default: current system time)
CANC_RECEIVE_DATE	Received cancellation date	(Default: current system date)
CANC_RECEIVE_TIME	Received cancellation time	(Default: current system time)

Example

You created a subscription contract with a subscription item. If the customer no longer needs this contract, you can cancel this contract immediately. The current system time stamp is set as the contract end date/time for this contract.

Related Information

[Cancellation](#)

[Revoke Cancellation of Contracts](#)

[Cancel Subitem](#)

Revoke Cancellation of Contracts

You use this change process to revoke the cancellation of a subscription contract if the requested cancellation date is in the future.

Prerequisites

This change process is not applicable in an immediate cancellation scenario, that is, you can revoke the contract cancellation only if the requested cancellation date is in the future.

Features

The following cancellation parameters of the main item of the contract are deleted while revoking the cancellation:

- Cancellation date (CANCDATE)
- Requested cancellation date (CANCREQUEST)
- Received cancellation date (CANCRECEIVE)
- Contract end date (CONTEND)

The cancellation set is also deleted. In case of subitems, the contract end date is deleted if the subitem does not have its own cancellation set. For example, if a subitem was canceled before the cancellation of the whole contract, the cancellation of this subitem cannot be revoked.

Technical Details

Process Name	Process Type	Process Class	Process View
REVOKE_CANCEL	ISTA	CL_CRM_BTMTF_TELCO_CANCEL_REVOK	None

To execute this change process using the [Subscription Contract Change Process](#) SOAP API, you must add the following parameters to the process container:

Parameter Name	Description
SubscriptionContract	Subscription contract number
SubscriptionContractItem	Subscription contract item number
SubscriptionContractChangeProcessType	Subscription contract change process type In this case, the value would be REVOKE_CANCEL
SubscriptionContractActivationDate	Activation of the change process

To execute this change process using the [Solution Quotation \(A2X\)](#) OData API, you must add the following parameters to the process container:

Parameter Name	Description
CAPProviderContract	Subscription contract number
SubscriptionContractItem	Subscription contract item number
ChangeProcessGroup	Subscription contract change process group

Example

A contract has been canceled, applicable by the end of the month. You can revoke the cancellation of this contract if the customer wants to use it again.

i Note

This process has no process view. A confirmation is not required.

Extend

Use

Use this change process to extend subscription contracts. You can select predefined extension terms from the product master data or define your own extension terms by specifying the **Duration** and **Unit**.

Prerequisites

To use predefined extension terms, define the extension terms in the product master data first.

i Note

If you haven't defined the extension period in the product master data, the system displays an error message during auto extension.

Activities

If you define your own extension terms using **Other**, the system prompts you to enter a duration and unit. You can submit the change only after entering suitable values.

When you implement the **Extend** change process, the system sets the **Start of Contract Term** (ISTRUNTMBEG) in the change order to the **End of Contract Term** (ISTRUNTMEND) in the previous contract time slice. The **Contract Validity Period** (CONTDURA) is set to the selected extension term. The system determines the **End of Contract Term** according to the date rules and selected extension term.

i Note

If you extend phased contracts using the **Extend** change process, you must ensure that the last phase of the contract is active.

Technical Details

Process Name	Process Type	Process Class	Process View
EXTENSION	ISTA	CL_CRM_BTMF_TELCO_EXTENSION	PROVDLPR/ExtensionVS

To execute this change process using the **Subscription Contract Change Process (Asynchronous)** SOAP API, add the following parameters to the process container:

Parameter Name	Description
SubscriptionContract	Subscription contract number
SubscriptionContractItem	Subscription contract item number
SubscriptionContractChangeProcessType	Subscription contract change process type
SubscriptionContractActivationDate	Activation of the change process
DURATION	Duration by which the contract is extended
TIME_UNIT	For example, month and year

i Note

When the implement change process on a bundle product using the [Subscription Contract Change Process \(Asynchronous\)](#) SOAP API, the supplied extension period is applied to the main and subitems **only** if they have the same end date.

To execute this change process using the [Solution Quotation \(A2X\)](#) OData API, add the following parameters to the process container:

Parameter Name	Description
CAPProviderContract	Subscription contract number
SubscriptionContractItem	Subscription contract item number
ChangeProcessGroup	Subscription contract change process group

You can add or remove specific subitems from a subscription contract when you extend the contract using the [Solution Quotation \(A2X\)](#) OData API. Consider the following important points:

- All subitems in the change quotation that reference an existing contract item are extended.
- All subitems in the change quotation that reference the contract but don't refer to a contract item are added as new subitems on the contract.
- All subitems that exist in the contract but do not exist in the change quotation are not be extended.

Example

If a contract with an initial duration of 24 months is extended by 12 months, the dates are as follows:

- **Dates before extension:**

Start of Contract Term (ISTRUNTMPEG): 15.06.2015

End of Contract Term (ISTRUNTMEND): 15.06.2017

Contract Validity Period (CONTDURA): 24 months

- **Dates after extension:**

Start of Contract Term (ISTRUNTMPEG): 15.06.2017

End of Contract Term (ISTRUNTMEND): 15.06.2018

Contract Validity Period (CONTDURA): 12 months

Related Information

[APIs and Events for Subscription Order Management](#)

[Revoke Contract Extension](#)

[Phased Contracts](#)

Extend (Expired Contracts)

You can extend an expired subscription contract that is still within the grace period.

Use

You use this change process to extend a subscription contract that has crossed its end date (expired) but is still within the grace period.

Prerequisites

- The subscription contract must have already expired, that is, the contract end date must be in the past.
- The contract must be within the **Grace Period** (GRACE_PERIOD). If the **End of Grace Period** (GRACE_END) is in the past, the contract can no longer be extended.

In case a grace period is not provided, you can maintain it with the **Maintain Grace Period** change process before the contract end date is reached.

Activities

The change process for extension of expired contracts is based on the change process type ISTE. For more information about this, refer to **Process Types**.

Since the validity of the subscription contract extension begins right after the contract end date, the **Activation Date** of the change process is set to the initial contract end date by default. If you try to modify the **Start of Contract Time Slice** in the change order, it will revert back to the contract end date with a warning. The new subscription contract item will be created with status **Technically Active = Yes**.

For subscription contracts with main and subitems, the extension change process is only applicable to the main item. You cannot extend a subitem individually after it has expired. If subitems have the same contract end date as the main item, they will be extended along with the main item. If the subitems expire before the main item, they will not be extended.

You can also configure whether or not the subscription contract will be active during the grace period, using the **Actv Grace** flag in the Customizing **►Service ► Transactions ► Settings for Subscription Transactions ► Define Settings for Item Categories**. If this flag is checked, the subscription contract will remain Technically Active during the grace period. Here, the end date for provider contract in FICA = end of grace period. If the flag is unchecked, the subscription contract will be Technically Inactive during the grace period and no operations will be possible unless the contract is extended. In this case, the end date for provider contract in FICA = end of subscription contract.

		Contract is inactive during grace period	Contract is active during grace period
Subscription Contract	Contract Start Date	Jan 1st	Jan 1st
	Contract End Date	June 30th	June 30th
	End of Grace Period	July 15th	July 15th
FICA Provider Contract	Valid From	Jan 1st	Jan 1st
	Valid To	June 30th (mapped from CONTEND)	July 15th (mapped from GRACEEND)
	Term Start	Jan 1st	Jan 1st
	Term End	June 30th	June 30th
CI Billing Plan	Valid From	Jan 1st	Jan 1st
	Valid To	June 30th	July 15th

		Contract is inactive during grace period	Contract is active during grace period
	Term Start	Jan 1st	Jan 1st
	Term End	June 30th	June 30th (mapped from CONTENTD)
CC Contract	Valid From	Jan 1st	Jan 1st
	Valid To	June 30th	July 15th (mapped from GRACEEND)

Technical Details

Process Name	Process Type	Process Class	Process View
EXTEND_EXPIRED	ISTE	CL_CRMS4_BTMF_EXTEND_EXPIRE	PROVDLPR/ExtensionVS

To execute this change process using the **Solution Quotation (A2X)** OData API, you must create a change process group that includes the **EXTEND_EXPIRED** change process.

To execute using the **Subscription Contract Change Process** SOAP API, you must add the following parameters to the process container:

Parameter Name	Description	Value
DURATION	Duration by which the contract is extended	
TIME_UNIT	For example, month and year	

Related Information

[Process Types](#)

[Maintain Grace Period](#)

Revoke Contract Extension

Use

You can use this function to revoke the extension of a subscription contract.

Features

You can revoke a contract extension and reset the end of the contract term to the original date. You can only execute this process in the same change order in which the extension was executed. If the contract extension has been scheduled for the current date, it can only be revoked within the change order. Once the extension process has been submitted, it can only be revoked if the start date for the extension is in the future.

Technical Details

To revoke the extension, the date ISTRUNTMPEG and the contract duration CONTDURA are set to the values of the previous contract time slice.

Process Name	Process Type	Process Class	Process View
REVOKE_EXTENSION	ISTA	CL_CRM_BTMF_TELCO_EXTENSION_RV	None

No parameters are required when executing the change process using the [Subscription Contract Change Process](#) SOAP API, no parameters are required.

i Note

This process has no process view. A confirmation is not required.

Related Information

[Extend](#)

[APIs and Events for Subscription Order Management](#)

Cancel Subitem

Use

A subscription contract may contain several subitems, where each subitem can have a separate runtime and can be canceled individually. Typically, you can define a subitem as a dependent component.

Prerequisites

To use this function, the main item in a contract must be active. For example, if the option is a subscription contract item, the main item must be active and not canceled (also applies if the cancellation date is in the future).

Features

The following three types of cancellations exist, based on your entries in the [Requested Cancellation Date](#) and [Requested Cancellation Time](#):

- Immediate: If you enter the current date and time, the option is canceled immediately. You cannot revoke it.
- Deferred: If you enter a date or time in the future, the cancellation of selected options is deferred. You can revoke it if the requested cancellation date/time has not been reached.
- Backdated: If you enter a date or time in the past, the selected options are canceled immediately with the entered requested cancellation date. You cannot revoke it.

When the option is canceled, all subitems in this option are also canceled. The contract end date (appointment type CONTENTD, precisely timed) for the subitem is set. The requested cancellation time stamp and activation time stamp of the cancellation must be different.

Activities

You must enter the following parameters:

- Requested cancellation date and time
- Cancellation reasons

You must also enter the activation date for this change process. The default setting is that the current system time stamp is set for cancellation dates and the default cancellation procedure 'ISTC', cancellation party 'CNPR', and cancellation reasons 'K004' are used.

Technical Details

Process Name	Process Type	Process Class	Process View
CANCEL_OPTION	ISTA	CL_CRM_BTMF_TELCO_CANCEL_OPT	PROVDLPR/CancelOptionViewSet

To execute using the Subscription Contract Change Process SOAP API, you must add the following parameters to the process container:

Parameter Name	Description	Value
CANC_REASON	cancellation reason	from Customizing (default: K004)
CANC_PARTY	cancellation party	from Customizing (default: CNPR)
CANC_PROC	cancellation procedure	from Customizing (default: ISTC)
CANC_REQUEST_DATE	requested cancellation date	default: current system date
CANC_REQUEST_TIME	requested cancellation time	default: current system time
CANC_RECEIVE_DATE	received cancellation date	default: current system date
CANC_RECEIVE_TIME	received cancellation time	default: current system time

Example

You create one subscription contract with two rate plan items. The second item (subitem) is an option, such as pay TV charges. The customer no longer requires this option. If you cancel this option immediately, the current system time stamp is defined as the contract end date/time for this option.

The following variants are available:

- Item
- Dependent component

If an option is dependent on an item, the item is defined accordingly.

Related Information

[Cancellation](#)

Revoke Cancellation of Subitem

You use this change process to revoke the cancellation of a subscription contract item if the requested cancellation date is in the future.

Prerequisites

This change process is not applicable in an immediate cancellation scenario, that is, you can revoke the contract item cancellation only if the requested cancellation date is in the future.

Features

The following parameters in the subitem are deleted while revoking the cancellation:

- Cancellation date (CANCDATE)
- Requested cancellation date (CANCREQUEST)
- Received cancellation date (CANCRECEIVE)
- Contract end date (CONTEND)

The cancellation set is also deleted.

Technical Details

Process Name	Process Type	Process Class	Process View
REVOKE_CANCEL_OPTION	ISTA	CL_CRM_BTMF_TELCO_CANC_OPT_REV	None

To execute this change process using the [Subscription Contract Change Process](#) SOAP API, you must add the following parameters to the process container:

Parameter Name	Description
SubscriptionContract	Subscription contract number
SubscriptionContractItem	Subscription contract item number
SubscriptionContractChangeProcessType	Subscription contract change process type In this case, the value would be REVOKE_CANCEL_OPTION
SubscriptionContractActivationDate	Activation of the change process

To execute this change process using the [Solution Quotation \(A2X\)](#) OData API, you must add the following parameters to the process container:

Parameter Name	Description
CAPProviderContract	Subscription contract number
SubscriptionContractItem	Subscription contract item number
ChangeProcessGroup	Subscription contract change process group

Example

A subitem, for example Pay TV, is set to be canceled at the end of the month. You can revoke the cancellation of this subitem if the customer wants to use it again.

i Note

This process has no process view. A confirmation is not required.

Change Contract Account

Use

You can use this function to change the contract account for an active subscription contract by assigning a new contract account to the change order.

Activities

You can implement this process in the class CL_CRM_BTMF_TELCO_ASSIGN_BUAG.

i Note

You can assign only one contract account ID per item.

Technical Details

Process Name	Process Type	Process Class	Process View
ASSIGN_BUAG	ISTA	CL_CRM_BTMF_TELCO_ASSIGN_BUAG	PROVDLPR/AssignBuAgVS

To execute using the Subscription Contract Change Process SOAP API, the following parameters must be added to the process container:

Parameter Name	Description	Value
BUAG_ID	Contract account ID	Existing contract account

i Note

In the API, the contract account entered is assigned to the main item and the subitems under the main item. If you want to assign another contract account to the sub item, you must create your own new process class.

Related Information

[APIs and Events for Subscription Order Management](#)

Change Technical Resources

Use

You can use this change process to create and assign a new technical resource, or change/delete the existing technical resource for a subscription contract.

i Note

You can also modify the technical resource assignment in inactive subscription contracts. For more information about this, refer to **Change Technical Resources**.

Prerequisites

- You can only use this change process if the contract item uses a technical resource.
- You can update the technical resource in accordance with your Customizing settings for the technical resource schema. If, for example, the product only allows for one phone number, you cannot assign two numbers.

Activities

In the UI, you only need to enter the new technical resource ID.

If you are executing the change process via API, you must also enter the:

- Technical resource type
- Slot number
- Value/ID for technical resource

You must define the activation date to validate the start of the new technical resource ID. If billable items exist after the provided activation date, the change process is stopped, and an error message is displayed. You can only proceed with the change process if you change the activation date such that no billable items exist after it.

Technical Details

Process Name	Process Type	Process Class	Process View
CHANGE_TR	ISTA	CL_CRM_BTMF_TELCO_CHANGE_TR	PROVDLPR/TechResVS

To execute using the Subscription Contract Change Process API, you must enter the following parameters to the process container:

Parameter Name	Description
TR_TYPE	Technical resource type
TR_SLOT	Slot number
TR_OBJ_ID	ID of technical resource
TR_OBJ_KEY	Technical resource key

If you want to change several technical resources in one step, you must add numbers beginning with 1 as a suffix to all fields, such as TR_TYPE1. If you want to delete one existing technical resource, you must add all fields to the API. The field TR_OBJ_ID (ID of technical resource) must remain blank.

Example

You have created a subscription contract with two phone numbers, for example 33543 and 5678, but the customer wants to have specific numbers, such as 1234 and 2222. You can change the phone numbers using this process.

Parameter Name	Description	Value
TR_TYPE1	Technical resource type	PN (phone number)
TR_SLOT1	Slot number	0001
TR_OBJ_ID1	ID of technical resource	1234
TR_OBJ_KEY1	Technical resource key	
TR_TYPE2	Technical resource type	PN (phone number)
TR_SLOT2	Slot number	0002
TR_OBJ_ID2	ID of technical resource	2222
TR_OBJ_KEY2	Technical resource key	

Related Information

[APIs and Events for Subscription Order Management](#)

[Change Technical Resources](#)

Change Configuration

Use

You can use this contract change process to change the product configuration for a contract.

Prerequisites

This process is only available for products that can be configured using Variant Configuration Modeling Environment.

Features

Once you have started the configuration change process, the system creates a change order, and displays the following information in the process execution dialog box:

- The item tree for the change order, including the contract rate plan items and, if applicable, activation fees for the current process
 - The configuration status for configurable products is displayed in the **Configuration** column
- Activation date and time
 - You can plan the change for a future point in time
- An overview of all configuration attributes and their values
- Recurring charges and the change (in comparison to the current configuration)
- One-off charges

Activities

Choose the one-click configuration button or **Config** in the item details to navigate to the configuration maintenance UI. Change the configuration, then return to the process UI by choosing **Done**. The changed attributes and the price changes are now transferred to the process execution UI. Choose **Cancel** to exit the process without applying your changes to the characteristics.

Technical Details

Process Name	Process Type	Process Class	Process View
CONFIG_CHANGE	ISTA	CL_CRM_BTMF_TELCO_CONFIG_CHANG	PROVDLPR/ConfigurationVS

To execute using the **Subscription Contract Change Process** SOAP API, you must enter the following parameters in the process container:

Parameter Name	Description
CHARACTERISTIC	Name of the characteristic
CHARC_VALUE	Value of the characteristic
ASSIGNMENT_TYPE	Usage of the characteristic The value is blank for normal characteristics and VARCOND for reference characteristics for SD pricing
VARIANT_CONDITION_FACTOR	Factor used for SD pricing

To send multiple characteristics in one step or one characteristic with multiple values, you must add numbers beginning with 1 as a suffix to all fields, such as CHARACTERISTIC1.

All attributes transferred in the parameter container are set to the new value, regardless of their previous values. Those attributes not in the container will be deleted.

Example

You have created a subscription contract and want to update the configuration to the following values:

Characteristic	Value
BACKUP	'Y'
STORAGE_SPACE	10
VARCOND	VP_B_Y; VP_SP_10

The container must have the following parameters:

Parameter Name	Parameter Value
CHARACTERISTIC1	BACKUP
CHARC_VALUE1	Y
ASSIGNMENT_TYPE1	
VARIANT_CONDITION_FACTOR1	

Parameter Name	Parameter Value
CHARACTERISTIC2	STORAGE_SPACE
CHARC_VALUE2	10
ASSIGNMENT_TYPE2	
VARIANT_CONDITION_FACTOR2	
CHARACTERISTIC3	VARKOND
CHARC_VALUE3	VP_B_Y
ASSIGNMENT_TYPE3	VARCOND
VARIANT_CONDITION_FACTOR3	1
CHARACTERISTIC4	VARKOND
CHARC_VALUE4	VP_SP_10
ASSIGNMENT_TYPE4	VARCOND
VARIANT_CONDITION_FACTOR4	1

Upgrade Cross-Catalog Mapping Version

Use

You can change the cross-catalog mapping version of existing contracts, for example, you may need to upgrade if a new cross-catalog mapping version of a product is created that uses different charge plans.

An ISTF contract change process is performed for each selected contract and the subscription contract is updated accordingly. For more information, see **Contract Change Processes**. The updated contract is then replicated to FI-CA by performing a specific **Order Distribution Infrastructure** (ODI) step. For more information, see **Document Distribution Categories**.

This change process is only available if a new cross-catalog mapping version exists for the product and if the new version supports the same sharing feature, that is, if the old version supported shared allowances, the new version must also support shared allowances. Otherwise, an upgrade is not possible. If a CCM version is updated, redefined parameters and tables are taken over to the new version as long as the **Charge Plan Assignment ID** and **Parameter ID** are the same in both versions.

Features

Charge Plan Where-Used List

If a charge plan has been changed in the CC system, it might be required to know which products are using this charge plan. The report CRM_ISX_CHPLAN_WHERE_USED is available to select the products containing a specific charge plan. The charge plan ID and the status of the cross-catalog mapping version can be selected from the selection screen. All products with their cross-catalog mapping versions containing the selected charge plan are displayed in the results list.

Related Information

[Contract Change Processes](#)

[Document Distribution Step Categories](#)

Create Follow-Up Order

Use

You can use this function to create a new change order for contract changes that may not be possible with the existing change processes.

Prerequisites

No open change orders should exist while executing this change process.

Activities

Technical Details

Process Name	Process Type	Process Class	Process View
NEW_CHANGEORDER	ISTA	CL_CRM_BTMF_CONTENT_TELCO	

i Note

This process has no process view. A confirmation is not required.

Refill Allowance

Use

You can perform an allowance refill by executing the **Refill Allowance** change process on the subscription contract. You can use this process to trigger the creation of an allowance in **SAP Convergent Charging** (SAP CC) for the current subscription contract. A contract history record is stored when the process is executed.

i Note

You can also execute an allowance refill from the **Display Allowances** app, under the **Allowance Definitions** section.

Prerequisites

You must consider the following prerequisites during the change process for refill of allowances:

- This process can only be selected from the process list, when charge plans assigned in the subscription contract product contain allowances. You can see the process list under the assignment block **Processes**.
- This change process is available only if the manual refill option for the contained allowance definition is marked as **Yes**.

- The refill of allowances can only be processed if a list of available allowances has been maintained in Cross-Catalog Mapping for the corresponding product. You can do so by activating all allowance definitions of the allowance definition table assigned to a charge plan or by the assignment of an allowance definition group.

Features

Process Description

The process Refill of Allowances is implemented by class CL_CRM_BTMF_TELCO_ALLOW_REFILL and is dependent on the implementation of the allowance logic in SAP CC. It uses the following assumptions:

- The available allowance definitions are determined using the consumption date, which is the current time stamp.
- An activation date (start date of the newly created allowance) that is different from the consumption date can be maintained in the WebClient UI. Therefore, the chargeable item class in SAP CC needs a property ALLOW_START_DATE and must be considered when creating the allowance.
- There is no check of the activation date against the validity of the contract or the allowance definition. The activation date is defaulted to the consumption date but can be changed to a date either in the past or in the future.

If this process does not fulfill your requirements, you can either adjust it using the process enhancements (discussed below) or you can use this process as a template to create your own process.

Process Enhancements

The refill of allowances is triggered using the [SAP Convergent Charging](#) (SAP CC) web service `ChargeableItemChargingServices`. The operation that is used (`chargeableItemCharge`) contains user properties in its input payload. Initially, only the allowance activation date is transferred as a user property. To add more properties to the web service call you can enhance the process by executing the following steps:

1. Enhance the UI value node with the necessary parameters and transfer the parameters to the view `ISX_ALLOW_CREAT/AllowParameters`, context node `PARAMETERS`
2. Redefine the component controller of the UI component `ISX_ALLOW_CREAT` in order to fill the process context with the new fields in method `set_parameters_to_context`
3. The implementation for the `ALLOWANCE_REFILL` process (`CL_CRM_BTMF_TELCO_ALLOW_REFILL`) then takes all parameters found in the process context and adds them to the user properties section of the web service

Activities

Execute the [Refill Allowance](#) process on the [Processes](#) assignment block. The system shows a guided two-step procedure. In the first step, you select an allowance definition ID from a list of available allowances. Once you have selected an allowance, the details for this allowance definition are read from SAP CC and displayed in the selection table. You can also enter an activation date in order to activate the allowance at a later point in time.

If the selected allowance is available for more than one service provided in this contract, you select the technical resource ID of the corresponding service in the second step. When the allowance is available for a single service only, you can finish the process after the first step.

This list of available allowances depends on the assignment of allowance definitions in the Cross-Catalog Mapping of the contract's product and consists of the cumulative allowance definitions for all active charge plans of the contract. For each charge plan, the available allowances are determined as follows:

- If you have selected [No assignment](#), no allowances are available.

- If you have selected **Full Table**, the allowance definitions are taken from SAP CC by reading all entries from the corresponding allowance definition table.
- If you have selected an allowance definition group, all allowances in this group marked for manual refill in this group are available.

Technical Details

Process Name	Process Type	Process Class	Process View
ALLOWANCE_REFILL	ISTF	CL_CRM_BTMF_TELCO_ALLOW_REFILL	ISX_ALLOW_CREAT/MainWindow

To execute using the Subscription Contract Change Process SOAP API, you must add the following parameters to the process container:

Parameter Name	Description
ALLOWANCE_ID	Allowance ID of the allowance to be created.
ACTIVATION_DATE	Activation date (optional, Format YYYYMMDDHHMMSS)
CHARGE_PLAN_ID	Charge plan ID (optional)
TECH_RES_ID	Technical resource ID (optional)

If the parameter ACTIVATION_DATE is initial, the allowance is created at the current day and time.

i Note

Multiple assignments of the same charge plan are not considered.

Example

Example for One-Step Procedure

List of available allowances:

Allowance Definition ID	Charge Plan Assignment ID	Technical Resource ID	Technical Resource Type
B2C_MOBILE_ALLO_01	B2C_MOBILE_ALLO_USG_CP_01	12345	Phone Number
B2C_MOBILE_ALLO_02	B2C_MOBILE_ALLO_USG_CP_01	12345	Phone Number

The allowance definition IDs are unique. No second step is required.

Example for Two-Step Procedure

List of available allowances:

Allowance Definition ID	Charge Plan Assignment ID	Technical Resource ID	Technical Resource Type
B2C_MOBILE_ALLO_01	B2C_MOBILE_ALLO_USG_CP_01	12345	Phone Number
B2C_MOBILE_ALLO_01	B2C_MOBILE_ALLO_USG_CP_02	67890	Phone Number

The allowance definition IDs are not unique. In a second step you must select for which service ID the allowance should be created.

Change Quantity

You use this change process to change the quantity of the items and subitems within the subscription contract.

Use

This change process is available only if the item category is configured for item quantity functionality in the Customizing ► [Service ► Transactions ► Settings for Subscription Transactions ► Define Settings for Item Categories](#) ►.

i Note

- In case of a preconfigured product bundle, you can only change the quantity for the main item using this change process. The quantity for the subitems would be updated as per the product bundle configuration.

In case of manually added subitems, you can use this change process to update the quantity of the relevant subitems as well.

- If the item category supports object list, you can use the [Scale Up/Down Equipment](#) change process to increase/decrease the quantity as the [Change Quantity](#) change process is not supported in this case.

Activities

In the WebClient UI, you can enter the new quantity under the [Items](#) header.

In the API, enter the quantity using the [QUANTITY](#) parameter in the process container.

Technical Details

Process Name	Process Type	Process Class	Process View
CHANGE_QUANTITY	ISTA	CL_CRMS4_BTMF_CHANGE_QTY	PROVDLPR/ChangeQuantityVS

i Note

This process has no process view. A confirmation is not required.

Related Information

[Scale Up/Down Equipment](#)

Lock/Unlock

Use

You use this function to lock/unlock the subscription contract for a specific duration. Once you have started the process, you can select the lock/unlock reasons, the lock strategy, and set the relevant flags to set the status for the provider contracts in FI-CA and CC.

Prerequisites

Following are the prerequisites for this changes process:

- You have defined the lock/unlock reasons and set the lock flags in the Customizing for **Service** under ► **Transactions** ► **Settings for Subscription Transactions** ► **Contract Changes** ► **Lock Processing** ► **Define Lock/Unlock Reasons** ►.
- The ODI lock/unlock schema definition **PCLK** is activated in the schema determination in the Customizing for **Service** under ► **Transactions** ► **Settings for Subscription Transactions** ► **Document Distribution** ► **Define Settings for Document Distribution** ►.

Features

You can use the following flags to set the provider contract status in FI-CA and CC:

- A lock **Origin** to determine if the lock is set by a Financial system or by a Customer (for example, in case of loss of device or a planned vacation).
- **Lock Strategy** to decide whether the lock is set directly on the contract or via a change order.
- **Lock CC** flag, which sets the status of the provider contract in CC to locked when checked.
- **Lock FI-CA** flag, which maintains the mapping between subscription contract lock reasons and the following FI-CA lock reasons:
 - Invoicing lock reason to determine which provider contracts are excluded during invoicing in Convergent Invoicing.
 - Incoming and Outgoing Payments lock reason to determine which individual item is excluded from a settlement in the payment run. The locked items are added to an exception list, and you can also see the lock reason here.
 - Dunning lock reason to determine which individual item to exclude during a dunning run.
 - Billing Plan lock reason to determine if the billing plan is locked at a header level or an item level.

The lock can either be applied immediately or scheduled for future dates. Locks set for the future are processed using the batch program CRMS4_SOM_LOCK_LIST_PROCESSING.

Technical Details

Process Name	Process Type	Process Class	Process View
LOCK_UNLOCK	ISTC	CL_CRM_BTMF_TELCO_LOCK_UNLOCK	

To execute this change process using the **Subscription Contract Change Process (Asynchronous)** SOAP API, you must add the following parameters to the process container:

Parameter Name	Description	Expected Value	Necessity
LOCK_GUID	Lock GUID	Enter the value generated by the CRM_GUID_CREATE function module. In case no value is entered here, a GUID will be generated automatically. The LOCK_GUID must also be entered when unlocking a subscription contract.	Mandatory while locking and unlocking a subscription contract.

Parameter Name	Description	Expected Value	Necessity
LOCK_REASON	Lock reason	Relevant lock reason defined in the Customizing, Service Transactions Settings for Subscription Transactions Contract Changes Lock Processing Define Lock/Unlock Reasons	Mandatory while locking a subscription contract.
VALID_FROM	Lock timestamp	YYYY-MM-DDThh:mm:ss	Mandatory while locking and unlocking a subscription contract.
UNLOCK_REASON	Unlock reason	Relevant unlock reason defined in the Customizing, Service Transactions Settings for Subscription Transactions Contract Changes Lock Processing Define Lock/Unlock Reasons	Mandatory while locking a subscription contract for a predefined period or while unlocking a locked subscription contract.
VALID_TO	Unlock timestamp	YYYY-MM-DDThh:mm:ss	Mandatory while locking a subscription contract for a predefined period or while unlocking a locked subscription contract.

i Note

You can use the Subscription Contract Change Process SOAP API to create multiple locks on a subscription contract, as long as they are assigned unique LOCK_GUIDs, that must be added after the respective attribute name.

Example

Locking a Contract for a Predefined Period

You are going on a vacation for 6 months and would like to suspend your subscription:

Parameter Name	Description	Expected Value
LOCK_GUID	Lock GUID	A123BZ
LOCK_REASON	Lock reason	Vaca6M
VALID_FROM	Lock timestamp	2023-01-01T00:00:00
UNLOCK_REASON	Unlock reason	UL_Vaca6M
VALID_TO	Unlock timestamp	2023-06-30T23:59:59

Locking a Contract Indefinitely

You have misplaced your device and would like to suspend your subscription until it is found or replaced:

Parameter Name	Description	Expected Value
LOCK_GUID	Lock GUID	A123BC
LOCK_REASON	Lock reason	Indefinite
VALID_FROM	Lock timestamp	2023-01-01T00:00:00
UNLOCK_REASON	Unlock reason	
VALID_TO	Unlock timestamp	

Processing BITs During Contract Lock/Unlock

The following ODI steps are associated with the lock/unlock change process:

1. **PBVC** (Call FICA for BITs Reversal)
2. **PCEA** (Call FICA Message Contract Activation)
3. **PBRC** (Call FICA for BITs Rerequest)

The following scenarios discuss the relevance of these ODI steps:

Scenario	Example	Status of the ODI Steps
Subscription contract is locked via the Lock/Unlock change process	A contract is created on 1st January 2023 10:00:00. The BITs are requested on contract creation for the entire month. Now, consider that the contract is locked on 10th January 2023, 10:00:00.	1. PBVC (Call FICA for BITs Reversal): Executed successfully and the BIT is reversed for the entire month. 2. PCEA (Call FICA Message Contract Activation): Executed successfully. The provider contract and billing plan are locked. 3. PBRC (Call FICA for BITs Rerequest): Irrelevant because the billing plan is locked.
Subscription contract is unlocked via the Lock/Unlock change process	The same contract is now unlocked on 15th January 2023, 10:00:00.	1. PBVC (Call FICA for BITs Reversal): Irrelevant because no BITs exist as the contract was previously locked. 2. PCEA (Call FICA Message Contract Activation): Executed successfully. The provider contract is unlocked and a lock end date is set. Since the CI billing plan considers only the date of unlock and not the timestamp, the billing plan is considered to be locked until end of day (that is, 15th January 2023 23:59:59). 3. PBRC (Call FICA for BITs Rerequest): Irrelevant because the billing plan is considered locked

Scenario	Example	Status of the ODI Steps
		until end of day. Hence, the BITs cannot be re-requested until the day after the lock end date (16th January 00:00:00). Thus, you must manually re-request the BITs on the day after the lock end date.

Change Auto Renew Indicator

You can use this process to change the auto renewal indicator within the subscription contract. This process is available only if the value maintained on the product allows for change. The process updates the subscription contract directly without the creation of a change order. The change is relevant only to the subscription contract and no changes are done to downstream objects.

Activities

- In the UI, you can change the indicator and the extension period.
- In the API, enter the indicator and extension period values using the parameters [SUBSCRPNCONTRAUTORNWLISACTV](#), [SUBSCRPNCONTRITEMEXTNDURATION](#), and [SUBSCRPNCONTRITEMEXTNDURNUNIT](#) in the process container.

Technical Details

Process Name	Process Class	Process View
CHNG_AUTO_RENEW	CL_CRMS4_BTMF_CHNG_AUTO_RENEW	BT156_ISTF/usageARISTF.CRMS4_SOM_AR/MainWindow

Replace Equipment

You can use this process to change the equipment ID(s) within the subscription contract. This process is available only if the item category is marked as relevant for equipment link under [►Service ► Transactions ► Settings for Subscription Transactions ► Define Settings for Item Categories \(flag Obj List\) ►](#).

i Note

This process is only replacement of equipment ID's with no process behind it. Triggering of Return process in SD is not supported.

Activities

In the UI, you can change the list of equipment IDs in the Object List view.

This process can be executed via API only if there is mapping between the Equipment ID(s) and Technical Resources.

In the API, enter the equipment ID(s) as you would Technical Resources in the process container.

Technical Details

Process Name	Process Type	Process Class	Process View
REPLACE_EQUIPMENT	ISTA	CL_CRMS4_BTMF_REPLACE_EQUI	PROVDLPR/ChangeObjListVS

Scale Up/Down Equipment

You can use this process to increase or decrease quantity and correspondingly the equipment within the subscription contract. In the case of scale up, if the product bundle contains a relevant sales product, it will be added along with the delta quantity in the change order. This process is available only if the item category is marked as relevant for equipment using the flag **Obj List** in the Customizing [►Service►Transactions►Settings for Subscription Transactions►Define Settings for Item Categories►](#).

Activities

You can change the quantity and list of equipment IDs on the UI.

This change process can be executed via API only if there is mapping between the equipment IDs and technical resources. In the API, enter the equipment ID(s) as you would technical resources in the process container.

Technical Details

Process Name	Process Class	Process View
SCALE_EQUIPMENT	CL_CRMS4_BTMF_SCALE_EQUI	PROVDLPR/ ChangeQuantityVS PROVDLPR/ChangeObjListVS

Change Price

Use

You use this change process to update the manual/external prices of an active subscription contract via an API.

i Note

You cannot change variant prices using this change process.

Prerequisites

The subscription contract must be technically active, and a manual change of price is allowed for the respective condition type.

Activities

Technical Details

Process Name	Process Type	Process Class	Process View
CHANGE_PRICE	ISTA	CL_CRMS4_BTMF_CHNG_PRICE	PROVDLPR/ChangePriceVS

To execute this change process using the [Subscription Contract Change Process \(Asynchronous\)](#) SOAP API, you must add the following parameters to the process container:

Parameter Name	Description
KSCHL	Condition Type

Parameter Name	Description
KBETR	Condition Rate Value
WAERS	Condition Currency
KPEIN	Condition Quantity
KMEIN	Condition Quantity Unit

i Note

If you are executing this change process using the [Solution Quotation \(A2X\)](#) OData API, you must ensure that you pass values for all the manual pricing conditions present in the subscription contract. If you do not do so, the pricing conditions for which values are not provided on the change quote will be deleted from the change order and from the updated subscription contract as well.

Example

To change multiple prices in a single step, you can distinguish between them by adding serial numbers (starting with 1) as a prefix to all the fields. For example, KSCHL1 and KSCHL2.

Change Sold-To Party

Prerequisites

The following are the prerequisites for this change process:

- Active subscription contracts exist in the subscription contract document.
- No open change order exists.
- Document flow strategy is 'N' (New contract in new contract document)

Use

You can use this change process to change the sold-to party of an existing contract document. This is a header change process (BTMF process type ISTA) that will create new contracts in a new subscription contract document. The contract information is copied over from the corresponding old contracts, which are then terminated in the old subscription contract document. Currently, this change process is only supported from the UI. After you select a new product, the master agreement is validated against it in the change order. In case of an error, you must either remove the existing master agreement or manually assign a relevant one in the [Subscription-Specific Data](#) assignment block in item details.

You can implement this change process on multiple subscription contracts at the same time using either mass runs or mass processing runs. You can also change the related organizational data if you are using this change process in a mass processing run.

i Note

If you are using allowances, the ODI schema should also include the step type ALCT (Transfer Allowance and Counter), which must be configured as a subsequent step to the PCEA step.

Activities

Technical Details

Process Name	Process Type	Process Class	Process View
CHANGE_SOLDTO_PARTY	ISTA	CL_CRMS4_BTMF_SOLD_TO_PARTY	CRMS4_SOM_HDRCP/SoldToPartyView

Related Information

[Contract Transfer](#)

Add New Contract

Use

You use this header change process (BTMF process type ISTG) to add a new main item (a new contract) to an existing contract document.

Once the change process is executed, you are redirected to a new change order without any items. Here, you can enter details such as the product, contract account, and dates, to create a contract in an existing contract document.

Prerequisites

The following are the prerequisites for this change process:

- No open change order exists.

Activities

Technical Details

Process Name	Process Type	Process Class	Process View
ADD_NEW_CONTRACT	ISTG	CL_CRMS4_BTMF_ADD_CONTRACT	

For executing the change process using the Business Solution Quotation API (via the Change Process Group), you must add the following parameters to the process container:

Parameter Name	Description
ChangeProcessGroup	Change process group
SubscriptionContractReference	Reference to the contract document where the contract is to be added

Example

You wish to add a related product as a main item and not a subitem into a contract document.

Add Subitem

Activities

Technical Details

Process Name	Process Type	Process Class	Process View
ADD_SUBITEM	ISTA	CL_CRMS4_BTMF_ADD_SUBITEM	

You can execute this change process using the [Business Solution Quotation](#) API. The change quotation must contain the original main item (with the change process group) on which a subitem can be added.

Withdraw

You use this change process to withdraw from a contract within the withdrawal period, with the possibility of a refund on the original amount.

Prerequisites

- The subscription contract must be within the [Withdrawal Period](#) (CW_PERIOD).
- You cannot execute a future dated withdrawal on a subscription contract.

Activities

The [Withdraw](#) change process enables you to withdraw from a subscription contract as long as the end of withdrawal period is in the future. The date rule for the end of withdrawal period (WITHDRAW_END), date type [End of Withdrawal Period](#) (WITHDRAW_END), and the duration [Withdrawal Period](#) (CW_PERIOD) are defined in the Customizing [Service > Basic Functions > Date Management > Define Date Types, Duration Types and Date Rules for Service Transactions](#). These are then added to the date profiles [PROVITEM](#) and [PROV_QUTIM](#) in the Customizing [Define Date Profile for Service Transactions](#). The end of withdrawal period is calculated as the contract start date + withdrawal period. In case of a contract with main and subitems, the end of withdrawal period is only set on the main item as you can only execute withdrawal on the main item. The subitems are withdrawn automatically along with the main item.

Depending on when during the withdrawal period you have withdrawn from the contract, you may be eligible for a refund on the original amount. To determine the refund code, you use the [Withdrawal Reason](#) and the time periods based on the number of days from contract start when the withdrawal was executed. If the refund determination criteria is anything other than the time period, you must maintain a refund code determination class for which you must implement the interface [IF_CRMS4_SOM_REFUND_DETERM](#). If a determination class is maintained, the refund code is determined based on the [Withdrawal Reason](#) and the refund code determination class. Depending on the refund code, pricing determines a refund condition record that calculates the refund amount as a percentage of the total contract value. This then becomes the amount for the new contract item.

When you execute the change process, the refund amount is automatically determined based on the withdrawal reason and the refund code. However, after selecting a withdrawal reason, you may also enter a refund amount manually. In this case, the manually entered refund amount will be considered for price calculation of the new contract item.

The refund pricing condition type (PSRF) is a one-time discount/surcharge price whose condition base value is the sum of the net value of the contract items (calculated by the pricing routine 30). This relationship is defined in Customizing under [Service > Transactions > Settings for Subscription Transactions > Contract Changes > Withdrawal > Maintain Withdrawal Reasons and Refund Codes](#).

If you withdraw a subscription contract with future dated billing schedule lines (BSLs), the refund amount is calculated based on the total billing value of those BSLs that have their billing date before the withdrawal date. Once you execute the change process, in the change order, go to the **Billing Schedule** tab in the item details and ensure that the settlement start and end dates are the same as the activation date of the change process. Once contract withdrawal is triggered, an item is created with a one second duration where the refund amount is calculated. You can see the estimated refund amount (condition type PSRF) in the **Price Details** tab in the BSL details. The net value of the subscription contract is adjusted based on the refund amount.

i Note

- You can change the withdrawal reason in the change order under the **Withdrawal** tab at item level details. Based on the selected withdrawal, the refund code and estimated refund amount are determined from the Customizing and pricing.
- You cannot withdraw a subscription contract with an adhoc billing schedule.

❖ Example

Contract Start Date = 1st January

Withdrawal Period = 30 days

End of Withdrawal Period = 30th January

If date of withdrawal = 15 January, number of days from contract start when contract is withdrawn = 15

Keeping the above information in mind, consider a contract with a fixed price of \$1000 (item 10), on which a **Withdraw** is executed and a 75% refund of total contract value (- \$750 for item 20) is provided. The estimated total contract value will be \$1000 - \$750 = \$250. The CI billing plan is adjusted to process this refund.

Technical Details

Process Name	Process Type	Process Class	Process View
WITHDRAW	ISTA	CL_CRMS4_BTMF_CONT_WITHDRAWAL	PROVDLPR/WithdrawalVS

To execute using the **Subscription Contract Change Process** SOAP API, you must add the following parameters to the process container:

Parameter Name	Description
WITHDRAWAL_REASON	Withdrawal reason

Maintain Grace Period

You can maintain a grace period on an active subscription contract.

Use

You use this change process to maintain a grace period on an active subscription contract, which does not have a grace period initially.

Prerequisites

This change process is applicable only for the main item of a subscription contract. If subitem(s) in the subscription contract have the same contract end date as of the main item, they will inherit the grace period from the main item.

Activities

The date rule for the end of grace period (GRACE_END_RL), date type **End of Grace Period** (GRACE_END), and the duration **Grace Period** (GRACE_PERIOD) are defined in the Customizing **Service > Basic Functions > Date Management > Define Date Types, Duration Types and Date Rules for Service Transactions**. These are then added to the date profile **PROVITEM** and **PROV_QUTITM** in the Customizing **Define Date Profile for Service Transactions**. The end of grace period is calculated as the contract end date + grace period. In case of a contract with main and subitems, the end of grace period is only set on the main item as you can only extend the main item.

An **Actv Grace** flag is available in the Customizing **Service > Transactions > Settings for Subscription Transactions > Define Settings for Item Categories**. If this flag is checked, the subscription contract will remain active during the grace period. Here, the end date for provider contract in FI-CA = end of grace period. If the flag is unchecked, the subscription contract will be Technically Inactive, and no operations will be possible unless the contract is extended.

Technical Details

Process Name	Process Type	Process Class	Process View
MAINTAIN_GRACE_PERIOD	ISTF	CL_CRMS4_BTMF_ADD_GRACE_PERIOD	BT156_ISTF/CUAddGrace.CRMS4_SOM_GRACE/Ma

Related Information

[Extend \(Expired Contracts\)](#)

Change Billing Cycle

You can change the billing cycle in an active subscription contract.

Use

You use this change process to change the billing cycle in a subscription contract where the Convergent Invoicing billing plan is used for recurring charges. You can change the billing cycle any time during the billing period.

The billing cycle for a subscription item is determined based on one of the following:

- the Billing cycle maintained on the product

If a default billing cycle is maintained on the product, you cannot change this either in the initial order/quote or via a contract change process.

- the Billing Cycle Determination Rule ID maintained on the product

If a default billing cycle determination rule ID is maintained on the product, you can only change to a billing cycle that is associated with this determination rule ID.

- the default rule maintained in the Customizing **Service > Transactions > Settings for Subscription Transactions > Recurring Fees > Configure Settings using Billing Plans with Convergent Invoicing > Define Billing Cycle Determination Rules**.

The default Billing Cycle Determination Rule is considered when neither a billing cycle nor a billing cycle determination rule ID is associated with the selected product. This billing cycle can be changed, only if the determination rule is configured to allow this change.

Prerequisites

You can configure the baseline date to determine when the billing items will be generated during the billing cycle period in the Customizing [Contract Accounts Receivable and Payable](#) [Convergent Invoicing](#) [Enhanced Functions](#) [Billing Plans](#) [Define Billing Plan Item Types](#).

Activities

i Note

The [Change Billing Cycle](#) is only enabled for products with a CI billing plan.

Technical Details

Process Name	Process Type	Process Class	Process View	Text
CHANGE_BILL_CYCLE	ISTA	CL_CRMS4_BTMF_CHANGE_BILLCYCLE	PROVDLPR/ChangeBillCycleVS	Change Billing Cycle

To execute this change process using the Subscription Contract Change Process API, you must enter the following parameters in the process container:

Parameter Name	Description
BILLCYCLE	New billing cycle

Example

Consider a subscription contract with a monthly billing cycle where you are billed at the first of every month. After being billed for two months (say, 1st January 2023 and 1st February 2023), you switch to an annual billing cycle using the [Change Billing Cycle](#) change process. Once executed, you will be billed for an entire year on 1st March 2023 and the next bill will be generated on 1st March 2024.

Change Business Validity

Use

You use this change process to modify the business validity of a subscription contract as long as the start of business is in the future.

Prerequisites

The updated start of business must be in the future and after the contract start date.

Activities

Technical Details

Process Name	Process Type	Process Class	Process View
CHNG_BUSINESS_VAL	ISTF	CL_CRMS4_BTMF_CHNG_BUSI_DATES	BT156_ISTF/CUBusinessDates.CRMS4_SOM_BUSDT/Main

Example

A subscription contract is created such that the start of business is not yet confirmed but an active contract is required for collecting payments.

In this example, consider a student who registers for a course on January 1st and pays the entire fee at the time of registration. The course is scheduled to start on February 1st. The dates would be as below:

- **Technical Validity** = Contract Start (01.01.2022 08:00:00)
- **Billing Validity** = Start of Billing (01.01.2022 08:00:00)
- **Business Validity** = Start of Business (01.02.2022 08:00:00) to End of Business (01.02.2023 08:00:00)

The start of the course, however, is delayed by 10 days and you use the **Change Business Validity** change process to move start of business from February 1st to February 10th. The business validity would thus be updated as Start of Business (10.02.2022 08:00:00) to End of Business (10.02.2023 08:00:00).

Change WBS Element

You can change the account assignment object (WBS element) on an active subscription contract.

Use

You use this change process to either assign a new WBS element or replace an existing WBS element on an active subscription contract.

Activities

Technical Details

Process Name	Process Type	Process Class	Process View	Text
CHANGE_WBS_ELEMENT	ISTA	CL_CRMS4_BTMF_CHANGE_WBS_ELE	PROVDLPR/ChangeAcAssignVS	Change WBS Element

To execute this change process using the Subscription Contract Change Process API, you must enter the following parameters in the process container:

Parameter Name	Description
WBS_ELEMENT	ID of the new WBS element

To execute this change process using the Solution Quotation API (via the change process group), you must enter the following parameters in the process container:

Parameter Name	Description
ChangeProcessGroup	Change process group
WBSElementExternalID	ID of the new WBS element

Reactivate

You use this change process to **Reactivate** a canceled subscription contract as long as it is within the contract grace period.

If your subscription contract contains product bundles, then the reactivation is applicable for the main item and all valid subitems. That is, on reactivating the main item, all the valid subitems are also reactivated. Those subitems that had already expired at the time of cancellation will not be reactivated. You can also reactivate a subitem individually after its cancellation. However, you can do so only if the main item is still active.

While executing the reactivation, the start of contract start timeslice for all the reactivated items is set to end of contract timeslice for the previous item. The cancellation parameters for the reactivated items are deleted during the reactivation process.

Activities

Technical Details

Process Name	Process Type	Process Class	Process View
REACTIVATE_CONTRACT	ISTE	CL_CRM_BTMF_REACTIVT_CANC_CONT	None

To execute this change process using the **Subscription Contract Change Process** SOAP API, you must enter the following parameters in the process container:

Parameter Name	Description
SubscriptionContract	Subscription contract number
SubscriptionContractItem	Subscription contract item number
SubscriptionContractChangeProcessType	Subscription contract change process type In this case the value would be REACTIVATE_CONTRACT
SubscriptionContractActivationDate	Activation of the change process

To execute this change process using the **Solution Quotation (A2X)** OData API, you must enter the following parameters in the process container:

Parameter Name	Description
CAPProviderContract	Contract number
ChangeProcessGroup	Change process group
SubscriptionContractItem	Contract item number

Constraints

The **Reactivate** change process has the following constraints:

- You cannot reactivate subscription contracts that have a multi-level configurable product with a bill of materials (BOM).
- You cannot reactivate phased contracts.

Update Price

Use

You use this change process to trigger the repricing of subscription contracts within a mass processing based on the master data pricing. The **Update Price** change process is available only while creating a mass processing using either the **Mass Changes of Subscriptions** or the **Manage Mass Processing of Subscriptions** Fiori apps.

i Note

You cannot change variant prices using this change process.

Prerequisites

The subscription contract must be technically active and a manual change of price is allowed for the respective condition type.

Activities

Technical Details

Process Name	Process Type	Process Class	Process View
UPDATE_PRICE	ISTA	CL_CRMS4_BTMF_UPDATE_PRICE	None

To execute this change process using the **Subscription Contract Change Process** SOAP API, you must enter the following parameters in the process container:

Parameter Name	Description	Value
REDETERMINE_PRICING_TYPE	This parameter in the Complete Reprice field determines whether a repricing or a complete repricing is triggered when the change process is implemented.	If the value is set to B = Complete Repricing, including manual condition values (Carry Out New Pricing) If the value is set to C = Repricing, excluding manual condition values (Copy Manual Pricing Elements and Redetermine the Others)

Change Billing Schedule

You use the **Change Billing Schedule** change process to implement changes on a subscription contract with periodic or adhoc billing schedule lines (BSLs).

In case of a contract with an adhoc billing schedule, all BSLs with billing date before the activation date of the change are retained on the previous contract item and all BSLs with billing date after the activation date of the change are copied over to the change order with the status **Brought Forward**. In the new contract item, the total billing value of these future dated BSLs is added as a surcharge with the pricing condition VRM1 (**Remaining Value**). In the previous contract item, the status of these future dated BSLs is set to **Carried Forward** and their total billing value is added as a negative surcharge to ensure that the contract net value remains consistent.

Activities

Technical Details

Process Name	Process Type	Process Class	Process View
CHANGE_BILL_SCHD	ISTA	CL_CRMS4_BTMF_CHANGE_BILL_SCHD	None

To execute this change process using the **Solution Quotation (A2X)** OData API, you must add the following parameters to the process container:

Parameter Name	Description	Comments
BillgReqItemDescription	Description	
BillgReqItemBillingDateTime	Billing date	For subscription contracts with an adhoc billing schedule
BillgReqItemNetAmount	Net value	
SettlementStartDateTime	Settlement start date	For subscription contracts with a periodic billing schedule
SettlementEndDateTime	Settlement end date	

Change CC Data

You use this change process to modify SAP Convergent Charging (SAP CC) parameters (table, string, date, or numeric) in Subscription Order Management.

Use

The **Change CC Data** change process can only be executed using the API scenario and enables you to support usage-based scenarios in Subscription Order Management. You can do the following changes with this change process:

- Change only the cross-catalog mapping version
- Change SAP CC parameters using the existing cross-catalog mapping version
- Change SAP CC parameters using a new cross-catalog mapping version

i Note

If you execute this change process using the **Solution Quotation (A2X)** OData API, then all the SAP CC parameters in the change quotation that are different from the parameters in the subscription contract will be updated. For example, if a parameter is redefined in the contract and the default value is used for this parameter in the change quotation then the value of this parameter will be reset to its default value. If you want to retain any redefined values for the SAP CC parameters, you must provide these values in the change quotation.

Activities

Technical Details

Process Name	Process Type	Process Class	Process View
CHANGE_CC_DATA	ISTA	CL_CRMS4_BTMF_CC_DATA	None

To execute this change process using the [Solution Quotation \(A2X\)](#) and the [Subscription Contract Change Process](#) OData APIs, you can add the following parameters to the process container:

Parameter Name	Description
CCM_VERSION	Here, you provide the cross-catalog mapping version (CCM version) that is currently in use. If no data is provided then it is assumed that the current CCM version is still valid.
KEEP_REDEFINITION	This attribute is required only if a CCM_VERSION value has been provided and the redefined parameters should be retained. That is, if the CCM version is changed and the parameter ID remains the same between the old and new versions, you can decide whether the existing redefined parameters should be retained by providing an 'X'.
The following attributes always belong together and represent the change of a single parameter. - CC_PAR_ID: SAP Convergent Charging (SAP CC) parameter ID - CC_ASSI_GUID: Charge plan assignment GUID - CC_VAL_STA: Type of parameter value (such as Default, Redefined, or BRF+) - CC_PAR_VALUE: New parameter value	For each changed parameter, all the four values must be provided. To know which attributes belong together, a suffix with a consecutive number must be provided.

Example

The following example will help clarify the container attributes and their values:

Name	Value	Comment
CCM_VERSION	0003	New CCM version number
KEEP_REDEFINITION	X	Retain the redefined values from the previous CCM version
CC_PAR_ID - 1	Contract_Date	Date Parameter ID
CC_ASSI_GUID - 1	B7036E6334A61EEE94F5111A67EE022A	Assignment UUID
CC_VAL_STA - 1	2	Use redefined value
CC_PAR_VALUE - 1	20250101	Redefined value

Name	Value	Comment
CC_PAR_ID - 2	Item_Quantity	Numeric parameter ID
CC_ASSI_GUID - 2	B7036E6334A61EEE94F511A67EE022A	Assignment UUID
CC_VAL_STA - 2	7	Use BRF+ function
CC_PAR_VALUE - 2	FUNC_NAME_ITEM_QTY	BRF+ function name

Repair

Use

You use this change process to fix errors in inactive subscription contracts by creating a repair order that acts as a preceding document to the contract.

Prerequisites

The [Repair](#) change process is available only if the following prerequisites are met:

- Contract is currently valid (that is, the current date is after the start date but before the end date) but inactive.
- The preceding document does not exist. For example, in migration scenarios when only the subscription contract is migrated and the preceding document is not.
- Contract distribution has either not begun yet or has ended in error.

Activities

Technical Details

Process Name	Process Type	Process Class	Process View
CREATE_REPAIR_ORDER	ISTG	CL_CRMS4_BTMF_REPAIR_ORDER	

Request Change

Use

This change process is used to make changes to a subscription contract that is created from a solution quotation and is not yet active. The item number on which the change process is executed remains unchanged even after the changes are implemented.

If [Request Change](#) was the last executed change process, executing it again will lead you to the already open change order where you can make the required changes.

Prerequisites

The following are the prerequisites for this change process:

- Contract is inactive
- Preceding document (solution quotation) exists but is completed
- Distribution of the contract has not started (activation date is in the future)

Activities

Technical Details

Process Name	Process Type	Process Class	Process View
CUST_CHNG_REQUEST	ISTG	CL_CRMS4_BTMF_CHNG_REQUEST	

Change Technical Resources

Use

You use this change process to create and assign a new technical resource, or change/delete the existing technical resource for an inactive subscription contract.

i Note

This change process is used for inactive subscription contracts without reference to a subscription order. For example, if the subscription contract is created using a solution quotation. In all other scenarios, the [Change Technical Resources](#) (CHANGE_TR) change process is applicable.

Prerequisites

The following are the prerequisites for this change process:

- Contract is inactive
- Preceding document exists but is completed (solution quotation scenario)
- Distribution of the contract has not started (activation date is in the future)
- The contract item must use a technical resource
- The technical resource assignment can be modified as per the Customizing settings for the technical resource schema. For example, if the product allows only a single phone number, you cannot assign more than one number.

Activities

In the UI, you only need to enter the new technical resource ID.

If you are executing the change process using an API, you must also provide the:

- Type
- Slot number
- Value/ID for technical resource

Technical Details

Process Name	Process Type	Process Class	Process View
INACTIVE_CHANGE_TR	ISTH	CL_CRMS4_BTMF_INAC_TR_CHANGE	BT156_ISTH/CUISXTR.ISXTR/MainWindow

To change the technical resource using the Subscription Contract Change Process SOAP API, you must enter the following parameters in the process container:

Parameter Name	Description
TR_TYPE	Technical resource type
TR_SLOT	Slot number
TR_OBJ_ID	ID of technical resource
TR_OBJ_KEY	Technical resource key

If you want to change several technical resources in one step, you must add numbers with 1 as a suffix to all fields, such as TR_TYPE1. If you want to delete one existing technical resource, you must add all fields to the API. The field TR_OBJ_ID (ID of technical resource) must remain blank.

Related Information

[Change Technical Resources](#)

Correct Errors

Use

This change process is used to make changes in a subscription contract when the preceding document is complete, and the distribution has ended in error.

Prerequisites

The following are the prerequisites for this change process:

- Contract is inactive
- Preceding document exists but is completed (solution quotation scenario)
- Distribution of the contract has ended in error

Activities

Technical Details

Process Name	Process Type	Process Class	Process View
GEN_ERROR_CORR	ISTG	CL_CRMS4_BTMF_GEN_ERR_CORR	

Reject Items

Use

You use the following change processes to reject future dated, technically inactive items in a subscription contract:

- **Reject All Items** is a header change process that is available **only** as a button on the footer of the UI. You use this change process to reject all the items within a subscription contract document and set the status of the contract document is set to 'Completed'.
- **Reject Item** is an item level change process that is available as a button in the UI footer and a change process in the **Processes** assignment block. You use this change process to reject a selected item. In a product bundle scenario, you can only reject manually added subitems.

Once the change process is completed, the status of the rejected items is set to **Rejected** in Subscription Order Management. Also, the rejection date and time is updated in the **Withdrawn On** time stamp for the provider contract in FI-CA.

If you've assigned transaction-specific rejection reasons using the rejection profile, then only these transaction-specific reasons are displayed while rejecting a subscription order or a subscription contract. This means that if you're rejecting the items in a subscription order or a subscription contract and a rejection profile is assigned to the transaction type, then you can choose only from the profile-specific rejection reasons. If a rejection profile isn't assigned, then you can choose from all the rejection reasons available in the system. You can make the relevant settings for your rejection reasons in the following Customizing activities:

- **Define Rejection Reasons**
- **Assign Rejection Reasons to Rejection Profiles**
- **Assign Rejection Profiles to Service Transactions** where you can assign the rejection profile to a transaction type or a transaction type and a sales organization

under **►Service ► Transactions ► Settings for Service Transactions ► Settings for Rejection Reasons for Service Transactions ►**.

Prerequisites

The following are the prerequisites for this change process:

- The preceding document is complete.
- The item to be rejected must be future dated and inactive.
- If all items are to be rejected at the contract document level, they must all be future dated and inactive.

Activities

Technical Details

Process Name	Process Type	Process Class
REJECT	ISTH	CL_CRMS4_BTMF_REJECT
REJECT_ALL	ISTH	CL_CRMS4_BTMF_REJECT_ALL

Processing Billable Items During Contract Changes

You can execute certain change processes on a subscription contract either with an activation date in the past or with advance payment. Any billable items that may have already been generated, billed, and invoiced for the relevant contract, may block this contract change. Convergent Invoicing provides the ability to reverse billable items, even if they are billed or invoiced, to allow the change process to execute successfully. Further, it is possible to automatically reverse the billable items that are valid for the period of change, make changes to subscription contract, and re-request BITs.

For example, if a subscription contract is valid for 1 year and a change is requested after 6 months, the billable item for the whole year will be reversed. They are then requested again for the period before and after change, based on the change process.

Prerequisites

- Reversing and recreating billable items from billing plan can be done automatically, if the ODI steps (PBRC and PBVC) are set up accordingly in the Customizing, [▶Service ▶ Transactions ▶ Settings for Subscription Transactions ▶ Document Distribution ▶ Define Settings for Document Distribution ▶](#).
- Customizing for the reversal of billable items must be maintained here, [▶Contract Accounts Receivable and Payable ▶ Convergent Invoicing ▶ Basic Functions ▶ Billable Items ▶ Reversal of Billable Items ▶](#).

You must ensure that the ODI steps for reversal and re-requesting must be part of active schema:

- New ODI steps of category P1: PBRC and PBVC
- New ODI steps of category P6: PBRD and PBVD

i Note

This feature is not supported with Recurring Charges in SAP Convergent Charging

Contract Transfer

Use

Contract transfer enables customers to choose how their requested contract changes are updated. The possible options are:

- Updating an existing contract in an existing contract document
- Updating a new contract in an existing contract document
- Updating a new contract in a new contract document

This feature is supported by the document flow strategy, which is a BTMF enhancement that helps customers trigger changes to multiple contracts within a single subscription contract document, either via header level change processes (like the [Change Sold-to Party](#) change process) or item level change processes (like the [Change Product](#) change process).

Prerequisites

You must set the document flow strategy in the scenario-specific attributes of change process. You can do so in the Customizing for [Service](#) under [▶Transactions ▶ Settings for Subscription Transactions ▶ Contract Changes ▶ Settings for the Maintenance Framework \(BTMF\) ▶](#).

Possible values for document flow strategy:

- ' ' (space) - Update original contract in original contract document

- N - Create new contract in new contract document
- U - Create new contract in original contract document

Description

During the execution of certain change processes (process type ISTA), business or legal requirements are met by transferring data from an old subscription contract into a new contract. The requested change data is adapted either in the existing contract or in a new contract, in which case the old contract is then terminated. The behavior of the change process is determined by the document flow strategy.

Restrictions

The following are the restrictions for the document flow strategy:

- If you are executing the **Change Sold-to Party** change process, you must ensure that there is no open change order.

Related Information

[Maintenance Framework for Business Transactions](#)

[Change Sold-To Party](#)

[Change Product](#)

Document Distribution

Document distribution of subscription contract documents is critical in fulfilling the terms of subscription orders and subscription contracts across multiple systems for billing, charging, and other fulfillment requirements.

Use

The document distribution of subscription documents is implemented by the **Order Distribution Infrastructure** (ODI) framework and follows the basic process outlined below:

1. Create a subscription order or solution quotation.
2. A subscription contract is created when the subscription order is completed or submitted, or the solution quotation is accepted. The distribution for this subscription contract is triggered
3. Once the subscription order is successfully distributed, the subscription contract is created.
4. Distribution of the subscription contract is triggered when the subscription contract is released (which in most scenarios is done automatically).
5. Once the subscription contract distribution is successful, the subscription contract status is set to **Active**. In case of contract termination, the deactivation is also a part of the distribution.

You cannot make any changes to the subscription contract while the document distribution is in progress. Once the distribution is completed, control is passed back to the contract document. Depending on the final status of the distribution, you can either execute a change process on an active contract, or implement data correction for any distribution errors. Submitting the change order triggers the document distribution again.

Features

- Scheduling, which allows fast processing for the end user and postpones the actual execution of the distribution to a delayed batch process
- Storing the result of an executed action in case of a redistribution
- Definition of complex distribution models with branching and merging options
- Error handling during the distribution process

ODI Framework

Use

The **Order Distribution Infrastructure** (ODI) framework offers a flexible way to set up a document-specific, customizable distribution that can be influenced by custom exits and scheduling mechanisms. It provides a layer in which you can define the actions that need to be taken in which order to fulfill or deliver services according to the subscription order and subscription contract.

The main elements of the framework are:

- Document Distribution Categories
- Document Distribution Step Type
- Document Distribution Schema
- Document Distribution Schema Determination

The behavior of each element is controlled by an implementation class which can be replaced by a customer-specific implementation class. However, we recommend using the standard implementation classes for distribution categories and distribution schema.

More Information

For more information, especially regarding the implementation of classes for the elements, see Customizing for **Service** under **Transactions** > **Settings for Subscription Transactions** > **Document Distribution** > **Define Settings for Document Distribution**.

Further Hints

The usage of the document distribution in the ODI is only possible if no ISTB (change technical objects) processes are used, as ISTB processes are not supported. The same applies for ISTC (set, reset, or remove locks) processes where lock fees are included.

Related Information

[Document Distribution Step Categories](#)

[Document Distribution Step Type](#)

[Document Distribution Schema](#)

[Document Distribution Schema Determination](#)

Document Distribution Step Categories

A step category represents a phase in the document distribution process, for example contract activation. Document distribution step categories are a central element of the **Order Distribution Infrastructure** (ODI). Even if additional step categories can be

defined in the Customizing, there are six step categories that are mandatory for use with subscription documents.

Step Category	Description
O1	Order distribution
P0	Subscription contract waiting for activation
P1	Subscription contract activation
P5	Subscription contract waiting for deactivation
P6	Subscription contract deactivation
C1	Subscription contract update

Step category O1 is used for the activation of subscription orders whereas P0 and P1 are used for the activation, and P5 and P6 are used for the deactivation of the subscription contract. The deactivation is only scheduled if the contract end date has already been set.

Each document distribution schema used with subscription contracts needs steps with the categories P0, P1, P5, and P6 to ensure that status handling works correctly. It is not mandatory to have document distribution schema for subscription orders.

Step category C1 is only used to update the contract (change contract without using a change order).

Document Distribution Step Type

A document distribution step type defines a single distribution step within document distribution. It represents a single distribution function, for example sending a notification message to an external system or triggering an action in a fulfillment system.

Each document distribution step type has to be assigned to a document distribution category. Documents distribution steps can depend on various criteria, such as:

- The executed process (sales, product change, cancellation, change of technical details)
- The type of service (for example, GSM, data, IP)
- The result of other steps (for example, only execute step 2 if step 1 is successful)
- The requested activation dates

The step type can be defined as an asynchronous distribution step type. This setting prevents the direct execution of the distribution step. The distribution step is registered, the distribution status is **Open**, and the distribution step is executed with the next batch run. For more information, see **Batch Report**.

The implementation class at document distribution step type level is used to implement the actual distribution step execution logic. For each new step type, an implementation class must be created. The class must implement interface `IF_CRM_ISX_ORDER_MSG_DIST_TYPE`. The methods of this interface are called at runtime from the framework and allowed to react at a dedicated point of time to influence the execution of the distribution step.

The main methods to be implemented are the following:

- **EXECUTE**
Execution of the fulfillment logic, such as calling a web service of an external fulfillment system

- CANCEL

Called when the distribution is canceled and actions in an external system have to be reverted

- HANDLE_EXTERNAL_STATE_CHANGE

Handle changes of the step status triggered from the distribution monitor application or from a program

- TRANSFER_PREVIOUS_STATE

Handle re-execution of steps

Related Information

[Batch Report](#)

Document Distribution Schema

Document Distribution Schema

The document distribution schema is the model of document distribution that consists of document distribution schema steps. Each distribution step within a schema can have an individual description (which is displayed in the end user UI) and is linked to a distribution type. You can define the same distribution type several times within a schema using different distribution step IDs.

Within the distribution schema, the execution sequence of the different distribution steps can be defined using the [Step Sequence](#) field. You can also define dependencies between distribution steps using the [Prerequisite Step](#) field. The distribution step then is only executed if the distribution step with the distribution step sequence defined in the [Prerequisite Step](#) field have all been executed completely. This can be used to branch the distribution process and merge it later.

Distribution steps can be marked as inactive to deactivate them for a certain time period or for testing purposes. If the distribution step is marked as inactive, it is ignored at runtime. If the step is used as a prerequisite step for another distribution step, this step is also ignored.

If the assigned distribution type of a distribution step is defined as an asynchronous (batch) step, the [Asynchronous](#) field is also marked at distribution step level and the field is disabled. You cannot mark a distribution step as synchronous if the corresponding distribution type is already marked for asynchronous execution. Nevertheless, you can mark a distribution step as asynchronous if its distribution step type is marked for synchronous execution. With this mechanism, distribution types can be registered for batch processing for certain distribution schema only.

The distribution schema contains all existing distribution steps for a document item. The current execution of the single distribution steps can be controlled using the distribution category implementation class or the distribution schema implementation class. For example, you can exclude some steps depending on the document status and reschedule them later when the document status has changed. The implementation class for the document distribution schema is used to implement logic similar to the distribution category but for the whole distribution schema. This means you can set a document status that is dependent on all distribution steps or influence the distribution in addition to the distribution type or category implementations.

Standard Schema

The standard schema for the subscription contract is PERP. It contains the categories and steps that replicate the subscription contract to SAP FI-CA. For more information, see [Distribution of Data to SAP FI-CA](#).

Related Information

Document Distribution Schema Determination

After defining a distribution schema, its determination at runtime can be controlled using the document process type and/or document item category. If a document is saved, the distribution step registration procedure checks the Customizing settings for each document item and checks if it is relevant for distribution.

For each document item, the header process type and item category are determined, and the Customizing settings are evaluated. If no matching entry is found for the exact combination of process type and item category, the entries for the empty process type and the exact item type are determined. This means it is possible to define default process-type-independent entries. The item category must be maintained for every item category for which a distribution schema is to be determined. It is not possible to define a default distribution schema determination using an initial item category.

For every combination of process type and item category it is possible to assign a distribution schema determination implementation class. With this class, the distribution schema determination can be influenced at runtime.

The class has to implement interface `IF_CRM_ISX_ORDER_MSG_DIST_DET`.

The distribution schema assignment allows multiple entries. One of the assigned distribution schemas must be marked as default if no determination implementation class is defined. The default distribution schema is determined at runtime if no implementation class is defined. If no default distribution schema is chosen, the implementation class has to evaluate the actual distribution schema at runtime.

For each assigned distribution schema, it is possible to deactivate single steps of the distribution schema or to activate asynchronous (batch) execution for the specific process type and item category combination. The same distribution schema can be used for different combinations but can be handled differently depending on the individual assignment settings.

The asynchronous execution can be scheduled using an appointment type of the document item. If an appointment type is assigned to the distribution schema step, this appointment is determined at runtime from the document item and the batch report only executes the distribution step if the actual date and time is reached. The determination of the scheduled date and time can also be handled within the distribution-type-dependent implementation class.

Document Distribution Status

Definition

The document distribution status defines the status of a single distribution step within a distribution schema. It is used to control the execution of the distribution schema. The status provides information about the current phase of distribution.

The following statuses exist:

	Status ID	Status Description	Meaning
1.	A	Scheduled	Step is scheduled for execution with the next batch run at the scheduled date
2.	B	Executed	Step successfully completed. Final status.

3.	C	Failed	Step execution failed. Final status.
4.	D	Irrelevant	Step is not relevant for current schema execution. Step is ignored.
5.	E	Waiting	Step is waiting for an external response (for example, from an external system).
6.	F	Canceled	Step canceled due to rerun of step or schema.
7.	G	Technically Failed	Step execution temporarily failed. Rerun possible.
8.	H	Open	Step registered and ready for execution.

After determining relevant steps for an item, the **Order Distribution Infrastructure** (ODI) sets the **Open** status for each step. During the execution of a schema, the ODI calls the EXECUTE method of each step, which has to return one of the status IDs as result.

Example

- Distribution step is successful

After a successful action in a fulfillment system, the status must be set to **Executed** (B).

- Distribution step failed

An action in a fulfillment system could not be completed due to incomplete order data. Therefore, the subscription order must be changed, and the distribution must be triggered again. In this case, the status has to be set to **Failed** (C).

- Distribution step failed due to technical problems

An action in a fulfillment system could not be completed, for example because the connection for the service call could not be established. The status must be set to **Technically failed** (G). The distribution can be restarted, and the step will be executed again.

- Asynchronous communication with an external system

There are the following two options to identify the subscription items when processing the confirmation from the external system, if the communication with an external system is asynchronous:

- Item GUID + step GUID
- Contract number + step GUID

This option is only valid for main items.

If only the status of the distribution step has to be updated, the step GUID is sufficient.

The following steps are part of the basic processing:

1. During step execution, the step implementation calls the external system to trigger an asynchronous action. The step GUID is passed to the external system and the status is set to **Waiting** (E).

2. ODI stops the distribution process and waits for the confirmation of the step implementation.
3. Later on, the external system confirms the execution of the action and passes along the item GUID + step GUID. The program that handles the confirmation calls the update service of the ODI framework and reports the new status of the step:

```
cl_crm_isx_order_msg_dist_srv=>get_instance()
->get_update_service( iv_trx_handling = abap_true )
->update_msg_state_by_step_guid(
  iv_step_guid = step_guid "needs to be passed
  iv_msg_state = msg_state "new state
).
```

4. ODI continues the distribution process by calling the method `HANDLE_EXTERNAL_STATE_CHANGE` of the step implementation class first.

Step Reference

Use

You can use the step reference to save a specific reference that is required for follow-up processes or can be used by the agent to understand the step result. A step reference can be part of the result of a step execution (method `EXECUTE`) in the subscription order or subscription contract.

Integration

This functionality is integrated with both subscription contract creation and activation, and the subscription order.

Prerequisites

- A step is executed.
- The executed step creates a step reference.

Features

Steps are displayed in a table in the **Fulfillment Status** assignment block on the subscription order or subscription contract item overview page or in the **Fulfillment Steps** assignment block on the distribution monitor overview page. A step can have a reference which is the result of the step execution. The reference can have a reference type. Four SAP standard reference types are available. The references for these known reference types are available using hyperlinks. When a hyperlink is chosen, the reference is either displayed on a dialog box or the navigation is triggered to the reference application. If the reference type is initial or is not an SAP standard type, no hyperlink is displayed.

For long references there is an additional field called `MSG_REFERENCE_LONG`. Using this field, long values can be stored as XML, such as URLs or tables. The delivered document distribution type **Call Allowance Creation (ALLW)** uses such a reference.

For more information, see Allowance Activation.

SAP Standard Reference Types

- URL

- A reference can be displayed in a specific application that is accessed using a URL link.
- The URL is opened in a dialog box.
- You must implement the method `IF_CRM_ISX_ORDER_MSG_DIST_TYPE~GET_REFERENCE_URL` in the **Document Distribution Step Type** class. This method returns the URL for the step reference.
- Transaction launcher
 - A reference can be displayed in a specific application that is accessed using the transaction launcher (such as an SAP GUI FI-CA transaction that displays a reference object).
 - When you choose the link, the navigation to the transaction launcher is triggered.
 - You must implement the method `IF_CRM_ISX_ORDER_MSG_DIST_TYPE~GET_REFERENCE_TRANSACTION` in the **Document Distribution Step Type** class. This method returns the transaction launcher details for the step reference (see details below).
- Generic
 - You can develop your own response using the **Reference** button (see details below).
 - This type should only be used if the other types are not sufficient for customer requirements.

Transaction Launcher Details

To navigate to the reference using a transaction launcher, the step type method `IF_CRM_ISX_ORDER_MSG_DIST_TYPE~GET_REFERENCE_TRANSACTION` must return a specific UI object type and the object key of the reference. If the UI object type for the specific reference “transaction type” does not exist, customers should create a new one.

The following prerequisites apply when launching the SAP FI-CA transaction in the WebClient UI:

1. Enter a transaction launcher ID (BOR Type). For more information, see **Transaction Launcher**.
2. Create a logical link.

Perform the following steps in Customizing for **Service** under **UI Framework** > **Technical Role Definition** > **Define Navigation Bar Profile**:

In the **Define Logical Links** sub dialog, create a new logical link or select an existing one and assign your launch transaction to it by defining the details of the logical link as follows:

- In the **Type** field, select the **Launch Transaction** option.
- In the **Target ID** field, enter or select EXECLTX.
- In the **Parameter** field, enter or select the name of the launch transaction that you defined.
- In the **Parameter Class** field, enter or select CL_CRM_UI_LTX_NAVBAR_PARAM.

3. Create a generic outbound plug.

In the **Define Generic Outbound Plug Mappings** sub dialog for the selected navigation bar profile, select the row that contains the UI object type and object action of the work area toolbar item that you want to configure or create a new one.

In the details of the selected row, make the following settings:

- a. Choose **Use Link ID**.
- b. In the **Logical Link ID** field, enter or select the ID of the logical link to which you assigned your launch transaction (see previous step).

c. In addition, enter or select the ID of the logical link in the **Highlighted Link ID** field.

d. Save your settings.

Implementation for the “Generic” Reference Type

Customers can develop their own response when the reference link is clicked. Therefore, the UI component ISXDMON must be enhanced. The view ISXDMON/MsgList must be enhanced and the method EH_ONGENERIC should be redefined. This method is processed when the agent clicks the hyperlink of a step reference of the **Generic** type. An implementation can be introduced in this method, which:

- Opens a dialog box or
- Raises an info, warning, or error messages
- Navigates to a different application

See also the source comments in method CL_ISXDMON_MSGLIST_IMPL -> EH_ONGENERIC in the system.

Perform the same extension for the UI components and views: ISXDMON/StepList and PROVICDE/MessageDistribution.

Related Information

[Transaction Launcher](#)

[Allowance Activation](#)

Repeat Execution

Use

In most cases, document distribution can be processed without any issues:

- All steps are successfully processed.
- Distribution is successful and completed.
- Contract is technically active.

However, in some cases document distribution is unsuccessful:

- Not all steps can be executed successfully.
- Distribution is canceled in the distribution monitor.
- Distribution has not started yet, for example when the contract has a start date in the future.

In this situation, the subscription order (sales or change order) can still be changed. Once changes have been made to this order, it needs to be released again (by choosing **Submit**). This results in the document distribution steps being executed again.

Features

When the execution of document distribution is repeated, the system also determines the steps again. If the same steps are determined as those in the previous document distribution, the implementation of these steps can decide whether the previous execution result is still valid and can be used or whether a new execution is needed (for example, a call to the external system).

The previous step results are stored in following fields in the step structure:

- PREV_MSG_STATE – Previous step status (**Executed**, **Failed**....)
- PREV_MSG_REF_TYPE – Previous step reference type (Undefined, URL, Transaction Launcher)
- PREV_MSG_REFERENCE – Previous reference
- PREV_MSG_EXECUTED_TS – Previous execution time
- PREV_MSG_REFERENCE_LONG – Previous reference (long)

These fields are also available in the **Distribution Monitor**. You might need to personalize the **Fulfillment Steps** view first to display them. During the transition from the previous result to the new result state, entries are made in the previous result fields. The current step result fields are initialized:

- MSG_STATE – Step status is set to **Open**
- MSG_PLANNED_TS – **Scheduled For** is set to the current time

The step implementation method IF_CRM_ISX_ORDER_MSG_DIST_TYPE~TRANSFER_PREVIOUS_STATE is also called. Customers can use this method to handle the transition. They can do this by adjusting or deleting the current reference fields, copying the execution status, and so on.

The new status is **Open** as a default setting. This means that this step is executed again, but during this execution the previous result fields are filled.

If new steps are determined, the system does not process a transition and the steps are processed as usual.

If previous steps are not determined during the repeat execution, these steps are canceled:

- **Canceled** status is set.
- The IF_CRM_ISX_ORDER_MSG_DIST_TYPE~CANCEL method in the step implementation is called. The customer can handle this cancellation in their external system, for example.

Related Information

[Distribution Monitor](#)

Document Distribution in Subscription Order

Use

The document distribution in the subscription order provides an additional functionality to process and execute actions for the subscription order items before the contract is created. Only items with the **Subscription Order** item category are supported for document distribution.

All document distribution step types relevant for order distribution must be assigned to category “O1”. We recommend that you define dedicated distribution schema for the order distribution.

The document distribution is triggered when the order is released (by choosing **Submit**). At the beginning of the distribution, system status I2350 (**Order Distribution in Process**) is set for the subscription order item. After the successful execution of all distribution steps, the system status I2352 (**Order Distribution Completed**) is set. If a distribution step has the status **Failed** or if the distribution is canceled, the system status I2351 (**Order Distribution Failed**) is set for the item.

The subscription order is not changeable if even one of the subscription items has the status I2350 ([Order Distribution in Process](#)). The distribution for the order is completed once all subscription items have the system status I2352 ([Order Distribution Completed](#)). The subscription contract is then created.

Example

You can use this function to create and execute actions based on business requirements such as those listed below:

- Update the technical resource for the item:

During order distribution, an external system reserves a technical identifier (such as a port number, SIM card number, phone number) and returns this information within a distribution step. The information is stored in the technical resource for the corresponding item.

- Activate two services before the contract is replicated to SAP FI-CA:

The service belonging to a subscription product must be activated in a provisioning system. During order capturing, two subscription items are part of the contract, but the contract must only be created when both services (one for each subscription product) are successfully activated.

- Admission process:

Each business partner needs an admission license before they can place an order. After capturing the order, the system checks whether the partner has a license. If not, the licensing process is started. Once the license is available and linked to the partner, the license is also linked to the order (and then to the contract). The contract must only be created after the license is available.

Subscription Order Copy

Use

You can use this functionality to create a subscription order by copying an existing order. This can be useful if a subscription order has a large volume.

Prerequisites

If you are using your own item category, you must define the copying control for the item category in Customizing for [Service](#) under [Transactions](#) > [Basic Settings](#) > [Copying Control for Business Transactions](#) > [Define Copying Control for Item Categories](#) .

SAP delivers the copying control for the provider item category PROP (for the provider item). It can be used as an example for copying another item category.

Features

- You can copy standard subscription orders

Here, the system does not copy the document flow.

- You cannot copy change orders

Here, the [Copy](#) button is not visible when you access a change order or if you call the search view for orders and select the change order. The system displays an error message.

Data level:

- Business Partner: Transferred
- Employee : Transferred
- Status: Reset: the order obtains the status **Open**
- Dates: recalculated dates match those in the standard order created
- Document flow: Does not get copied
- Item level:
 - Technical resources are deleted
 - Contract number is deleted

How to copy set types:

- Discounts: Transferred; but only those discounts that are valid for the new time period
- Mapping table
 - Assigned table: Transferred
 - Redefined table: Deleted and warning message issued
 - Parameters: Transferred

Example

There is already a subscription order for a specific business partner, and you want to create a similar order without having to enter all of the data again. You can use this function and only maintain certain data.

Contract Distribution

Once a subscription contract is released, the system triggers the distribution of the new or changed contract data to the external systems. You cannot change the contract data as long as the document distribution is active. Once the document distribution is finished successfully, you can change the contract using the change processes. If the document distribution has not been successful, there are several reprocessing activities available.

For more information, see **Contract Distribution and Contract Status**.

Related Information

[Contract Distribution and Contract Status](#)

Contract Distribution and Contract Status

You can change an active contract by changing the contract items. Each item has a status that indicates whether the distribution of a change to downstream systems has already been started. The status also displays the result of the distribution. The following statuses exist:

- Deferred

The distribution of contract changes will be executed in the future.

- To Be Distributed

The distribution process for contract changes has not yet started.

- In Distribution

The distribution of contract changes is in process.

- Successful

The distribution of contract changes has been successfully completed. This status allows you to change the contract again.

- Failed

The distribution of contract changes failed. You can rerun the distribution at a later point of time. This status allows you to correct the change order and to trigger the distribution of the contract again.

Only one of these statuses can be active for each contract item. These statuses indicate whether you can define additional changes for an item. You can only make additional changes once the distribution of the preceding change has been finished.

There exist the following options for finishing the distribution:

- Successful confirmation of the completed activation flow
- Error during activation
- Manual cancellation of the activation

When the activation has been successful, the **Business Transaction Maintenance Framework** (BTMF), which is used for modeling change processes, allows further change processes to be executed.

Immediate Change

After Process Execution

If you want to change a contract immediately and the change order has been triggered, the system copies the current contract time slice into a new change order. In the back-office UI, this is done by choosing the assignment block **Processes**. Depending on the Customizing for the process, either the configured process-specific UIs are displayed with the option to apply or cancel the contract, or the process is applied automatically to the change order.

At this point of time, the current change order will not be released before the first change process was successfully finished. You can stop the transaction retaining this status. The change order of the contract then has the status **Open**, and no follow-up actions are triggered. The contract is not updated and there is no activation in downstream systems. The **Activation Date** field is technically stored in the appointment ISTCONTSRTTS in the change order item.

Additional Change Process in the Same Order

Change processes are still available for an open change order. The list of allowed processes is the same, regardless of whether you are currently viewing the contract or the change order on the UI.

After the Release of the Change Order

You can reject or release open change orders. A rejection sets the status **Rejected** in the change order. Since no change of data has so far been recorded in the subscription contract, no additional action is required at this level.

If the change order is released by choosing **Submit**, it is copied back to the contract creating a new time slice. The new status of the contract is copied to a new item in the contract. All data from the contract is copied here. Exceptions to this are the statuses that are not transferred, because the statuses track the different follow-up processing.

Statutes After a Successful Distribution

Once all relevant ODI steps for the activation phase have been processed successfully, the status of the new contract time slice is transferred to the item. The change order now has the status **Completed**, assuming that no other item exists that could block this (for example, a hardware that needs to be shipped). You can no longer execute any change processes at the level of the existing change order.

Change processes are available at contract level. The relevant time slice that is displayed fulfills the requirement of having the status **Active**, and no follow-up has yet been created. Therefore, triggering the next change process will result in a new change order.

Statutes After a Failed Distribution

If one of the relevant ODI steps for the activation phase fails, the status of the new contract time slice is set to **TS Start Advanced Dist. (XI)**. The status **Technically Active** remains for the item.

At the same time, processes are available again at both contract and order levels enabling the correction of the problem or capturing additional change requests.

Planned Change

After Process Execution

There is no difference to the immediate start date mentioned above except that the activation date must be in the future. Nevertheless, further change processes can be triggered. The same change order is used for storage.

After the Release of the Change Order

The change order is copied back to the contract and a new time slice is created.

Activation Date Reached

Once the activation date has been reached, the contract switches to the same status that is given when releasing an order containing an immediate change. The change order status and the status of contract item are changed to **In Activation** and **In Distribution** respectively. Change processes are not available.

Statutes after Finishing a Distribution

The same logic applies as with immediate changes.

Subscription Contract Update

When updating a subscription contract without using a subscription order, the following statuses exist:

- Contract update distribution in process

The distribution of the contract update is in process. No contract processes are allowed.

- Contract update distribution completed

The distribution of the contract update has been successfully completed. This status allows you to change the contract again.

- Contract update distribution failed

The distribution of the contract update failed. You can correct the contract and trigger the distribution again.

Joint Distribution of Master Data and Subscription Contracts

Use

Once a subscription contract is distributed, master data is distributed **just-in-time** to SAP Convergent Charging (SAP CC). FI-CA does not transfer business partners and contract accounts to SAP CC, but triggers a replication of all required master data to SAP CC once a corresponding subscription contract is distributed to create a subscription contract in FI-CA. This keeps the master data payload in SAP CC to a minimum. The use of joint distribution for master data and subscription contracts is mandatory in SAP S/4HANA.

Integration

The main functionality for this feature is found in SAP FI-CA and triggers the distribution of master data and provider contracts to SAP CC.

Due to the fact that SAP FI-CA triggers the distribution of several master data objects, communication takes place asynchronously. For example, the distribution step in the subscription contract does not receive the result immediately, but a status update of the subscription contract is triggered after SAP FI-CA has completed the distribution of all necessary objects. This asynchronicity may lead to an issue where the provider contract is still or again already locked when the update arrives. Therefore, a customizable retry mechanism is available for the status update.

In effect, the distribution step type implementation must be altered to support this response behavior.

Prerequisites

You can activate this functionality by customizing your order distribution infrastructure (ODI) distribution schema accordingly. The standard **Call FICA Message contract activation** (PCEA) and **Call FICA Message contract deactivation** (PCED) step types call the SAP FI-CA contract update with the correct settings for joint distribution.

You can make the relevant settings and define distribution schema in Customizing for **Service** under **Transactions > Settings for Subscription Transactions > Document Distribution > Define Settings for Document Distribution**.

Activate the joint distribution of master data in SAP FI-CA according to the corresponding documentation. Maintain the settings that specify how often you can retry locking the contract for the asynchronous status updates from SAP FI-CA in Customizing for **Service** under **Transactions > Settings for Subscription Transactions > Integration > Integration with Convergent Charging > Settings for Joint Distribution**.

Features

You access the provider contract distribution monitor in SAP FI-CA using the **FICA Distribution Log** link in the **Reference** column of the corresponding distribution step in the **Fulfillment Status** assignment block of the subscription contract. The distribution monitor can also be started from the **Fulfillment Status** assignment block using the **Monitor** button.

Use Cases in Contract Distribution

In the provider contract distribution, the following use cases exist:

Use Case – Immediate Activation

If a subscription contract is created with an activation date of today or earlier, the system skips category P0. The system schedules the steps assigned to category P1 directly.

Use Case – Deferred Activation

If a subscription contract is created with an activation date in the future, the system first processes the steps of category P0. The other steps are not scheduled before the wait step is finished. Directly after creation, only the step PCWA (Wait for activation) is scheduled. The default setting is scheduled date = contract activation date (ISTCONTSRTTS).

This step is processed by the batch report CRM_ISX_MSG_START_PLANNED. After execution of the status of the step, the contract is set to status In activation and the execution of steps assigned to category P1 is triggered. This is comparable to the immediate activation described above.

Use Case - Early Distribution

For a subscription contract with a future activation date, the distribution of the contract is typically executed only on the activation date. In certain cases, it may be required that the contract be distributed at the time of creation or change itself, such as when Revenue Accounting and Reporting is involved. For this early distribution, an additional distribution step with the category P0 must be configured.

Directly after contract creation or contract change:

- The step PCER (Call Message for Early Distribution) of category P0 is executed. This creates or updates the FI-CA provider contract (CC contract is not created here).
- The step PCWA (Wait for Activation) of category P0 is scheduled. The default setting is scheduled date = contract activation date (ISTCONTSRTTS).

On contract activation date:

- The step PCWA (Wait for activation) of category P0 is processed by the batch report CRM_ISX_MSG_START_PLANNED, which results in the start of processing of category P1 steps.
- The step PCEA (Call FICA message contract activation) of category P1 is executed, which updates the FI-CA provider contract and creates the CC contract.

In an early distribution scenario, you cannot request billable item (BITs) as the provider contract is not yet active. If you provide the early billing date in such a scenario, the provider contract will be activated even if the contract start date is in the future, thus enabling you to request BITs based on the early billing date.

Use Case – Activation and Deactivation at the Same Time

If you want a running contract to be deactivated immediately, the system does not execute the activation logic, but only the deactivation steps.

This is different from the situation where the deactivation is scheduled for a specific date in the future. In this case, once the new time slice has been created, it is activated. The deactivation steps are only executed once the deactivation date is reached.

Modeling Contract Deactivation Steps for Subitems

You can model contract deactivation steps for subitems using the following options:

- Dependent components
- Manual subitems

If these subitems are not mandatory, you have the following options for canceling or deselecting them:

- Execute the **Cancel Subitem** process: The subitem still exists but it has a contract end date (CONTENTD), similar to manually setting the contract end date of the subitem.
- Deselect the subitem: The subitem is deleted immediately.
- Deselect the subitem by changing the condition for the item availability (BRF event in dependent components): The subitem is deleted immediately.

In the Customizing for document distribution, you can also define steps for subitems (in addition to those for main items). However, there is a constraint for the steps for contract deactivation (category P6) at subitem level. These steps cannot be processed for subitems that were deleted immediately in the change order. There are two options for overcoming this constraint:

1. Do not configure deactivation steps for subitems but only for main items. Also, model activation steps for subitems at main-item level.
2. Do not use processes where items are deleted immediately in the change order. Use the **Cancel Subitem** process or set the end date for the subitem. In the standard integration to SAP FI-CA and SAP CC, the subitems are distributed as part of a step at main-item level.

Standard Steps

In the **Order Distribution Infrastructure** (ODI), the following standard document distribution steps exist:

Type	Description
CL_CRM_ISX_MD_EXAMPLE_TRUE	Simple example: always true
CL_CRM_ISX_ORDER_MD_WAIT_P0	Message Distribution Wait for Activation (P0) Set status To be Distributed for contract item. The corresponding order is then locked for processing as the status In Distribution is assigned to the order item. Additionally, the determination of possible steps is triggered again so that steps of other categories can now start. Mostly intended for category P1 at this stage.
CL_CRM_ISX_ORDER_MD_WAIT_P5	Message Distribution Wait for Deactivation (P5) This step waits until the contract end date is reached. After that it assigns the status To be Distributed at Contract End to the contract item. Additionally, the determination of possible steps is triggered again so that steps of other categories can now start. Mostly intended for category P6 at this stage.
CL_CRM_ISX_ORDER_MD_ERP_MCC	Call SAP FI-CA message contract (de)activation P1(P6)

	Provides the Common Object Layer (COL) with the current contract status to create or update the contract in Convergent Invoicing and SAP Convergent Charging (SAP CC). For more information, see Joint Distribution of Master Data and Subscription Contracts.
CL_CRM_ISX_ORDER_MD_ALLOW	Call allowance activation (P1) Activates SAP CC allowances that have been marked to be automatically activated in an allowance definition group for the current distribution reason (sales or change). This step should be scheduled once contract activation has been completed successfully.
CL_CRM_ISX_MD_TECH_RES_CHECKS	Technical resource checks before activation This step checks whether all technical resources assigned to the contract contain a value. This step processed in order distribution represents the check moment Before Activation. The check moment is maintained in Customizing for the technical resource type. For more information, see Technical Resource.
CL_CRMS4_SOM_LOCK_CC_WS	Lock CC webservice (P1) This step will lock/activate the CC contract through the CC webservice charging Contract OperationStatechange. For more information about webservice documentation, see Charging Contract Operational Status Management.
CL_CRMS4_SOM_LOCK_SET_STATUS	Lock status set (P1) This step will set the status of the lock/unlock record. The status of the old contract items is set to Inactive. For more information, see Lock/Unlock.

Distribution of Data to SAP FI-CA

Use

A set of Customizing settings is provided with the implementation of the document distribution step types for the distribution to SAP FI-CA. You can use these settings to set up a typical scenario when you implement the distribution of contracts in a system landscape with an **SAP Convergent Charging** (SAP CC) system for rating and a Convergent Invoicing system for billing and invoicing.

For more information about these Customizing settings, see:

- Document Distribution
- Document Distribution Step Type.

Integration

Class CL_CRM_ISX_ORDER_MD_ERP_MCC

Validation

During validation, the system checks whether the current item is relevant for distribution. The system only creates a message for main items. Subitems are distributed with their corresponding main items.

Data Mapping

Data mapping consists of data from **Contract Accounts Receivable and Payable** (FI-CA) and SAP CC. The FI-CA data consist of the contract data (business partner, sales organization, appointments).

During SAP CC data mapping, the current cross-catalog mapping version of the product is evaluated first. Following this, the system selects the data for all active charge plans (parameters, technical resources, account assignments and shared counters) and maps this to the corresponding structures.

During the mapping, the system performs some consistency checks. If the data is not consistent, the ODI execution status is set to **Failed** and the message is not sent to the SAP FI-CA system.

Send Data to SAP FI-CA

The mapped data is sent to the **Common Object Layer** (COL) in SAP FI-CA by calling an RFC function module. In the COL, data is forwarded to FI-CA and to the SAP CC system. If an error occurs during this process, an error message is sent to the subscription contract. An error categorization is entered in the PARAMETER field in the error message. According to this categorization, the ODI distribution status is set as follows:

Error Category	ODI Status ID	Status Description
'alreadyExists'	C	Failed
'doesNotExist'	C	Failed
'invalid'	C	Failed
'prerequisiteMissing'	G	Technically failed
'incompatibleConfiguration'	G	Technically failed
'illegalState'	C	Failed
'temporaryIllegalState'	G	Technically failed

Schema Definition

The distribution categories P0, P1, P5, and P6 are mandatory for use with subscription contracts. Distribution steps for each of these categories are assigned to the new distribution schema PERP as follows:

- PCEA: step type PCEA, executed during contract activation (category P1)
- PCED: step type PCED, executed during contract deactivation (category P6)
- PCWA: step type PCWA, executed if the contract activation is deferred (category P0)
- PCWD: step type PCWD, executed if the contract deactivation is deferred (category P5)

i Note

PCER : step type PCER, special schema for early contract distribution to FI-CA (category P0)

Business Add-In (BAI)

The enhancement spot CRM_ISX_ERP_CONTRACT is used during distribution to SAP FI-CA to determine processes and to map the relevant data. The enhancement includes the BAdI CRM_ISX_ERP_CONTRACT, which you can use to control how data is replicated to SAP FI-CA. The BAdI is called when you execute the distribution step to SAP FI-CA using the Order Distribution Infrastructure (ODI). The interface IF_CRM_ISX_ERP_CONTRACT contains the following methods:

- MAP_CREATE: implements customer-specific mapping logic for the **Creation** process

You can also read the necessary data and forward it to SAP FI-CA.

- MAP_CHANGE: implements customer-specific mapping logic for the **Change** process

You can also read the necessary data and forward it to SAP FI-CA.

Document Distribution Customizing Settings

If you want to use the ODI for distribution to SAP FI-CA, the following Customizing settings are required:

- Schema Determination

Assign the distribution schema to the relevant combination of transaction type/item category, which is used to capture the subscription contract.

1. Call transaction CRMC_ISX_MSG_DIST.
2. Choose the **Schema Determination** folder in the dialog structure on the left-hand side of the screen.
3. Create entries for the transaction type PRVC (Subscription Contract) with different item categories for the items that have to be distributed. For example, define two entries with transaction type PRVC for the main item with item category PRCP and one entry for the subitem with the category PRCO.
4. Select the newly entry created, choose **Schema Assignment** in the dialog structure. Create a new entry using the schema PERP and set it as the default schema.
5. Save your entries.

- Scheduling

Step types PCWA and PCWD both implement waiting steps. The date on which the execution is scheduled is defined by the **Doc. Distribution Scheduled App. Type** assigned to the respective step.

- Step PCWA is executed when the contract start date is reached. This is defined by assigning date type ISTCONTSRTTS (Start of Contract Time Slice) to the step.
- Step PCWD is executed when the contract end date is reached. This is defined by assigning date type CONTENTD (Contract End) to the step.

Prerequisites

The following prerequisites are required:

- The Cross-Catalog Mapping version in the product master data must be maintained.

Activities

The distribution of contract data to SAP FI-CA is relevant for the **Creation**, **Change**, and **Cancellation** processes. When these processes are executed, the corresponding document triggers the ODI. Creating and changing contract data is relevant for the activation process (distribution category P1). The cancellation of contract data is relevant for the deactivation process

(distribution category P6). The step type class CL_CRM_ISX_ORDER_MD_ERP_MCC implements the following steps which are used for the activation and deactivation:

- Step type PCEA is assigned to category P1 and is executed during contract activation is performed
- Step type PCED is assigned to category P6 and is executed during contract deactivation

More Information

Handling of Technically Failed Distribution Steps

If the connection to the SAP FI-CA system is temporarily interrupted or a newly created contract account is not replicated to the SAP FI-CA system when the contract replication starts, the distribution step is assigned the status **Technically failed**.

In this case, we recommend the following procedure:

Start contract replication again using the batch report CRM_ISX_MSG_START_PLANNED.

1. Create a variant of the report with the following selection parameters:

- Date (From): current date
- Date (To): current date
- Time (From): for example, 20 minutes in the past (select the latest contracts only)
- Time (To): current time

Since you must determine the date and time parameters dynamically when the job starts, you must define them as **Variant attributes**.

Select the check box **Only technically failed**.

Select the check box **Only start if queues are empty** to prevent the report from using existing queues from other batch processes.

2. Schedule the report with the newly created variant to run as a batch job, for example, every 10 minutes. It collects all messages with the status **Technically failed** and executes them again. Unlock the report during the batch job.

A new batch job for the report CRM_ISX_MSG_START_PLANNED is run to perform contract activation and cancellation. This job is only run once a day with the default selection fields (execution without a variant).

Related Information

[Document Distribution](#)

[Document Distribution Step Type](#)

Allowance Activation

Use

When using allowances for subscription contracts, you can trigger the automatic activation of allowances if contract processing was successful. This can take place either after contract activation or a contract change process.

Integration

The activation of allowances is processed in a separate order distribution infrastructure (ODI) step when a subscription contract is distributed. During this step, the system evaluates cross-catalog mapping for the product. If allowance definition groups have been assigned to a charge plan that is currently active, the conditions maintained for the active process (contract creation or change process) are evaluated. The system evaluates an allowance for every entry in the allowance definition group that meets these conditions. For more information, see the section [►Master Data ► Allowance Definition Groups ►](#).

Allowances that have been activated are stored in a table as a step reference in the field CV_REFERENCE_LONG. The table contains the allowance definition ID and the mapping table ID and is stored in XML format on the database. If the ODI step is executed again, the system evaluates the reference and only activates those allowances that have not previously been activated.

Prerequisites

You must consider the following prerequisites during allowance activation:

- You have created an allowance definition group containing allowances that are to be automatically activated for newly created contracts or during change processes.
- You have defined a validity period that covers the relevant date to be used.
- The contract start date is relevant for contract activation.
- The activation date is relevant for the contract change process.
- You have assigned the allowance definition group to a contract product in Cross-Catalog Mapping.

Activities

You set up the automatic activation of allowances for new and changed contracts by

- making settings for document distribution to execute the allowance activation step for the relevant contracts

To do so, you can use the step type [Call Message for Allowance Creation](#) (ALLW). Assign this step type to the distribution schema used for the corresponding contracts. For more information, see **Document Distribution**. Execute this step once the [Contract Activation](#) step has been successfully executed.

- assigning allowance definition groups to Cross-Catalog Mapping in product maintenance
- maintaining conditions for each allowance in allowance definition groups using BRFplus functions

These conditions are evaluated, and allowances are activated automatically during the document distribution step. You cannot influence this while you create or change contracts on the WebClient. However, it is possible to simulate the allowance activation step during order capturing before you submit the process.

i Note

You find the list of automatically activated allowances in the details of the distribution step in the assignment block [Fulfillment Status](#). If an error occurs during allowance activation, the distribution status of the corresponding contract is set to [Technically Failed](#).

Related Information

[Document Distribution](#)

[Refill Allowance](#)

[BRFplus and BRFplus Functions](#)

Batch Report

The scheduled records are executed using report CRM_ISX_MSG_START_PLANNED. This report is to be included in a job and started periodically. Usually, the report can be started with the default parameters.

With default values, the report executes all planned entries with status **Open (H)**, **Technically failed (G)**, or **Scheduled (A)**.

The records are executed asynchronously. The queues can be found in the inbound queue. You can call up the inbound queue using the transaction SMQ2.

The queue name is CSA_ISX_ACTIVATE with a rolling number at the end. The report has the following import parameters:

- Date (From) and date (To)

These dates are defaulted automatically to executed past and present records. You may change the filter if you want to execute future activations even if the time is not reached.

- Number of queues: 5

Change this number to influence the number of parallel queues executing the records. A lower number will result in fewer queues and therefore blocks fewer resources on the server but increases the runtime. Increasing the number results in the opposite.

- Block size: 50

This parameter controls how many records are executed in one COMMIT WORK statement. During execution, this value can vary since records of one single order document are always executed in a COMMIT WORK statement.

Distribution Monitor

Use

The **Order Distribution Infrastructure (ODI)** distribution monitor is an application in which distribution steps for subscription contracts can be found, displayed, changed, rescheduled, canceled, or executed again. The UI for the distribution monitor consists of a search page and an overview page.

Some functionality is only available in expert mode of the distribution monitor (such as rescheduling of distribution steps).

Integration

The distribution monitor can either be accessed from the search page by selecting the relevant documents or from the document that has triggered the distribution. In the second case, the assignment block on which the fulfillment status is displayed must be available.

Features

Distribution Monitor Search Page

You can search for distribution records using various parameters, such as the subscription contract number and fulfillment status. You can use the search parameters in multiple combinations to narrow down the search results.

The distribution record is data aggregated from all distribution steps in a subscription contract. It contains some attributes for the related contract, a cumulated overall status of the distribution, and important dates.

Distribution Monitor Overview Page

The distribution overview page consists of the following views:

- Header view displaying the distribution record
- A list of fulfillment steps for the related subscription contract is displayed below the header view. The following fields are available:
 - Fulfillment steps

The steps displayed relate to the subscription order item or subscription contract item.
 - Displaying inactive steps

By default, only active steps (that is, steps that do not have the **Canceled** status) are displayed. You can see the canceled steps by choosing **Show Inactive**.
- Fulfillment messages are displayed in the final view displayed after the steps.

If the user has been granted the relevant authorizations, additional functionality for adjusting distribution data is also available (expert mode).

Header View

The overall status is set to **Finished**, **In Process**, or **Failed** according to the status of the related steps.

Refreshing Distribution Data

If you choose **Refresh**, all distribution data for the current document is reloaded from the database and displayed in monitor views. This is relevant if distribution is still ongoing and the results of the step, which is running asynchronously, are expected.

Navigation to Related Data

If some subscription contracts have subcontract(s), the **Related items** button is active and provides direct navigation to the relevant subcontract.

Retrigger Distribution

A step may have failed due to a technical failure (indicated by the status **Technically Failed**). Once the technical issue has been resolved, the distribution steps must be executed again. This is done by choosing **Rerun**.

Display Cross-Catalog Mapping of Product

The cross-catalog mapping may have been defined in the master data for the used product. For more information, see **Cross-Catalog Mapping**. Choosing **Cross-Catalog Mapping** takes you to a dialog box where the actual data for this mapping taken from the current item is displayed.

Process Flow

The **Order Distribution Infrastructure** (ODI) monitor in the subscription contract provides the **Process Flow** button to display single fulfillment steps when a contract has been changed.

Features only Available in Expert Mode

Cancel Activation

If any step with the category O1 (Order Distribution), P0 (Wait for Activation), or P1 (Activation) is still to be executed (status is not **Executed**, **Failed**, **Cancelled**, or **Irrelevant**), the **Cancel Activation** button is active. If you choose this, a dialog box appears on which you are asked which of the following two options is to be executed:

- Cancellation with Redetermination of Steps

After canceling all steps with the category P0 and P1, new steps are determined and available with the status **Open**. This should be done if a complete restart of the activation process is required, without adaption of the subscription document. Choosing **Rerun** starts execution of the steps.

- Cancellation without Redetermination of Steps

All steps which have not yet been executed (status is not **Executed**, **Failed**, **Cancelled**, or **Irrelevant**) are set to the status **Failed**. This causes the distribution process to be aborted and control is returned to the subscription document, which can be edited again. The required changes can be made here and when they are submitted, this triggers a new execution of the distribution process in which new steps are generated.

Cancel Deactivation

If any step with the category P5 (Wait for Deactivation) or P6 (Deactivation) is still to be executed (status is not **Executed**, **Failed**, **Cancelled**, or **Irrelevant**), the **Cancel Deactivation** button is active. If you choose **Cancel Deactivation**, all steps in this category are canceled and new steps are determined and available with the status **Open**. Choosing **Rerun** starts execution of the steps.

Changing Status of Single Step

The status of a single step can be changed by choosing **Change Status**. This status change not only sets the status but also executes the related contract processes.

Reschedule Step

If a step is scheduled to be run at a planned time, its status is set to **Scheduled**. The **Reschedule** button is active for these steps. If you choose this, you can change the planned time in the dialog box appears.

Execute Single or Multiple Steps

By choosing **Execute...step** the selected step is executed again. This is possible if the status is either **Scheduled**, **Technically Failed**, or **Open**. If the status is **Executed** or **Open** (Waiting for Prerequisite), an additional confirmation dialog box is shown, since executing steps with this status is more important.

By choosing **Execute...step and subsequent steps**, the selected step, and all steps for which the selected step is defined as a prerequisite step are executed again. This is possible if the status is either **Scheduled**, **Technically Failed**, or **Open**.

Fulfillment Messages

If only one step is selected, the system only displays messages for this particular step.

Activities

You must grant the authorization required in the expert mode for authorized users to change the step data for distribution.

Example

You can see some common examples here:

- You want to be informed about the processing status of an asynchronous step executed in a legacy system.

- Distribution process failed due to a technical failure. The problem has now been resolved and you want to start the process again.
- Distribution process failed due to invalid document data. You want to cancel distribution, adapt the document as required, and start the distribution process again.
- A certain step becomes obsolete – for example, you have already made the required changes in a legacy system manually – you can set the status to **Executed** or **Irrelevant** to prevent processing.

Related Information

[Cross-Catalog Mapping](#)

Contract Lifecycle Management

A subscription contract manages the long-term agreement for a service that a provider company offers to customers. Throughout its lifetime, a subscription contract may undergo multiple changes and hold multiple states, such as active and expired. You can track and maintain the lifecycle of a subscription contract using a variety of SAP Fiori apps.

More Information

Refer to the following points for more information:

- The overall subscription contract and using change processes on the subscription contract, refer to [Subscription Contracts](#) and [Contract Change Processes](#).
- Changing multiple subscription contract documents and contract items at the same time, refer to [Mass Changes of Subscriptions](#) and [Manage Mass Processing of Subscriptions](#).
- Viewing and managing the overall subscription contract and subscription contract document, refer to [Manage Subscription Contract Lifecycle](#) and [Manage Subscription Contract Documents](#).

For item level details, refer to [Manage Subscription Contract Item Details](#).

Manage Subscription Contract Lifecycle

App ID: F6193

With this app, you get an overview of a specific subscription contract and all the items within the contract. You can see the contract state for each contract time slice, represented by the items of the contract. That is, you can see whether the item is currently in an active, planned, expired, or rejected state.

Key Features

You can use this app to do the following:

- View the **Contract States** of each individual item, as well as the evolution of the contract from Initial state until the current state.

You can navigate between the time slices and items of the subscription contract using the  *hierarchical tree* navigation icon and navigate into the contract details using the  *chevron* navigation icon.

- View the estimated contract value and the contract time slice (item) value.

- Get an overview of the complete structure, item hierarchy, history, and evolution of the subscription contract (with respect to the contract states).
- Execute change processes.
- Navigate to [Manage Subscription Contract Item Details](#) app.
- View the [Fulfillment Status](#) in the [Distribution Monitor](#).
- Navigate to [Monitor Master Data Distribution](#) app.

You can enable insights cards in the [Manage Subscription Contract Lifecycle](#) app using the Collaboration Manager. For information about this, refer to [Insights Cards](#) and [Adding Insights Cards to Chats](#).

Situation Handling

This app uses Situation Handling, which automatically informs users responsible about matters that require their attention. For more information, see [Situation Handling](#).

Supported Device Types

- Desktop
- Tablet

Related Information

[Manage Subscription Contract Item Details](#)
[Subscription Contract Item Due to Expire](#)
[Subscription Contract Item Grace Period](#)
[Subscription Contract Item Distribution](#)

App Implementation: Manage Subscription Contract Lifecycle

Prerequisites

Before implementing the app, you must ensure the following:

- You must activate the OData service UI_SBSCRCONTRITEM_02_MON on the SAP Fiori front-end server.
- You must assign the role SAP_BR_INTRNAL_SALESREP_SOM on the SAP Fiori front-end server.
- If you have upgraded to SAP S/4HANA 2022 FPS00 or later, you must consider SAP Note [3151794](#)  as a prerequisite, for the app to work as expected.
- If you have upgraded to SAP S/4HANA 2022 FPS01 or later, you must consider SAP Note [3331714](#)  as a prerequisite, for the app to work as expected.

System Landscape Requirements

For information about the system requirements, see [SAP Fiori Overview](#).

Required SAP Notes

For this app, the following SAP Notes must be implemented:

Back end / front end Server	SAP Note	Description
Back-end	3151794 	Post Upgrade Report to Resolve Limitations for Fiori Apps - Manage Subscription Contract Lifecycle, Manage Subscription Contract Item Details, and Display Allowances

Implementation Tasks

The following sections list tasks that have to be performed to implement this app. The tables contain the app-specific data required for these tasks.

You can also find the data required to perform these tasks in the SAP Fiori apps reference library at [Manage Subscription Contract Lifecycle](#).

Back end Server: Create Connectors

This app is based on ABAP Core Data Services and does not require search models to work. However, if you want to start the app from the search results list, you must activate the underlying search connectors. You can create connectors in transaction ESH_COCKPIT.

For more information, see [Creating Connectors](#).

i Note

Note the following system behavior:

Search software components build a stack. Different layers can be installed on top of the basis component.

Search models can be available in different components - in their original component, but also in higher layer components (extension components).

Once you create a search connector in an extension component, all search models from the original component are transferred into the extension component. You will therefore no longer find the search models in the original component, as is shown in the table below.

Front end Server: Activate OData Services

Component	Technical Name
UI_SBSCRCONTRITEM_02_MON	For the Manage Subscription Contract Lifecycle app, the following OData service always needs to be activated: UI_SBSCRCONTRITEM_02_MON (0001)

For more information about activating OData services, see [Activating OData Services](#).

Front end Server : Activate SAPUI5 Application

Component	Technical Name
SAPUI5 Application	CONTRLC_MONS1

For more information about how to activate the ICF service, see [Activate ICF Services of SAPUI5 Application](#).

Front end Server: Activate ICF Service

Component	Technical Name
ICF Service	/sap/bc/ui5_ui5/sap/contrlc_mons1

For more information about how to activate the ICF service, see [Activate ICF Services of SAPUI5 Application](#).

Front end Server : Enable App for Access in SAP Fiori Launchpad

The SAP Fiori launchpad is the entry point to apps. In the launchpad, the user can see and access those apps that have been assigned by an administrator to the catalog designed for this user's role.

The administrator makes these assignments in the launchpad app manager. For more information, see [Setting Up Technical Catalogs with the SAP Fiori Launchpad App Manager](#).

SAP delivers technical catalogs for groups of apps as repositories to create your own catalogs in the launchpad app manager or content manager (see [Tools for Setting Up Launchpad Content](#)). Along with these catalogs, more technical content is delivered for each SAP Fiori app. You can find the delivered technical content for each SAP Fiori app in the SAP Fiori apps reference library.

Front end- Server and back end- Server: Assign OData Service Authorizations to Users

You must assign OData service authorizations for the app to your users.

Caution

Several authorization default values are connected to the OData service. To ensure that all these default values are assigned to a user, you have to follow the instructions given under the documentation links provided.

Make the assignment on the back-end server and on the front-end server:

- On the **back-end server**, you have to assign the OData service authorization to a PFCG role. For more information, see [Creating Authorization Roles for Catalogs](#).
- On the **front-end server**, you must assign the OData service authorization to a new or existing role, such as a business role that has been adjusted according to your needs. For more information, see [Creating Roles on Front-End and Assigning Launchpad Catalogs and Spaces/Pages](#) and [Assigning Roles to Users on Front-End](#).

OData Service (Version Number)	Back end Server: Authorization Role (PFCG Role)	Front end Server: Assignment to Authorization Role
UI_SBSCRCONTRITEM_O2_MON (2)	You can use the example business catalog SAP_COM_BC_SUBSCR_ORDER or your custom catalog to create the back-end PFCG role.	You can use the existing business role, SAP_BR_INTRNAL_SALESREP_S0M as a template to create your own business role.

Configuration Tasks

To be able to implement the app, you must perform the following configuration tasks in the back-end system:

- Create and define change processes such that they are available in the **Change Contract** quick access menu as per your requirement.

Related Information

[Manage Subscription Contract Lifecycle](#)

[App Details: Manage Subscription Contract Lifecycle](#)

[Define Change Processes](#)

App Details: Manage Subscription Contract Lifecycle

Manage the contract state of a subscription contract as represented by its items.

Searching for Subscription Contracts

You can search for the relevant subscription contract using a wide variety of filter criteria, such as the **Contract**, **Document Number**, and other contract attributes such as the technical resource identifier, auto renewal indicator, or the **Contains Situations** indicator.

i Note

There is a predefined filter that, by default, excludes rejected contract items from the search results.

The list is categorized based on the contract state (current, planned, or expired) and you can further filter it to display either the **Contract Main Item** or subitems, and to include **Rejected Items**.

There are additional links that enable you to navigate to related apps where you can work on the product, business partner, subscription contract, and the subscription contract document. Further navigation options in the **Open In** menu lead you to other applications such as **Display Provider Contract** and **Display Subscription Contract**. In case an entitlement management system is available, you can also navigate to the **Display Entitlements** app from this menu.

General Information

You find the most important attributes of the subscription contract per item, such as contract and time slice dates, system status, partner, product, and organization information under **General Information**. Navigation to related apps is available for common objects, such as partners, product, subscription product-specific data, and contract account. For subscription products that are enabled for allowances, the navigation to the app **Display Charging Data for Subscription Contracts** is enabled as well.

Contract States and Contract State Sequence

The contract state is determined based on the actual date and the contract time slice start and end dates. These states are:

Section	Contract States
Contract States on the Page Header	Current, Planned, Expired, Rejected
Contract State and Contract State Sequence	Initial, Current, Planned, Expired

Section	Contract States
Contract State	Rejected items are represented in a separate tab

In the **Contract States** section, the complete contract lifecycle is visible over all the contract item time slices and subitems, from the initial item to the current or planned state. Within the contract hierarchy, you can navigate between the timeslices of the subscription contract using the  navigation icon and to the **Manage Subscription Contract Item Details** Fiori app using the  navigation icon. In both cases, a page with details of the specific item is displayed. Within the item hierarchy, you can only navigate between the main items of the contract.

In the **Contract State Sequence** section, only the most important contract items are displayed for a quick overview. Here, you see the initial item and the most current items from the last expired to the current to the planned.

i Note

These states are only for the main items. No subitems are displayed.

You can navigate to the **Manage Subscription Contract Item Details** app from both the sections. You can also navigate between the states of the contract using the item number in the **Contract States** section. Further, you are able to check the **Fulfillment Status** by navigating to the distribution monitor.

Estimated Contract and Item Values, and Prices

The **Contract Net Value (Estimated)** is the aggregated estimated value across all items of the subscription contract. The **Item Net Value (Estimated)** is the value calculated for one item during its lifespan. In case of a time slice without end date, the contract end date is used or, if no contract end date is given, the end of contract term. Rejected items retain their item values but do not contribute to the overall contract value. You can get a detailed overview of the item value per contract time slice in the **Contract States** section.

The net value is displayed at the header KPIs and the **Contract State Sequence** section. In the **Prices** section, the **Net Value**, **Gross Value**, and **Tax Value** per item is displayed. Depending on whether the contract is a recurring or a one-time contract, the **Recurring** or the **One-Time** values are displayed here as well.

Technical Activation Status

The **Technically Active** status indicates whether the subscription contract was distributed without errors and can be fulfilled successfully. You see this indicator at the header level, and it is color coded depending on the contract state, to represent the contract fulfillment status.

	Expired	Current	Planned	Rejected
Technically Active				
Technically Inactive				

Technical Activation Status

Contract History

The **Contract History** section shows the contract changes, the change process that triggered the changes, and the completion status of these changes. You can also navigate to the related change document, in addition to the initial subscription order or solution quotation document, as applicable.

Contract Change Processes

You can access and execute specific change processes using the **Change Contract** quick access menu. Only those change processes that are allowed for the specific item and have been enabled for the Fiori menu are visible here. There may be additional change processes that have not been enabled for the Fiori menu but can still be implemented on the contract. For a view of the overall change processes for the subscription contract, refer to the **Change Process** section.

i Note

You can enable or disable a specific change process to appear in the quick access menu in the Customizing ► **Service** ► **Transactions** ► **Settings for Subscription Transactions** ► **Contract Changes** ► **Settings for the Maintenance Framework (BTMF)** ►. Here, you set the check box for **Fiori Menu Enabled**, in the **Define Scenario Dependent Attributes** view.

When executing a change process, you are navigated directly to a window where you can input the change values. For change processes with change process type ISTA/ISTE, you can either:

- Directly **Submit** your change request **OR**
- Navigate to the open change order using the **Change Order** button where you can make any further changes.

Special Cases

Phased Contracts

The **Phased Contract: Yes**, indicator is visible in the header when you open a phased contract. In other cases, this indicator is not available on the UI. In case of phased contracts or contracts with product bundles, this section shows you the entire item hierarchy, grouped by the item state and ordered as **Planned**, **Current**, **Rejected**, and **Expired**.

Master Agreements

When contracts are called off from a master agreement, the **Master Agreement** section is displayed, which you can use to navigate to the master agreement.

Related Apps

The following apps are accessible from the **Manage Subscription Contract Lifecycle** app:

- [Manage Subscription Contract Item Details](#)
- [Manage Subscription Contract Documents](#)
- [Mass Changes of Subscriptions](#)
- [Manage Mass Processing of Subscriptions](#)
- [Manage Subscription Product-Specific Data](#)
- [Display Charging Data for Subscription Contracts](#)
- **Monitor Master Data Distribution**

Subscription Contract Item Due to Expire

Technical Name: SOM_EXPIRING_SUB_CONTR_ITEM

Consuming App: [Manage Subscription Contract Lifecycle](#)

Template used for Situation Handling that informs the user about a subscription contract item due to expire, that is, the upcoming contract end date.

For background information about the Situation Handling framework, see the Product Assistance for [Situation Handling](#).

Business Value

As a configuration expert for business processes (SAP business role SAP_BR_BPC_EXPERT), you can use this template for the following situation:

The responsible employee is informed about the end of a subscription contract using the automatic notifications, which allow them to navigate directly to the subscription contract and take appropriate action. Subscription contracts that are technically inactive, setup for auto renewal, rejected, or are phased contracts are excluded from these automatic notifications.

Default Settings

The template comes with predefined settings. For the settings that aren't visible or self-explanatory, you can find information in the following sections. For generic information about how to configure situations based on this template, refer to the documentation of the [Manage Situation Types](#) app with which you can display and use the template.

For more information, see [Manage Situation Types](#).

Prerequisites

If you have upgraded to SAP S/4HANA 2022 FPS01, you must consider SAP Note [3331714](#)  as a prerequisite, for the situation to work as expected.

Situation Trigger Object and Anchor Object

These technical settings define for which object a situation is displayed (anchor object) and the object based on which a situation is triggered (trigger object):

Object Type	Technical Name	Description
Trigger Object	SOM Subscription Contract Item (SubscriptionContractItemUUID)	Subscription contract item
CDS View for Trigger Object	C_SubscriptionContractItemSitn	CDS view that contains the conditions required for the creation of the situation
Trigger Type	Batch	Type of trigger
Anchor Object	SOM Subscription Contract Item (SubscriptionContractItemUUID)	Subscription contract item
CDS View for Anchor Object	I_SubscriptionContractItemSitn	CDS view that contains information about the subscription contract item

Conditions

The condition in this template defines that a situation is created by a situation batch run if conditions for end of the subscription contract item are met.

The preconfigured conditions in this template are defined for processing in this order:

Fixed conditions for this situation are:

Condition	Condition Description
Subscription Contract	The subscription contract item: • is the main item • has no auto renewal indicator • is technically active • is not rejected • is not a phased contract • has an end date

Changeable conditions for this situation are:

Condition	Condition Description
Contract Number	Subscription contract number
Distribution Channel	Organizational data
Division	Division
Employee Responsible	Employee responsible
Order Quantity & Product	Subscription contract item data
Sold-to Party	Sold-to party
Subscription Contract Item End Date	Time period is defined using the condition Days before subscription item end date
Transaction Type	Business transaction type

For more information, see [Conditions](#).

Situation Display

When situations occur, users are informed about them by various texts, such as in-app situation messages or notifications on SAP Fiori launchpad. In this section, you can see all preconfigured texts.

For more information, see [Situation Display](#).

i Note

When you copy the template and adapt it to your needs, you can define the notification settings and change all text elements in this section. After making the changes in the original language, please remember to translate these changes into the other languages that you want to use. The template comes with translations that can be changed directly when you copy the template.

For more information, see [Translate Situation Types](#).

Recipients

In this section, you see the predefined settings for determining who is responsible for a situation instance. Based on these, you can define who receives a notification when a situation instance is triggered (if enabled) or who sees the instance in the [My Situations](#) app.

For more information, see [Recipients](#).

Situation Monitoring

Monitor Instances: Monitoring (the tracking of changes to situation instances) isn't enabled by default for situation templates. When you copy the template to create a situation type, you need to enable this option if you want to capture data for this situation type and enable the creation of [business situation events](#). You can view the data in the [Monitor Situations](#) app.

Capture Data Context: You can capture the data associated with the situation type that you define with this template. With this feature you can track the data listed below.

i Note

When the template offers the **Data Context** capturing feature and you've selected it, you also have the option to enable **Situation Automation** for your own automation solution. Please note that if a situation instance is handled automatically, the end user is not informed. If you haven't implemented an automation solution, do not select this option.

Notification Recipients: For this Situation, the recipients of the notifications can be determined by the agent rule ID SOMSITMRELEMPRES (Notify employee responsible for subscription contract (User from Partner Employee Responsible)).

Monitor Instances: Monitoring (tracking changes on situation instances) is not enabled for situation templates by default. When you copy the template to create a situation type, you need to select this checkbox to use the [Monitor Situations](#) app and enable the creation of business situation events.

With the [Business Situation – Read](#) API, you can use this data for further processing on a platform of your choice. You can also use the API to import automation rules for the situation type.

For more information, see [Business Situation – Read](#).

i Note

Customers are responsible for the data, including the adherence to legal restrictions in their judicial area. Neither SAP nor its associated companies assume any responsibility for the adherence to authorizations or data protection rules for data captured and processed by its customers. This means that customers alone must ensure adherence to these rules and regulations.

Related Information

[Situation Handling](#)

Subscription Contract Item Grace Period

Situation Template ID: SOM_GRACE_PERD_SUB_CONTR_ITEM

This is a situation template delivered by SAP for [Situation Handling](#).

Consuming App: [Manage Subscription Contract Lifecycle](#)

The consuming app is a business app for which this template is designed and which displays a situation message.

Business Value

As a configuration expert for business processes (SAP business role SAP_BR_BPC_EXPERT), you can use the this situation template for the following situation:

The responsible employee is informed when a subscription contract item is past its end date but still within the grace period using automatic notifications. These allow them to navigate directly to the subscription contract and take appropriate action. Subscription contracts that are setup for auto renewal, rejected, or are phased contracts are excluded from these automatic notifications.

Default Settings

The template comes with predefined settings. For the settings that aren't visible or self-explanatory, you can find information in the following sections. For generic information about how to configure situations based on this template, refer to the documentation of the [Manage Situation Types](#) app with which you can display and use the template.

Prerequisites

If you have upgraded to SAP S/4HANA 2022 FPS01, you must consider SAP Note [3331714](#)  as a prerequisite, for the situation to work as expected.

Situation Trigger Object and Anchor Object

These technical settings define for which object a situation is displayed (anchor object) and the object based on which a situation is triggered (trigger object):

Trigger Object	SOM Subscription Contract End Situation SubscriptionContractItemUUID
CDS View for Trigger Object	C_SubscriptionContractItemSitn
Anchor Object	SOM Subscription Contract End Situation SubscriptionContractItemUUID
CDS View for Anchor Object	I_SubscriptionContractItemSitn
Trigger Type	Batch

The situation is triggered when the grace period of the subscription contract item is nearing its end date.

i Note

These settings aren't visible in the template in the [Manage Situation Types](#) app and cannot be changed when you create a situation type based on this template. All of the following settings are visible in the template and you can adapt them when you copy the template to create a situation type.

Conditions

The condition in this template defines that a situation is created by a situation batch run if conditions for end of grace period of the subscription contract item are met.

The preconfigured conditions in this template are defined for processing in this order:

Condition	Description
1	The subscription contract item:

Condition	Description
	• is the main item • has no auto renewal indicator • is technically active • is not rejected • is not a phased contract • has an end date

Changeable condition for this situation are:

Condition	Condition Description
Contract Number	Subscription contract number
Distribution Channel	Organizational data
Division	Division
Employee Responsible	Employee responsible
Order Quantity and Product	Subscription contract item data
Sold-to Party	Sold-to party
Subscription Contract Item End Date	Time period is defined using the condition Days before subscription item end date
Transaction Type	Business transaction type

For more information, see [Conditions](#).

Recipients

In this section, you see the predefined settings for determining who is responsible for a situation instance. Based on these, you can define who receives a notification when a situation instance is triggered (if enabled) or who sees the instance in the [My Situations](#) app.

- **Responsibility by Teams**

The predefined settings in this template determine that the notification addresses only members of the **SOM Subscription Order Management** team category.

- **Responsibility by Rules**

The determination of the notification recipients is predefined by a rule for this template. The rule is: If the user in **Partner Employee Responsible** is responsible for the subscription contract, they should be informed when the end date of the grace period is coming up .

For more information, see [Recipients](#).

Situation Monitoring

Monitor Instances: Monitoring (the tracking of changes to situation instances) isn't enabled by default for situation templates. When you copy the template to create a situation type, you need to enable this option if you want to capture data for this situation type and enable the creation of [business situation events](#). You can view the data in the [Monitor Situations](#) app.

Related Information

[Situation Handling](#)

Subscription Contract Item Distribution

Situation Template ID: SOM_DISTR_ISSUE_SUB_CONTR_ITEM

This is a situation template delivered by SAP for [Situation Handling](#).

Consuming App:[Manage Subscription Contract Lifecycle](#)

The consuming app is a business app for which this template is designed and which displays a situation message.

Business Value

As a configuration expert for business processes (SAP business role SAP_BR_BPC_EXPERT), you can use this template for the following situation:

The responsible employee is informed when the distribution of a subscription contract item has either failed or has entered a waiting state. In such a situation, the distribution status of the subscription contract item is either **Waiting**, **Failed**, or **Technically Failed**.

Default Settings

The template comes with predefined settings. For the settings that aren't visible or self-explanatory, you can find information in the following sections. For generic information about how to configure situations based on this template, refer to the documentation of the [Manage Situation Types](#) app with which you can display and use the template.

Situation Trigger Object and Anchor Object

These technical settings define for which object a situation is displayed (anchor object) and the object based on which a situation is triggered (trigger object):

Trigger Object	SOM Subscription Contract Item (SubscriptionContractItemUUID)
CDS View for Trigger Object	C_SUBSCRPNCONTRITEMDISTRSITN
Anchor Object	SOM Subscription Contract Item (SubscriptionContractItemUUID)
CDS View for Anchor Object	I_SUBSCRPNCONTRITEMDISTRSITN
Trigger Type	Batch

The situation is triggered when the distribution of a subscription contract item has either failed or has entered a waiting state.

i Note

These settings aren't visible in the template in the [Manage Situation Types](#) app and cannot be changed when you create a situation type based on this template. All of the following settings are visible in the template and you can adapt them when you copy the template to create a situation type.

Conditions

The condition in this template defines that a situation is created when the distribution of the subscription contract item has either failed or is in waiting state for a long time.

The preconfigured conditions in this template are defined for processing in this order:

Condition	Description
Last contract timeslice	Flag to indicate that the last timeslice is to be used

Changeable conditions for this situation are:

Condition	Description
Employee Responsible	Employee responsible
Status	Distribution status of the subscription contract item
Transaction Type	Business transaction type
Category	Category of the Document Distribution Step
Distribution Channel	Distribution Channel
Division	Division
Executed At	Timestamp Execution Time
Item Category	Item Category
Product	Product
Sales Organization	Sales Organization
Schema	Document Distribution Schema
Seconds in waiting state	Time duration (in seconds) for how long the contract distribution has been in waiting state
Step ID	Document Distribution Step ID
Step Type	Document Distribution Step Type

For more information, see [Conditions](#).

Recipients

In this section, you see the predefined settings for determining who is responsible for a situation instance. Based on these, you can define who receives a notification when a situation instance is triggered (if enabled) or who sees the instance in the [My Situations](#) app.

- **Responsibility by Teams**

The predefined settings in this template determine that the notification addresses only members of the **SOM Subscription Order Management** team category.

- **Responsibility by Rules**

For this Situation, the recipients of the notifications can be determined by the agent rule ID SOMSITMRELEMPRES (**Notify employee responsible for subscription contract (User from Partner Employee Responsible)**). That is, the user for the subscription contract should be informed if the distribution of the subscription contract item has failed or has entered a waiting state.

For more information, see [Recipients](#).

Situation Monitoring

Monitor Instances: Monitoring (the tracking of changes to situation instances) isn't enabled by default for situation templates. When you copy the template to create a situation type, you need to enable this option if you want to capture data for this situation type and enable the creation of [business situation events](#). You can view the data in the [Monitor Situations](#) app.

Manage Subscription Contract Item Details

App ID: F6487

With this app, you can view specific details for a subscription contract item. You can use the  *chevron* navigation to access this app from the **Manage Subscription Contract Lifecycle** app for a more detailed view of a single contract state.

Key Features

You can use this app to do the following:

- View all details of the selected subscription contract item, such as price details, technical resource, and related documents.
If the item is a recurring sales item, you can view delivery plan details like the delivery date of the first delivery plan line and the recurring period based used to generate the delivery plan lines. If a sales document has been generated for any of the delivery plan lines, you can click on the document number to navigate into the document details.
- View the estimated contract value and the contract item value.
- Implement change process on the subscription contract.
- View the **Fulfillment Status** in the **Distribution Monitor**.
- Navigate to the **Monitor Master Data Distribution** app.
- Navigate to the **Mass Processing Information** where you can check if the subscription contract has been included in a mass processing operation and the relevant details. If not, you can create a mass processing from this page.

You can enable insights cards in the **Manage Subscription Contract Item Details** app using the Collaboration Manager. For information, refer to [Insights Cards](#) and [Adding Insights Cards to Chats](#).

Supported Device Types

- Desktop
- Tablet

Related Information

[Manage Subscription Contract Lifecycle](#)

App Implementation: Manage Subscription Contract Item Details

Prerequisites

Before implementing the app, you must ensure the following:

- You must activate the OData service UI_SBSCRCONTRITEM_02_MON on the Fiori front-end server.
- You must assign the role SAP_BR_INTRNAL_SALESREP_S0M on the Fiori front-end server.
- If you have upgraded to SAP S/4HANA 2022 FPS00 or later, you must consider SAP Note [3151794](#)  as a prerequisite, for the app to work as expected.
- If you have upgraded to SAP S/4HANA 2022 FPS01 or later, you must consider SAP Note [3331714](#)  as a prerequisite, for the app to work as expected.

System Landscape Requirements

For information about the system requirements, see [SAP Fiori Overview](#).

Required SAP Notes

For this app, the following SAP Notes must be implemented:

Back-End / Front-End Server	SAP Note	Description
If you have upgraded to SAP S/4HANA 2022 FPS00 or later, this SAP Note is a prerequisite.	3151794 	Post Upgrade Report to Resolve Limitations for Fiori Apps - Manage Subscription Contract Lifecycle, Manage Subscription Contract Item Details, and Display Allowances

Implementation Tasks

The following sections list tasks that have to be performed to implement this app. The tables contain the app-specific data required for these tasks.

You can also find the data required to perform these tasks in the SAP Fiori apps reference library at [Manage Subscription Contract Item Details](#).

Back-End Server: Create Connectors

This app is based on ABAP Core Data Services and does not require search models to work. However, if you want to start the app from the search results list, you must activate the underlying search connectors. You can create connectors in transaction ESH_COCKPIT.

For more information, see [Creating Connectors](#).

i Note

Note the following system behavior:

Search software components build a stack. Different layers can be installed on top of the basis component.

Search models can be available in different components - in their original component, but also in higher layer components (extension components).

Once you create a search connector in an extension component, all search models from the original component are transferred into the extension component. You will therefore no longer find the search models in the original component, as is shown in the table below.

Front-End Server: Activate OData Services

Component	Technical Name
UI_SBSCRCONTRITEM_02_MON	For the Manage Subscription Contract Item Details app, the following OData service always needs to be activated: UI_SBSCRCONTRITEM_02_MON (0001)

For more information about activating OData services, see [Activating OData Services](#).

Front-End Server : Activate SAPUI5 Application

Component	Technical Name
SAPUI5 Application	CONTRITM_MONS1

For more information about how to activate the ICF service, see [Activate ICF Services of SAPUI5 Application](#).

Front-End Server: Activate ICF Service

Component	Technical Name
ICF Service	/sap/bc/ui5_ui5/sap/contritm_mons1

For more information about how to activate the ICF service, see [Activate ICF Services of SAPUI5 Application](#).

Front-End Server : Enable App for Access in SAP Fiori Launchpad

The SAP Fiori launchpad is the entry point to apps. In the launchpad, the user can see and access those apps that have been assigned by an administrator to the catalog designed for this user's role.

The administrator makes these assignments in the launchpad app manager. For more information, see [Setting Up Technical Catalogs with the SAP Fiori Launchpad App Manager](#).

SAP delivers technical catalogs for groups of apps as repositories to create your own catalogs in the launchpad app manager or content manager (see [Tools for Setting Up Launchpad Content](#)). Along with these catalogs, more technical content is delivered for each SAP Fiori app. You can find the delivered technical content for each SAP Fiori app in the SAP Fiori apps reference library.

Front-End Server and Back-End Server: Assign OData Service Authorizations to Users

You must assign OData service authorizations for the app to your users.

Caution

Several authorization default values are connected to the OData service. To ensure that all these default values are assigned to a user, you have to follow the instructions given under the documentation links provided.

Make the assignment on the back-end server and on the front-end server:

- On the **back-end server**, you have to assign the OData service authorization to a PFCG role. For more information, see [Creating Authorization Roles for Catalogs](#).
- On the **front-end server**, you must assign the OData service authorization to a new or existing role, such as a business role that has been adjusted according to your needs. For more information, see [Creating Roles on Front-End and Assigning Launchpad Catalogs and Spaces/Pages](#) and [Assigning Roles to Users on Front-End](#).

OData Service (Version Number)	Back-End Server: Authorization Role (PFCG Role)	Front-End Server: Assignment to Authorization Role
UI_SBSCRCONTRITEM_02_MON (2)	You can use the example business catalog SAP_SOM_BC_SUBSCR_ORDER or your custom catalog to create the back-end PFCG role.	You can use the existing business role SAP_BR_INTRNAL_SALESREP_SOM as template to create your own business role.

Configuration Tasks

You can perform the following configuration tasks in the back-end system:

- Create and define change processes such that they are available in the **Change Contract** quick access menu as per your requirement.

Related Information

[Manage Subscription Contract Item Details](#)

[App Details: Manage Subscription Contract Item Details](#)

[Define Change Processes](#)

App Details: Manage Subscription Contract Item Details

The **Manage Subscription Contract Item Details** app allows you to check all the relevant details for that subscription contract item.

You use this app along with the **Manage Subscription Contract Lifecycle** app. It gives a comprehensive overview on the specific attributes and entities of a subscription contract item, while the **Manage Subscription Contract Lifecycle** provides a holistic picture about the whole subscription contract. Together, both the apps provide a common set of attributes and information, such as the header, general information, prices, and master agreement. They also have the same functions, such as the change process quick access menu and the change process execution.

The following information and functions are available at the item details only

- Price Details or Billing Schedule

i Note

If the selected item category has an associated billing plan item type, the **Billing Schedule** section is displayed at contract level, instead of the **Price Details** section. However, you can see the **Price Details** for the individual billing items in the billing schedule using **>**. For more information about this, refer to **Billing Schedule for Periodic Subscriptions**.

- Billing
- Delivery Plan

i Note

If the selected item is a recurring sales item, you can view delivery plan details such as the delivery date of the first delivery plan line and the recurring period used to generate the delivery plan lines. If a sales document has been generated for any of the delivery plan lines, you can click on the document number to navigate into the document details.

- Master Agreement
- Related Documents (Covers information about Sharing, External Purchase Order and Contract Transfer)
- Automatic Renewal
- Technical Resources
- Cancellation (Covers information about Cancellation and Rejection)
- Withdrawal
- Discounts
- Charges
- Lock History
- Equipment
- Partners
- Dates
- Duration
- Status (Displays the User Status)
- Configuration
- Account Assignment object (only WBS elements are supported currently)

This section is displayed only if a WBS element is assigned to the contract item.

i Note

Since you would primarily access this app by navigating from the **Manage Subscription Contract Lifecycle**, for the details view on a subscription contract item, this app is not included as tile into the standard configuration of the Fiori Launchpad. You can adapt this easily by adding the app to your personal Fiori Launchpad configuration by changing **►User Settings > Edit Home Page >**

Searching for Subscription Contracts

If you access this app directly from the Fiori tile, you can search for the relevant subscription contract using a wide variety of filter criteria, such as the [Contract](#), [Document Number](#), and other contract attributes such as the technical resource identifier, auto renewal indicator, or the [Contains Situations](#) indicator.

i Note

There is a predefined filter that, by default, excludes rejected contract items from the search results.

The list is categorized based on the contract state (current, planned, or expired) and you can further filter it to display either the [Contract Main Item](#) or subitems, and to include [Rejected Items](#).

There are additional links that enable you to navigate to related apps where you can work on the product, business partner, and the subscription contract document. Further navigation options in the [Open In](#) menu lead you to other applications such as [Display Provider Contract](#) and [Display Subscription Contract](#). In case an entitlement management system is available, you can also navigate to the [Display Entitlements](#) app from this menu.

Special Cases

Phased Contracts

The [Phased Contract: Yes](#), indicator is visible on the header when you open a phased contract. In other cases, this indicator is not available on the UI.

Configuration

Here, you can view the configuration details of the subscription contract.

i Note

This is available only if the contract is using configurable subscription products.

Master Agreements

When contracts are called off from a master agreement, the [Master Agreement](#) section is displayed, which you can use to navigate to the master agreement.

Related Documents

This section is displayed only if related documents are available. Here, you can see information about [Sharing](#), [External Purchase Order Number](#), and [Contract Transfer](#).

Auto Renewal

This section is displayed only if a subscription contract is enabled for auto renewal.

Cancellation

This section shows information on cancellation and rejection and is available only if a contract is cancelled or the respective item was rejected.

Withdrawal

This section shows information on withdrawal and is available only if a contract is withdrawn.

Related Apps

The following apps are accessible from the **Manage Subscription Contract Item Details** app:

- **Manage Subscription Contract Lifecycle**
- **Manage Subscription Contract Documents**
- **Mass Changes of Subscriptions**
- **Manage Mass Processing of Subscriptions**
- **Manage Subscription Product-Specific Data**
- **Display Allowances**
- **Monitor Master Data Distribution**

Related Information

[Manage Subscription Contract Lifecycle](#)

[Manage Subscription Contract Documents](#)

[Mass Changes of Subscriptions](#)

[Manage Mass Processing of Subscriptions](#)

[Manage Subscription Product-Specific Data](#)

[Display Charging Data for Subscription Contracts](#)

[Billing Schedule for Periodic Subscriptions](#)

Display Charging Data for Subscription Contracts

App ID: F5597

With this app, you can view subscription contract data from SAP Convergent Charging (SAP CC), refilled and charged values for allowances from Convergent Invoicing, as well as allowances and the valid allowance definitions for the contract. You can refill allowances in SAP Convergent Charging from this app.

Key Features

This app provides the following key features:

- Display overall SAP CC data relevant for the subscription contract, such as technical resources, tables, counters, and parameters.
- Track your aggregated monetary allowance values on different granular levels using key figures.
- Display allowances, allowance definitions, and the related charge plans that are assigned to the subscription contract.
- Trigger an allowance refill in SAP Convergent Charging.
- Navigate to SAP Convergent Invoicing using **►Open In► Display Billable Items◀** to view the creation and usage billable items for an allowance.
- Navigate from the **Display Charging Data for Subscription Contracts** app to other related apps such as the **Manage Subscription Contract Lifecycle** app, the **Manage Subscription Contract Documents** app, and the **Manage Subscription Contract Item Details** app.

These Fiori apps are tightly integrated with each other to enable easy navigation as well as access to view and manage objects such as the subscription contract, subscription contract document, subscription product-specific data, allowance definition table, and business partner.

You can enable insights cards in the **Display Charging Data for Subscription Contracts** app using the Collaboration Manager. For information about this, refer to [Insights Cards](#) and [Adding Insights Cards to Chats](#).

Supported Device Types

- Desktop
- Tablet

Related Information

[Allowances in Subscription Order Management](#)

[Refill Allowance](#)

[Allowance Assignment](#)

[Allowances in Convergent Invoicing](#)

App Implementation: Display Charging Data for Subscription Contracts

Prerequisites

To use the full functionality of the app, the following prerequisites must be fulfilled in the backend:

- SAP Convergent Charging must be enabled and connected to your backend system.
- The following SAP Convergent Charging services must be enabled:
 - `v1/allowanceManagement` by defining a logical port for proxy `CO_ISX_ALLOWANCE_MANAGEMENT_V1`
 - `chargeableItemCharging` by defining a logical port for proxy `CO_CRM_ISX_CC_CHARGEABLE_ITEM`
 - `v3/catalog` by defining a logical port for proxy `CO_CRMS4_SOM_CC_V3_CATALOG_SER`
 - SAP Convergent Charging service `contractProvisioning` by defining a logical port for the proxy `CO_ISX_CC_CTR_PROV_SERVICES`.
 - SAP Convergent Charging service `contractstate` by defining a logical port for proxy `CO_ISX_CC_CONTRACT_STATE_MANAG`.
- The optional integration with Convergent Invoicing must be configured so that aggregated allowance values are available in the **Allowances** section.
- In Convergent Invoicing, a billable item class containing the interface component `ALLOWANCE` must be defined and the corresponding mapping to SAP Convergent Charging must be established.

For more information about connecting SAP Convergent Charging and enabling the service connections, refer to the [Integration Guide for SAP S/4HANA BRIM Using Subscription Order Management](#).

- If you have upgraded to SAP S/4HANA 2022 FPS00 or later, you must consider SAP Note [3151794](#)  as a prerequisite, for the app to work as expected.

System Landscape Requirements

The app consists of front-end components (such as the user interfaces) and back-end components (such as the OData service). These must be installed in a system landscape that is enabled for SAP Fiori in an ABAP environment.

Implementation Tasks

The following sections list tasks that must be performed to implement this app. The tables contain the app-specific data required for these tasks.

You can also find the data required to perform these tasks in the SAP Fiori apps reference library at [Display Allowances](#).

Front-End Server: Activate OData Services

Component	Technical Name
OData Service (Version Number)	UI_CNVCHRGALLWNC_MON (0001)

For more information about activating OData services, see [Activating OData Services](#).

Front-End Server: Activate SAPUI5 Application

Component	Technical Name
SAPUI5 Application	ALLWNC_MONS1
Generic Drill-Down Application	/default_host/sap/bc/ui5_ui5/sap/ALLWNC_MONS1

For more information about how to activate the ICF service, see [Activate ICF Services of SAPUI5 Application](#).

Front-End Server: Activate ICF Service

Component	Technical Name
ICF Service	/sap/bc/ui5_ui5/sap/ALLWNC_MONS1

For more information about how to activate the ICF service, see [Activate ICF Services of SAPUI5 Application](#).

Front-End Server: Enable App for Access in SAP Fiori Launchpad

The SAP Fiori launchpad is the entry point to apps. In the launchpad, the user can see and access those apps that have been assigned by an administrator to the catalog designed for this user's role.

The administrator makes these assignments in the launchpad app manager. For more information, see [Setting Up Technical Catalogs with the SAP Fiori Launchpad App Manager](#).

SAP delivers technical catalogs for groups of apps as repositories to create your own catalogs in the launchpad app manager or content manager (see [Tools for Setting Up Launchpad Content](#)). Along with these catalogs, more technical content is delivered for each SAP Fiori app. You can find the delivered technical content for each SAP Fiori app in the SAP Fiori apps reference library.

Front-End Server and Back-End Server: Assign OData Service Authorizations to Users

You must assign OData service authorizations for the app to your users.

Caution

Several authorization default values are connected to the OData service. To ensure that all these default values are assigned to a user, you have to follow the instructions given under the documentation links provided.

Make the assignment on the back-end server and on the front-end server:

- On the **back-end server**, you have to assign the OData service authorization to a PFCG role. For more information, see [Creating Authorization Roles for Catalogs](#).
- On the **front-end server**, you must assign the OData service authorization to a new or existing role, such as a business role that has been adjusted according to your needs. For more information, see [Creating Roles on Front-End and Assigning Launchpad Catalogs and Spaces/Pages](#) and [Assigning Roles to Users on Front-End](#).

OData Service (Version Number)	Back-End Server: Authorization Role (PFCG Role)	Front-End Server: Assignment to Authorization Role
UI_CNVCHRGALLWNC_MON (0001)	You can use the example business catalog SAP_SOM_BC_SUBSCR_ORDER or your custom catalog to create the back-end PFCG role.	You can use the existing business role SAP_BR_INTRNAL_SALESREP_SOM as template to create your own business role.

Configuration Tasks

You can configure the [Display Charging Data for Subscription Contracts](#) app using specific BAdIs in the Customizing  [Service Transactions](#)  [Settings for Subscription Transactions](#)  [Integration](#)  [Integration with Convergent Charging](#)  [Business Add-Ins \(BAdIs\)](#) .

Related Information

[Display Charging Data for Subscription Contracts](#)
[App Details: Display Charging Data for Subscription Contracts](#)

App Details: Display Charging Data for Subscription Contracts

The [Display Charging Data for Subscription Contracts](#) app provides a complete overview of SAP Convergent Charging data with respect to a specific subscription contract. All the data displayed on this app is derived directly from SAP Convergent Charging. This means that the values created from BRFplus functions or characteristics of a configurable product are shown by their actual value, in addition to the redefined values from subscription product-specific data, a subscription order, solution quotation, or via call-off from a master agreement. The following information is collected from SAP Convergent Charging and Convergent Invoicing (if the optional integration is configured):

- [Technical Resources](#)
- [Parameters](#)
Default and contract-specific parameters, as well as price keys
- [Counters](#)

Internal and external counters

- **Tables**

Default and contract-specific mapping and range tables, and price tables

- **Allowance**

- **Aggregated values**

These values are available only when the integration with Convergent Invoicing is configured.

- **Allowances**

- **Allowance Definitions**

You can trigger a **Refill** of allowances from an allowance definition.

Prerequisites

To access this app via the SAP Fiori Launchpad, you must assign the business role Internal Sales Representative (SOM) (SAP_BR_INTRNAL_SALESREP_SOM) to your user. You can also create your own role using this role as template. For more information about the role, business catalog, and the technical catalog, refer to the [SAP Fiori Apps Reference Library](#).

Navigation

Navigation from the **Display Charging Data for Subscription Contracts** app is enabled such that:

- From the result list, you can navigate directly into the details page to view convergent charging data relevant for the subscription contract. Here, you can further head to the allowance definitions and Convergent Invoicing, for viewing billable items transferred from SAP Convergent Charging for allowance creation and usage.
- From the result set, you can also navigate to multiple Fiori apps based on the authorizations assigned to your user. These apps enable you to view details and manage information for the subscription contract document, the subscription contract, and the subscription contract item.
- The app is closely integrated with the **Manage Subscription Contract Lifecycle** Fiori app, which serves as a single point of entry to a subscription contract. Hence, you can move seamlessly between the two apps.

Search and Filter

You can search for subscription contracts using multiple search criteria, such as the contract or contract document number. In the overview, you see the active and completed subscription contracts (that are not canceled), and which have a cross-catalog mapping version assigned.

i Note

By default, the **Contract Main Item** filter is set to exclude subitems from the search results.

SAP Convergent Charging-Specific Details

SAP Convergent Charging Overview

The overview page combines information from the subscription contract, the relevant charging data from SAP Convergent Charging, as well as the aggregated billable item amount from Convergent Invoicing.

There are additional links that allow you to navigate to related apps where you can work on the product, business partner, subscription contract, and the subscription contract document. Further navigation options in the [Open In](#) menu lead you to other specific applications such as [Display Provider Contract](#) and [Display Subscription Contract](#).

Technical Resources

In this section, you see information about the assigned technical resources, their definition, and their source (for example, parent item, child item, own item, or a BRFplus function).

Parameters

In this section, you see the SAP Convergent Charging parameters used in the subscription contract. All the values are derived and resolved from SAP Convergent Charging, and include the actual values for BRFplus functions and characteristic values set by configurable products. In the parameter details, if the [Contract Specific](#) indicator is set to Yes, it means that a parameter value has been redefined specifically for the contract. You can see the different types of parameter values below:

- Default parameter values (if [Contract Specific](#) = No)
- Redefined parameter values
- Values set by a [BRF+ Function](#)
- Values set by a [Characteristic](#) of a configurable product
- Price keys

These allow you to view the used prices in accordance to the price table. For more information, see [Tables](#).

- For the parameter [Bill Cycle Date](#) (SAP_BILL_CYCLE_DATE), you see the date/time calculated in FI-CA during contract distribution to SAP CC.

You use the table configuration () to display further attributes of the parameters, such as the [Assignment ID](#) that you can use to distinguish between the parameters when several charge plans are assigned, information about contract-specific parameter redefinitions, and the [BRFplus Function](#) or [Characteristic](#) currently in use.

By default, all parameters that are created by the selected contract are displayed here. You can also filter out the parameters that should not be displayed here, using the BAdI: Display SAP CC Data for Contracts (CRMS4_SOM_CONTR_CC_CT_BADI).

Counters

In this section, you see the current value of all external and internal, shared and non-shared counters that are available for the subscription contract:

- Actual External Counter Value (Visibility External)
- Actual Internal Counter Value (Visibility Internal)
- Actual Shared Counter Value (Namespace, Type)

The counter [Type](#) indicates whether a counter is shared and which [Namespace](#) has been used for the counter sharing.

The numeric [Value](#) for each counter, derived from SAP Convergent Charging, shows 6 decimal places by default. However, this can be configured to a custom representation, for example, to two decimal places. The setting can be specified using the BAdI CRM_ISX_PROV_CC_COUNTER ([Subscription Contract SAP CC Counters](#)).

You use the table configuration (⚙️) to display further attributes of the counters, such as the **Assignment ID** that you can use to distinguish between the counters when several charge plans are assigned, the namespace and type to identify shared counters, and the visibility indicator to identify internal and external parameters.

Tables

In this section, you see all the mapping tables, range tables, and price tables used in the subscription contract. In the table details, if the **Contract Specific** indicator is set to Yes, it means that an object-specific table has been used, and has been redefined specifically for the contract. You can see the different types of tables below:

- Default mapping or range table (if **Contract Specific** = No)
- Object-specific mapping or range table
- Price table (can be either a mapping or a range table)

The Table ID enables you to navigate to the specific table definition or to adapt the table values. You can use the price key from the **Parameters** section to search for the relevant entry in the price table.

You use the table configuration (⚙️) to display further attributes of the tables, such as the **Assignment ID** that you can use to distinguish between the tables when several charge plans are assigned, the class ID, and information about contract-specific usage of the object-specific tables.

Allowances

For more information about the allowances section, refer to [Allowance-Specific Details](#).

Allowance-Specific Details

Allowances

In this section, you see the information about allowances that is collated from SAP Convergent Charging and Convergent Invoicing (if the optional integration is configured). You can also navigate to ► **Open In ► Display Billable Items** ► Fiori app to view further details of the allowance in Convergent Invoicing. All billable items for allowance creation and usage are displayed here, in addition to the aggregated amount of these billable items from Convergent Invoicing.

For more information about the integration with Convergent Invoicing, refer to **Technical Details**.

Aggregated Values

Allowance details can be linked to monetary information retrieved from Convergent Invoicing. Once configured, you can see the following aggregated values for all the allowances collected from Convergent Invoicing:

- **Consumed**: Aggregated usage BITs restricted by BIT Type Allowance Usage (U)
- **Refilled**: Aggregated creation BITs restricted to BIT Type Allowance Creation (C)
- **Balance**: Calculated as Refilled-Consumed

For more information, see **Key Figures** and **Technical Details**.

Allowance and Allowance Details

In the **Allowances** section, you find information about the current state of the allowances and allowance definitions from SAP Convergent Charging. You can navigate into the allowance details using the  *chevron*. From here, you can use  **Open In**  **Display Billable Items**  to view further details of the allowances in Convergent Invoicing. All billable items for allowance creation and usage are displayed here. For more information about this, refer to **Technical Details**.

By default, all allowances that are created by the selected (or any related) contract are displayed here. This could either be a shared contract, shared allowances (if they are used by the selected contract), or another contract that uses the same sharing group. The allowance also displays the contract from where it has been created. However, you can also filter out the allowances that should not be displayed here, using the BAdI: Display Allowances from SAP Convergent Charging (CRMS4_SOM_CC_ALLOWANCE_BADI).

Allowance Definition and Allowance Definition Details

Here, you see an overview of all valid allowance definitions. You can create and adapt these allowances definitions by navigating to the **Manage Mapping Tables** app via the table ID.

You can view further details of the allowance definitions by navigating to the **Allowance Definition Table**. If you have defined the description ALLOW_DEF_DESC in your allowance definition table, this column will be automatically mapped as a description for the allowance definition ID. On the **Allowance Definition Details** page, you can view the assigned charge plan and its technical resource(s) used for the specific charge plan.

You also see the definition details that will be used during allowance creation. You can use the BAdI: Display Allowance Definitions (CRMS4_SOM_CC_ALLOW_DEF_BADI) for managing the display of allowance definitions.

For more information about this, refer to **Allowance Definitions**.

Refill

You can refill a specific allowance from the allowance definition overview. Technically, a refill creates a new allowance in SAP Convergent Charging. As the service calls are buffered, the newly created allowance will become visible only if the buffer expires or is reset, and a new data read is executed. A refill action resets the buffer, and the new allowance becomes visible after the manual refresh is executed. For more information about this, refer to **Technical Details**.

i Note

In case of a newly created allowance, values from Convergent Invoicing are only displayed when the usage billable items are configured to hold an amount (BIT_AMOUNT) and transferred from SAP Convergent Charging to Convergent Invoicing.

In case multiple charge plan assignments exist for the subscription product or the charge plans are configured to hold multiple technical resources, a refill can be triggered only at the allowance definition details level, by choosing the charge plan assignment with the appropriate technical resource.

While executing a refill, you must specify the date and time when the allowance will be activated. The allowance can be created only if this date/time is within the validity period of the allowance definition.

Related Information

[Technical Details](#)

[Key Figures](#)

Key Figures

Key Figures Calculated from the Allowance

Allowance Expires in (Days): This key figure is available in the [Allowances](#) section and the allowance details page. It calculates the number of days until the allowance expires as allowance's Valid From Date - Valid To Date. If the allowance has already expired, this key figure is not displayed.

Key Figures Calculated from Billable Items

The billable item (BIT) amounts from Convergent Invoicing are aggregated at different granularities for the allowance creation BITs and the allowance usage BITs:

- For all allowances available for the contract
- For all allowances of a specific allowance definition
- For each allowance

Further details for these measures are available on the detail pages per allowance and per allowance definition. The aggregation is done as per the BIT amount, by its currency, taken from Convergent Invoicing. From the contract document, deviating currencies are translated into the document currency.

The BIT amount is restricted to the billable item types [Allowance Usage](#) and [Allowance Creation](#) for the calculation:

- **Consumed:** Aggregated usage BITs restricted to BIT Type Allowance Usage (U)
- **Refilled:** Aggregated Creation BITs, restricted to BIT Type Allowance Creation (C)
- **Balance:** Calculated as Refilled – Consumed

BADIs for Key Figure Color Configuration

The graphical representation on the UI illustrates the consumption progress by setting the total refill amount with respect to the total consumed amount. You can adapt the colors on this representation, using specific BADIs. For more information about these BADIs, refer to **Technical Details**.

Related Information

[Technical Details](#)

Technical Details

Prerequisites

Allowance Integration with Convergent Invoicing

In an integrated scenario with Convergent Invoicing, you must activate the interface component [Data for Creating an Allowance](#) (ALLOW_CREATE) in the [Display Charging Data for Subscription Contracts](#) app (F5597).

To use the BIT amount, the BIT class must contain the interface component ALLOWANCE ([Allowance](#)) and the charge plan must send the BIT for creating the allowance and the BIT for using the allowance with a BIT amount from SAP Convergent Charging to Convergent Invoicing.

If the interface component for handling the deferred revenue (DEF_REV_BY_E) is in use, a mapping to the structure is required by the allowances and can be established via the [BAdI: Select Billable Items for Allowance Display](#) (CRMS4_SOM_CC_ALLOW_BIT_BADI). This BAdI also allows determination of the BIT status for the billable items that are used for the key figure aggregation. For more information about handling allowances in Convergent Invoicing, refer to [Allowances in Convergent Invoicing](#).

SAP Convergent Charging

To retrieve data for technical resources, tables, parameters, and counters as well as for allowances and allowance definition data, the following services from SAP Convergent Charging must be enabled:

- v1/allowanceManagement
- chargeableItemCharging
- v3/catalog
- contractProvisioning
- contractstate

General Operation on the Key Figures

The color coding of the key figures can be adapted by using the [BAdI: Set Semantic Colors for Contract Item/Allowance UI Application](#) (CRMS4_SOM_ALLOW_SEM_COLOR_BADI).

Buffer for Refreshing Service Calls

There are specific buffers that enable a better performance when accessing SAP Convergent Charging multiple times, by persisting the results of the web service calls. You can influence the behavior of these buffers, by changing or bypassing the refresh interval, with the following BAdIs:

- BAdI: Display SAP CC Data for Contracts (CRMS4_SOM_CONTR_CC_CT_BADI), Method: SET_SECONDS_FOR_BUFFER_REFRESH
- BAdI: Display Allowances from SAP CC (CRMS4_SOM_CC_ALLOWANCE_BADI), Method: SET_SECONDS_FOR_BUFFER_REFRESH

By default, a buffer refresh is triggered after 300 seconds, for reading allowances from SAP Convergent Charging.

- BAdI: Display Allowance Definitions (CRMS4_SOM_CC_ALLOW_DEF_BADI), Method: SET_SECONDS_FOR_BUFFER_REFRESH

By default, a refresh is triggered after 3600 seconds, for reading allowance definitions from SAP Convergent Charging.

You can define the time period (in seconds) after which the data is outdated. Once the time limit is reached, the data will be read again from SAP Convergent Charging by the next access to this specific contract or allowance. You can also clear the buffer entries using the reports CRMS4_SOM_ALLOW_DB_DELETE and CRMS4_SOM_CC_DATA_DELETE.

Formatting Counter Values for Display

By default, the numeric values from SAP Convergent Charging are configured to display 6 decimal places (99.999999). You can use the [Formatted Value](#) to configure a custom representation for the number of decimal places. You can do so using the [BAdI: Display Allowances from SAP CC](#) (CRMS4_SOM_CC_ALLOWANCE_BADI), method CHANGE_ALLOWANCES.

Extending Allowances using Counters and Parameters

As a key user, you can extend the [Display Charging Data for Subscription Contract](#) app, to display additional counters and parameters for allowances. You do so by adding custom fields and mapping them via a custom logic using the [BAdI: Mapping Counter and Parameter Extension Fields into Allowances](#) (BAdI: CRMS4_SOM_CC_MAP_ALLOW_BADI). For more information, refer to [App Extensibility: Display Charging Data for Subscription Contracts \(Key User\)](#).

Available BAdIs

You can find the available BAdIs for the [Display Charging Data for Subscription Contracts](#) app in Customizing under [Service Transactions](#) [Settings for Subscription Transactions](#) [Integration](#) [Integration with Convergent Charging](#) [Business Add-Ins \(BAdIs\)](#):

- [BAdI: Display Allowances from SAP CC](#)
- [BAdI: Display Allowance Definitions](#)
- [BAdI: Select Billable Items for Allowance Display](#)
- [BAdI: Set Semantic Colors for Allowance UI Application](#)
- [BAdI: Mapping Counters and Parameters Extension Fields into Allowances](#)
- [BAdI: Display SAP CC Data for Contracts](#)

Related Information

[App Extensibility: Display Charging Data for Subscription Contracts \(Key User\)](#)

App Extensibility: Display Charging Data for Subscription Contracts (Key User)

As a key user, you can extend the [Display Charging Data for Subscription Contracts](#) app to add further counters or parameters for allowances, according to your business needs.

i Note

You can only extend the allowances.

Custom Fields

You can create and edit custom fields to enhance applications using the [Custom Fields](#) app. You can add fields to the following UI elements using key user adaptation:

Business Context	UI Technology	UI Element Type
Subscription Contract Item - Allowance (CRMS4_SOM_ALLNC)	SAP Fiori	Field

For more information, see:

- [Custom Fields](#)

- [Adding Extension Fields to Apps with Different UI-Technologies](#)

Business users can display fields that you have enabled in the **Custom Fields** Fiori app. The fields that you enable for this app are included in the personalization options for business users for the following elements:

UI Element	Business Context
Table - Allowances of Convergent Charging	Subscription Contract Item - Allowance (CRMS4_SOM_ALLNC)

Custom Logic

With the **Custom Logic** app, you can create and maintain implementations of custom logic that can be used to enhance applications and change application behavior. You can implement custom logic for the following business add-ins (BAdIs):

BAdI Description	Business Context	Business Use
BAdI: Field mapping of counters and parameters into allowances (CRMS4_SOM_CC_MAP_ALLOW_BADI)	Subscription Contract Item – Allowance (CRMS4_SOM_ALLNC)	You use this BAdI to map the values of counters and parameters to extension fields at the allowance level.

For more information, see:

- [Custom Logic](#)
- For more information about the individual BAdIs, check the BAdI-specific documentation: [Custom Logic](#) > [your implementation](#) > [BAdI Documentation](#) > [View BAdI Documentation](#)

Activities

To map the newly created custom fields via custom logic, you must first implement the BAdI: Mapping Counter and Parameter Extension Fields into Allowances (CRMS4_SOM_CC_MAP_ALLOW_BADI), and then map the field from the allowance web service to the custom field. The right allowance can be determined by the Allowance Plan ID (ALLOWANCE_PLAN_ID), which specifies the relevant allowance for the counter and parameter mapping.

You can find the SAP sample coding for the BAdI implementation in the **Custom Logic** app.

Additional Standard Fields

You can add standard fields that are not shown on the UI by default, using key user adaptation. To check which standard fields you can add to the app, start adding a field to the UI element that you want to enhance. You can then view a list of all available fields, both standard and custom).

For more information on how to adapt SAP Fiori at runtime, see [Making UI Changes \(Object Pages\)](#).

For a general description of the extensibility options and procedures of SAP Fiori apps, see [Extend SAP Fiori Apps](#).

Related Information

[Extensibility](#)

Manage Subscription Contract Documents

With this app, you can view details for a subscription contract document and execute header level change processes on the entire document. While you get an overview of all the subscription contracts contained within the document, you can also navigate to the [Manage Subscription Contract Lifecycle](#) and [Manage Subscription Contract Documents](#) apps for getting detailed information.

Key Features

You can use this app to do the following:

- View all contracts contained within a subscription contract document
- View the most recent contract items, ordered by their state (current, planned, expired), and further attributes
- View the overall subscription contract document value and the value of each contract contained within the document

This app provides the following key features:

- Hierarchical view of all the contracts within the contract document and the current, planned, or the last expired items, whichever is most relevant
- The item hierarchy is also available if your subscription contract document contains product bundles or phased contracts.
- Navigation to the [Manage Subscription Contract Lifecycle](#) and [Manage Subscription Contract Item Details](#) apps
- Net value of the entire subscription contract document
- Header level change processes for the subscription contract document

You can enable insights cards in the [Manage Subscription Contract Documents](#) app using the Collaboration Manager. For information about this, refer to [Insights Cards](#) and [Adding Insights Cards to Chats](#) .

Supported Device Types

- Desktop
- Tablet

App Implementation: Manage Subscription Contract Documents

Prerequisites

Before implementing the app, you must ensure the following:

- You must activate the OData service UI_SUBSCRPNCONTRACT_02_MON on the Fiori front-end server.
- You must assign the role SAP_BR_INTRNAL_SALESREP_SOM on the Fiori front-end server.

System Landscape Requirements

For information about the system requirements, see [SAP Fiori Overview](#).

Implementation Tasks

The following sections list tasks that have to be performed to implement this app. The tables contain the app-specific data required for these tasks.

You can also find the data required to perform these tasks in the SAP Fiori apps reference library at [Manage Subscription Contract Documents](#).

Back-End Server: Create Connectors

This app is based on ABAP Core Data Services and does not require search models to work. However, if you want to start the app from the search results list, you must activate the underlying search connectors. You can create connectors in transaction ESH_COCKPIT.

For more information, see [Creating Connectors](#).

i Note

Note the following system behavior:

Search software components build a stack. Different layers can be installed on top of the basis component.

Search models can be available in different components - in their original component, but also in higher layer components (extension components).

Once you create a search connector in an extension component, all search models from the original component are transferred into the extension component. You will therefore no longer find the search models in the original component, as is shown in the table below.

Object pages are based on search models. To be able to use an object page, you must activate the underlying search models by creating connectors. Create connectors in transaction ESH_COCKPIT.

You can find the search models for this app in the SAP Fiori apps reference library.

For more information, see [Creating Connectors](#).

i Note

Note the following system behavior:

Search software components build a stack. Different layers can be installed on top of the basis component.

Search models can be available in different components - in their original component, but also in higher layer components (extension components).

Once you create a search connector in an extension component, all search models from the original component are transferred into the extension component. You will therefore no longer find the search models in the original component, as is shown in the table below.

Front-End Server: Activate OData Services

Component	Technical Name
UI_SUBSCRPNCONTRACT_02_MON (1)	For the Manage Subscription Contract Item Details app, the following OData service always needs to be activated:

Component	Technical Name
	UI_SBSCRCONTRITEM_02_MON (0001)

For more information about activating OData services, see [Activating OData Services](#).

Front-End Server : Activate SAPUI5 Application

Component	Technical Name
SAPUI5 Application	CONTRDOC_MONS1
Generic Drill-Down Application	/default_host/sap/bc/ui5_ui5/sap/sbrt_appss1

For more information about how to activate the ICF service, see [Activate ICF Services of SAPUI5 Application](#).

Front-End Server: Activate ICF Service

Component	Technical Name
ICF Service	/sap/bc/ui5_ui5/sap/contrdoc_mons1

For more information about how to activate the ICF service, see [Activate ICF Services of SAPUI5 Application](#).

Front-End Server : Enable App for Access in SAP Fiori Launchpad

The SAP Fiori launchpad is the entry point to apps. In the launchpad, the user can see and access those apps that have been assigned by an administrator to the catalog designed for this user's role.

The administrator makes these assignments in the launchpad app manager. For more information, see [Setting Up Technical Catalogs with the SAP Fiori Launchpad App Manager](#).

SAP delivers technical catalogs for groups of apps as repositories to create your own catalogs in the launchpad app manager or content manager (see [Tools for Setting Up Launchpad Content](#)). Along with these catalogs, more technical content is delivered for each SAP Fiori app. You can find the delivered technical content for each SAP Fiori app in the SAP Fiori apps reference library.

Front-End Server and Back-End Server: Assign OData Service Authorizations to Users

You must assign OData service authorizations for the app to your users.

Caution

Several authorization default values are connected to the OData service. To ensure that all these default values are assigned to a user, you have to follow the instructions given under the documentation links provided.

Make the assignment on the back-end server and on the front-end server:

- On the **back-end server**, you have to assign the OData service authorization to a PFCG role. For more information, see [Creating Authorization Roles for Catalogs](#).

- On the **front-end server**, you must assign the OData service authorization to a new or existing role, such as a business role that has been adjusted according to your needs. For more information, see [Creating Roles on Front-End and Assigning Launchpad Catalogs and Spaces/Pages](#) and [Assigning Roles to Users on Front-End](#).

OData Service (Version Number)	Back-End Server: Authorization Role (PFCG Role)	Front-End Server: Assignment to Authorization Role
UI_SUBSCRPNCONTRACT_02_MON (1)	You can use the example business catalog SAP_SOM_BC_SUBSCR_ORDER or your custom catalog to create the back-end PFCG role.	You can use the existing business role SAP_BR_INTRNAL_SALESREP_SOM as a template to create your own business role.

Configuration Tasks

You can perform the following configuration tasks in the back-end system:

- Create and define change processes such that they are available in the **Change Contracts** quick access menu as per your requirement.

Related Information

[Manage Subscription Contract Documents](#)

[App Details: Manage Subscription Contract Documents](#)

App Details: Manage Subscription Contract Documents

Manage the subscription contract document and get an overview of all subscription contracts contained within the document, which are represented by the document items.

Searching for Documents

You can search for the relevant subscription contract document using multiple filter criteria, such as the **Document Number**, **Sold-To Party**, and other document attributes such as the **Posting Date** and the **Employee Responsible**.

i Note

The filter attributes **Product** and **Contract** are enabled to search over the underlying contract items. For technical reasons, they must always be present in the result list. Since multiple products or contracts may exist within the contract document, these fields will show no content in the search results.

Additional links allow you to navigate to related apps where you can manage the business partner and the subscription contract document. Further navigation options exist via the **Open In** menu , which is leads you to specific applications in the back end for the subscription contract document.

General Information

The most important attributes of the document, such as partners and organization information, are available under **General Information**. You can navigate to further details, such as for the **Payer Party**, **Bill-To Party**, and **Employee Responsible**. The **Error Status** is only visible in case of any errors in the contract document.

Contracts

The contract hierarchy displays all the contracts and underlying contract items and subitems contained in the subscription contract document. For active or future dated contracts, the items and subitems from the current and planned contract state are shown. For contracts that are inactive because they have expired or have been rejected, the expired/rejected items and subitems are displayed. For phased contracts with multiple future dated items, all planned items are shown in the hierarchy in addition to the current items.

At the contract level of the hierarchy, only common attributes are displayed, such as the estimated contract net and gross values, contract start and end dates. At the item and subitem level, item-specific attributes are available such as the contract state, product, quantity, and status. The **Net Value (Estimated)** and the **Gross Value (Estimated)** of the contract are calculated as an aggregation of all contract item values.

From the contract hierarchy, you can navigate to the **Manage Subscription Contract Lifecycle** app using the  navigation icon and to the **Manage Subscription Contract Item Details** app using the  chevron navigation icon. In both cases, a page with details of the specific item is displayed.

Since a specific contract item cannot be selected at the contract level, the resultant search list will contain all the contract items and the relevant item can be selected for navigation to the details.

Estimated Document Values and Prices

The **Document Net Value (Estimated)** is the aggregated estimated net value across all contracts of the document. It is displayed at the header KPIs. In the **Prices** section, the **Net Value**, **Gross Value**, and **Tax Value** are displayed per document.

Document History

In the **Document History** assignment block, you see all change-related documents (such as change orders) with respect to the subscription contracts within the subscription contract document.

Contract Document Change Processes

You can access and execute specific header change processes using the **Change Contracts** quick access menu. In this menu, the change processes are available only if they have been enabled for the SAP Fiori apps. To enable these change processes for the apps, you must check the **Fiori Menu Enabled** flag in Customizing for **Service** under  **Transactions**  **Settings for Subscription Transactions**  **Contract Changes**  **Settings for the Maintenance Framework (BTMF)**  **Define Scenario Dependent Attributes**  view.

Related Apps

The following apps are accessible from the **Manage Subscription Contract Document** app:

- **Manage Subscription Contract Lifecycle**
- **Manage Subscription Contract Item Details**
- **Manage Subscription Product-Specific Data**

Related Information

[Manage Subscription Contract Lifecycle](#)

[Manage Subscription Contract Item Details](#)

[Manage Subscription Product-Specific Data](#)

Mass Changes of Subscriptions

App ID: F7405

With this app, you can choose all the subscription contracts or subscription contract documents that you want to change and take them over into a mass processing.

Key Features

This app provides the following key features:

- A create and configure wizard that enables you to conveniently configure the mass processing.
- Create a mass processing by using the available subscription contracts or contract documents and change processes (item level and header level). All subscription contracts or contract documents that don't already belong to a mass processing with either **Open**, **In Process**, or **Error** status are eligible for a new mass processing.
- Since only certain contracts or contract documents are eligible for mass processing, the app provides multiple filters to enable a simple search and select experience using the **Contract Items** and **Contract Documents** tabs.
- The app is tightly integrated with the following contract lifecycle apps on the SAP Fiori launchpad:
 - **Manage Subscription Contract Lifecycle**
 - **Manage Subscription Contract Documents**
 - **Manage Mass Processing of Subscriptions**
- While some predefined and standard change processes, such as **Cancel** and **Change Sold-to Party**, are delivered, you can also integrate own change processes of type ISTA via configuration and app extensibility.

Supported Device Types

- Desktop
- Tablet

Related Information

[Manage Subscription Contract Lifecycle](#)

[Manage Subscription Contract Documents](#)

[Manage Mass Processing of Subscriptions](#)

App Implementation: Mass Changes of Subscriptions

Prerequisites

Before implementing the app, you must ensure the following:

- You have activated the OData service `UI_SUBSCRPNMASSPROCG_MANAGE` on the SAP Fiori front-end server.
- You have assigned the role `SAP_BR_INTRNAL_SALESREP_SOM` on the SAP Fiori front-end server.

System Landscape Requirements

This is custom documentation. For more information, please visit [SAP Help Portal](#).

For information about the system requirements, see [SAP Fiori Overview](#).

Implementation Tasks

You have activated the OData service. The following sections list tasks that have to be performed to implement this app. The tables contain the app-specific data required for these tasks.

You can also find the data required to perform these tasks in the SAP Fiori apps reference library at [Mass Changes of Subscriptions](#).

Back-End Server: Create Connectors

This app is based on ABAP Core Data Services and does not require search models to work. However, if you want to start the app from the search results list, you must activate the underlying search connectors. You can create connectors in transaction ESH_COCKPIT.

For more information, see [Creating Connectors](#).

i Note

Note the following system behavior:

Search software components build a stack. Different layers can be installed on top of the basis component.

Search models can be available in different components - in their original component, but also in higher layer components (extension components).

Once you create a search connector in an extension component, all search models from the original component are transferred into the extension component. You will therefore no longer find the search models in the original component, as is shown in the table below.

Front-End Server: Activate OData Services

Component	Technical Name
S4CEXT	For the Mass Changes of Subscriptions app, the following OData service must always be activated: UI_SUBSCRPNMASSPROCG_MANAGE

For more information about activating OData services, see [Activating OData Services](#).

Front-End Server : Activate SAPUI5 Application

Component	Technical Name
SAPUI5 Application	MASSCHNG_MANS1

For more information about how to activate the ICF service, see [Activate ICF Services of SAPUI5 Application](#).

Front-End Server: Activate ICF Service

Component	Technical Name
ICF Service	/sap/bc/ui5_ui5/sap/masschnng_mans1

For more information about how to activate the ICF service, see [Activate ICF Services of SAPUI5 Application](#).

Front-End Server : Enable App for Access in SAP Fiori Launchpad

The SAP Fiori launchpad is the entry point to apps. In the launchpad, the user can see and access those apps that have been assigned by an administrator to the catalog designed for this user's role.

The administrator makes these assignments in the launchpad app manager. For more information, see [Setting Up Technical Catalogs with the SAP Fiori Launchpad App Manager](#).

SAP delivers technical catalogs for groups of apps as repositories to create your own catalogs in the launchpad app manager or content manager (see [Tools for Setting Up Launchpad Content](#)). Along with these catalogs, more technical content is delivered for each SAP Fiori app. You can find the delivered technical content for each SAP Fiori app in the SAP Fiori apps reference library.

Front-End Server and Back-End Server: Assign OData Service Authorizations to Users

You must assign OData service authorizations for the app to your users.

Caution

Several authorization default values are connected to the OData service. To ensure that all these default values are assigned to a user, you have to follow the instructions given under the documentation links provided.

Make the assignment on the back-end server and on the front-end server:

- On the **back-end server**, you have to assign the OData service authorization to a PFCG role. For more information, see [Creating Authorization Roles for Catalogs](#).
- On the **front-end server**, you must assign the OData service authorization to a new or existing role, such as a business role that has been adjusted according to your needs. For more information, see [Creating Roles on Front-End and Assigning Launchpad Catalogs and Spaces/Pages](#) and [Assigning Roles to Users on Front-End](#).

OData Service (Version Number)	Back-End Server: Authorization Role (PFCG Role)	Front-End Server: Assignment to Authorization Role
UI_SUBSCRPNMASSPROCG_MANAGE	You can use the example business catalog SAP_COM_BC_SUBSCR_ORDER or your custom catalog to create the back-end PFCG role.	You can use the existing business role, SAP_BR_INTRNAL_SALESREP_SOM as a template to create your own business role.

Configuration Tasks

To be able to implement the app, you must perform the following configuration tasks in the back-end system:

- Enable number range object for creating a mass processing ID in the Customizing  **Service**  **Transactions**  **Settings for Subscription Transactions**  **Mass Processing**  **Define Number Range for Mass Processing** .

- Configure change process groups for your own change processes that are to be included into the mass processing. You can do so in the Customizing ► [Service](#) ► [Transactions](#) ► [Settings for Subscription Transactions](#) ► [Contract Changes](#) ► [Settings for the Maintenance Framework \(BTMF\)](#) ►

Related Information

[Mass Changes of Subscriptions](#)

[App Details: Mass Changes of Subscriptions](#)

App Details: Mass Changes of Subscriptions

The [Mass Changes of Subscriptions](#) app is a list report app for selecting and taking over subscription contracts or contract documents relevant for mass processing.

This app is tightly integrated with the following SAP Fiori apps:

- [Manage Subscription Contract Lifecycle](#)
- [Manage Subscription Contract Documents](#)
- [Manage Mass Processing of Subscriptions](#)

Prerequisites

You must define number ranges before using mass processing for the first time. You can do so in the Customizing ► [Service](#) ► [Transactions](#) ► [Settings for Subscription Transactions](#) ► [Mass Processing](#) ► [Define Mass Processing Type](#) ►.

Activities

Searching for Subscription Contracts and Documents for Mass Processing

You can search for subscription contracts or contract documents that are relevant for a mass processing using multiple search filters. As this app handles subscription contracts and contracts documents, the given filter criteria is typically effective for only one of both. For example, some filters such as [Contract Main Item](#), [Contract Start](#), and [Contract End](#) are not relevant for the [Contract Documents](#) tab. If you try to filter using any of the irrelevant filter values, an error message informs you of the same.

However, a specific set of the most important filters are enabled to work on both the [Contract Items](#) and [Contract Documents](#) views:

- [Contract](#) number
- [Document](#) number
- [Sold-To Party](#)
- [Product](#)
- [Sales Organization](#) (along with the [Distribution Channel](#), [Division](#), [Sales Area](#), and [Sales Office](#))
- [Posting Date](#)

In case of combined filters, the result set is restricted for both views, otherwise only one of both is restricted, while the other view continues to show the total number of available records. For combined selection fields, the number of contracts and contract documents must be in sync. That is, the number of documents is either less than or equal to the number of contracts since a document can hold more than one contract item.

In the search results, there are additional links that enable you to navigate to related apps where you can work on the product, business partner, subscription contract, and the subscription contract document.

To take over relevant subscription contracts or contract documents into a mass processing, you select them in the search results and start the wizard using [Create Mass Processing](#).

i Note

In case a subscription contract or contract document is locked in a mass processing with the status [Open](#), [Scheduled for Processing](#), [In Process](#), or [Processed with Errors](#), it cannot be included in another mass processing. Such a contract or contract document will have an associated [Mass Processing ID](#) and the [Create Mass Processing](#) wizard will be disabled if these are selected from the search list. Additionally, no manual change processes can be executed on such a subscription contract either until the mass processing is completed successfully or the contract is deleted from the mass processing. In such a case, you can use the BAdI CRMS4_SOM_MP_CP_CHECK in Customizing under [Service > Transactions > Settings for Subscription Transactions > Mass Processing > BAdI: Additional Checks for Change Processes](#) to ensure that change processes are hidden in the [Processes](#) section as well.

This is because a change process, when executed, sets an end date on the current time slice and creates a new time slice (item) as per the activation date. Due to this, the current time slice (that is included in the mass processing) is invalidated and the new time slice becomes relevant (but is not included in the mass processing). Additionally, the new item may also lie in the future (depending on the provided activation date) and therefore, may also not meet the conditions for a contract in the mass processing. To avoid this issue, a set of check methods have been provided and they suppress the change processes on the UI when required. Similar checks already exist in case a contract is contained in a change quote.

Create and Configure Mass Processing

You can configure relevant subscription contracts or contract documents using the mass processing wizard, which allows you to determine the change data and then hand over the contracts or contract documents to the mass processing using the following procedure:

1. Choose the type of change you want to execute on the subscription contracts using [Mass Processing Type](#) and then provide a description for the mass processing.

i Note

You can only trigger the header change process [Change Sold-to Party](#) from the [Documents](#) tab in the search list.

2. Depending on the selected mass processing type, you can provide values for specific fields required for implementing the underlying change processes. Once these values are taken over into a mass processing, you can still adapt them for the individual subscription contracts or contract documents. For example, you may set different activation dates for one or more contracts or contract documents.
3. Once you review and finalize the provided details, you are navigated to the mass processing details on the [Manage Mass Processing of Subscriptions](#) Fiori app.

Mass Processing Type

The mass processing type determines the contract changes that can be combined into a single mass processing. The mass processing types depends on the Customizing of the change process groups under [Service > Transactions > Settings for Subscription Transactions > Contract Changes > Settings for Maintenance Framework \(BTMF\)](#). This allows you to choose between executing a single change process per mass processing execution or multiple change processes combined into one change process group. Here, you can also define own combinations of change processes, which can then be integrated into a mass processing run.

❖ Example

You have defined a change process group that combines the change processes for contract renewal (extend) and quantity change, where both change processes are executed in a specific sequence within the same mass processing run. In the mass processing wizard, the fields required for both change processes are determined based on the mass processing type Customizing.

i Note

Only change processes of type ISTA are supported currently for the mass processing

You can see the existing mass processing types and create a new one in the Customizing [►Service►Transactions►Settings for Subscription Transactions►Mass Processing►Define Mass Processing Types](#).

Standard Item-Level Change Process Groups

SAP provides a set of preconfigured change process groups that work out of the box.

i Note

Not all change processes can be combined within a single change process group. For example, you cannot combine the Cancel change process with any other change processes.

❖ Example

- You must not combine change processes that are specific to main and subitems, especially [Cancel](#) and [Cancel Subitem](#). This is because the [Cancel](#) change process, when implemented on the main item, includes the subitems implicitly. A combination of both the change processes will trigger an error during subitem cancellation as this would already be done during cancellation of the main item.

Typically, cancellation is considered to be a standalone change process.
- The [Change Product](#) change process has certain constraints with respect to configurable products. Additionally, you cannot change to a product that references a master agreement.

The following change processes are currently available for mass processing:

Technical Name	Type	Description
CANCEL	ISTA	Cancel
CANCEL_OPTION	ISTA	Cancel Subitem
CHANGE_BILL_CYCLE	ISTA	Change Billing Cycle
CHANGE_PROD_QTY	ISTA	Change Product and Quantity
CHANGE_QUANTITY	ISTA	Change Quantity
EXTENSION	ISTA	Extend
PROD_CHANGE	ISTA	Change Product
		i Note

Technical Name	Type	Description
		This change process has some constraints in the mass processing framework.
CHANGE_WBS_ELEMENT	ISTA	Change WBS Element
CHANGE_PRICE_AND_QTY	ISTA	Change fixed price (condition type PSFX) and quantity
UPDATE_PRICE	ISTA	Execute a repricing (either a complete repricing or keep the manual changes)
CCM_VERSION_UPGRADE	ISTA	Upgrade a cross-catalog mapping version to current version

You can refer to the sample change process types for some information on the combination of change processes.

Standard Header-Level Change Process

The following header-level change process is currently available for mass processing:

Technical Name	Type	Description
CHANGE_SOLDTO_PARTY	ISTA	Change Sold-to Party i Note This is a header change process that is available only if the mass processing is created for subscription contract documents. You can use this change process to modify organizational data in the contract document.

You can modify the organizational data when executing the **Change Sold-To Party** change process. The change of organizational data is supported in the following organizational models:

- SD Sales Area (Sales Organization, Distribution Channel, Division, Sales Office, Sales Group)

It is assumed that the SD Sales Area is used by default.

- CRM Sales Organizational Unit

In case an organizations unit is used, you must consider the following prerequisites:

- If the sales organizational unit is used, it must be configured in the Customizing under **Service > Transactions > Settings for Subscription Transactions > Mass Processing > Define Mass Processing Type**.
- The buffering of the organizational data must be activated. You can do so in the Customizing under **Service > Master Data > Organizational Management > Tools > Organizational Management Buffer Setting**.

The change process is configured to provide different fields based on organizational model in use.

Change Process Fields

The relevant change process fields are defined for specific change processes and can be seen in the Customizing ► [Service Transactions](#) ► [Settings for Subscription Transactions](#) ► [Mass Processing](#) ► [Define Mass Processing Type](#) .

Activation Date

The activation date is a mandatory field for all change process types. When you [Create Mass Processing](#) via the configuration wizard, you can only provide a single activation date for all the contracts or contract documents included in the mass processing. However, you can use the [Manage Mass Processing of Subscriptions](#) app to provide separate activation dates for the contracts or contract documents. You can also use the mass edit feature to set the activation date for multiple contracts or contract documents at the same time.

i Note

All items of a specific subscription contract must have the same activation date.

❖ Example

Consider a mass processing which contains subscription contracts from different business partners. You may choose to set the same activation date for all contracts belonging to the same business partner.

Related Information

[Maintenance Framework for Business Transactions](#)

[App Details: Manage Mass Processing of Subscriptions](#)

App Extensibility: Mass Changes of Subscriptions (Key User)

As a key user, you can extend the [Mass Changes of Subscriptions](#) app according to your business needs.

i Note

The SAP Fiori app [Mass Changes of Subscriptions](#) (F7405) is closely related to the [Manage Mass Processing of Subscriptions](#) (F7406) app, and they share the same business context. The activities and specific logic for adding change process related fields are shared and apply to both apps in general.

Custom Fields

You can create and edit custom fields to enhance applications using the [Custom Fields](#) app. You can add fields to the following UI elements using key user adaptation:

Business Context	UI Technology	UI Element Type
Subscription Mass Processing - Elements of Contract Items (CRMS4_SOM_MPEE)	SAP Fiori	Create Mass Processing - Wizard Popup

For more information, see:

- [Custom Fields](#)
- [Adding Extension Fields to Apps with Different UI-Technologies](#)

Activities

If the newly created custom fields need to be displayed on the Fiori app, they must be enabled in the Customizing of the corresponding mass processing types for the UI. You do so in the Customizing [►Service►Transactions►Settings for Subscription Transactions►Mass Processing►Define Mass Processing Type](#) . This ensures that all relevant fields for a change process are determined during runtime from the Customizing and displayed dynamically. This procedure is only applicable for those fields that are relevant for change processes.

Related Information

[Extensibility](#)

[App Extensibility: Manage Mass Processing of Subscriptions \(Key User\)](#)

Manage Mass Processing of Subscriptions

App ID: F7406

With this app, you can manage mass processing functionalities such as triggering a mass processing execution and monitoring the results.

Key Features

This app provides the following key features:

- Create a mass processing by using the available subscription contracts or contract documents and change processes (item level for subscription contracts and header level for contract documents). Only those subscription contracts or contract documents that don't belong to a mass processing with status **Open**, **In Process**, or **Error** can be added to a new mass processing.
- Delete an existing mass processing.
- Manage existing mass processing runs by:
 - Triggering the mass processing either to run immediately or by scheduling it to run at a later date and time
 - Modifying the list of subscription contracts or contract documents either by adding or deleting contracts or contract documents
 - Configuring the change process on an individual level for each contract or contract document, for example by having separate activation dates per line
 - Modifying the prices for subscription contracts within a mass processing based on the relevant pricing conditions using the **Change Price** change process.
 - Updating the prices (repricing) for subscription contracts with the option of retaining manual conditions using the **Update Price** change process.
 - Updating the pricing condition type and additional related details such as the ratio, price, and quantity for the selected subscription contracts.

i Note

This feature is available only if the mass processing is created for a price-relevant process type.

The existing condition types are always deleted before the new condition types are created.

- Viewing the application logs for subscription contracts or contract documents for error determination
- Rejecting any open changes created during the mass run and associated with subscription contracts or contract documents

Caution

We recommend that you do not add more than 1000 elements in a single mass processing run.

You can enable insights cards in the [Manage Mass Processing of Subscriptions](#) app using the Collaboration Manager. For information about this, refer to [Insights Cards](#) and [Adding Insights Cards to Chats](#).

Supported Device Types

- Desktop
- Tablet

App Implementation: Manage Mass Processing of Subscriptions

Prerequisites

Before implementing the app, you must ensure the following:

- You have activated the OData service UI_SUBSCRPNMASSPROCG_MANAGE on the SAP Fiori front-end server.
- You have assigned the role SAP_BR_INTRNAL_SALESREP_SOM on the SAP Fiori front-end server.

System Landscape Requirements

For information about the system requirements, see [SAP Fiori Overview](#).

Implementation Tasks

The following sections list tasks that have to be performed to implement this app. The tables contain the app-specific data required for these tasks.

You can also find the data required to perform these tasks in the SAP Fiori apps reference library at [Manage Mass Processing of Subscriptions](#).

Back-End Server: Create Connectors

This app is based on ABAP Core Data Services and does not require search models to work. However, if you want to start the app from the search results list, you must activate the underlying search connectors. You can create connectors in transaction ESH_COCKPIT.

For more information, see [Creating Connectors](#).

i Note

Note the following system behavior:

Search software components build a stack. Different layers can be installed on top of the basis component.

Search models can be available in different components - in their original component, but also in higher layer components (extension components).

Once you create a search connector in an extension component, all search models from the original component are transferred into the extension component. You will therefore no longer find the search models in the original component, as is shown in the table below.

Front-End Server: Activate OData Services

Component	Technical Name
S4CEXT	For the Manage Mass Processing of Subscriptions app, the following OData service must always be activated: UI_SUBSCRPNMASSPROC_G_MANAGE

For more information about activating OData services, see [Activating OData Services](#).

Front-End Server : Activate SAPUI5 Application

Component	Technical Name
SAPUI5 Application	MASSPROC_MANS1

For more information about how to activate the ICF service, see [Activate ICF Services of SAPUI5 Application](#).

Front-End Server: Activate ICF Service

Component	Technical Name
ICF Service	/sap/bc/ui5_ui5/sap/massproc_mans1

For more information about how to activate the ICF service, see [Activate ICF Services of SAPUI5 Application](#).

Front-End Server : Enable App for Access in SAP Fiori Launchpad

The SAP Fiori launchpad is the entry point to apps. In the launchpad, the user can see and access those apps that have been assigned by an administrator to the catalog designed for this user's role.

The administrator makes these assignments in the launchpad app manager. For more information, see [Setting Up Technical Catalogs with the SAP Fiori Launchpad App Manager](#).

SAP delivers technical catalogs for groups of apps as repositories to create your own catalogs in the launchpad app manager or content manager (see [Tools for Setting Up Launchpad Content](#)). Along with these catalogs, more technical content is delivered for each SAP Fiori app. You can find the delivered technical content for each SAP Fiori app in the SAP Fiori apps reference library.

Front-End Server and Back-End Server: Assign OData Service Authorizations to Users

You must assign OData service authorizations for the app to your users.

Caution

Several authorization default values are connected to the OData service. To ensure that all these default values are assigned to a user, you have to follow the instructions given under the documentation links provided.

Make the assignment on the back-end server and on the front-end server:

- On the **back-end server**, you have to assign the OData service authorization to a PFCG role. For more information, see [Creating Authorization Roles for Catalogs](#).
- On the **front-end server**, you must assign the OData service authorization to a new or existing role, such as a business role that has been adjusted according to your needs. For more information, see [Creating Roles on Front-End and Assigning Launchpad Catalogs and Spaces/Pages](#) and [Assigning Roles to Users on Front-End](#).

OData Service (Version Number)	Back-End Server: Authorization Role (PFCG Role)	Front-End Server: Assignment to Authorization Role
UI_SUBSCRPNMASSPROCG_MANAGE	You can use the sample business catalog SAP_COM_BC_SUBSCR_ORDER or your custom catalog to create the back-end PFCG role.	You can use the existing business role, SAP_BR_INTRNAL_SALESREP_SOM as a template to create your own business role.

Configuration Tasks

To implement the app, you must perform the following configuration tasks in the back-end system:

- Enable number range object for creating a mass processing ID in the Customizing [►Service►Transactions►Settings for Subscription Transactions►Mass Processing►Define Number Range for Mass Processing](#).
- Configure change process groups for your own change processes that are to be included into the mass processing. You can do so in the Customizing [►Service►Transactions►Settings for Subscription Transactions►Contract Changes►Settings for the Maintenance Framework \(BTMF\)](#).

Related Information

[Manage Mass Processing of Subscriptions](#)

[App Details: Manage Mass Processing of Subscriptions](#)

App Details: Manage Mass Processing of Subscriptions

The [Manage Mass Processing of Subscriptions](#) app enables you to maintain and execute mass processing runs.

i Note

While you can create a mass processing from here as well, the recommendation is to use the mass processing configuration wizard from the [Mass Changes of Subscriptions](#) app, which is tightly integrated with this app.

Prerequisites

You must consider the following prerequisites before adding a subscription contract or contract document to a mass processing:

- A subscription contract or contract document can only be included once in a single mass processing run.

As long as a contract or contract document is included in a mass processing with status **Open**, **Scheduled for Processing**, **In Process**, or **Processed with Errors**, it cannot be added into another mass processing. Such a contract or contract document can also not be changed manually via a regular change process as this would generate a new time slice, thus invalidating the item contained in the mass change process.

- When a contract or contract document is deleted from a mass processing, any associated open changes must be rejected beforehand. You can do so from within the app itself.

In case a change is created during the mass processing and the contract or contract document change fails, this change must be rejected before the contract or contract document can be deleted from the mass processing.

- Only the current and technically active items can be included in a mass processing run.
- You can include main items and subitems in the same mass processing if they support all the change processes included in the change process group. In such a case, the main item and subitems must always have the same activation date.
- Phased contracts are not supported.
- No open change order or open change quote must exist with respect to the selected contract or contract document.

Activities

You can search for existing mass processing runs using multiple search filters, such as the **Type** of change process, **Status** of the mass processing run, or the **Scheduled Execution Date**. Once you navigate into a mass processing, you can see further details in the header and the **General Information** section. Depending on the status of the mass processing, you can see unprocessed or in process contracts in the **Processing** section and fully processed contracts in the **Processed** section.

There are additional links that enable you to navigate to related apps where you can work on the product, business partner, subscription contract, and the subscription contract document. Further navigation options in the **Open In** menu lead you to other applications such as **Display Provider Contract** and **Display Subscription Contract**. If your mass processing has item level change processes, you can navigate into the item details page to access the **Open In** menu.

Creating a New Mass Processing

From the list view page, the **Create** button allows you to choose a mass processing type. An empty mass processing is created with status **Open**. Depending on the **Processing Type** that you have chosen, you can add either subscription contracts (for item level changes such as **Cancel**) or subscription contract documents (for header level changes such as **Change Sold-To Party**) to the mass processing. Each processing type has some mandatory information that must be provided before the mass processing can be created, for example, the **Activation Date**. Once all the required contracts or contract documents are added and the mandatory details are supplied, you can **Create** the mass processing. As long as a mass processing is not **Processed Successfully**, you can make changes to it in the **Edit** mode.

Executing a Mass Processing from the List View

You can select one or more mass processing runs on the list overview page, to run or schedule them without navigating into the mass processing details. This is possible only if the mass processing status is **Open** or is not **Processed Successfully**. You can also reset an existing schedule for one or more mass processing runs as long as the status is **Scheduled for Processing**.

Processing of Change Process Groups

A change process contained in a change process group is executed only if the value that is changed is not initial and it differs from the existing value in the subscription contract. If either of these criteria is not met, the change process is ignored when the group is processed. Those change processes within the same change process group that fulfill these criteria are still processed. The change order is submitted if all relevant change processes are executed successfully while the irrelevant change processes are ignored.

❖ Example

An existing contract has the quantity 1. If it is now configured in the processing with no new value (that is, quantity is left empty) or the new value is again set to 1, this change process within the change process group is ignored during execution, an error is raised, and the application log is updated with the reason for non-execution, which can later be analyzed.

Adding Contracts or Contract Documents to a Mass Processing

You can [Add](#) subscription contracts or contract documents in the [Processing](#) section as long as the overall status of the mass processing is either [Open](#) or [Processed with Errors](#).

i Note

You can only add those contracts or contract documents that are not already included in an open, scheduled, in process, or erroneous mass processing. This condition is already considered when filtering for the list of eligible contracts or contract documents.

Editing Subscription Contracts

You can edit the change processing details for each contract or contract document in a mass processing, either one at a time or for multiple lines at once. The fields that are enabled for input are specific to the selected mass processing type.

❖ Example

You have contracts belonging to different sold-to parties that will undergo the same changes but will have a different activation date per business partner. You can now filter for all the contracts belonging to one business partner and apply one activation date. You can then filter for all contracts belonging to another business partner and apply a second activation date to these contracts at once.

Repricing Subscription Contracts

You can modify the prices for the subscription contracts within a mass processing in one of the following ways:

- Reprice the subscription contracts within a mass processing based on the pricing master data of the product. The new change process [Update Price](#) is used for this repricing. You can check the [Complete Reprice](#) flag while creating a mass processing from the [Mass Changes of Subscriptions](#) Fiori app or directly within the mass processing itself. If you execute a complete repricing, then the manually maintained pricing conditions on the contracts will be deleted.
- Manually update the price for the given subscription contracts within a mass processing based on the pricing condition assigned manually to the mass processing type in the Customizing node [Define Mass Processing Type](#). For this functionality, the [Price Relevant Change](#) flag has been introduced in Customizing [Service](#) > [Transactions](#) > [Settings for Subscription Transactions](#) > [Mass Processing](#) > [Define Mass Processing Type](#) . Only if this flag is set, can you navigate to the [Pricing Conditions](#) page to define the relevant pricing conditions for the mass processing type.

i Note

Variant pricing conditions are out of scope.

The assigned pricing condition must be a part of the pricing procedure.

If this flag is checked, then you can navigate to the item details page in the mass processing where you can select the condition type in the [Change Price Condition](#) section and provide the required change details, for example, price.

Removing Contracts or Contract Documents From a Mass Processing

You can **Delete** selected contracts or contract documents from the mass processing altogether. In some cases, if the execution of a specific contract or contract document has triggered an error and an open change already exists, the change order must be rejected before the contract or contract document can be deleted from the mass processing.

Executing the Mass Processing

Once you create a mass processing, you can either **Run** it immediately or **Schedule** it to run at a later date and time. The status of the mass processing becomes either **In Process** or **Scheduled for Processing** accordingly. To schedule a mass processing, you must first schedule the report CRMS4_SOM_MP_EXECUTE (Execute Report for Mass Processing of Subscriptions). The report executes all those mass processing runs that have their scheduled date on or before the report run date and the status as **Scheduled for Processing**. You find this report in the Customizing ► **Service** ► **Transactions** ► **Settings for Subscription Transactions** ► **Mass Processing** ► **Execute Report for Mass Processing of Subscriptions** ►.

You can also **Reset Schedule** for a schedule mass processing as long as the processing has not yet been triggered. This removes the predefined schedule and resets the mass processing status to **Open** or **Processed with Errors**, depending on the execution status.

Manual Execution

You can execute a mass processing manually using **Run**. The **Refresh** button enables you to update the mass processing status and check the number of contracts or contract documents with status **Processing**, **Processed Successfully**, or **Processed with Errors**. All successfully processed contracts or contract documents are displayed in the **Processed** section.

Error Detection and Correction

A mass processing is **Processed Successfully** only if all the subscription contracts or contract documents have been processed successfully. If even one has execution errors, the status will be **Processed with Errors**. Erroneous contracts or contract documents remain in the **Processing** section for correction and error analysis. For more information about the error, you can access the **Application Log** using .

Corrections of Change Orders

There may arise a situation where a change order is already created during change processing but it is not submitted due to some execution errors. In such a case, the application log is updated with a description and reason for error. The change order is displayed as a link in the **Processing** section within the mass processing and you can use this link to directly navigate to the change order. Although you can fix the errors in the change order directly, you cannot submit it. Instead, you must re trigger the mass processing using **Run**. If you choose to delete the erroneous contract from the mass processing, you must first reject the open change order.

Rejection of Change Orders

When you reject a change order from the mass processing details, you will be able to see all the available rejection reasons. The rejection profile that is defined in Customizing under ► **Service** ► **Transactions** ► **Settings for Service Transactions** ► **Settings for Rejection Reasons for Service Transactions** ► is not considered in the value help. Once a mass processing is executed, the system checks whether the selected rejection reason is allowed based on whether a rejection profile is assigned to the transaction.

Constraints

There are some constraints to using the mass processing functionality:

- Only ISTA change processes are supported.
- No change processes for inactive subscription contracts are supported.
- Phased contracts are not supported.

- The **Change Product** change process has limitations for contracts with reference to a master agreement. For configurable products, the configuration must be adapted in the change order once it is created.

Related Information

[Maintenance Framework for Business Transactions](#)

[App Details: Mass Changes of Subscriptions](#)

App Extensibility: Manage Mass Processing of Subscriptions (Key User)

As a key user, you can extend the **Manage Mass Processing of Subscriptions** app according to your business needs.

i Note

The SAP Fiori app **Manage Mass Processing of Subscriptions** (F7406) is closely related to the **Mass Changes of Subscriptions** (F7405) app, and they share the same business context. The activities and specific logic for adding change process related fields are shared and apply to both apps in general.

For more information, refer to **App Extensibility: Mass Changes of Subscriptions (Key User)**.

Custom Fields

You can create and edit custom fields to enhance applications using the **Custom Fields** app. You can add fields to the following UI elements using key user adaptation:

Business Context	UI Technology	UI Element Type
Subscription Mass Processing (CRMS4_SOM_MP)	SAP Fiori	Mass Processing - List, Details - section General Information
Subscription Mass Processing - Elements of Contract Items (CRMS4_SOM_MPEE)	SAP Fiori	Mass Processing - Details, Contract Items section Contract Item - Details - section Change Data

For more information, see:

- [Custom Fields](#)
- [Adding Extension Fields to Apps with Different UI-Technologies](#)

Business users can display fields that you have enabled in the **Custom Fields** app. The fields that you enable for this app are included in the personalization options for business users for the following elements:

UI Element	Business Context
Mass Processing - list	Subscription Mass Processing (CRMS4_SOM_MP)
Contract Item - Details - section change data, table	Subscription Mass Processing - elements of Contract Items (CRMS4_SOM_MPEE)

i Note

Depending on the mass processing type, if the custom fields are to be shown or hidden dynamically, they must be added in the [►Change Data►Contract Changes►Contract Item Detail►](#).

Activities

If the newly created custom fields need to be displayed on the Fiori app, they must be enabled in the Customizing of the corresponding mass processing types for the UI. You do so in the Customizing [►Service►Transactions►Settings for Subscription Transactions►Mass Processing►Define Mass Processing Type►](#). This ensures that all relevant fields for a change process are determined during runtime from the Customizing and displayed dynamically. This procedure is only applicable for those fields that are relevant for change processes.

Additional Standard Fields

You can add standard fields that are not shown on the UI by default, using key user adaptation. To check which standard fields you can add to the app, start adding a field to the UI element that you want to enhance. You can then view a list of all available fields (standard and custom).

For more information on how to adapt SAP Fiori at runtime, see [Make UI Changes \(Object Pages\)](#).

For a general description of the extensibility options and procedures of SAP Fiori apps, see [Extend SAP Fiori Apps](#).

Integration of Own Change Processes

The mass processing framework is designed to combine change processes into groups so that the change processes can be executed together. You can also include your own change processes and easily enable them for mass processing using the following procedure:

1. Extend the mass processing and apply the necessary fields required for the change process. For more information about this, refer to **App Extensibility: Manage Mass Processing of Subscriptions (Key User)**.
2. Adapt the necessary fields for handling the change process in a change process group in **Define Change Process Group**. You do so in the Customizing under [►Service►Transactions►Settings for Subscription Transactions►Contract Changes►Settings for the Maintenance Framework \(BTMF\)►](#).
3. Extend the interface-method IF_CRM_BTMF~PREPARE_CONTAINER to integrate the change process and necessary checks for the processing execution.

Related Information

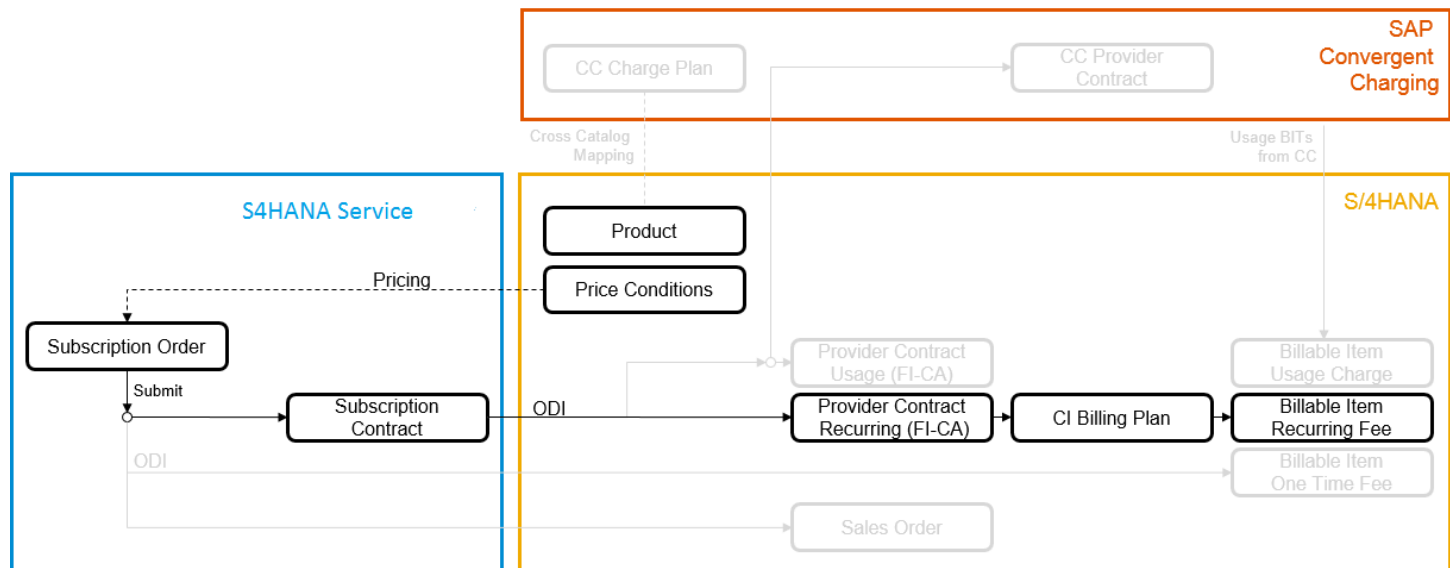
[Extensibility](#)

[App Extensibility: Mass Changes of Subscriptions \(Key User\)](#)

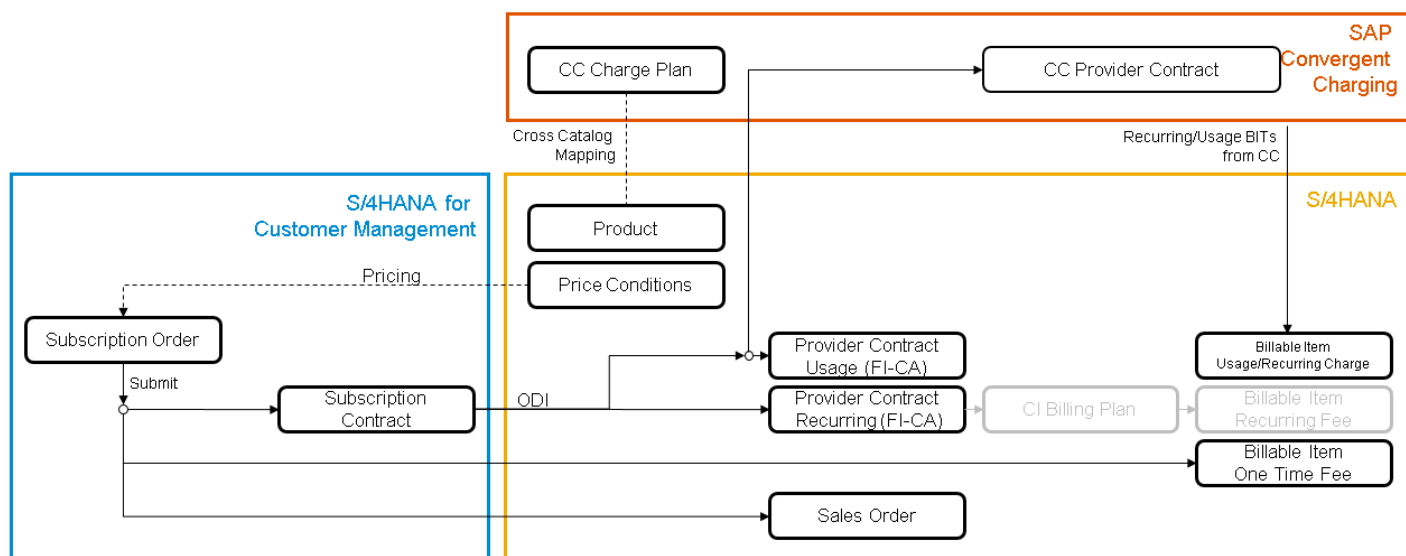
Price Modelling

Price modelling enables you to use multiple types of fees and charges, such as recurring fees, usage and one-off charges, and allowances, for price determination in Convergent Invoicing and SAP Convergent Charging.

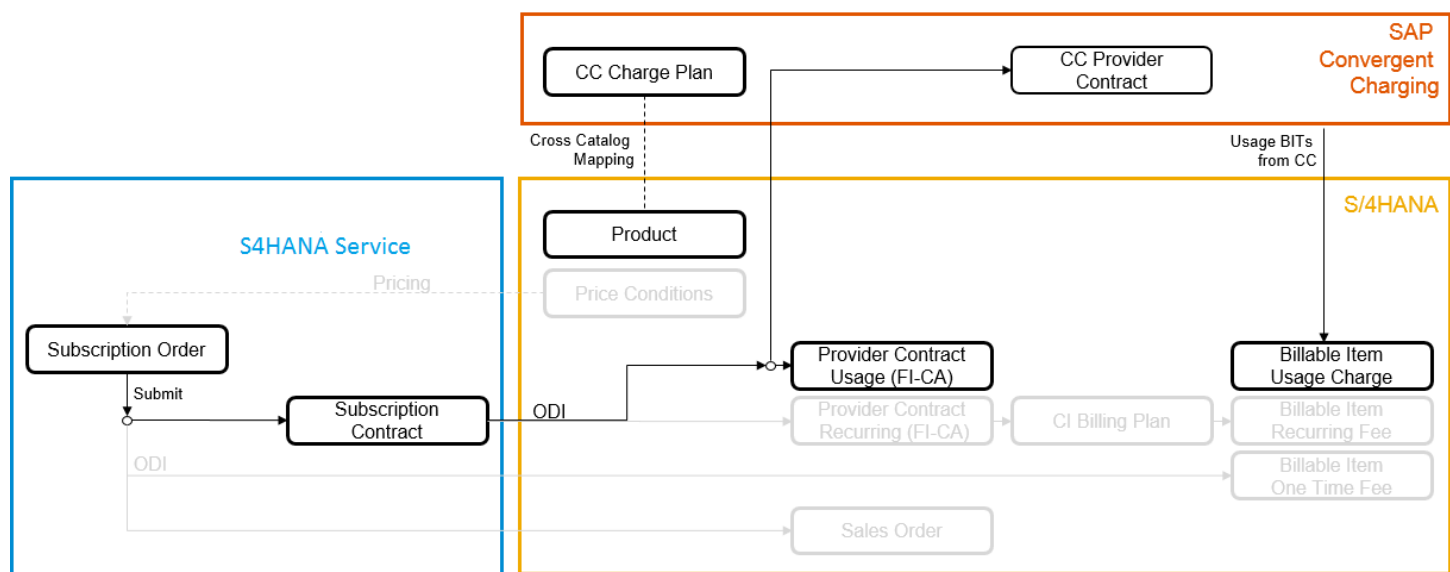
For more information about these fees and charges, refer to the figures below.



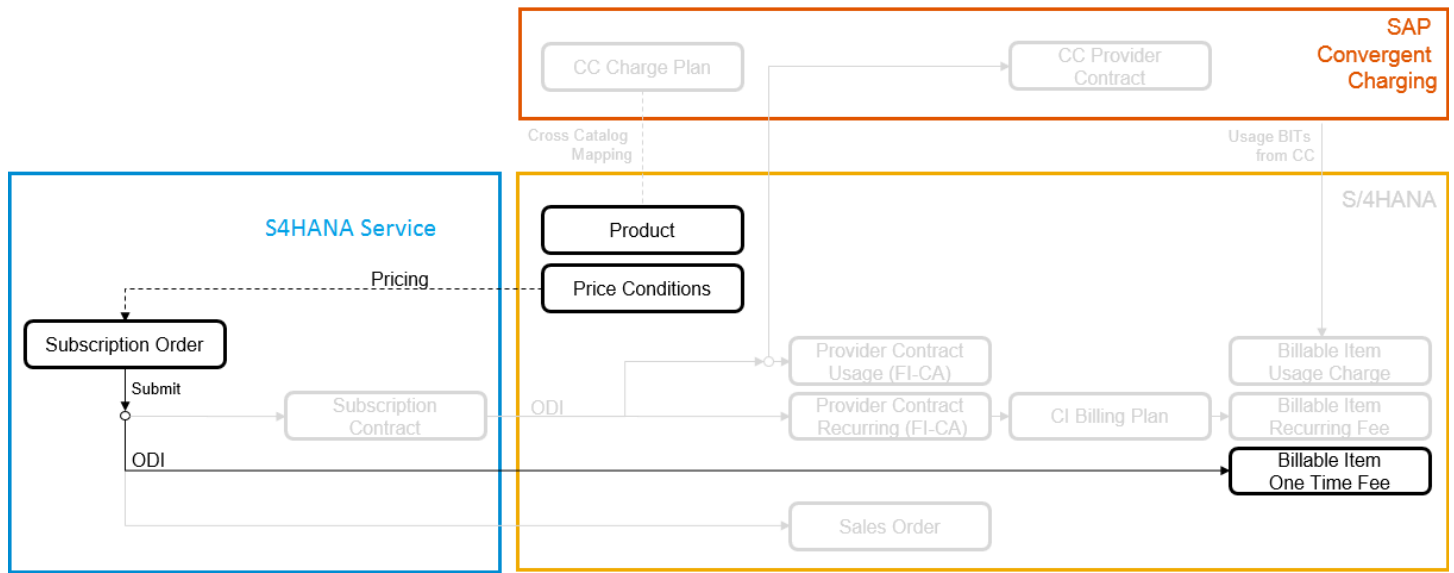
Recurring Fees with CI Bill Plans in Convergent Invoicing



Recurring Fees in SAP Convergent Charging



Usage-Based Charges



One-Off Charges

In Subscription Order Management, you can also manage free usage or discounts using allowances and counters in subscription contracts. For more information about this, refer to **Allowances** and **Counters**. You may choose to share these allowances and counters between multiple subscription contracts. For more information, refer to **Sharing**.

Related Information

[Recurring Fees](#)

[Usage-Based Charges](#)

[One-Off Charges](#)

[Counters](#)

[Sharing](#)

Recurring Fees

Subscription products that are used in subscription orders, may need to be invoiced periodically like a monthly fee for a cloud subscription.

Related Information

[Product Master](#)

[Document Distribution](#)

Recurring Fees in Convergent Invoicing

Integrating into Billing Plans in Convergent Invoicing

When a subscription order with a recurring fee is submitted (released), a subscription contract is created. During contract distribution to FI-CA, contract items for which recurring fees exist in the pricing elements are identified. A dependent billing plan is created in Convergent Invoicing for these items, in addition to the provider contract item.

This billing plan holds the information about the billing cycle, reference to the provider contract item, payment amount, and payment currency. You can now use this billing plan to request the recurring fees using the corresponding features in Convergent Invoicing.

Billing Cycle

The billing cycle used for subscription order is either derived from the billing cycle or based on the billing cycle determination rule in the product. Billing cycle can be replaced manually in subscription order. This billing cycle is copied to subscription contract and provider contract in FICA and is used for scheduling the billing dates. It can also be used to schedule the invoicing in Convergent Invoicing.

This billing cycle is also copied from the provider contract to the assigned CI billing plan by default.

Sample Scenario

You have defined a product with a monthly recurring price of \$10/month and assigned a yearly billing cycle (YEAR) to your product. The following steps occur during order creation:

1. The billing cycle is derived either directly from the product or based on a billing cycle determination rule in the product. YEAR is found in this scenario.
2. Pricing is executed and the monthly recurring fee is derived (PSMB monthly subscription fee \$10)
3. Integrating with Billing Plans in Convergent Invoicing: Recurring value on item is calculated based of the billing cycle frequency (\$10 per month * 12 months = \$120/year)
4. Since the frequency of the recurring price condition (Monthly) is not identical with the frequency assigned to the billing cycle (Yearly), the billing plan cycle field to maintain an alternative billing cycle for the billing plan is editable in the subscription order item.

Scenario 1: Use the Yearly Billing Cycle

The billing plan inherits the billing cycle from the provider contract and is created with the amount for the yearly fee (\$120) and a yearly billing cycle (YEAR). Once a year you will receive one BIT with the yearly amount (\$120) if your Convergent Invoicing is set up accordingly.

Scenario 2: Use the Billing Plan Cycle

Your order is created as described in the scenario above. However, you want to have one billable item for your recurring fee per month in Convergent Invoicing (CI). In this case, the billing plan cycle for the CI billing document will not be inherited from the contract but can be explicitly overwritten. You can set an alternative billing cycle (MONTH) in the billing plan cycle field in the subscription order item. The recurring fee for the item is recalculated (\$10) based on this billing plan cycle (MONTH). The billing plan will be created with the amount for the monthly fee (\$10) and the billing plan cycle (MONTH), instead of the one still used in the contract (YEAR). Once a month you will receive BIT with the monthly amount (\$10) if your Convergent Invoicing is set up accordingly.

Change Processes

You can request billable items based on future billing plans. If you try to run a change process that affects contracts items with a billing plan for which such billable items are created, the contract activation fails with an error in the contract distribution fulfillment messages. BITs have to first be reverted based on the corresponding process steps in Convergent Invoicing. Post that, the contract distribution can be re triggered.

Calculation of Prorated Values in CI Billing Plan

Since the calculation of BIT values in CI differs from that in Subscription Order Management, there is a chance of mismatch between the BIT total amount and the total estimated contract value.

Sample Scenario

Consider a subscription contract with a duration for 5 months starting from 01.01.2023 to 31.05.2023. There is a price of \$100 per month and a quarterly billing cycle.

In Subscription Order Management, the contract net value is calculated as $\$100 * 5 \text{ months} = \500 and the recurring charge is \$300 per quarter.

In Convergent Invoicing, the provider contract is created with a recurring billing plan item with amount \$300 per quarter. The BIT values are calculated as below:

Quarter 1 01.01.2023 to 31.03.2023 = \$300

Quarter 2 01.04.2023 to 31.05.2023 (This is a prorated period of 2 months in Q2) = $\$300 / 91 \text{ days (total number of days in Q2)} * 61 \text{ (actual number of days billed in the quarter)} = \201.10

Thus, the BIT total amount becomes \$501.10, which does not match the contract net value in Subscription Order Management.

To avoid such a rounding issue, you must check the **Sync Prorated Values in SOM and CI** flag in the Customizing under **►Service ►Transactions ►Settings for Subscription Transactions ►General Settings for Subscription Transactions ►**.

Activities

You can perform the following functions for recurring fees:

1. Calculate a price for your recurring fee based on your pricing procedure and the maintained price conditions.
2. You can change this price manually.
3. You can maintain a billing plan cycle if the frequency of your price condition is not the same as the frequency in your billing cycle.
4. When the subscription order is submitted, the subscription contract is distributed and the FI-CA provider contract with a billing plan in Convergent Invoicing is created.

More Information

The following features give you an overview of the recommended Customizing for this function, delivered as standard by SAP:

- Use condition type PSMB (Monthly Subscription Fee) used in pricing procedure A17023 Subscription Order (US) to create a price for your product.
- Use a subscription product and assign a billing cycle to this subscription product and item category group PRRP (Subscription Order Item) in product master. If the product does not have usage-based price components that are managed in SAP Convergent Charging, a cross catalog mapping in the Manage Subscription Product-Specific app is not required.
- Object Type is BUS2000155 Subscription Order Item
- Use transaction type PRVO
- Item category PROP is customized as a subscription item
- The document distribution step to create a provider contract in FI-CA is PCEA. With BADI CRM_ISX_ERP_CONTRACT, you can implement specific logic during document distribution.

The system identifies recurring fees based in the calculation type in the condition type. The following calculation types are allowed for recurring fees:

- M Quantity - Monthly Price
- N Quantity - Year Price
- O Quantity - Daily Price
- P Quantity - Weekly Price

For more information about this, refer to

- **Billing Plan in Convergent Invoicing**
- Customizing for **Service** under **Basic Functions > Pricing > Configure Pricing > Define Condition Types**
- Customizing for **Service** under **Transactions > Settings for Subscription Transactions > Recurring Fees**

Discounts and Surcharges

The following sample scenarios discuss the possible ways you could model your discounts, surcharges, and condition records in Subscription Order Management:

Scenario	Example	
Absolute one-time discount	Consider a subscription contract with: Duration: 3 years Price: \$1000/year Billing cycle: Yearly Discount: One-time overall discount of \$300 to be billed in the first year	Since discount is set to one-time, the Condition Type is setup with Calculation Type = C (Quantity) or B(Fixed). With detailed billing, the condition type must be mapped to a one-time CI billing plan item type. The invoices would be generated as below: Invoice for Year 1 = \$1000 – \$300 = \$700 Invoice for Years 2 and 3 = \$1000 (discount applied one time in the first year)
Absolute recurring discount	Consider a subscription contract with: Duration: 1 year Price: \$100/month Billing cycle: Monthly Discount: \$10/month	Since discount is set to recurring, the Condition Type is setup with Calculation Type = M (Quantity - Monthly Price), N (Quantity - Year Price), O (Quantity - Daily Price), or P(Quantity - Weekly Price). With detailed billing, the condition type must be mapped to a recurring CI billing plan item type. The invoices would be generated as below: Invoice generated every month = \$100 – \$10 = \$90
Percentage one-time discount i Note This scenario is similar to the first scenario which discusses absolute one-	Consider a subscription contract with: Duration: 3 years Price: \$1000/year Billing cycle: Yearly	Since discount is set to one-time and calculated in percentage, the Condition Type is setup with Calculation Type = A (Percentage).

Scenario	Example	
time discounts	Discount: One-time overall discount of 10% (= \$300) to be billed in the first year	With detailed billing, the condition type is mapped to a one-time CI billing plan item type. The invoices would be generated as below: Invoice for Year 1 = \$1000 – \$300 = \$700 Invoice for Years 2 and 3 = \$1000 (discount applied one time in the first year)
Percentage recurring discount	Consider a subscription contract with: Duration: 1 year Price: \$100/month Billing cycle: Monthly Discount: 10%/month	It is not possible to setup a Condition Type which is a % discount with Calculation Type = recurring (M,N,O, or P), since the calculation type itself = A (Percentage). Hence, this scenario can only be modelled using one of the following workarounds: With quantity based pricing You check the Quantity Based Calculation flag in the Customizing under ► Service ► Transactions ► Settings for Subscription Transactions ► General Settings for Subscription Transactions ►. For more information, refer to SAP Note 3022091 (Switching from Quantity time based pricing to quantity based pricing). When this flag is checked, the End Value for an order item is not based on duration but on the calculation of price*quantity. In this example, since the condition type is mapped to a recurring CI billing plan item type, the discount item in the billing plan is calculated as -\$10 per month. Thus, the invoice generated every month = \$100 - \$10 = \$90. With quantity-duration based pricing You uncheck the Quantity Based Calculation flag. When this flag is unchecked, the End Value for an order item is calculated based on the duration and is mapped to the discount item in the billing plan and the division is not done as this condition type is not of type M,N,O and P. To use this method, you use the BAdI <code>CRM_ISX_ERP_CONTRACT</code> for a custom implementation. i Note You must ensure that you cater for any possible rounding off issues in your custom implementation.

Detailed Billing

Use

It is now possible to create a billing plan with one detailed billing plan item for a recurring condition type in Convergent Invoicing. This enables generation of a detailed invoice to customers with details of the recurring fee, discounts, and surcharges applied.

Prerequisites

The following are the prerequisites for detailed billing:

- Detailed billing must be enabled for a specific transaction type item category in the Customizing for [Service](#) under [► Transactions ► Settings for Subscription Transactions ► Recurring Fees ► Configure Settings using Billing Plans with Convergent Invoicing ► Assign Billing Plan and Billing Plan Item Types ►](#).
- If detailed billing is enabled, the billing plan item category for condition type must be maintained in the Customizing for [Service](#) under [► Transactions ► Settings for Subscription Transactions ► Recurring Fees ► Configure Settings using Billing Plans with Convergent Invoicing ► Define Price Conditions for Distribution to Convergent Invoicing ►](#).
- If detailed billing is marked relevant for a subscription contract, the billing plan item categories must be maintained for all possible recurring condition types relevant for that subscription contract.
- No billing plan item is created if the recurring charge is 0. However, if there is a recurring charge, a billing plan item is created for the relevant condition types even with 0 value discounts and surcharges. These can be filtered by implementing the BAdI `CRM_ISX_ERP_CONTRACT`.

Recurring Fees in SAP Convergent Charging

Recurring fees can also be managed via SAP Convergent Charging (SAP CC) as an alternate to billing plans in Convergent Invoicing.

This is applicable for scenarios with a complex logic for the recurring fee. You can use this functionality to replicate relevant price conditions to SAP CC using the Data Replication Framework (DRF). Condition records for recurring fees must be replicated from SAP S/4HANA to SAP CC.

You can maintain price conditions in the SAP S/4HANA central price master data, where some price conditions represent recurring fees. While processing the solution quotation, subscription order, or subscription contract, pricing determines the (gross or net) prices, discounts and surcharges, and scale prices using the condition technique.

The condition technique can be used to define prices according to marketing requirements. These marketing requirements are driven by SAP S/4HANA central pricing and do not need to be modeled in SAP CC. Marketing-driven pricing criteria such as campaigns or customer groups can determine the price conditions for a product. The pricing conditions must be replicated from Subscription Order Management to SAP CC in this scenario since recurring fees are processed in SAP CC. In SAP S/4HANA central pricing, master data can be used for price determination. Since not all this master data is available in SAP CC, the SAP S/4HANA pricing functionality is used to determine the recurring prices. All the relevant price conditions including base price, variant price, surcharge and discounts, and scale-based prices are replicated to SAP CC using the price table, price table percentage, scale-based price tables (base scale, to scale, and graduated scale tables in SAP CC). These are then considered during calculation of recurring fees in SAP CC.

Features

- Replicate price master data, including base price, variant price, discounts, and surcharges, and scale-based prices in SAP CC.
- Create and update price table, price table percentage and scale-based table in SAP CC.

i Note

Replicate price master data, including discounts and surcharges to SAP CC using Data Replication Framework (DRF)

- Price Tables, Price Table Percentage, and Range Tables (Price Table Base Scale, Price Table To Scale, and Price Table Graduated Scale) are agreement tables.

Price tables are created using the name of the subscription product and price table percentage are created using the name of the product along with the suffix '_PERCENTAGE'.

Price table scales are created using the name of the subscription product along with the suffix '_BSCALE' (for base scale), '_TSCALE' (for to scale), and '_GSCALE' (for graduated scale).

- A price key in a range table will be suffixed with either _A, _B, or _D depending on whether it is stored in a base scale table, to scale table, or a graduated scale table respectively.

If the range tables contain percentage values, the price keys will be prefixed with '%_'.

- Replicate transaction-specific manual prices, including discounts and surcharges, to SAP CC using the ODI step **PRCC**.

i Note

Manual prices or contract-specific prices are not supported for scale-based prices.

- Replicate aggregated surcharge and discounts to SAP CC.

i Note

The aggregated surcharge and discounts would be considered as manual fees because no condition records exist and they are determined in the transaction. A condition type must be maintained for replicating the aggregate values.

- In case of manual fees, only a manual price table/price table percentage is derived or created in SAP CC during contract distribution.
- SAP CC creates recurring fees and passes them to Convergent Invoicing for further processing.

Price Keys in Recurring fees

All fields in a condition table that influence a price condition are grouped into an ID known as the price key. The price key ID is used to replicate condition records from Subscription Order Management to SAP CC. You can define the condition types that represent recurring fees in the Customizing [►Service ► Transactions ► Settings for Subscription Transactions ► Recurring Fees ► Configure Settings Using SAP Convergent Charging ► Define Price Conditions for Distribution to SAP Convergent Charging ►](#).

When a condition record is created for the relevant condition type, a price key is generated for this condition record and replicated to SAP CC using Data replication Framework (DRF). The condition record access fields are not replicated. The system only transfers the price key, validity, amount, and currency to SAP CC.

If you create a new price or change an existing one, which means the system adds a new condition record with identical access fields to an existing condition record, it automatically assigns the price key used in the existing record to the new condition record during replication. In SAP CC, replicated pricing data is entered in agreement tables.

Assigning Price Key Mapping in Cross Catalog Mapping of a Subscription Product

The definition of recurring fees is also part of the subscription-specific data of subscription product. When you assign charge plan handling with recurring fees, a price key parameter is available and can be linked to the condition types or variant condition key aliases that define the condition record referenced during contract creation. Only the price keys maintained in the subscription product-specific data, under [Cross Catalog Mapping > Charge Plans > Price Keys](#) are available as parameters in the SAP CC provider contract. The price key parameters can be defined as a part of the charge plan in SAP CC and condition types can be assigned. Using price versions, you can retain identical prices across multiple mapping versions. For more information about using price versions in cross catalog mapping, see [Price Version](#).

i Note

When working with manual prices in combination with list prices, you must model the different recurring fees in two charge plan mappings in Cross-Catalog Mapping. Nevertheless, you can use the same charge plan. In the case of Variant Condition Key alias, the alias name must be defined such that a unique variant condition key can be determined during contract creation.

For more information about using recurring fees in product master data, see [Defining Recurring Fees from SAP CC for Subscription Products](#).

Price Replication

The Data Replication Framework (DRF) is used to replicate price condition records, prices, and percentage values (if any) from SAP S/4HANA to SAP CC. The report DRFOUT can be used for Initial or delta replication of prices. The following settings are required for initial or delta replication using DRFOUT.

1. Define technical settings for business systems under (transaction) [DRFIMG > Data Replication > Define Custom Settings for Data Replication > Define Technical Settings > Define Technical Settings for Business Systems](#)
2. Define replication models under (transaction) [DRFIMG > Data Replication > Define Custom Settings for Data Replication > Define Replication Models](#). The replication model and outbound implementation is used in transaction [DRFOUT](#) to replicate pricing records to SAP CC.
3. The relevant price table classes must be manually created in SAP CC for absolute values, scale-based prices, and for discounts and surcharges. The following classes can be created depending on the scenario: Price Table, Price Table Percentage, and Range Tables.
4. You use the transaction DRFLOG to monitor the replication status of condition records in either one of these cases:
 - automatic replication during condition record maintenance from price maintenance APP
 - transaction [VK11](#) or [VK12](#)

You use the transaction DRFOUT to monitor status in case of initial or delta data replication.

Activities

1. Set the distribution scenario as FI-CA and SAP CC in the Customizing [Service > Transactions > Settings for Subscription Transactions > Document Distribution > Define Distribution Scenario](#)
2. Define condition types relevant for replication to SAP CC in the Customizing [Service > Transactions > Settings for Subscription Transactions > Recurring Fees > Configure Settings using SAP Convergent Charging > Define Price Conditions for Distribution to SAP Convergent Charging](#)
3. Set the [Recurring in CC](#) flag as relevant item categories in the Customizing [Service > Transactions > Settings for Subscription Transactions > Define Settings for Item Categories](#)
4. To activate the charging contract in SAP CC for recurring only scenario, the ODI step [PCCA](#) must be part of the relevant schema.

5. Define number of agreement tables in case manual price scenario is relevant in the Customizing [Service Transactions Settings for Subscription Transactions](#) [Recurring Fees](#) [Configure Settings using SAP Convergent Charging](#) [Maintain Manual Price Table Count in Convergent Charging](#).

6. Create a mapping table class in SAP CC with following attributes:

Price Table Classes

Name	Input Columns	Output Columns
Price Table	price.key (Type - String) currency (Type currency)	amount (Type - number)
Price Table Percentage	price.key (Type - String)	price (Type - Number) unit (Type - String)

This mapping table class is used to create a price table and/or price table percentage and must exist exactly with that definition.

7. Create a range table class in SAP CC with the following attributes:

Range Table Classes

Name	Range	Input Columns	Output Columns
Price Table Base Scale	Exclusive...Unbounded	price.key (Type - String)	price (Type - Number) unit (Type - String)
Price Table To Scale	Inclusive...Bounded	price.key (Type - String)	price (Type - Number) unit (Type - String)
Price Table Graduated Scale	Inclusive...Bounded	price.key (Type - String)	price (Type - Range Size Cumulative Number) unit (Type - String)

This range table class is used to create a Price Table Base Scale, Price Table To Scale, and Price Table Graduated Scale and must exist exactly with that definition.

Manual Contract Price Replication

The ODI step **PRCC** is part of the relevant schema and checks if the determined recurring prices have been manually updated in the subscription item. During contract replication, this step automatically determines or creates a price table and replicates the new contract specific recurring prices to SAP CC.

i Note

Manual contract replication is only relevant for price tables and price tables percentage.

Related Information

[Price Version](#)

[Defining Recurring Fees from SAP CC for Subscription Products](#)

Replicating Scale Based Prices to SAP Convergent Charging

Replicate scale based prices and scale based discounts and surcharges to SAP CC for consideration during price calculation.

Features

For a scale condition type, if regular prices (not scale prices) are maintained in SAP S/4HANA, the price will be replicated to SAP CC as a price table and not as a range table.

If you choose to maintain scale values later on, the scale prices will still be replicated to SAP CC as range tables.

Replicating Quantity to SAP CC

Scale-based prices are calculated based on the item quantity defined in the subscription contract. Due to this, the item quantity must also be replicated to the SAP CC. To do so, a mandatory parameter must be created in the charge and the charge plan must be created so that the quantity is replicated to the provider contract in SAP CC via the BRFplus function Item Quantity (CRMS4_SOM_ITEM_QUANTITY). For more information about this, refer to **BRFplus and BRFplus Functions**.

The charge and charge plan can contain a mix of price tables, price tables percentage, and range tables of type price and/or surcharge and discounts. If there are range tables and range table price keys in the SAP CC parameters, the quantity parameter must also exist.

Revision in Range Tables in SAP CC

The range tables Price Table Base Scale, Price Table To Scale, and Price Table Graduated Scale can now store prices, and surcharge and discount in absolute and percentage values.

Since multiple condition records can be created for each of the condition types, revision of the range table is independent of the validity period of the condition record validity. When the range table is created, the version is Initial. The version then updated along with the system date every time an existing record is updated or a new record is inserted. Since the condition record for the same condition type is different when the validity is different, it results in different price keys and both price keys are visible after the revision. Since the correct condition record is determined based on the contract creation, the correct price key is identified and picked up. If a particular record/scale is updated, the price key for the new version holds the updated values, along with the unchanged records of the other condition types.

Constraints

- Manual scale prices or contract-specific scale prices are not supported.
- It is recommended that From Scales should start with the lowest value = 0 due to technical reasons in SAP CC where the supported lowest value = 0.

Related Information

[Recurring Fees in SAP Convergent Charging](#)

[BRFplus and BRFplus Functions](#)

Price Tables and Related Classes in SAP Convergent Charging

There are 5 sets of price table classes in SAP Convergent Charging (SAP CC), each serving a specific purpose - Price Table, Price Table Percentage, Price Table Base Scale, Price Table To Scale, and Price Table Graduated Scale. When dealing with condition types for pricing adjustments in SAP SD (such as surcharge, discount, absolute values, and scales), it's essential to understand which class creates which type of table in SAP CC.

To enable use of a combination of manual and conditional prices, you must model your charge plan with a minimum of one and a maximum of four price tables. Refer to the possible combinations of Price Tables and Price Tables Percentage, and how these combinations influence the pricing:

Combination	Pricing Option
2 Price Tables (PT) and 2 Price Tables Percentage (PTP)	When surcharge and discount are in percentage and recurring price is absolute, then the recurring price can be fetched either from the product PT or manual PT. The surcharge and discount will be fetched from the product PTP or Manual PTP.
1 Price Table (PT) and 1 Price Table Percentage (PTP)	If a single PT is modeled in the charge plan, then the recurring price, variant price, and other absolute prices must either all be fetched from the product PT or must all be manually updated so that they are fetched from the manual PT. A mixed condition scenario, that is, if only some are manually updated is not supported and will trigger an error. Similarly, if only a single PTP is modeled in the charge plan then all the percentage prices should either exist in the product PTP or manual PTP. A mixed scenario is supported only when more than one PTs or PTPs are modeled in the charge plan.
2 Price Tables (PT) and 0 Price Table Percentage (PTP)	If no percentage values need to be replicated to SAP CC, the charge plan can be modeled with 2 PTs only. Absolute prices can reside either in the product PT or manual PT.
1 Price Table (PT) and 0 Price Table Percentage (PTP)	If no percentage values need to be replicated to SAP CC, the charge plan can be modeled without a PTP. If only a single PT is modeled in the charge plan, then the recurring price, variant price, and other absolute prices must either all be fetched from the product PT or must all be manually updated so that they are fetched from the manual PT. A mixed condition scenario, that is, if only some are manually updated is not supported and will trigger an error.

i Note

Additional combinations of PT and PTP are also supported but if only a single PT or PTP is modeled in the charge plan then all the records must be fetched either from condition records or from manual price tables.

Example

Condition Type	Condition Class	Calculation Type	Scale Type
Base scale	B - Prices	M - Quantity Monthly	A - Base scale
To scale	B - Prices	M - Quantity Monthly	B - To scale
Graduated To scale	B - Prices	M - Quantity Monthly	D - Graduated To scale

Condition Type	Condition Class	Calculation Type	Scale Type
Recurring price	B - Prices	M - Quantity Monthly	
Absolute surcharge	A Disc/Sur	M - Quantity Monthly	
Discount percentage	A Disc/Sur	A - Percentage	

Base Scale

Consider the following scale prices for a product with reference to the condition type Base-Scale.

- **Price Configuration**
 - Amounts
 - 1 to 100 : \$3
 - 101 to 200 : \$2
 - 201 to infinity: \$1
- **Agreement Table in CC**
 - Created under Price Table Base Scale class
 - Table Name: <product name>_BSCALE

This ensures that the product has defined scale prices and the associated agreement table follows the naming convention with the _BSCALE suffix.

To Scale

Consider the following scale prices for a product with reference to the condition type To_Scale_Product.

- **Price Configuration**
 - Amounts:
 - 0 to 100: \$30
 - 101 to 200: \$20
 - 201 to 300: \$10
- **Agreement Table in CC**
 - Created under Price Table To Scale class
 - Table Name: <product name>_TSCALE

This ensures that the product has defined scale prices and the associated agreement table follows the naming convention with the _TSCALE suffix.

Graduated Scale

Consider the following scale prices for a product with reference to the condition type Graduated_Scale_Product

- **Price Configuration**

- Amounts
 - 0 to 100: \$30
 - 101 to 200: \$20
 - 201 to 300: \$10

- **Agreement Table in CC**

- Created under Price Table Graduated Scale class
- Table Name: <product name>_GSCALE

This ensures that the product has defined scale prices and the associated agreement table follows the naming convention with the _GSCALE suffix.

Recurring Price

Consider the following recurring price for a product with reference to the condition type Recurring Price.

- **Price Configuration**

- Amount: \$100

- **Agreement Table in CC**

- Created under Price Table class
- Table Name: <product name>

This table name class will have the same name as the product (without any suffix) and will refer to the Price Table Class.

Absolute Surcharge

Consider the following absolute surcharge for a product with reference to the condition type Absolute Surcharge.

- **Price Configuration**

- Amount: \$50

- **Agreement Table in CC**

- Created under Price Table class
- Table Name: <product name>

This table name class will have the same name as the product (without any suffix) and will refer to the Price Table Class.

Discount Percentage

Consider the following discount percentage for a product with reference to the condition type Discount Percentage.

- **Price Configuration**

- Amount: \$50

- **Agreement Table in CC**

- Created under Price Table Percentage class

- Table Name: <product name>_Percentage

Since the maintained discount is in percentage value, the associated agreement table follows the naming convention with the _PERCENTAGE suffix.

In summary, this documentation provides a generic understanding of how different condition types contribute to the creation of agreement tables in SAP CC.

Price Version

This function provides the price version as a reference during mapping of cross-catalog products. To model a cross-catalog product, data is mapped to a charge plan data in the SAP Convergent Charging system. Each mapping version has a validity period and is identified by a unique mapping ID. Recurring charges are relevant for each mapping version. Since one price is often valid for several mapping versions, you can now use the price version as a reference within your mapping.

❖ Example

Maintain the same price for new product version

You have modeled an unlimited cloud usage product. The current cross-catalog mapping references the charge plan data in SAP Convergent Charging. With the beginning of the next period, this mapping needs to be adapted since additional features are available for the product. However, you do not want to define new recurring charges for this mapping version. For this purpose, the new mapping version references the existing price version. Both mapping versions reference the same price master data.

❖ Example

Price increase for new customers

You want to increase the price of an existing product and keep the price stable for existing customers, in the respective contracts. You create a new version of the cross-catalog mapping for this product and assign a new price version to this mapping. For this price version, you adapt the recurring charges in pricing master data. Since the price version is an attribute of the subscription contract, old and new customers are charged differently.

Prorated Fees in SAP Convergent Charging

Example

Assume you have a recurring fee that is calculated as price * quantity, and the quantity is changed in the middle of a billing cycle. In such a case, the calculated price for the first half of the period must be prorated. Additionally, a new price must be calculated for the second half of the period using the new quantity. These calculations are performed in SAP Convergent Charging (SAP CC) using prorata parameters on the recurring rate. For more information about this, refer to [Recurring Rate](#). If a change in the quantity or any other parameters with an impact on the calculation of the recurring fee triggers this proration in SAP CC, a new item must be created as a successor of the previous item. The previous item is delimited and an end date is set such that its validity ends once the change is implemented.

Prerequisites

To implement proration from Subscription Order Management, the following settings are required:

- In the Cross-Catalog Mapping version of your subscription product, you must define that a charge plan assignment is relevant for proration using the **Proration** indicator. For more information about this, refer to **Charge Plans**.
- If you use counters for such a charge plan, they must be defined as **External** in SAP CC so that the counter can be shared between the original and successor items (created during proration). If you assign no namespace to such a counter in the **Cross-Catalog Mapping Version**, a unique namespace is assigned when the subscription contract is distributed to SAP CC.
- If the quantity is relevant for calculation of recurring fees, your charge plan needs a numeric parameter for the quantity. You can use the standard BRFplus function, Item Quantity, to do the mapping.
- If the billing cycle period is relevant for calculation of recurring fees, your charge plan needs a string parameter for the bill cycle period. You can use the standard BRFplus function, Bill Cycle Period, to map this parameter.
- If the billing cycle is relevant for calculation of recurring fees, your charge plan needs a string parameter for the billing cycle. You can use the standard BRFplus function, Bill Cycle, to map this parameter.
- Your item categories in subscription orders and subscription contracts must have the **Recurring in CC** and **Enable Quantity** flags set.
- It is not recommended to use one-time fees, like activation or suspension fees, for such charge plans. Otherwise, they will always be triggered when one item has ended and the new item is created. You must model one-time fees as separate charge plans.
- It is not recommended to set **Visibility Level = External (Mandatory unmodifiable)** for the standard parameter SAP_BILL_CYCLE_DATE in SAP CC, because this parameter is changed every time a new item is created in the FI-CA provider contract. As the change may not be relevant for proration, a new item is not required in SAP CC.

In the standard integration, a change of quantity and a change of billing cycle are considered to be relevant for proration. If one of these attributes is changed, a new charging item is created in SAP CC. If there are other parameters that impact the calculation of a recurring fee, you must implement the BAdI CRMS4_SOM_CC_ITEM_SPLIT_BADI to check if such a parameter has been changed.

i Note

SAP CC sends such prorated BITs to SAP Convergent Invoicing with the highest precision. This means that if your currency (for example, USD) only supports up to two decimal points, you must define an internal currency with 6 digits (for example, USD6) in SAP S/4HANA to avoid replication errors.

The integration feature **D Support unmodifiable charge plan parameters for proration** has been available in SAP CC from release 2023 FPS00. You can activate this feature in the Customizing under **Service > Master Data > Settings for Subscription Master Data > Convergent Charging > Activate Integration Features**. If you have activated this feature, you can already define the charge plan parameters that are relevant for proration using **Visibility Level = External (Mandatory unmodifiable)** in SAP CC. In this case, you cannot set the proration indicator on a charge plan assignment manually. It will be set automatically when reading a charge plan from SAP CC. If this feature is active, the BAdI CRMS4_SOM_CC_ITEM_SPLIT_BADI is no longer be called for the CCM versions that are created after the feature has been activated. However, the BAdI is still called for the older CCM versions.

Related Information

[Cross-Catalog Mapping Versions](#)

[Charge Plans](#)

Billing Cycle and Determination Rules

Use

You can use this function to automatically assign billing cycles to subscription orders. The assignment is determined by making settings for the corresponding product and can be a direct assignment of a billing cycle or a complex determination based on a class implementation.

For more information about displaying billing cycles, refer to Customizing under ► [Service](#) ► [Transactions](#) ► [Settings for Subscription Transactions](#) ► [Recurring Fees](#) ► [Configure Settings using Billing Plans with Convergent Invoicing](#) ► [Configure Settings for Recurring Fees with Billing Plans](#) ►.

You can define rules to determine billing cycles for a product when capturing a sales order or a change order. These rules can be used in the product master data.

You can specify that one of these rules is used as the default setting. The system uses this rule if a product has not been assigned a specific rule.

Features

Determination Rules

The definition of billing cycle determination rules has three main purposes:

1. You can define a default rule that applies to all products that require a billing cycle, but do not have one specified.
2. You can assign a determination rule to a group of products and change the billing cycle without maintaining the products.
3. You can apply complex rules instead of simple assignments to the determination, by implementing determination logic.

You can make these settings in Customizing under ► [Service](#) ► [Transactions](#) ► [Settings for Subscription Transactions](#) ► [Recurring Fees](#) ► [Configure Settings using Billing Plans with Convergent Invoicing](#) ► [Define Billing Cycle Determination Rules](#) ►.

Maintaining a Determination Rule

The following general types of determination rules exist:

- Rules that assign a single billing cycle

Enter an ID, a description and, a billing cycle. When you assign a rule to a product, you can maintain it centrally.

- Rules that assign a billing cycle determination class

An implementation class using the interface IF_CRM_ISX_CYCLE_DETERMINATN is required. Enter an ID and description for the rule and select your implementation class using the F4 Help in the column **Billing Cycle Determination Class**. The F4 Help returns classes that implement the required interface.

Billing Cycle Determination Class

Implement the method IS_CYCLE_MANUALLY_SELECTABLE for the interface IF_CRM_ISX_CYCLE_DETERMINATN in a new implementation class to allow the manual selection of billing cycles in the subscription order.

The method interface includes the following fields:

- IV_RULE_ID: Use this ID if you want to use the same implementation in different determination rules
- IV_ITEM_GUID: Use this ID to read the appropriate information, for example product or appointments, if the result depends on an order attribute

- RV_IS_MANUALLY_SELECTABLE: Returns ABAP_TRUE to return a list of several billing cycles

Return ABAP_FALSE if you want to determine a single billing cycle automatically. In case you do not implement this method, the manual selection of billing cycles is turned off

Implement the method IS_CYCLE_AND_RULE_ALLWD_IN_PRD for the interface IF_CRM_ISX_CYCLE_DETERMINATN to allow assignment of a billing cycle and a billing cycle determination rule, in the product master, at a time.

The method interface includes the following fields:

- IV_RULE_ID: ID rule that is assigned in the product master
- IV_CYCLE: Billing cycle that is assigned in the product master
- RV_ALLOWED: Set this indicator = 'X' if the combination of billing cycle and billing cycle determination rule is allowed together in the product master

Implement the method GET_CYCLE_LIST for the interface IF_CRM_ISX_CYCLE_DETERMINATN in a new implementation class to dynamically determine a billing cycle.

The method interface includes the following fields:

- IV_RULE_ID: Use this ID if you want to use the same implementation in different determination rules.
- IV_ITEM_GUID: Use this ID to read the appropriate information, for example product or appointments, if the available billing cycles depend on an order attribute.
- ET_CYCLES

Add the billing cycle determined to this table. You can return several billing cycles for manual selection. In case the method IS_CYCLE_MANUALLY_SELECTABLE is not implemented or returns FALSE, the system uses either the billing cycle marked as default or simply the first one from the list.

- ET_MESSAGES: Add error or warning messages if problems occur.

For more information, you can see one of these sample implementations - CL_CRM_ISX_BC_DETERM_EXAMPLE, CL_CRMS4_SOM_BC_RULE_EXAMPLE.

Default Determination Rule

You can define one of the determination rules as the default rule. The default rule is evaluated when no billing cycle settings have been maintained for a product that requires a billing cycle. You can only define one entry as a default.

Billing Cycle in Contracts

During order capturing, a billing cycle is determined, stored with the contract, and replicated to Convergent Invoicing. This happens at order item level. Only main items have their own bill cycle. Subitems inherit the billing cycle from their corresponding main item, but the billing cycle is not stored in subitems.

If a contract change process is executed, the billing cycle is copied from the contract into the change order and is not determined again.

Determination Triggers

During order capturing, the billing cycle determination is carried out twice:

- Directly after entering a product

- Before the order is released

This is done because the determined billing cycle may change due to attribute changes in the order (if a determination rule with a billing cycle determination class implementation is used).

Determination Logic

In Subscription Order Management, the system searches for a billing cycle determination method using a certain pattern. Once a determination method (or a fixed billing cycle) is found, the system does not continue to search, even if the determination method did not return a bill cycle or the billing cycle found is no longer valid (usable to field).

The following search patterns are available for the determination logic:

- Billing cycle determination rule for the product. The rule is evaluated in the following order:
 - Implementation class
 - Billing cycle
- Fixed billing cycle for the product
- Default billing cycle determination rule in Customizing. The rule is evaluated in the following order:
 - Implementation class
 - Billing cycle

The default determination rule is only used if data is not maintained for the product.

Billing Cycle Storage

The billing cycle is stored in the extension table CRMS4D_SERV_I, field CI_BILLING_CYCLE.

Billing Cycle in Contract Processes

The shipped processes are designed to keep the billing cycle stable. One reason for this is that billing cycle changes are not allowed during the billing period. Therefore, the billing cycle is transferred when contracts are copied to change orders and vice versa, or if no process implements are determination of the billing cycle.

Usage-Based Charges

When a subscription order with a usage-based fee is submitted (released), a subscription contract is created. During contract distribution, the contract items for which a charge plan is assigned are identified. The provider contract in FI-CA and the provider contract in SAP Convergent Charging are created.

Activities

You can perform the following functions for usage-based charges:

1. Enter a product for which a cross-catalog mapping is defined in the subscription order.
2. The cross-catalog mapping version for your charge plan will be derived.
3. Maintain subscriber mapping tables for your subscription order.

4. When the subscription order is submitted, the resulting subscription contract is distributed. The FI-CA and CC provider contracts are created.

More Information

The following features give you an overview of the recommended Customizing for this function, delivered as standard by SAP:

- Products that are only priced in SAP Convergent Charging do not need a price condition. In this case, item price will be 0. Here, you can also combine a recurring fee with a usage-based fee in a single product.
- You can use a subscription product and assign a billing cycle to this subscription product and item category group PRRP (Subscription Order Item) in product master.
- You can maintain cross-catalog mapping data in the Manage Subscription Product-Specific Data app.
- Object type is BUS2000155 subscription order item
- Use transaction type PRVO
- Item category PROP is customized as a subscription item
- The document distribution step to create a provider contract in FI-CA is PCEA. With the BAdI CRM_ISX_ERP_CONTRACT, you can implement customized logic during document distribution.

Related Information

[Product Master](#)

[Manage Subscription Product-Specific Data](#)

[Document Distribution](#)

One-Off Charges

Use

A one-off charge is a charge which has only to be paid once, for example, an activation fee or suspension fee. The one-off charges are delivered to Convergent Invoicing in the form of billable items and, later provided either on the customer's invoice, either together with usage charges or on a separate invoice.

You can use the following functions:

- Distribution of Relevant One-Off Charges from Subscription Order
- Distribution of Relevant One-Off Charges from Subscription Contract

i Note

The reversal of one-off charges is not supported as a standard and must be individually implemented by the customer.

More Information

Define Item Categories

The item category **PROF (Subscr. One Off Item)** is customized as a one-off charge. Every one-off item category must have the following details:

- Object Type = BUS2000155 Subscription Order Item
- Bill. Relevance = Not Relevant for Billing

Subscription-Specific Customizing of Item Categories

You can mark item categories as type one-off in this Customizing path, [Service](#) > [Transactions](#) > [Settings for Subscription Transactions](#) > [Define Settings for Item Categories](#).

Default Tax Code

You can assign a default tax code for each country in this activity in the Customizing path, [Service](#) > [Transactions](#) > [Settings for Subscription Transactions](#) > [One-Off Charges](#) > [Default Tax Code](#).

Contract Accounts Receivable and Payable

The following settings must be executed in the customizing for [Contract Accounts Receivable and Payable](#) under [Financial Accounting](#) > [Contract Accounts Receivable and Payable](#) > [Integration](#) > [Customer Relationship Management](#) > [Transfer of One-Off Charges](#):

- [Define Billable Item Classes for One-Off Charges](#)
- [Define Account Assignments for One-Off Charges](#)
- [Define Transactions for Posting One-Off Charges](#)

Related Information

[Distribution of One-Off Charges from Subscription Order](#)

[Distribution of One-Off Charges from Subscription Contract](#)

Distribution of One-Off Charges from Subscription Order

User

Depending on how products are modeled and used in subscription transactions, one-off charges can be treated as either relevant for distribution from subscription order or from subscription contract. The one-off charges that are modeled as independent of a contract item (such as main one-off items or one-off items directly under a sales package) can be distributed directly from the subscription order.

You can consider the following scenarios as examples:

- **Scenario 1**

Item 10: Subscription Product

Item 20: One-Off Product

- **Scenario 2**

Item 10: Sales Package

Item 20: Subscription Product

Item 30: One-Off Product (Parent 10)

In the above scenarios, the one-off charge is not dependent on the subscription item that will be carried forward to the subscription contract. Such a one-off charge can be distributed directly from the subscription order once it is submitted.

i Note

BITs that are created for one-off charges and are distributed directly from the subscription order will not have the subscription contract reference.

More Information

You must Customize this activity in the following customizing path: [Service](#) > [Transactions](#) > [Settings for Subscription Transactions](#) > [Document Distribution](#) > [Define Settings for Document Distribution](#).

Standard step type ONEF, schema definition POOF, and schema determination as shown below are available for use with the item category PROF for distribution of relevant one-off charges directly from the subscription order.

- Step Type: ONEF
- Schema Definition: POOF
- Schema Determination
 - Transaction Type: PRVO
 - Item Category: PROF
 - Schema Assignment: POOF

Distribution of One-Off Charges from Subscription Contract

Use

One-off charges that are modeled as dependent on a contract item can be distributed along with the distribution of the subscription contract.

You can consider the following scenarios as examples:

- **Scenario 1**
 - Item 10: Subscription Product
 - Item 20: one-off Product (Parent 10)
- **Scenario 2**
 - Item 10: Sales Package
 - Item 20: Subscription Product
 - Item 30: one-off Product (Parent 20)

In the above scenarios the one-off charge is dependent on the subscription item that will be carried forward to the subscription contract. Such one-off charges are not to be distributed directly from the subscription order and can, instead, be distributed along with the distribution of the subscription contract.

More Information

You have to customize this activity in the following Customizing path ► [Service](#) ► [Transactions](#) ► [Settings for Subscription Transactions](#) ► [Document Distribution](#) ► [Define Settings for Document Distribution](#) ►.

Standard step type PCOF included in the schema definition PERP is available for use for distribution of a specific contract relevant one-off charge from the subscription contract.

Step Type: PCOF

Schema Definition: PERP

Pre-requisite Step: PCEA

You can also have a combined scenario as below:

Scenario 3

Item 10: Subscription Product 1

Item 20: One-Off Product (Parent 10)

Item 30: Subscription Product 2

Item 40: One-Off Product (Parent 30)

Item 50: One-Off Product

In this scenario:

- Item 50 one-off charge will be distributed directly from the subscription order on submission.
- Item 20 one-off charge will be distributed from the subscription contract item corresponding to item 10.
- Item 40 one-off charge will be distributed from the subscription contract item corresponding to item 30.

Manual Fixed Price

A subscription contract processes recurring and one-time charges. You may need to maintain the one-time charge manually, as a fixed price for the subscription contract, irrespective of the quantity or duration of the contract. Such a fixed price is implemented using the condition type PSFX (calculation type = B), maintained in the Customizing ► [Sales and Distribution](#) ► [Basic Functions](#) ► [Pricing](#) ► [Pricing Control](#) ► [Define Condition Types](#) ►.

The pricing procedure for subscription order (A17023) is enhanced to accommodate the fixed price condition type in the Customizing ► [Sales and Distribution](#) ► [Basic Functions](#) ► [Pricing](#) ► [Pricing Control](#) ► [Define Condition Types](#) ► [Define and Assign Pricing Procedures](#) ►. The PSFX condition type can be added to the solution quotation pricing procedure (A17019), where it can be adapted for specific pricing requirements.

The fixed price can be maintained in the [Price Details](#) assignment block at the item level. If a manual fixed price is maintained, the recurring price (if any) is automatically deactivated and the fixed price is considered. This results in a Convergent Invoicing billing plan one-time item.

i Note

You cannot maintain a fixed price for a recurring product in SAP Convergent Charging.

If you are using Revenue Accounting, you must decide how to calculate the total transaction price sent to SAP Revenue Accounting and Reporting, based on your business process. To do so, you must implement the relevant BAdI in this Customizing, [Service > Transactions > Settings for Subscription Transactions > Integration > Integration with Provider Contract > BAdI: Replication to Provider Contract](#).

Change Processes on Contracts with Fixed Price

While executing a change process on a fixed price contract, the fixed price is not copied over to the change order. If an additional price is to be charged, you can manually maintain the relevant price in the change order. The delta fee you enter here will be created as a one-time bill plan item, like the original contract amount. The original and delta billing items are considered during the invoicing run in Convergent Invoicing.

Billing Schedule for Periodic Subscriptions

Model price variations over the course of the contract duration. Typically, these prices are already planned at the time of contract creation.

Prerequisites

- The following item categories are delivered to support the scenarios relevant for periodic billing schedule in the Customizing [Service > Transactions > Basic Settings > Define Item Categories](#):
 - PROB Subscription Order Item for Billg Sched
 - PRCB Subscription Contr Item for Billg Sched
 - PRQB Subscription Item in Qtn for Billg Sched
- The following settings are delivered in the Customizing [Service > Transactions > Basic Settings > Billing Plan](#):
 - **Define Billing Plan Types:**
 - **Billing Plans:** Billing Plan Type '91' of type **Periodic Billing**
 - **Plan Lines:**

Usage is set to **Periodic Billing**

The date rule selected in [Billing Data: Date Proposal > Period](#) generates the periods of the billing schedule

The value selected in [Billing Data: Date Proposal > Pricing Date](#) controls the date that is used in the pricing of the billing schedule period
 - **Assign Billing Plan Type to Item Category:** The Billing Plan Type '91' is assigned to the newly delivered item categories for periodic billing schedule scenarios.
 - In case you wish to enable the detailed billing scenario, you can do so by checking the **Detailed Billing** flag in the Customizing [Service > Transactions > Settings for Subscription Transactions > Recurring Fees > Configure Settings using Billing Plans with Convergent Invoicing > Assign Billing Plan and Billing Plan Item Types](#).

Use

In case of a billing schedule scenario, the **Total Value (Estimated)** is calculated as the sum of all billing values of all the billing schedule lines and the **Recurring Value for Billing Period** is not calculated at the item level.

i Note

The **Value One-Time** label has been renamed to **Total Value (Estimated)**. Consider the following scenarios at an item level:

- In case of a product with a billing schedule, this field displays the sum of billing values of each billing schedule line
- In case of a product with a recurring charge, this field displays the recurring charge * contract duration
- In case of a product with a one-off charge, this field displays the one-off charge

At the header level, the **Total Value (Estimated)** is the sum of all the **Total Value (Estimated)** figures at item level.

Also, the **Price Details** assignment block is not available on the subscription order item or the subscription contract. Instead, you can see the billing schedule lines in the **Billing Schedule** assignment block. Within this assignment block, you may also select a **Settlement Period** that controls the billing schedule settlement period. By default, the settlement period is the same as the billing cycle.

❖ Example

In a subscription contract with a validity of 2 years and a monthly billing cycle, the default billing schedule settlement period is also monthly. In this case, 24 billing schedule lines are generated.

However, if you select a yearly settlement period instead, the number of billing schedule lines would change to 2 and the billing value would be adjusted so that the sum of all the billing values matches the predetermined total contract value.

Once the subscription order is submitted and the subscription contract is distributed successfully, the billing schedule lines are distributed to create individual CI billing plan items and the status is updated to **Transferred to CI**. The term dates (**Term To** and **Term From**) of the CI billing plan item are derived from the settlement start and end dates in the billing schedule.

The invoicing cycles in CI are determined based on the configuration assigned to the billing cycle. The date rule FICA_BILLCYC (Billing Cycle (Settlement Period)) is provided to ensure that, by default, the billing schedule periods in Subscription Order Management are the same as the invoicing periods generated by CI.

⚠ Caution

You may also assign other date rules that are a part of the date profile. In such a case, however, the billing schedule periods in Subscription Order Management may not match the invoicing periods generated later on in CI.

Sample Billing Schedule Scenarios

Scenario	Example	
A discount is offered on the subscription contract but the duration of discount is shorter than the contract duration	Consider a subscription contract with a validity of 1 year and a quarterly billing cycle. A 50% discount is offered in Q1 and a 25% discount in Q2. Thus, the total contract value is calculated as below: Quarter 1 Price \$3,000 with 50% discount = \$1,500 Quarter 2 Price \$3,000 with 25% discount = \$2,250 Quarter 3 Price \$3,000 = \$3,000 Quarter 4 Price \$3,000 = \$3,000	The billing schedule lines are initially created with a billing value per of \$3000 per billing cycle (quarterly) in the Billing Schedule assignment block. You navigate into the Details for the Q1 and Q2 billing cycles and apply the relevant discount. Once the discount is applied, the billing value is updated as per the applied discount and the Item Net Value (Estimated) is calculated accordingly.

Scenario	Example																																																									
	Total contract value = \$9,750																																																									
The total contract value is broken into uneven installments on even dates	Consider a subscription contract with a validity of 1 year and a monthly billing cycle with the monthly billing value as \$40000 per month. The total contract value is \$480000. Thus, the total contract value is calculated as below: <table><tr><th>Month</th><th>Billing Value</th></tr><tr><td>January</td><td>\$40,000</td></tr><tr><td>February</td><td>\$40,000</td></tr><tr><td>March</td><td>\$40,000</td></tr><tr><td>April</td><td>\$40,000</td></tr><tr><td>May</td><td>\$40,000</td></tr><tr><td>June</td><td>\$40,000</td></tr><tr><td>July</td><td>\$40,000</td></tr><tr><td>August</td><td>\$40,000</td></tr><tr><td>September</td><td>\$40,000</td></tr><tr><td>October</td><td>\$40,000</td></tr><tr><td>November</td><td>\$40,000</td></tr><tr><td>December</td><td>\$40,000</td></tr><tr><td>Total contract value</td><td>\$480,000</td></tr></table>	Month	Billing Value	January	\$40,000	February	\$40,000	March	\$40,000	April	\$40,000	May	\$40,000	June	\$40,000	July	\$40,000	August	\$40,000	September	\$40,000	October	\$40,000	November	\$40,000	December	\$40,000	Total contract value	\$480,000	The billing schedule lines are initially created with a billing value of \$40000 per billing cycle (monthly) in the Billing Schedule assignment block. You can manually update the billing value for each billing schedule line so it looks as below: <table><tr><th>Month</th><th>Billing Value</th></tr><tr><td>January</td><td>\$200,000</td></tr><tr><td>February</td><td>\$0</td></tr><tr><td>March</td><td>\$0</td></tr><tr><td>April</td><td>\$0</td></tr><tr><td>May</td><td>\$0</td></tr><tr><td>June</td><td>\$200,000</td></tr><tr><td>July</td><td>\$0</td></tr><tr><td>August</td><td>\$0</td></tr><tr><td>September</td><td>\$0</td></tr><tr><td>October</td><td>\$0</td></tr><tr><td>November</td><td>\$80,000</td></tr><tr><td>December</td><td>\$0</td></tr><tr><td>Total contract value</td><td>\$480,000</td></tr></table>	Month	Billing Value	January	\$200,000	February	\$0	March	\$0	April	\$0	May	\$0	June	\$200,000	July	\$0	August	\$0	September	\$0	October	\$0	November	\$80,000	December	\$0	Total contract value	\$480,000
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The total contract value is predetermined and the amount is evenly distributed across the billing cycles, with rounding off adjustments. This ensures that the contract net value is accurate and customer is invoiced for the correct amount.	Consider a subscription contract with a validity of 1 year and a monthly billing cycle. The total contract value is \$500,000 and distributed as below: <table><tr><th>Month</th><th>Billing Value</th></tr><tr><td>January</td><td>\$41,666.67</td></tr><tr><td>February</td><td>\$41,666.67</td></tr><tr><td>March</td><td>\$41,666.67</td></tr><tr><td>April</td><td>\$41,666.67</td></tr><tr><td>May</td><td>\$41,666.67</td></tr><tr><td>June</td><td>\$41,666.66</td></tr><tr><td>July</td><td>\$41,666.67</td></tr><tr><td>August</td><td>\$41,666.66</td></tr></table>	Month	Billing Value	January	\$41,666.67	February	\$41,666.67	March	\$41,666.67	April	\$41,666.67	May	\$41,666.67	June	\$41,666.66	July	\$41,666.67	August	\$41,666.66	You can manually update the billing value for each billing schedule line so it looks as below: <table><tr><th>Month</th><th>Billing Value</th></tr><tr><td>January</td><td>\$41,666.67</td></tr><tr><td>February</td><td>\$41,666.67</td></tr><tr><td>March</td><td>\$41,666.67</td></tr><tr><td>April</td><td>\$41,666.67</td></tr><tr><td>May</td><td>\$41,666.67</td></tr><tr><td>June</td><td>\$41,666.67</td></tr><tr><td>July</td><td>\$41,666.67</td></tr><tr><td>August</td><td>\$41,666.67</td></tr></table>	Month	Billing Value	January	\$41,666.67	February	\$41,666.67	March	\$41,666.67	April	\$41,666.67	May	\$41,666.67	June	\$41,666.67	July	\$41,666.67	August	\$41,666.67																				
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Scenario	Example			
	Month	Billing Value	Month	Billing Value
	September	\$41,666.67	September	\$41,666.67
	October	\$41,666.66	October	\$41,666.67
	November	\$41,666.67	November	\$41,666.67
	December	\$41,666.66	December	\$41,666.63
	Total contract value	\$500,000	The amount for the last billing cycle is rounded off to ensure that the total contract value is correct.	
			Total contract value	\$500,000
The total contract value is calculated based on a recurring value that increases or decreases over the duration of the contract (also known as the Ramp Up/Down Scenario or the Uplift Scenario).	Consider a subscription contract with a validity of 3 years, a monthly billing cycle, and a ramp-up pricing that shows a 10% increase every year. The total contract value is calculated as below: Year 1 \$100/month Year 2 \$110/month Year 3 \$121/month The total contract value (12*100 + 12*110 + 12*121) is expected to be visible when the contract is created. Since the billing cycle is monthly, the invoicing would look like this: Invoice 1 to 12 - \$100 Invoice 13 to 24 - \$110 Invoice 25 to 36 - \$121		By default, the billing schedule settlement period is the same as the billing cycle (in this case, monthly). However, if the billing value is not changing frequently, you can switch to a billing schedule settlement period to reduce the number of billing schedule lines. In this case, there would be 36 billing schedule lines even though the billing value is only changing every 12 months. Switching to a yearly billing schedule settlement period would create only 3 billing schedule lines such that the billing value for each would be as below: Year 1 12*\$100/month = \$1,200 Year 2 12*\$110/month = \$1,320 Year 3 12*\$121/month = \$1,452 Total contract value = \$3,972	

Adhoc Billing Schedule for Fixed Price Subscriptions

The adhoc billing schedule enables you to split the total contract value over the contract duration such that it is distributed over multiple billing schedule lines with different billing dates and billing values.

In case of an adhoc billing schedule scenario, you define the **Total Value (Estimated)** using the fixed price condition type, PSFX. While defining the billing schedule lines (BSLs), you must ensure that the sum of the billing values must equal the **Total Value (Estimated)**. The **Billing Schedule** assignment block is displayed along with the **Price Details** assignment block at item level. You can see the billing schedule lines in the **Billing Schedule** assignment block. Once the subscription order is submitted and the subscription contract is distributed successfully, the BSLs are distributed to create individual CI billing plan items and the status is updated to **Transferred to CI**. The **Valid From** date is derived from the billing date. The term dates (**Term To** and **Term From**) will remain blank.

While creating a subscription contract with an adhoc billing plan, you must ensure that:

- You specify unique billing dates for each item in the adhoc billing plan and that these dates lie within the contract item validity period.
- The total billing value for all the billing schedule lines must match the contract net value.

When you implement a change process on a subscription contract with an adhoc billing plan and future dated BSLs, that is, BSLs with a start date later than the activation date of the change, then the change process must make some specific adjustments to the price of the previous item and the current item to ensure that the contract net value remains consistent.

When such a change process is executed:

- All BSLs with billing date before the activation date of the change are retained on the previous contract item.
- All BSLs with billing date after the activation date of the change are copied over to the change order with the status **Brought Forward**.

The total billing value of these future dated BSLs is added to the new contract item as a surcharge with the pricing condition VRM1 (Remaining Value). This surcharge cannot be changed manually. You can, however, enter a delta price for the change in the new contract item using the condition type PSFX.

- When you submit the change order, a new contract item is created with the BSLs in the change order with the status **Transferred to CI**.

The BSLs from the previous contract item that have been sent to the new contract item are updated with the status **Carried Forward**. The total value of the carried forward BSLs is added as a negative surcharge in the previous contract item to ensure that the contract net value remains consistent.

- If you reject the open change order, the status of the future dated BSLs is set to **Not Transferred to Billing**.

When a contract with future dated BSLs is canceled, a change order is created with the error that billing date must lie within the validity period of the item. To submit the change order, you must first adjust the BSL date to the cancellation date in the change order.

Configuration

- The following item categories are delivered to support the scenarios relevant for adhoc billing schedule in the Customizing **►Service►Transactions►Basic Settings►Define Item Categories►**:
 - PROA Subscription Order Item for Adhoc Billg
 - PRCA Subscription Contr Item for Adhoc Billg
 - PRQA Subscription Item in Qtn for Adhoc Billg
- The Billing Plan Type '92' of type **Adhoc Billing** is delivered in the Customizing **►Service►Transactions►Basic Settings►Billing Plan►Define Billing Plan Types►Billing Plans►**.

Sample Billing Schedule Scenarios

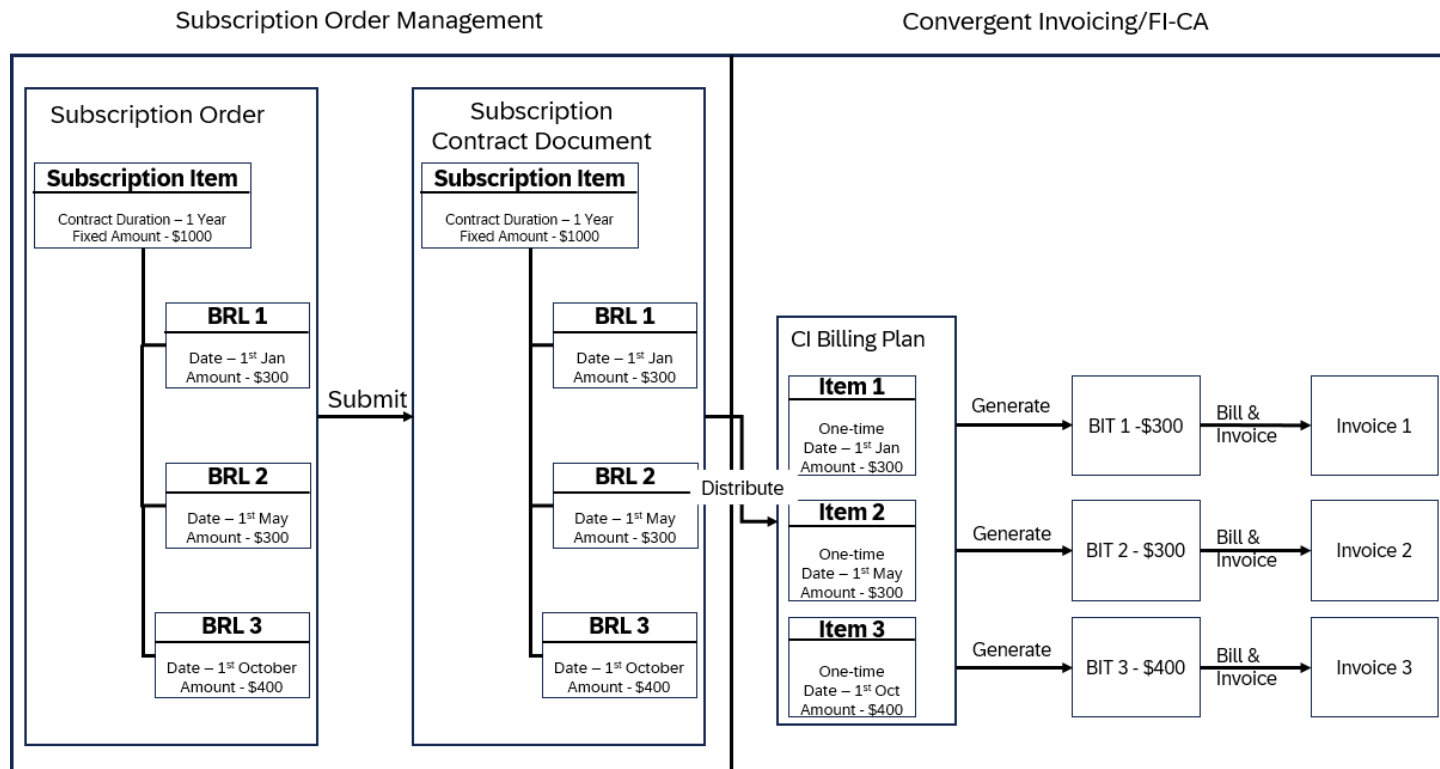
Scenario: The total contract value is calculated based upon a fixed price. This is broken up into uneven installments on uneven billing dates.

Example: Consider a subscription contract with validity of 1 year (contract start date 1st January 2024) and a manual fixed price of \$1000.

In the **Billing Schedule** assignment block, you add the billing schedule lines where you can manually add the billing date and the billing value.

Month	Billing Value
1st January 2024	\$300
1st May 2024	\$300
1st October 2024	\$400

You can see how the adhoc billing plan flows from Subscription Order Management to Convergent Invoicing in the figure below:



Constraints

- The **Change Product** and **Withdraw** change processes are not supported.
- Detailed billing is not supported.
- You cannot create and update adhoc billing schedule lines via API.

Status Management

System statuses are used to control the subscription order processing.

Enhanced Status Management

The system statuses provide information about the processing status of the respective subscription order items or subscription contract items. You control the subsequent processes in the processing of subscription orders and subscription contracts, for example, the distribution of data to external systems, or contract change processes, using these statuses. Each status informs you at a main item level and a subitem level that a certain state has been reached. This determines the next steps that can be performed from here.

Lifecycle Status

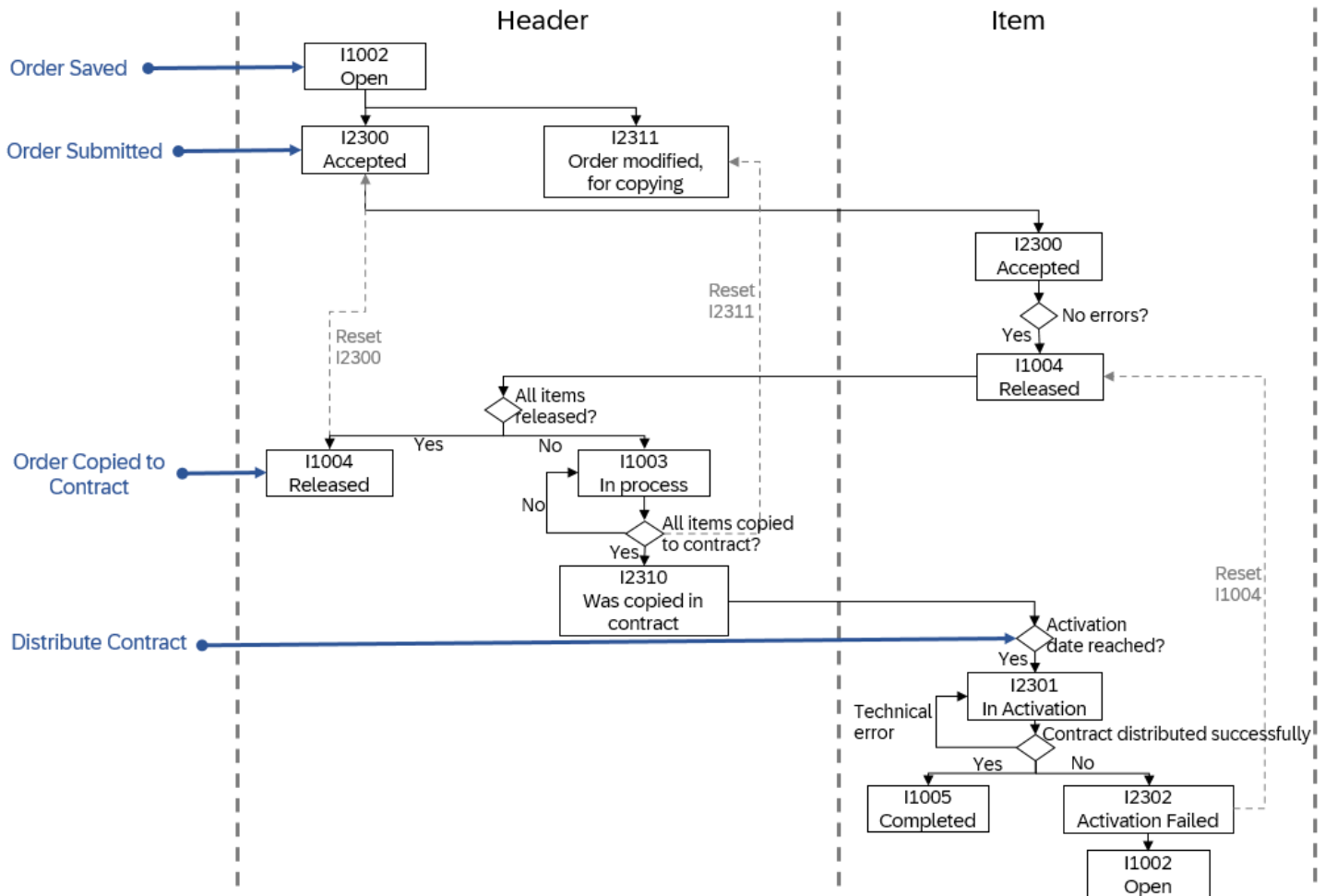
The following lifecycle statuses are available in the subscription order:

- I1002 - Open
- I1003 - In Process
- I1004 - Released
- I1005- Completed

Only one status is active at any time. If a subsequent status is set, the previous status is deleted. The higher-level status is used for the processes in the subscription order and subscription contract, to determine admissible follow-up transactions.

Subscription Order

You can see the status flow of a subscription order from the figure below:



Status Flow in a Subscription Order

Order Saved

If the order is saved, it has the status I1002 "Open". If a subscription order has this status or the status I1003 "In Process", you can still change it. These two statuses are the same. The setting of the status I1003 "In Process" is described below.

Order Submitted

When a subscription order is released to be copied into a contract, the statuses I2300 "Accepted" and I2311 "Order Modified, For Copying" are set.

If the status I2300 "Accepted" is set at header level and the order contains no errors and passes the credit check, the status I1004 "Released" is set for all subscription order items. If all subscription order items are released, the header status is set to I1004 "Released" and the status I2300 "Accepted" is reset. If all items have not been released, the header status is set to I1003 "In Process".

Order Copy to Contract

If the subscription order has been released and the status I2311 "Order Modified, For Copying" is set, the system automatically creates a subscription contract, based on the information in the subscription order. All subscription order items are copied to the subscription contract. For more information, see **Copying Logic**.

Later, the status I2311 "Order Modified, For Copying" is reset and the status I2310 "Was Copied in the Contract" is set.

Distribute Contract

If the contract start date is the current date, the subscription order items are simultaneously assigned the status I2301 "In Activation". Once the subscription order has this status, it cannot be changed.

If the contract start date is in the future, the status of the subscription order items does not change.

Once the start date of the current contract time slice has been reached, the contract is distributed using the **Order Distribution Infrastructure** (ODI).

If the subscription order was distributed successfully, for example, if an external system returns the message that the SIM card was activated, the subscription order items are assigned the status I1005 "Completed".

If distribution of the subscription contract fails, the status of the subscription order items is set to I1002 "Open" and I2302 "Activation failed", the statuses I1004 "Released" and I2301 "In Activation" are deleted. The user can then correct the subscription order document and resubmit it.

If distribution of the contract fails due to a technical error (for example the system connection was broken), the subscription order items have still the status I2301 "In Activation".

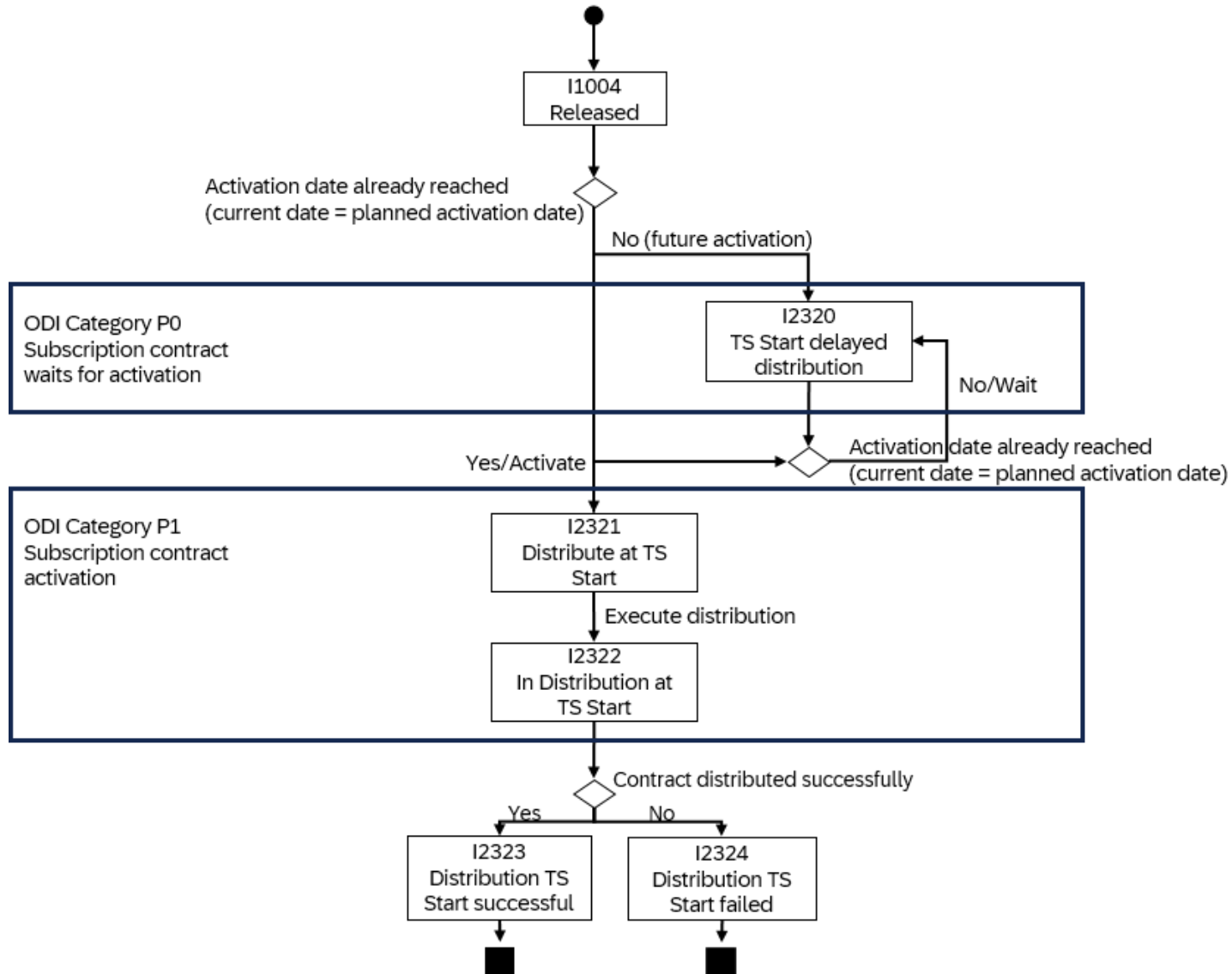
These statuses are also available for sub-items. Even if only one sub-item has the status I2301 "In Activation", you can no longer change the subscription order.

A higher-level status is also displayed at main item level which is determined according to the status of the sub-items. A higher-level status is always determined for the different statuses of the individual sub-items. This higher-level status provides information about the general activation process.

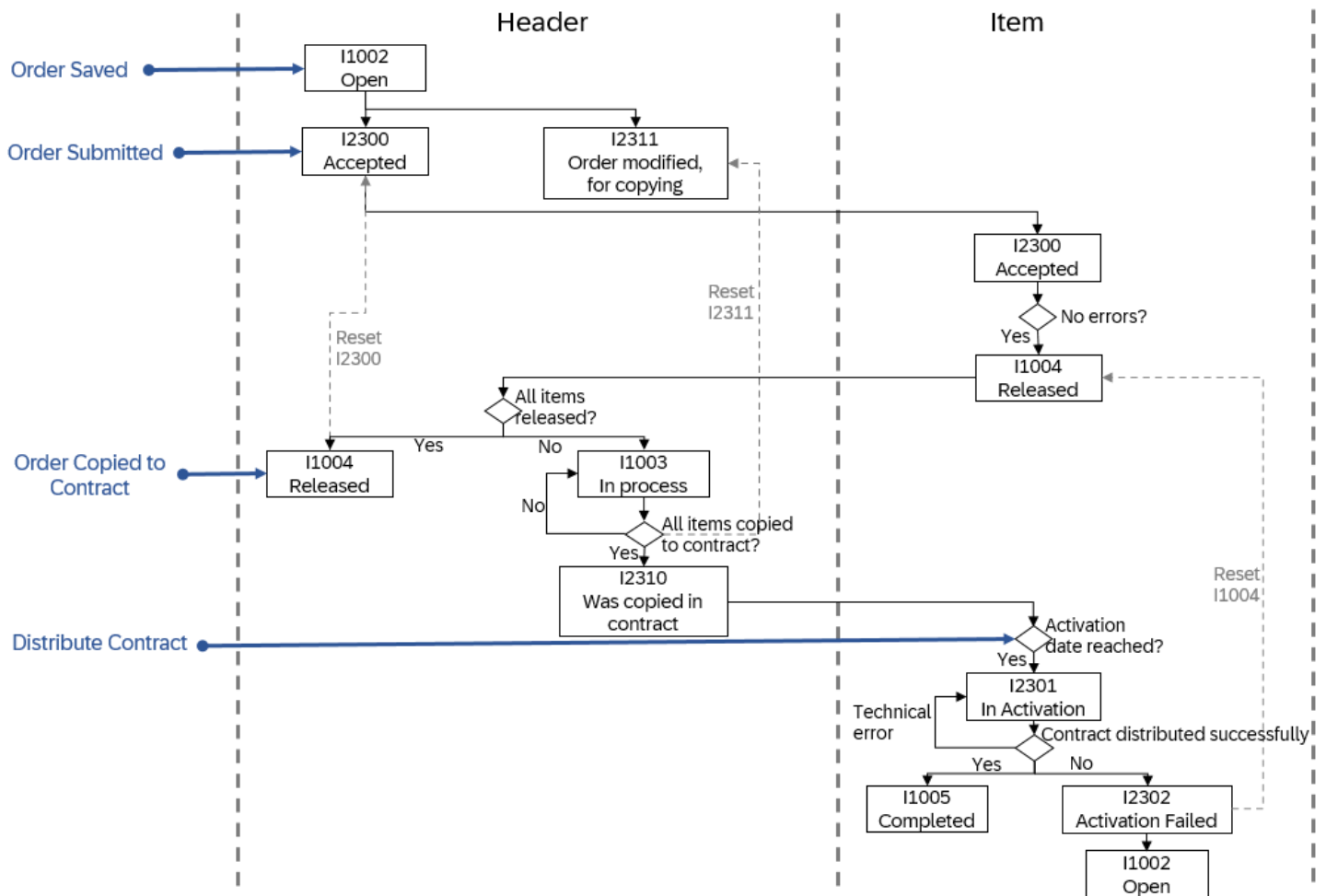
Subscription Contracts

You can see the status flow of a subscription contract activation/deactivation from the figure below:

Contract Activation



Activation



Deactivation

Copy Order to Contract

When a subscription contract item is copied from an order item, the status I2314 “Last contract time slice” is set in the new contract item. If a previous contract item exists, this status will be deleted for the previous contract item at the same time.

Contract Activation

The following cases have to be distinguished:

- Activation date is the current date

The statuses I1004 “Released” and I2321 “Distribute at TS Start (XI)” are set.

- Activation date is in the future

The status I2320 “TS Start Delayed Distrib.(XI)” is set.

- If the activation of the contract item is successful, the status I2313 “Techn.Activ.” is set.

If a previous contract item exists, the status I2313 “Techn.Activ.” is deleted in this contract item.

Changes on Contract Items

The statuses I2321 and I2322 are used to indicate if additional changes can still be made in the system. If they are set, the “point-of-no-return” has been reached, meaning that since the activation flow is currently being executed, no further changes can be captured. The system only allows you to capture additional changes once distribution has finished.

Contract Deactivation

Contract deactivation can only be performed if the CONTEND date is maintained in the new contract item.

If the deactivation date is the current date, the statuses I1004 "Released" and I2326 "Distribute at TS End (XI)" are set.

If the deactivation date is in the future, the status I2325 "Delayed Distributn TS End (XI)" is set.

Related Information

[Copying Logic](#)

[ODI Framework](#)

Date Management

Use the date profile to define the characteristics for date types and durations.

Use

The date profile controls which date types, time durations, reference objects, and date rules can be used in a specific transaction type or item category. Two date profiles are defined for subscription orders and subscription contracts, one at a header level and the other at an item level.

Prerequisites

You must maintain the date profiles in the Customizing [►Service ►Basic Functions ►Date Management ►](#).

Additionally, you must assign your date profile to your action profile. You can do so in the Customizing [►Service ►Basic Functions ►Actions ►Actions in Transaction ►Change Actions and Conditions ►Define Actions Profiles and Actions ►](#).

Contract Validity

The validity of a contract can be defined using multiple dates. These dates are also used to represent very specific types of contract validity.

Technical Validity of Subscription Contract

The overall validity (technical validity) of a subscription contract is defined by the Contract Start Date (CONTSTART) and the Contract End Date (CONTEND). CONTSTART defines the start date of the contract, and this cannot be changed during future processing.

❖ Example

You create a contract on 1st January 2019 and you choose to:

- Activate the contract immediately (CONTSTART = 1st January 2019)
- Activate the contract in the future (CONTSTART = 1st February 2019, for example)
- Activate the contract in the past (CONTSTART = 1st January 2018, for example)

The Contract End Date (CONTEND) can be changed when you execute an extension, or a cancellation change process. You may also choose not to maintain a contract end date for an evergreen contract.

Validity of Contract Time Slice

When an order/quotation is initially created, the start of contract time slice (ISTCONTSRTTS) is the same as the contract start date and the end of contract time slice (ISTCONTENDTS) is initial.

❖ Example

You create a contract on 1st January 2019 with a validity of one year. The contract start date (CONTSTART) = the start of contract time slice (ISTCONTSRTTS) = 1st January 2019. The end of contract time slice (ISTCONTENDTS) will be initial.

The start and end dates of a contract time slice become critical when you execute a change process. Most of the contract change processes are associated with an activation date that can either be set manually or is already predefined, depending on the scenario. Once the change order is submitted, the previous item ends with its end of contract time slice ISTCONTENDTS = **Activation Date** - 1 second. The new item begins with its start of contract time slice ISTCONTSRTTS = **Activation Date**.

❖ Example

You create a subscription contract with the following dates:

Contract Start	Start of Contract Time Slice	Contract End	End of Contract Time Slice
Original Dates			
1.1.2019 08:00:00	1.1.2019 08:00:00	1.1.2020 08:00:00	Initial
Change Quantity Executed on 1.2.2019 08:00:00			
1.1.2019 08:00:00	1.1.2019 08:00:00 (previous)		1.2.2019 07:59:59 (previous) One second before the activation date of the change process
	1.2.2019 08:00:00 (new) Activation date of the change process	1.1.2020 08:00:00	

Business and Billing Validity of Contracts

The business validity of a contract is the time period within the technical validity of the contract. This represents the period during which the provided service is consumed and when revenue is recognized in SAP Revenue Accounting and Reporting.

The billing validity is the time period during which you can be billed for the provided service.

The BUSINESSSTRT and BUSINESSSEND dates are used to maintain the business validity of a contract. The business end date (BUSINESSSEND) can be changed only when a contract is extended. The start of business can be changed using the **Change Business Validity** change process, only if the current timestamp is before the start of business. Once the current timestamp is beyond the start of business, it cannot be changed. You can also change a contract before the start of business. In such a case, the change will be applicable for the entire duration of the contract and the **Start of Contract Time Slice** (activation date of the change process) will automatically be adjusted to the contract start date. If the start of contract time slice is changed in a change order, it will be reset to the contract start date with a warning.

The business validity and billing validity enable you to maintain the following scenarios.

❖ Example

Scenario		Subscription Contract	FICA Provider Contract	CI Billing Plan
Scenario 1: The service is available for use and the customer is billed from the contract start date. This scenario requires the maintenance of Contract Start Date (CONTSTART) and Contract End Date (CONTEND)	Technical validity = 01.01.2019 08:00:00 - 01.01.2020 08:00:00 Business validity = 01.01.2019 08:00:00 - 01.01.2020 08:00:00 Billing validity = 01.01.2019 08:00:00 - 01.01.2020 08:00:00	Contract Start Date = 01.01.2019 08:00:00 and Contract End Date = 01.01.2020 08:00:00	Valid From = 01.01.2019 08:00:00 and Valid To = 01.01.2020 08:00:00 Term Start = 01.01.2019 08:00:00 and End of Duration = 01.01.2020 08:00:00 (Flows to SAP Revenue Accounting and Reporting)	Valid From = 01.01.2019 08:00:00 and Valid To = 01.01.2020 08:00:00 Term From = 01.01.2019 08:00:00 and Term To = 01.01.2020 08:00:00
Scenario 2: The availability of the service (contract activation) is dependent on the activation of a related device. This means that you will not be charged until the start of billing is reached. This scenario requires the maintenance of Contract Start Date (CONTSTART), Contract End Date (CONTEND), and Start of Billing (CONTBILS).	Technical validity = 01.01.2019 08:00:00 - 01.01.2020 08:00:00 Business validity = 01.02.2019 08:00:00 - 01.01.2020 08:00:00 Billing validity = 01.02.2019 08:00:00 - 01.01.2020 08:00:00 ❖ Note In this scenario, if we assume that the contract has a monthly billing cycle, you will be charged	Contract Start Date = 01.01.2019 08:00:00 and Contract End Date = 01.01.2020 08:00:00 Start of Billing = 01.02.2019 08:00:00	Valid From = 01.01.2019 08:00:00 and Valid To = 01.01.2020 08:00:00 Term Start = 01.02.2019 08:00:00 and End of Duration = 01.01.2020 08:00:00 (Flows to SAP Revenue Accounting and Reporting)	Valid From = 01.01.2019 08:00:00 and Valid To = 01.01.2020 08:00:00 Term From = 01.02.2019 08:00:00 and Term To = 01.01.2020 08:00:00
Scenario 3: A free period is provided at the start of the contract This scenario requires the maintenance of Contract Start Date (CONTSTART), Contract End Date (CONTEND), Start of Business (BUSINESSSSTRT), End of Business (BUSINESSEND), and Start of Billing (CONTBILS). ❖ Note The Start of Billing date is mapped to the CI Billing Plan Term From only. For usage charges from CC, the charge	Technical validity = 01.01.2019 08:00:00 - 01.01.2020 08:00:00 Business validity = 01.01.2019 08:00:00 - 01.01.2020 08:00:00 Billing validity = 01.02.2019 08:00:00 - 01.01.2020 08:00:00	Contract Start Date = 01.01.2019 08:00:00 and Contract End Date = 01.01.2020 08:00:00 Start of Billing = 01.02.2019 08:00:00 Start of Business = 01.01.2019 08:00:00 and End of Business = 01.01.2020 08:00:00	Valid From = 01.01.2019 08:00:00 and Valid To = 01.01.2020 08:00:00 Term Start = 01.01.2019 08:00:00 and End of Duration = 01.01.2020 08:00:00 (Flows to SAP Revenue Accounting and Reporting)	Valid From = 01.01.2019 08:00:00 and Valid To = 01.01.2020 08:00:00 Term From = 01.02.2019 08:00:00 and Term To = 01.01.2020 08:00:00

Scenario		Subscription Contract	FICA Provider Contract	CI Billing Plan
plan should be configured so it does not produce any charges for the 1st month.				
Scenario 4: Payment is done at contract start but the service will be available later The start of business is not confirmed when a contract is created. However, the contract must be active to collect payment. For example, a user registers and pays for an online course on Jan 1st that is to start on Feb 15th (tentatively). However, the course actually starts on Feb 20th.	Technical validity = 01.01.2019 08:00:00 - 01.01.2020 08:00:00 Business validity = 15.02.2019 08:00:00 (later 20.02.2019 08:00:00) - 01.01.2020 08:00:00 Billing validity = 01.01.2019 08:00:00	Contract Start Date = 01.01.2019 08:00:00 and Contract End Date = 01.01.2020 08:00:00 Start of Billing = 01.01.2019 08:00:00 Start of Business = 15.02.2019 08:00:00 (later 20.02.2019 08:00:00) and End of Business = 01.01.2020 08:00:00	Valid From = 01.01.2019 08:00:00 and Valid To = 01.01.2020 08:00:00 Term Start = 15.02.2019 08:00:00 (later 20.02.2019 08:00:00) and End of Duration = 01.01.2020 08:00:00 > Flows to SAP Revenue Accounting and Reporting	Valid From = 01.01.2019 08:00:00 and Valid To = 01.01.2020 08:00:00 Term From = 01.01.2019 08:00:00  Recommendation The billable item generated by default will not be relevant for SAP Revenue Accounting and Reporting as it is outside the business dates. To make such a billable item relevant, we recommend that you implement BITCAT Event 05 to enrich the billable item with the SAP Revenue Accounting and Reporting relevance flag.
Scenario 5: The customer has ordered a service with other related services that are bundled as subitems in the subscription contract. Some of these other services (subitems) are available only in the future. That is, the business validity and billing validity for these services (subitems) are in the future.	Main item Technical validity = 01.01.2019 08:00:00 - 01.01.2020 08:00:00 Business validity = 01.01.2019 08:00:00 Billing validity = 01.01.2019 08:00:00 Subitem Technical validity = 01.01.2019 08:00:00 - 01.01.2020 08:00:00 Business validity = 01.02.2019 08:00:00 Billing validity = 01.02.2019 08:00:00	Main item Contract Start Date = 01.01.2019 08:00:00 and Contract End Date = 01.01.2020 08:00:00 Start of Billing = 01.01.2019 08:00:00 Start of Business = 01.01.2019 08:00:00 and End of Business = 01.01.2020 08:00:00 Subitem Contract Start Date = 01.01.2019 08:00:00 and Contract End Date = 01.01.2020 08:00:00 Start of Billing = 01.02.2019 08:00:00 Start of Business = 01.02.2019 08:00:00	Main item Valid From = 01.01.2019 08:00:00 and Valid To = 01.01.2020 08:00:00 Term Start = 01.01.2019 08:00:00 and End of Duration = 01.01.2020 08:00:00 > Flows to SAP Revenue Accounting and Reporting Subitem Valid From = 01.01.2019 08:00:00 and Valid To = 01.01.2020 08:00:00 Term Start = 01.02.2019 08:00:00 and End of Duration = 01.01.2020 08:00:00 > Flows to SAP Revenue	Main item Valid From = 01.01.2019 08:00:00 and Valid To = 01.01.2020 08:00:00 Term From = 01.01.2019 08:00:00 and Term To = 01.01.2020 07:59:59 Subitem Valid From = 01.01.2019 08:00:00 and Valid To = 01.01.2020 08:00:00 Term From = 01.02.2019 08:00:00 and Term To = 01.01.2020 07:59:59

Scenario		Subscription Contract	FICA Provider Contract	CI Billing Plan
		and End of Business = 01.01.2020 08:00:00	Accounting and Reporting	

Early Billing Scenario

During early billing, the billing and invoicing for a contract is done before the contract start date. This is a special case for an early distribution scenario, where an early billing date is provided to create an active provider contract created in FI-CA and SAP Convergent Charging (SAP CC). Since the provider contract is already active even though the contract start date is in the future, you can generate billable items (BITs) from the billing plan based on the early billing date even before the contract start date is reached.

❖ Example

Case 1: Consider a subscription for an online education course which is to begin on 1st July 2024 with a fixed price of \$1000 that needs to be billed early at the time of creation of the subscription on 15 June 2024. Thus, **Posting Date = Early Billing Date = 15.06.2024 08:00:00**.

Case 2: Consider a 3 year subscription for software with a yearly recurring price of \$1000. The subscription is to begin on 1st July.2024 and the first year will be billed early at the time of creation of the subscription 15 June 2024. Thus, **Posting Date = Early Billing Date = 15.06.2024 08:00:00**.

In both the cases, since an early billing date has been provided, an active provider contract is created during early distribution even though the subscription contract start date = 01.07.2024 08:00:00. Thus, you can generate BITs on the early billing date, even before the contract start date is reached.

Validity of Contract Term

When a subscription order or solution quotation is initially created, the start of contract term (ISTRUNTMPEG) equals the contract start date. The end of contract term (ISTRUNTMEND) is used typically when the contract end is not maintained, that is, for evergreen contracts, to set the end of term for revenue accounting. The start and end of contract terms are used for calculating the updated end of contract term when a contract is extended.

❖ Example

You create a contract on 1st January 2019, with a validity of one year, with the following dates:

- Start of contract term (ISTRUNTMPEG) = Contract start date (CONTSTART) = 1st January 2019.
- End of contract term (ISTRUNTMEND) = Start of contract term (ISTRUNTMPEG) + Contract validity period = 1st January 2020.

You extend the contract by 6 months and the dates are updated as below:

- Contract start date (CONTSTART) = 1st January 2019.
- New start of contract term = Previous end of contract term = 1st January 2020.
- New end of contract term (ISTRUNTMEND) = New start of contract term + Extension duration = 1st July 2020.

Date for Price Determination

Since pricing condition records are bound by a validity period based on the **Valid From** and **Valid To** dates, you can maintain different prices of the same product over different time intervals.

❖ Example

Consider a product where the following table shows you the varying prices and their respective validity:

Amount	Valid From	Valid To
100,00 USD 1 EA	01.01.2023 00:00:00	31.12.2023 23:59:59
200,00 USD 1 EA	01.01.2024 00:00:00	31.12.2024 23:59:59
300,00 USD 1 EA	01.01.2025 00:00:00	31.12.2025 23:59:59

When you create a subscription order or a solution quotation using such a product, the relevant pricing condition record is determined based on the price date that is passed to the pricing engine. The **Requested Start** date is a header level date that is used for price determination. When you create a subscription order or solution quotation, the **Requested Start** date is blank and you can enter the required value here. Once updated, the **Requested Start** determines the relevant pricing condition record, and thus the relevant price. Since this is a header level date, if a value is provided here, it is inherited by the pricing date for all the order or quotation items. If the **Requested Start** date is not provided, the system considers the **Posting Date** for price determination.

To enable this date on the UI, you must assign the date type SRV_CUST_BEG (**Requested Start**) to the date profiles PROVHEADER for subscription orders and SRVP for solution quotations. You can do so in the Customizing under **Service > Basic Functions > Date Management > Assign Date Profile to Transaction Type**.

The **Requested Start** date enables you to maintain the following scenarios.

❖ Example

Scenario	Example
Scenario 1: Future dated contract	You create a subscription contract with the following dates: Posting Date = 01.06.2024 08:00:00 Contract Start Date (CONTSTART) = 01.08.2024 08:00:00 (in the future) For such contracts, it may be required that the pricing is determined based either on the Contract Start Date or any other criteria apart from the Posting Date. Here, you can manually set the Requested Start date to when the pricing condition should be determined.
Scenario 2: Scheduled contract extension	You create a subscription contract with the following dates: Posting Date = 01.01.2024 08:00:00 Contract Start Date (CONTSTART) = 01.01.2024 08:00:00 (in the past, that is, the subscription contract is currently active)

Scenario	Example
	Contract validity is 1 year so Contract End Date (CONTEND) = 01.01.2025 07:59:59 The subscription contract is extended for 1 year, and thus the updated Contract End Date (CONTEND) = 01.01.2026 07:59:59 In the change order, the price date = Posting Date of the change order, however, it should ideally be the Start of Contract Timeslice, that is, the date on which the extended period of the contract begins so that it can determine the price that is valid on that specific date.
Scenario 3: Phased contract	Consider a long running contract with multiple phases that contains items and successor items with different start and end of contract timeslice dates. Since each item is active over a different time period, the price determination for each item should be based on its validity period, instead of the posting date. Since you use an API to create a phased contract, different prices on different phases can be determined using the header level attribute RequestedServiceStartDateTime in the Subscription Order (Version 00002) and Solution Quotation (A2X) OData APIs.
Scenario 4: Subscription contract created using an API	If you create subscription orders or solution quotations using the Subscription Order (Version 00002) and Solution Quotation (A2X) OData APIs, you use the header level attribute RequestedServiceStartDateTime to determine the pricing date for the order and quotation items. With respect to CPQ integration, suppose a CPQ quote is created on 22-05-2024 with contract start from 01-06-2024 and the CPQ tiered price conditions are defined and replicated to SAP S/4HANA with valid from date as contract start date (01-06-2024). In such a case, if the price date is not passed from SAP CPQ (Configure, Price, and Quote) to SAP S/4HANA (from the API) when the quotation is created (replication), then by default the system sets the price date as 22-05-2024 and hence the system won't be able to determine the correct price conditions. To be able to determine the correct price condition(s), the intention is to pass the price date from SAP CPQ (Configure, Price, and Quote) to SAP S/4HANA (from the API).

Other Dates

The following dates are also available on subscription contracts:

- Date fields for cancellation are used to provide information about a contract cancellation.
- The bundle release date (IST_BOM_EXPL) is used to determine the cross-catalog mapping version.
- The frequency of billing plan (BILLFREQ) in the header is used to calculate a summarized value of the recurring item charges on header level. As the item charges may have different frequencies, this duration is used to convert the fees from item level to a common header frequency.

⚠ Caution

Some fees cannot be converted, such as week to month.

The calculation of the recurring fees on item level is not based on one of the date fields but uses the billing cycle in Convergent Invoicing. For more information about this, refer to **Recurring and One-Off Charges**.

- **Withdrawal Period** (CW_PERIOD) and **End of Withdrawal Period** (WITHDRAW_END) - You use these dates to define the period during which it is possible to withdraw from a subscription contract using the **Withdraw** change process. Once the end of withdrawal period is over, you can no longer withdraw from the subscription contract.
- **Grace Period** (GRACE_PERIOD) and **End of Grace Period** (GRACE_END) - You use these dates to define the period during which it is possible to extend a subscription contract using the **Extend** change process. Once the grace period is over, you can no longer extend the subscription contract. You can use specific Customizing settings to configure whether a contract will be active or inactive during the grace period.

Copying Logic

Use

Copying Order Items into the Contract

The subscription order items in the subscription order document are copied into a contract document. The system uses an action from the post processing framework that is predefined in Customizing (action profile IST_ACTION_ORDER_HEAD, action ORDER_FULFILLMENT) to transfer all order items (BUS2000155) to the contract document. For more information, see **Post Processing Framework (BC-SRV-GBT)** on the [SAP Help Portal](#).

i Note

You can use a user status profile to add an additional approval step to the release process. This can be achieved by:

- Setting a user status that blocks the business transaction **Release** (RELE).
- Implementing customer-specific logic for the approval that removes the user status.
- System starts contract creation after saving the order automatically.

Copy Contract to Change Order

If a change process is executed, the contract is copied into a change order using the **Maintenance Framework for Business Transactions** (BTMF framework).

Prerequisites

Prerequisites for Copying of Order Items into the Contract

You must maintain copying control for the transaction types and the item categories in the subscription order and the subscription contract document in Customizing for **Service** under **► Transactions ► Basic Settings ► Copying Control for Business Transactions ►**

- **► Define Copying Control for Transaction Types**
- **► Define Copying Control for Item Categories**

The orders must have the statuses **Released** or **Completed** and **Order Modified, For Copying**. They must also not contain any errors.

Prerequisites for Copying the Contract to the Change Order

In Customizing, copying control has to be maintained for the transaction types and the item categories in the subscription order and the subscription contract document. The copying method PRVCT0 has to be maintained in copying control for the item categories.

You can determine the transaction type for the change order in the Business Add-In (BAI) CRM_PROVIDER_COPY, in the Customizing structure for ► **Service** ► **Transactions** ► **Settings for Subscription Transactions** ► **Contract Changes** ► **Business Add-Ins (BAIs)** ► **BAI: Copy Subscription Contract to Subscription Chng Order** ►

Features

The following data in the order item is copied into the contract item:

- Dates
- Partner data
- Cancellation data
- Technical resources
- Prices
- Mapping tables
- Subscription-specific data

The contract number is generated during the copying process, document flow at header and item level is created, items are grouped by time, and contract activation is started according to the date and Customizing.

Archiving Subscription Contracts

Use

You can archive a contract document if all the subscription contracts it contains have been completed. The archiving object CRM_PRVC is available for doing so.

The system archives subscription contracts independently from subscription orders. This means that the system does not check whether all associated subscription orders for all contract time slices can be archived.

Prerequisites

If you want to archive a subscription contract document, the following prerequisites must be met:

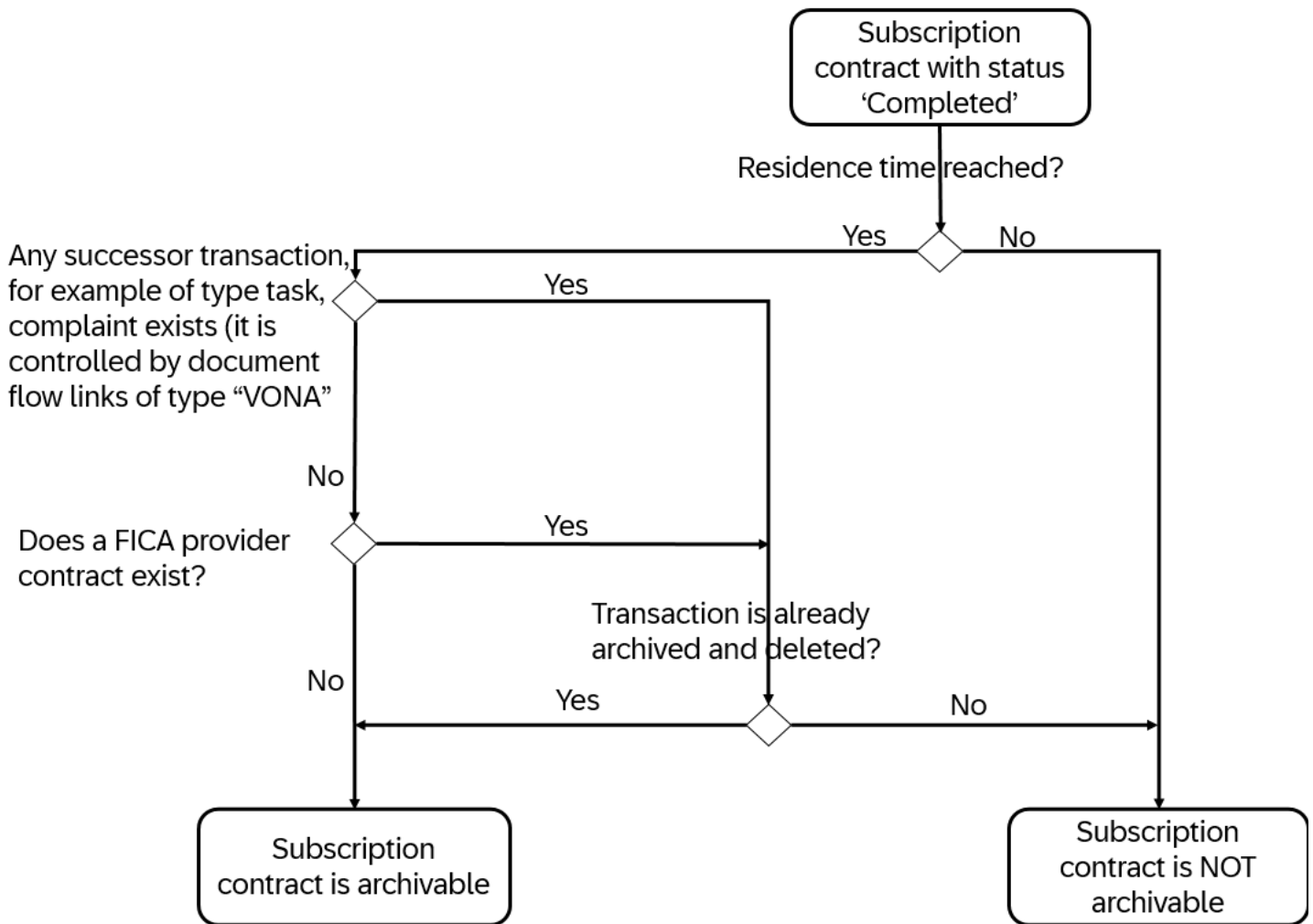
- The **Provider Contracts Finished** status must be set at header level.
- The residence time (number of days after the last change to the subscription contract document) must have elapsed.
- Subsequent documents (such as tasks or complaints) must already be archived and deleted from the system.

- If you have set up the ILM integration for the CRM_PRVC ILM object, your ILM rules must also allow the archiving. For more information, see **ILM Objects in Subscription Order Management**.
- Sharing Contracts can only be archived when no more linked Shared Contracts exist. In other words, Shared Contracts have to be archived and deleted before the referred Sharing Contracts.

More Information

The system sets the **Provider Contracts Finished** status automatically at header level in the subscription contract document if all contracts for this document have been completed. Locks for a future activation are not archived with the subscription contract and are deleted.

For more information, see **Data Management**.



Archiving Subscription Contracts

Related Information

[ILM Objects in Subscription Order Management](#)

[Data Management](#)

Revenue Sharing

In a service-based model, partner agreements are created to maintain terms of revenue sharing between service providers and content providers. When an end customer purchases a service, the involved content partner is given a part of the revenue based on the contract that the service provider finalizes with the end customer.

With traditional industries rapidly evolving their business models to service-based models, a revenue sharing business model is becoming increasingly important. In such a model, service providers who have contractual agreements with the end customers, either share part of their revenue with or pay certain royalties to their content providers. For this reason, a service provider enters into a partner agreement with the content provider. When an end customer purchases a service, the involved content partner is given a part of the revenue based on the contract that the service provider finalizes with the end customer. The provider that acts as billing operator, including invoicing and collections towards the end consumer, forwards a certain amount of money to the corresponding partners or third parties based on the partner agreement.

It is a common requirement to have one solution that supports the whole revenue and cost chain (revenue, partner settlement, royalty management, third-party sponsorship, or donations) in a seamless and a unique incoming event flow. For this reason, the subscription contract in Subscription Order Management is leveraged and reused to serve as a partner agreement. The partner agreement is first created in Subscription Order Management and then replicated to Convergent Invoicing and SAP Convergent Charging.

In a FI-CA only scenario (recurring) with CI billing plans, this functionality can be achieved without a partner product and partner agreement. In such a case, it is possible to handle part of a billable item separately, using dependent item types.

Partner Agreement Order

Use transaction types PRPO and PRPA for partner agreements.

Use

The transaction type PRPO creates a partner agreement order while the transaction type PRPA creates the partner agreement.

In Subscription Order Management, you can define products as **subscription partner products**, and these can be used in partner agreements.

i Note

We recommend creating a new product category for partner products.

Prerequisites

Customizing Settings

The following transaction type is available for subscription orders and subscription contracts in the standard system for partner agreement orders and partner agreements:

PRPO - Partner Agreement Order

You make general settings for subscription contracts in the Customizing for **Service** under **Transactions** > **Basic Settings** > **Define Transaction Types**.

The following transaction type is available for subscription orders and subscription contracts in the standard system:

Special settings for subscription transaction types can be made in the Customizing for **Service** under **Transactions** > **Basic Settings** > **Define Transaction Types**.

For partner agreements and partner agreement orders in the standard system, the following item categories are available:

- PRPO Partner Agreement Order Item

- PRPC Partner Agreement Item

You can only use products of a product type that is assigned to the product purpose "Partner Agreement Product". Transaction types that are used for partner agreements must have the document purpose "Partner Agreement Order"

Partner Settlement Rule

When processing partner agreements, Convergent Invoicing needs information about processing billable items. Therefore, partner agreements in SAP CI have the mandatory field **Partner Settlement Rule**. This partner settlement rule can be maintained in the partner agreement order on item level in the assignment block for subscription-specific data.

Related Information

[Product Modelling](#)

Dependent Item Types

When you create a contract using a subscription product, using an associated dependent item type, it is possible to handle a portion of a billable item amount differently from the rest.

i Note

This feature can be used for partner settlement, in case of recurring only scenario with billing plans.

Prerequisites

- The dependent item type and its related Customizing is maintained in Convergent Invoicing.
- The dependent item type is assigned to a subscription product.

Features

- A partner product and partner agreement are not required if you work with Convergent Invoicing billing plans.
- You can assign a dependent item type to a subscription product. For more information, see **Manage Subscription Product-Specific Data**.
- A dependent item types is assigned to the billing plan during contract distribution.
- Convergent Invoicing can now derive the partner billable item, based on the dependent item type.

Related Information

[Dependent Items in Convergent Invoicing](#)

[Manage Subscription Product-Specific Data](#)

Mass Processing

Subscription Order Management offers two solutions for bulk processing of a large number of subscription contracts, mass runs and mass processing. While both the solutions use the same framework at their core, each have separate focus areas and you can choose depending on the specific tasks you want to perform.

Mass Runs

The **Manage Mass Runs** SAP Fiori application is a long running solution that enables you to plan the creation and processing of a selected number of subscription orders, solution quotations, or subscription contracts using the mass processing framework. You can use mass runs to do any of the following tasks:

- Create a mass run for executing your own configured change processes on subscription contract items and contract headers for the selected subscription contracts.
- Execute changes for one sold-to party and for one change process within the same mass run.
- Integrate your own change processes with the **Manage Mass Runs** SAP Fiori app.
- Create subscription contracts from subscription orders or solutions quotations that can also be created as call-offs from master agreements.
- Upload files into the mass run.

Mass Processing

Mass processing is the solution that is based on SAP Fiori and the mass processing framework working seamlessly together and integrating into the subscription contract lifecycle management. You can select and change subscription contract items and documents, configure the change data, and create a mass processing from the selected data using the SAP Fiori app **Mass Changes of Subscriptions**. You then use the SAP Fiori app **Manage Mass Processing of Subscriptions** to edit, schedule, execute, and monitor the existing mass processing runs. You can also use this app to create new mass processing runs. You can use mass processing to do any of the following tasks:

- Filter for specific subscription contracts or contract documents that you want to change and add them into a new mass processing.
- Change contracts from different sold-to parties within a single mass processing run.
- Execute several change processes within a single mass processing run. A large set of existing change processes are already supported out of the box.
- You can also configure and integrate your own change processes and change process groups into the mass processing SAP Fiori app.
- Use the configuration wizard for configuring the change data within a mass processing.
- Customize unique change data for each record within a single mass processing.
- Extend an existing mass processing by adding more subscription contracts or contract documents.

With the SAP Fiori based mass processing, you can edit and adapt either single or multiple subscription contracts or contract documents within the mass processing, line by line. This enables you to conveniently adapt multiple subscription contracts at the same time. Along with the ability to combine multiple change processes, you can use mass processing for fast adaptation and change of multiple subscription contracts or contract documents at once.

Example

You have combined the **Change Quantity** and **Extend** change processes into a single change process group. Within a single mass processing, you can adapt and change multiple subscription contracts or contract documents by extending each by a different extension period, adding different quantities, and providing different activation dates for each of them.

Mass Runs

Use

You can use the mass run functionality to plan the creation or the processing of a preselected number of subscription orders, solution quotations, and subscription contracts, ensure high-performance execution, track the progress, and support monitoring of the results and further processing on the new or changed entities. You can trigger the mass run execution in the mass run maintenance or by planning appropriate jobs.

Prerequisites

Before you can create and process mass runs, make settings in Customizing for [Service](#) under [Transactions](#) [Settings for Subscription Transactions](#) [Mass Runs](#)

- **Define Number Range for Mass Runs**

Mass run IDs are taken from an internal number range interval. You must set up an interval to create mass runs.

- **Define Settings for Mass Runs**

You use this Customizing activity to define mass run types. Mass run types define the following characteristics of a mass run:

- Mass run category – either new contracts can be created, or existing contracts be changed.
- Process - the process to be executed when contracts are changed.
- Event handler implementation class – perform additional checks, enrich [Application Programming Interface](#) (API) calls or post process contracts.
- Execution profile – number of tasks that process the execution in parallel and number of elements in a processing block.

i Note

For execution profiles related to change processes, you may enter the value '0' here. If you do so, the SAP standard logic selects all subscription contracts belonging to the same subscription contract document and processes the change processes sequentially in a single work process. The processing time for such a work process depends on the number of subscription contracts within one subscription contract document.

Subscription contracts that belong to different subscription contract documents will still be processed in parallel depending on the parameter [Number of Parallel Tasks](#) in the execution profile.

- Spreadsheet upload settings – field mapping when uploading element data from a spreadsheet.
- Field settings – Define default values or switch fields to [read-only](#).
 - Default values can be defined for header. They are defined when a new mass run header is created or when elements are added to the mass run. Values are defined regardless of whether they are displayed on the UI.
 - Read-only fields cannot be changed on the UI. They are either defined by the default value or by the event handler implementation class.
 - One example of using default values for elements is, if there are only a few exceptions to be adapted manually, such as when you create a set of contracts, and the run time is only to be different for a small number. If the value for the element is always identical, it should be defined at header level.

Features

Structure

Mass runs consist of a header and elements. Mass run headers contain generic data for the mass creation; contract-specific data has to be maintained with the elements. Every element represents exactly one contract to be created or to be processed and has its own execution status.

Mass Run Header

Mass run header data is divided into two assignment blocks on the UI. Administration data is found in the assignment block [Mass Run](#) and mass run category-dependent data in the second assignment block. Since the process data view is dependent on the mass run category, it appears once the mass run type has been selected in the [Mass Run](#) view.

Mass Run Administration Data–View “Mass Run”

- Mass Run ID

Identifies a mass run and is taken from a number range specified in the mass run Customizing settings for [Service](#) under [Transactions](#) [Settings for Subscription Transactions](#) [Mass Runs](#) [Define Number Range for Mass Runs](#).

- Description

Descriptive text for the mass run.

- Type

The mass run type defines the behavior of the mass run. Starting from the type of activity to be processed (mass creation, mass change), the fields displayed on the maintenance UI, their default settings, the checks to be processed and the execution time (depending on how the execution jobs are planned). You can only define the type once in a mass run. It cannot be changed.

- Status

The status defines the next steps for a mass run. The following statuses are available:

Header Status	Description	Allowed Activities
Open	Initial status of a mass run	The mass run can be edited (with the exception of the 'mass run type' once it has been defined). The mass run is not observed by execution jobs. The mass run cannot be started. The mass run status can be manually switched to Released for Processing.
Released for Processing	This status can only be set manually. The setting can be reset by switching the status back to Open .	The mass run can no longer be edited. The status can be switched back to Open. The mass run can be started by choosing Run. The mass run is observed by execution jobs.
In Process	This status is set automatically once the elements of a mass run are processed manually or by an automatic job.	Mass run cannot be changed. This is a transient status. Once elements have been processed, the status switches to Processed Successfully or Processed with Errors, depending on the status of the elements.

		Since the processing of the elements happens asynchronously, the mass run display has to be refreshed by choosing Update to display the current status.
Processed Successfully	This status is set automatically once elements have been processed and all elements have the status Processed successfully or Processed Manually .	This is a final status.
Processed with Errors	This status is set automatically once elements have been processed and at least one element has the status Processed with Errors .	Elements that have been processed with errors can be manually adjusted or transferred to a follow-up mass run. Once all elements have been manually adjusted to Processed Successfully or taken over into a new mass run, the status switches to Processed Successfully.

- Changed by, changed on

Displays the user and date for the last change of the mass run.

- Planned execution

The earliest execution date for this mass run. When processing mass runs in an automatic job, the execution report only observes mass runs that are planned to be executed on or before the current date. This date is not evaluated when you start the execution of a mass run by choosing **Run**.

- Priority

Additional parameter to provide a more detailed plan for the execution of mass runs. You can divide the mass runs of one type into five priorities and select these accordingly for execution. The priority has no effect on processing but is an additional selection option for the execution report.

Mass Run Elements

Each mass run element represents a single contract to be processed. Depending on the category of the mass run, different fields are available for maintenance and for result-tracking after processing. The following fields are available in every mass run category:

- Actions

Contains a **Show Messages** link once messages are available after an element check or element processing. Use this action to open a dialog box showing the messages that have been reported during checking or processing of the element.

- Status

The status shows the processing status of the element. The following statuses exist:

Element Status	Description	Allowed Activities
Open	Initial status of an element.	The element can be edited (according to the field settings), but the status cannot be changed. The element can be deleted. The element can be checked.

		The element will be processed when the mass run is processed.
Successfully Processed	The element has been successfully processed without any errors being reported.	This is a final status.
Processed with Errors	Errors occurred during processing of the element.	Messages can be displayed choosing Show Messages in the Actions column. The element can be reprocessed, transferred to a new mass run or the status can be set to Processed Manually .
Processed Manually	This status is a final status that can only be set manually.	You can set this status when the contract of an element, processed with errors, has been corrected manually .

- Verification

The [Verification](#) status shows whether the element has been checked for validity and the check result.

The check implementation can be defined at mass run type level. It is available as input help. Therefore, the verification status does not affect processing of the mass run.

- Not checked: the check has not yet been processed.
- Successfully: the check has been successfully processed.
- Erroneous: the check reported errors. Display the errors choosing [Show Messages](#) in the [Actions](#) column.

In the standard configuration, elements do not have any process fields and therefore only allow a number of contracts to be processed according to the settings made in the header. Once you have adding elements, you can no longer make entries in any column. All these fields are filled automatically after checks have been made or after processing. You can easily add additional fields to the element data. For more information, see [Extensibility of Mass Runs](#).

Activities

Scheduled Processing of Mass Runs

In addition to running a mass run from the Web UI directly, you can also schedule a mass run to be executed at a certain point of time. This requires a report to be scheduled periodically as a background job (transaction SM36). The report CRM_ISX_MR_PROCESSING selects the relevant mass runs according to the selection parameters specified and triggers execution in each case. The selected set is sorted by the priority from the mass run header data. The mass run with the highest priority is executed first.

A mass run is selected, when the planned execution date is the earlier or the same as the time at which the report is executed and the current status is [Released for Processing](#).

Filtering is applied by using parameters in the report. The system only selects the specified mass run types or observes the specified range of priorities.

Mass Run Reprocessing

Once a mass run has been processed, several options are available for reprocessing elements with the status [Processed with Errors](#).

Reprocessing of Elements

If you have chosen **Edit**, you can change the input fields for the mass run element and correct invalid values. Once you have done so, select the elements that contain errors and are to be reprocessed. Choose **Reprocess and Save** to process the selected elements again.

Elements with the status **Successfully Processed** cannot be reprocessed. Successfully processed elements that are selected for reprocessing are skipped during reprocessing and an appropriate message is displayed on the UI. In this case, only the elements that contain errors are reprocessed. When all selected elements are reprocessed, the corresponding status is set at element and header level.

Manual Processing of Elements

You can mark erroneous elements for manual processing. This can be done in the **Status** column of the element assignment block. You can change the status from **Processed with Errors** to **Processed Manually**. The user must then ensure that the corresponding mass run element is corrected in another way.

Create Follow-up Mass Run

The elements that contain errors in a mass run with the status **Processed with Errors** can be copied into a new mass run by choosing **Follow Up**. In this case, all elements with the status **Processed with Errors** are then transferred to a new mass run.

When the new mass run is saved, all elements that have been copied have the status **Taken Over**. The status of the old mass run is switched to **Successfully processed**.

Upload Data to Mass Run

In the mass run process, you can add the elements to the mass run object by uploading files. This file can be created in a spreadsheet with the CSV format.

Customizing Settings for the Upload

You make the settings for this in Customizing for **Service** under **Transactions > Settings for Subscription Transactions > Mass Runs** .

Two fields can be defined:

- **Row Nr** at mass run type level. This field identifies the row of the spreadsheet from which the relevant data for the upload starts.
- **Column Nr** at field settings level. This field identifies which column of the spreadsheet includes the value for the corresponding element field.

Mass Runs

The diagram illustrates a spreadsheet layout for mass runs. It features a table with 5 rows and 3 columns. Callouts identify the columns as 'Column Nr = 1', 'Column Nr = 2', and 'Column Nr = 3'. A callout identifies the third row as 'Row Nr = 3'.

	A	B	C
1	Example File		
2	Service Recipient	Techn. Res. ID	Customer Field
3	2000026	092-3933-234	0
4	2000048	077-9637-849	3
5	2000059	068-2573-278	1

Spreadsheet File for Upload

Upload the Spreadsheet File to the UI

You can either call the existing mass run or create a new one. In both cases, there is only one way to add the data using the file-upload. The creation process is described here. Choose [Mass Run](#).

On the next view that appears, choose the mass run type. Choose [Upload](#). On the next dialog box that appears, select the corresponding spreadsheet file by choosing [Browse](#). Choose [Upload](#) to upload the data and add it to the elements. After maintaining the data, save the mass run.

You can manipulate the uploading process according to customer requirements, for example you can provide conversions and calculations of the original data in the file before the data is added to the mass run. You must format a technical resource ID here. For more information, see [Extensibility of Mass Runs](#).

Example

By now, mass runs are used for

- Executing mass creation of call-off documents from the master agreements.
- Executing mass creation of subscription orders and solution quotations.
- Executing mass changes for existing subscription contracts.

Related Information

[Master Agreement](#)

[Subscription Orders](#)

[Subscription Contracts](#)

Mass Creation of Subscription Orders and Solution Quotations

Prerequisites

Mass creation without a master agreement is possible using mass run category [Mass Creation Without Master Agreement](#). For more information, see **Mass Runs**.

Features

Mass Run Process Data

View [Process Data for Mass Creation](#)

Mass run process data gathers the transaction type and product ID, and applies the information to all the elements in a mass run. You must enter the required transaction type for the subscription orders or solution quotations. You can add additional fields to the mass run process data.

Mass Run Elements

In the standard configuration, elements with the mass run category [Mass Creation without Master Agreement](#) have fields on the header with default values for the contracts to be created. At the element level, you can provide values for the corresponding contract individually. If the same field is maintained on the header and element level, then the value at the element level takes precedence over the header. The following fields are only used in mass runs for the creation of subscription orders or solution quotations:

- Contract
 - Contains the contract ID of the created contract once the element has been successfully processed.
- Sales Order
 - Contains the document ID of the subscription sales order being created that is used to create the contract.

Activities

Check Mass Run Header

The check at mass run header level checks the following:

- Whether a transaction type has been entered.
- Whether the business partner entered exists.

Check Mass Run Elements

The default handler for mass runs with the category [Mass Creation without Master Agreement](#) checks if the number of provided technical resources exceeds the maximum limit defined in the Customizing. You can add additional fields to the element data. For more information, see **Extensibility of Mass Runs**.

Processing Results

You can use the links generated for the sales order and subscription contract to navigate directly to the corresponding transaction.

In case your mass run ends with the status [Processed with Errors](#), you can use the [Transfer Elements with Errors](#) button to move the erroneous elements to a new mass run where you can fix the errors and execute again.

Example

The mass run type **BULK_CREATE** is delivered as a sample implementation containing the sold-to party, ship-to party, and contract account.

- Handler class implementation

A sample handler class is available for processing the fields **Technical Resource ID** and **Ship-To** with the name **CL_CRM_ISX_MR_A_W_TECHRES_HDL**. You can enter this handler when defining a new mass run type. The handler class implements a check and mapping of the fields to API structures before processing an element.

- Technical enhancement of the element structure and UI

The fields **TECH_RES_ID** and **SHIP_TO** exist in the element structure.

In the case of the mass run type **BULK_CREATE**, elements have the additional field **Sold-To Party**, which can be edited. This field is mandatory for every element and determines the sold-to party, ship-to party, and payer for the subscription order and solution quotation to be created. Alternatively, you can also set a ship-to party in your elements. You can add additional fields to the element data.

For more information, see **Extensibility of Mass Runs**.

Related Information

[Mass Runs](#)

[Extensibility of Mass Runs](#)

Mass Creation of Call-Off Documents from Master Agreement

Use

You can use the mass creation of call-off documents from master agreement to create a large number of similar subscription orders and solution quotations from a single master agreement item. For example, a customer has a master agreement in place with a service provider to enable the process of ordering mobile services and phones. The enterprise customer wants to order 50 new mobile services in relation to their master agreement for 50 new employees. The mass creation of subscription contracts provides capabilities for such a bulk order.

Prerequisites

If you want to create mass runs for the mass creation of call-off documents from master agreement, you must define a mass run type with the mass run category **Mass Creation with Master Agreement**. For more information, see **Mass Runs**.

You must also define a master agreement with released items.

Features

Mass run process data gathers the following information that applies to all elements in a mass run:

- Master Agreement ID; Customer-Specific Product ID

Identification of the master agreement document and item to which the created orders and contracts refer.

- Contract Start Date

Defines the start of the contract validity period. It has no effect on the time at which the mass run is triggered.

- Transaction Type

Enter the required transaction type for the subscription orders or solution quotations.

- Sold-to; Ship-to; Payer; Employee

Business partners to be used in the contracts.

- Sharing Contract

Sharing contracts to be used in the contracts.

- Contract Account

Contract Account to be assigned to all items.

Fields that are available at the master agreement will be taken over in a read-only format after master agreement ID and customer-specific product ID are filled.

You can add more fields to the process data for a mass run. For more information, see **Extensibility of Mass Runs**.

Mass Run Elements

In the standard configuration, elements with the mass run category **Mass Creation with Master Agreement** use the default values for the header level fields to create contracts. At the element level, however, you can maintain separate values for the corresponding contract individually. If the same field is maintained on the header and element level, then the value at the element level takes precedence over the header value. While entering element-specific data, you can also choose a transaction type apart from what is already provided at the mass run header. This enables you to create a mix of subscription orders and solution quotations in a single mass run.

The following fields are only used in mass runs for the creation of subscription orders and solution quotations:

- **Contract Number:** The ID of the subscription contract that is created once the element has been successfully processed.
- **Sales Order:** The ID of the transaction, either subscription order or solution quotation, that is created in the mass run and is used to create the subscription contract.

You can add more fields to the element data. For more information, see **Extensibility of Mass Runs**.

Activities

Assisted Entry of Master Agreement Data

Enter a master agreement and open the search help for customer-specific product IDs. Search for customer-specific product IDs and select one from the result list. A dialog box appears on which you can select the missing fields according to the settings for the master agreement item.

Check Mass Run Header

The check at the mass run header level confirms whether

- A transaction type has been entered.
- The master agreement and customer-specific Product ID exist and are identical.
- the business partner entered exists.

Check Mass Run Elements

The default handler for mass runs with the category **Mass Creation with Master Agreement** checks if the number of provided technical resources exceeds the maximum limit defined in the Customizing.

You can add additional fields to the element data. For more information, see **Extensibility of Mass Runs**.

Processing Results

Once the mass run is completed successfully, you can use the links in the **Contract Number** and **Sales Order** to navigate directly to the corresponding transaction.

In case your mass run ends with the status **Processed with Errors**, you can use the **Transfer Elements with Error** button to move the erroneous elements to a new mass run where you can fix the errors and trigger the mass run.

Example

The mass run type **BULK_CALL_OFF_MA** is delivered as an example implementation containing the elements technical resource ID and ship-to. The extension consists of two parts: For more information, see **Extensibility of Mass Runs**.

- Handler class implementation

An example handler class is available for processing the fields **Technical Resource ID** and **Ship-to** with the name **CL_CRM_ISX_MR_A_W_TECHRES_HDL**. You can enter this handler when defining a new mass run type. The handler class implements a check and mapping of the fields to API structures before processing an element.

- Technical enhancement of the element structure and WebClient UI

The fields **TECH_RES_ID** and **SHIP_TO** exist in the element structure.

For more information, see **Extensibility of Mass Runs**.

Related Information

[Mass Runs](#)

[Extensibility of Mass Runs](#)

Mass Changes to Subscription Contracts

Use

Various business processes often require a mass change to be made to contracts. A typical business scenario in the subscription area is to update existing subscription contracts with adapted tariffs. You can schedule a mass run to adapt the cross-catalog mapping version assignment of pre-selected contracts at a certain point in time.

Mass changes use the mass run framework to implement bulk execution of the **Business Transactions Maintenance Framework (BTMF)** change process on existing subscription contracts. For more information, see **Maintenance Framework for Business Transactions**.

A specific set of changes is usually only intended to be made for execution for existing contracts. This set must first be modeled as change processes in the BTMF framework.

Processes modeled in the BTMF framework can be executed from the UI of a single document or for multiple documents by applying a mass run (mass change). To make the process work in the context of a mass run, the implementation of a change process has to fulfill specific requirements. For more information, see **Extensibility of Mass Runs**.

For details about process subscription contract change processes, see **Subscription Contract Changes**.

Prerequisites

You have defined the mass run type and the BTMF change process in Customizing for **Service** under **Transactions** > **Settings for Subscription Transactions** > **Mass Runs** > **Define Settings for Mass Runs**. The event handler class used in the mass run type has to implement an appropriate ABAP interface for mass changes (IF_CRM_ISX_MR_B_EVENT_HANDLER or IF_CRM_ISX_MR_D_EVENT_HANDLER for header change processes). The default class is CL_CRM_ISX_MR_B_EVENT_DFLT_HDL or CL_CRM_ISX_MR_D_EVENT_DFLT_HDL (for header change processes).

You have configured the BTMF framework. For more information, see **Maintenance Framework for Business Transactions**.

Features

Structure

According to the general structure of a mass run, data for a mass change is divided into three parts:

- Administration data
- Process data (such as the activation date of the mass run or the change process that is executed within the mass run)
- Element data

Elements in a mass change reference the subscription contracts or subscription contract documents (for header change processes) to be changed. This often applies to an extremely large number of subscription contracts/subscription contract documents. The element data is shown in a summary view that displays the number of elements with the following statuses:

- Total number of elements
- Number of elements successfully processed
- Number of elements not successfully processed

Process Data

A mass run has the following specific data:

- The BTMF change process, which is taken from the assigned mass run process type.
- The activation date of the change.

A set of specific data is also displayed for each change process to be used as input for the change to be executed.

Displaying Detailed Data of Elements

To display detailed data for elements, you have to navigate to the details screen where a subset of all elements is displayed. Enter the search criteria to define the result set according to your requirements.

The element data for a mass contains the following specific fields:

- Contract ID
The ID of the contract to be changed during a mass run.
- Transaction ID (for header change processes)

The ID of the contract document to be changed during a mass run.

- Change Order

The change of the subscription contract is triggered by a change order. The ID of this change order is available once an element has been successfully executed.

Adding Elements to a Mass Run

You can add the contracts to be changed in the mass run using the following actions:

- Search

The standard search for subscription contracts/subscription contract documents opens. Create new elements by selecting entries in the result set.

- Upload

For more information, see **Mass Runs**.

- Predefined selection

Predefined Selections

A predefined selection defines the set of subscription contracts/subscription contract documents to be changed by the mass run. When creating a mass run, a list of all predefined selections is available for adding contracts to be changed. Predefined selections available with a mass run are defined in Customizing for **Service** under **Transactions** > **Settings for Subscription Transactions** > **Mass Runs** > **Define Predefined Selections**.

A set of predefined selections is available by default. You can use it as an example to implement a customer-specific predefined selection:

- CL_CRM_ISX_MR_PS_MASS

Selects all subscription contracts in the system.

- CL_CRM_ISX_MR_PS_MY_THIS_MON

Selects all subscription contracts created within the current month.

- CL_CRM_ISX_MR_PS_TODAY

Selects all subscription contracts created on the current day.

- CL_CRM_ISX_MR_PS_CCM_VERSION

Selects all subscription contracts with a specified cross-catalog mapping version.

- CL_CRM_ISX_MR_PS_MASS_HDR

Selects 10000 subscription contracts

- CL_CRM_ISX_MR_PS_MY_TODAY

Selects all subscription contract documents that the user has created on the current day (today)

You add a customer-specific predefined selection by implementing the ABAP interface IF_CRM_ISX_MR_PRED_SELECTION or IF_CRM_ISX_MR_PRED_SELECTION_D (for header change processes). The method GET_CONTRACTS or GET_CONTRACT_DOCS (for header change processes) defines the set of subscription contracts/subscription contract documents to be changed by the mass run that returns a list of corresponding contract GUIDs (GUID of the main rate plan item or document

header). Finally, create a new entry in Customizing for the predefined selections and assign the name of the implementing ABAP class.

Example

If you want to cancel all related subscription contracts for an expiring master agreement using a mass run and the BTMF change process, click **Cancel**.

Preparation of the Mass Run

Make sure that the required input fields in the change process are available in the mass run data. When cancelling a contract, the reason and the requested date of cancellation must be provided. Fields CANCEL_REASON, CANCEL_REQUEST_DATE and CANCEL_REQUEST_TIME are already part of the header structure.

Create a new mass run type in Customizing and assign the change process **CANCEL**. Assign the handler class CL_CRM_ISX_MR_B_CANCEL_HDL. The handler provides data required by the cancellation process, for example the date when the cancellation was received and the procedure to be used.

To make the new fields visible in the maintenance UI for your mass run, maintain the header view by creating a new configuration for the mass run type. For more information, see **Extensibility of Mass Runs**.

Execution of the Mass Run

When creating a mass run for this new mass run type, additional input fields are displayed on the process data screen. Provide a cancellation reason and a requested date for cancellation. To apply the change to the relevant contracts, add elements by using the **Search** function. Use the master agreement ID as a search criterion. After entering all of the required data, release the mass run for processing by setting the header status.

Execute the mass change directly by using the **Run** function or use the scheduled processing function.

Related Information

[Maintenance Framework for Business Transactions](#)

[Extensibility of Mass Runs](#)

[Contract Change Processes](#)

[Mass Runs](#)

Extensibility of Mass Runs

Use

Mass runs for contract creation and maintenance are dependent on contract Customizing settings and product models and therefore come with a very basic standard setup. However, you can use highly flexible enhancement options to tailor mass run types to the required processes. Four elementary techniques can be used to extend the options for mass run processes and to customize the maintenance UI for each mass run type to ensure that maintenance remains targeted.

Prerequisites

You have made basic settings for mass runs in Customizing for **Service** under **Transactions** **Settings for Subscription Transactions** **Mass Runs**.

Mass changes use BTMF change processes. To handle process data provided from an Application Programming Interface (API) instead of UI, you have adapted the change process accordingly.

Both mass creation and mass change use the API for subscription documents when executing a mass run. Manipulation for the input data for the API is required for the extensibility of mass runs. For more information, see **Application Programming Interface (API) for Subscription Orders and Contracts**.

Features

UI Configuration

UI configurations for the views used in mass run maintenance use the mass run category and type as a key. This involves the following object types:

- ISX_MR_HEADER_B for mass change
- ISX_MR_HEADER_D for mass header change
- ISX_MR_HEADER_A for mass creation with master agreement
- ISX_MR_HEADER_C for mass creation without master agreement
- mass run Type ID as a sub-object type

Therefore, each mass run type can have its own UI configuration.

Use these configurations to either add newly created fields to the UI or to hide unwanted fields for the corresponding mass run type. Fields in form views can also be defined as read-only or as mandatory fields, which is not possible in table views. For more details see the **UI Configuration Tool**.

❖ Example

For a custom mass run type **XYZ_ABC** for a mass run category **Mass Creation Without Master Agreement**:

Header

- Component: ISXMR
- View: PROCESSMASSCREATE
- Object Type: ISX_MR_HEADER_C
- Sub-object Type: XYZ_ABC

Element

- Component: ISXMR
- View: ELEMENTLIST
- Object Type: ISX_MR_HEADER_C
- Sub-object Type: XYZ_ABC

Custom Field and Logic App

To add a field, you must select the business context. SAP offers the following business contexts with respect to mass runs:

- SOM Mass Run Elements for Create
- SOM Mass Run Header for Change
- SOM Mass Run Header for Create
- SOM Mass Run Header for Change Header

On the Tab UIs and reports, enable the proposed usage and publish the custom field. The field is available in the available fields list on the UI configuration. The field also appears in the mass run Customizing and handler classes.

Handler Implementation Class

Handler classes are intended to execute customer-specific processing steps, during UI maintenance or execution of mass runs. Therefore, corresponding methods in the customized ABAP handler classes are triggered at the relevant point in time.

A handler class is an ABAP class implementing interface IF_CRM_ISX_MR_EVENT_HANDLER. Derived from this interface, the following handler classes are available for adding specific additional functionality:

- Mass creation (IF_CRM_ISX_MR_A_EVENT_HANDLER or CL_CRMS4_SOM_MR_C_TECHRES_HDL)
- Mass changes (IF_CRM_ISX_MR_B_EVENT_HANDLER or IF_CRM_ISX_MR_D_EVENT_HANDLER)

A handler class is uniquely assigned to a mass run type in Customizing.

The following sample implementations are available:

- CL_CRM_ISX_MR_A_W_TECHRES_HDL or CL_CRMS4_SOM_MR_C_TECHRES_HDL
Handles the usage of technical resources when executing mass creation of contracts
- CL_CRM_ISX_MR_B_CANCEL_HDL
Handles required additional data when executing mass cancellation of contracts
- CL_CRM_ISX_MR_B_CCM_VERS_HDL
Handles an update of the cross-catalog mapping version
- CL_CRMS4_SOM_MR_D_CHNG_ST_HDL
Handles change of sold-to party header change process

The interface provides methods, for example for adapting the process data before a contract is created or changed or to execute checks of additional (customer-specific) fields.

Interface IF_CRM_ISX_MR_EVENT_HANDLER

Method Name	Execution Point of Time	Usage
CHECK_MASS_RUN_VALIDITY	When user triggers mass run checks. Before mass run is processed.	Checking validity of mass run data at header level. Check additional (customer-specific) fields here. Messages are visible in the header data.

Method Name	Execution Point of Time	Usage
CHECK_ELEMENT_VALIDITY	When user triggers element checks. Before single element is processed.	Checking validity of mass run data at element level. Check additional (customer-specific) fields here. The verification status is defined accordingly. Messages are available for each element.
DO_BEFORE_ELEMENT_CREATE	Before adding elements to the mass run	Add additional customer-specific data to mass run elements. Perform conversions of data, if required. Add new entries or remove existing entries, for example after adding contracts, remove instances with an invalid status according to customer requirements.
DO_BEFORE_ELEMENT_UPLOAD	Before adding uploaded data to the mass run	Customers can implement their own coding in this method that defines the handling of mass run elements before the file is uploaded.

Interface IF_CRM_ISX_MR_A_EVENT_HANDLER

Method Name	Execution Point of Time	Usage
BEFORE_ORDER_CREATE	When executing a mass creation for an element. Before the subscription order API for creating subscription contracts is called (Method CONTRACT_CREATE of ABAP class CL_CRM_ISX_PROV_ORDER_API_PRXY).	Provide additional parameters or adapt existing input parameters for the subscription contract API for creating contracts, for example by adding customer-specific fields.
UPDATE_ELEMENT_AFTER_CREATE	When executing a mass creation for an element. Before the subscription order API for creating subscription contracts is called (Method CONTRACT_CREATE of ABAP class CL_CRM_ISX_PROV_ORDER_API_PRXY).	Check the results of the subscription order API. Force rollback, if required. Define element data, for example by setting the status.

Interface IF_CRM_ISX_MR_B_EVENT_HANDLER or IF_CRM_ISX_MR_D_EVENT_HANDLER (for mass header changes)

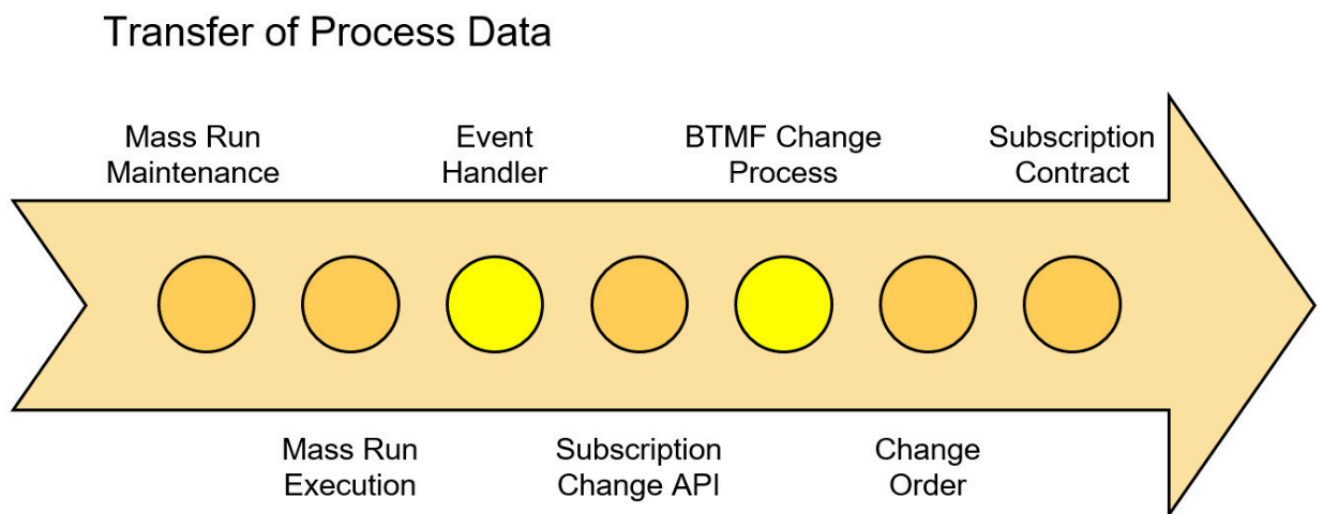
Method Name	Execution Point of Time	Usage
BEFORE_CONTRACT_CHANGE	When executing a mass change for a single contract/subscription contract document. Before the contract change API for subscription contracts/subscription contract documents is called (Method CONTRACT_CHANGE of ABAP class CL_CRM_ISX_PROV_ORDER_API_PRXY).	Provide additional input parameters or adapt existing input parameters in the subscription contract API for changing contracts, for example add customer-specific fields used by your BTMF change process.
UPDATE_ELEMENT_AFTER_CHANGE	When executing a mass change for a single contract/subscription contract document. After the subscription order API for changing subscription contracts/subscription contract documents is called (Method	Check the results of the subscription order API. Force rollback, if required. Set element data, for example set the status.

Method Name	Execution Point of Time	Usage
	CONTRACT_CHANGE of ABAP class CL_CRM_ISX_PROV_ORDER_API_PRXY) and before the element status is updated.	

Example

If you want to use mass changes, you cannot reuse any of the existing BTMF change processes. Most of the existing processes require input data from the UI. An exception is the BTMF process **CANCEL** that receives input data from a list of attributes that are entered before calling the process for changing contracts using the API.

Extensibility of Mass Runs



You want to extend the runtime of existing contracts. Instead of executing the change for every single document separately, you want to use a mass change and provide the requested period in the header data of the mass run.

Firstly, you enable the BTMF process to handle API data provided externally. The extension period is provided here from the mass run header. This example reuses the existing BTMF process **EXTENSION**. It is implemented by the ABAP class CL_CRM_BTMF_TELCO_EXTENSION. Extend the method START in this class by extracting the extension period from the BTMF container used to store global process data. Use this data to adapt the change order accordingly by calling the business transaction API (function module CRM_ORDER_MAINTAIN).

In the Customizing settings for the mass run, add a new mass run type to be used for the extension process, for example **ZEXTENSION**. Assign the BTMF process to be used, in this case **EXTENSION**. Create a new handler class by implementing the interface IF_CRM_ISX_MR_B_EVENT_HANDLER. Assign the handler class to your mass run type.

The extension period is to be available in the process data for the mass run. Since the process data is part of the header data, you have to add the new fields for the extension period to the header using the custom field and logic app. Create a UI configuration for the process data view with the object type **ISX_MR_HEADER_B** and sub-type **ZEXTENSION** (input help available).

Finally, the new data for the mass run has to be transferred to the BTMF change process. All fields of the header process data and the fields from the customer extension at header are available in the BTMF container as a default. If this is not sufficient, because a further calculation has to be made or data or field names have to be adapted according to the requirements of the BTMF process implementation, use the method BEFORE_CONTRACT_CHANGE in your handler class to extract the extension period from the mass run header and add it to the input data in the API for changing contracts. The process attributes are available in the BTMF container for the process.

Related Information

[Starting the Application Enhancement Tool](#)

[Application Enhancement Tool](#)

[UI Configuration Tool](#)

[APIs and Events for Subscription Order Management](#)

Deleting Mass Runs

Mass Run data can only be deleted, but it cannot be archived. This is because the Mass Run data is considered as control data.

Delete Options

There are two options for deleting mass runs:

- You can delete a Mass Run individually via the user interface. Click **Delete** in the action toolbar.
- You can use the ILM destruction object CRMS4_SOM_MR_DESTRUCTION to delete mass runs. You can use this option if your system contains a larger number of mass runs, and you want to delete them in a bulk and/or periodically in order to keep the system clean. This option is described in more detail in the following.

Mass Run Destruction

You can use destruction object CRMS4_SOM_MR_DESTRUCTION to delete Mass Run data in transaction ILM_DESTRUCTION. The details of the CRMS4_SOM_MR_DESTRUCTION destruction object can be found in transaction **DOBJ**.

Prerequisites

Mass Run data can only be deleted if the ILM retention rules allow it. The ILM rules as part of SAP Information Lifecycle Management are maintained in transaction **IRMPOL**. Here, you can create policies for retention rules.

The following condition fields are available:

- SERVICEMASSRUNTYPE – Mass Run Type
- SERVICEMASSRUNSTATUS – Mass Run Status

The following time references are available:

- CREATION_DATE - Created On
- LAST_CHANGE_DATE - Last Changed On

Deleting Data in ILM_Destruction

To use the ILM destruction object, proceed as follows:

1. Start transaction ILM_DESTRUCTION
2. Choose **Type of Data to Be Destroyed** as **Data from the Database**, enter ILM object CRMS4_SOM_MR_DESTRUCTION and click **Execute**.
3. Select the **Destroy** option

You can maintain your variant that contains the selection criteria for the data to be deleted. The available parameters are:

- Mass Run ID
- Mass Run Type
- Mass Run Header Status
- Created By
- Created At
- Changed At
- Planned Execution At

Additionally, you can choose if you want to really delete the data or if you want to test the destruction. In the latter case, the system simulates the deletion, but it will not delete the data eventually. You can use that option to check the destructibility of the Mass Run data or to test your selection criteria variant.

After you have defined and saved your variant, you can set the start date and the spool parameters before you can trigger the execution of the data destruction using the **Execute** option.

Authorizations for deleting Mass Runs

Activity	Required Authorization Object
ACTVT: 06 CRMS4_MR_T: * or the Mass Run Types the user shall be allowed to delete	CRMS4_MR
ACTVT: 16 DESTR_OBJ: CRMS4_SOM_MR_DESTRUCTION	S_ARCD_OBJ

Working With Mass Processing

Beneath the mass run framework, a new mass processing framework is available that is leveraged by certain SAP Fiori apps that are designed to integrate seamlessly into the Subscription Contract Lifecycle Management. The following apps are developed to simplify the workflow for setting up and executing a mass processing:

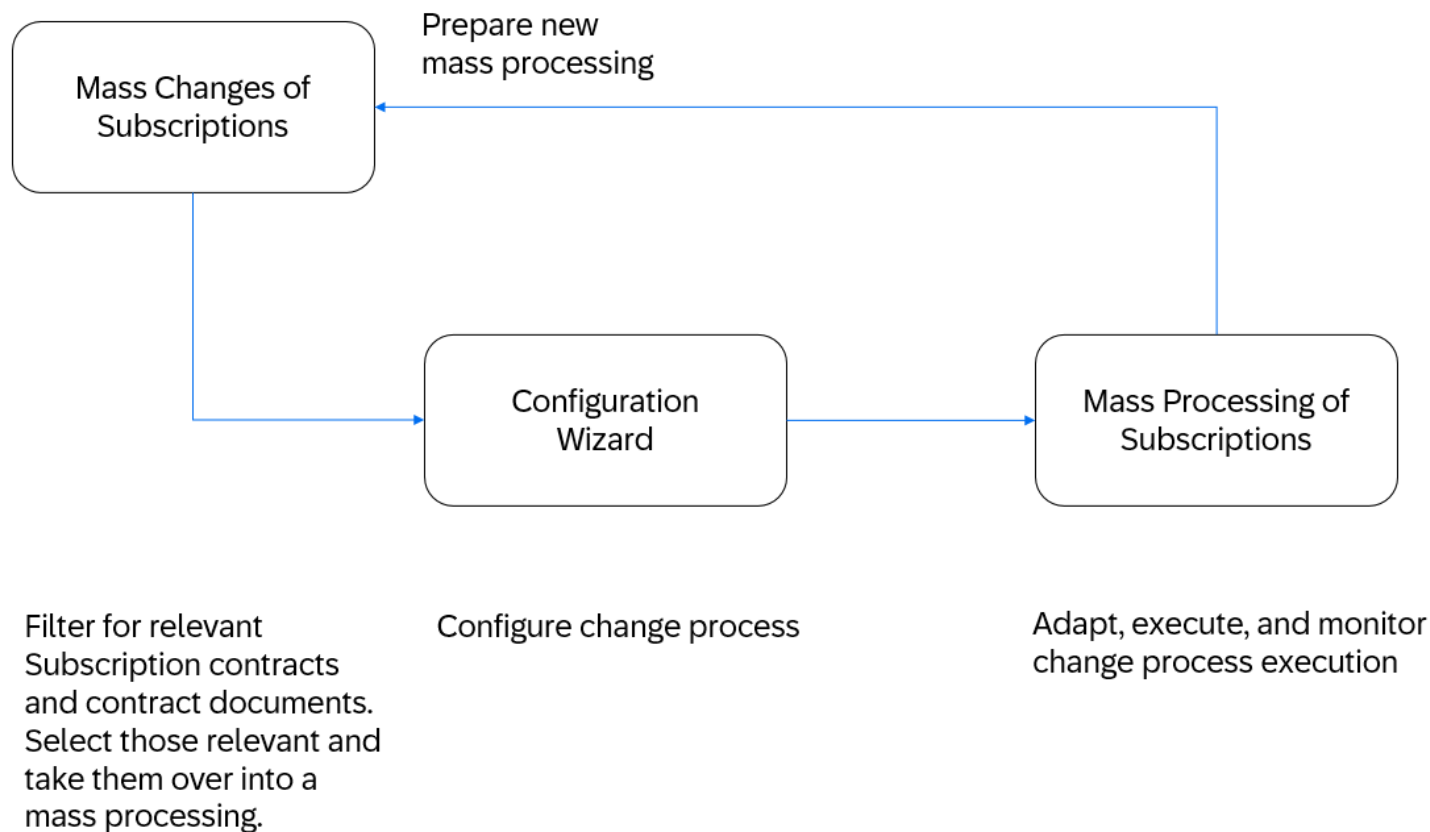
- **Mass Changes of Subscription** ([F7405](#))

You use this SAP Fiori app to filter for the relevant subscription contracts or subscription contract documents, and configure the mass process via a configuration wizard. Once the configuration is confirmed, a mass processing is created in draft status and you are directed into the open mass processing in the **Manage Mass Processing of Subscriptions** app ([F7406](#)).

- **Manage Mass Processing of Subscriptions** ([F7406](#))

You use this SAP Fiori app to access the mass processing framework and perform any or all of the following actions:

- Additional configuration of an existing mass processing
- Scheduling and executing a mass processing
- Analyzing any execution errors



Change Process Groups for Mass Processing

The following standard change process groups are delivered for use in the **Mass Changes of Subscriptions** and **Manage Mass Processing of Subscriptions** Fiori apps:

BTMF Process Group	Process Type	Change Process List
ADD_NEW_CONTRACT	ISTG	ADD_NEW_CONTRACT
ADD_SUBITEM	ISTA	ADD_SUBITEM
CANCEL	ISTA	CANCEL
CANCEL_OPTION	ISTA	CANCEL_OPTION
CHANGE_BILL_CYCLE	ISTA	CHANGE_BILL_CYCLE
CHANGE_PRICE_AND_QTY	ISTA	CHANGE_PRICE CHANGE_QUANTITY
CHANGE_PROD_QTY	ISTA	CHANGE_QUANTITY PROD_CHANGE
CHANGE_QUANTITY	ISTA	CHANGE_QUANTITY
CHANGE_SOLDTO_PARTY	ISTA	CHANGE_SOLDTO_PARTY
CHANGE_WBS_ELEMENT	ISTA	CHANGE_WBS_ELEMENT
EXTENSION	ISTA	EXTENSION
PROD_CHANGE	ISTA	PROD_CHANGE

BTMF Process Group	Process Type	Change Process List
REJECT	ISTH	REJECT
UPDATE_PRICE	ISTA	UPDATE_PRICE

i Note

You can also define your own customized change process groups for mass processing of subscription contracts using the extensibility capabilities.

Related Information

[App Extensibility: Manage Mass Processing of Subscriptions \(Key User\)](#)

[Mass Changes of Subscriptions](#)

[Manage Mass Processing of Subscriptions](#)

Sharing

Overview

With this functionality, multiple sharing options are available. Sharing allows you to share counters or allowances during a rating process in SAP CC, within a contract or across multiple contracts.

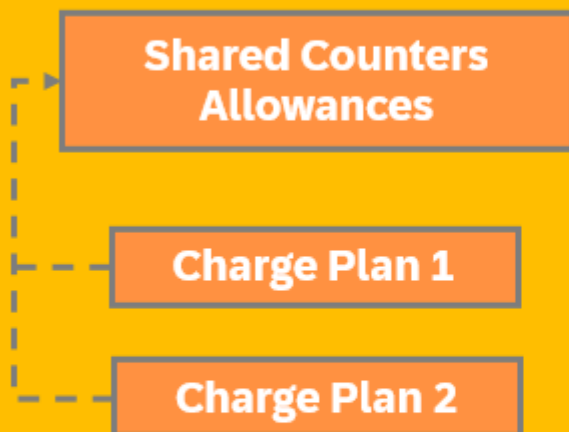
The following sharing options are available:

Sharing within a contract	You can share counters and allowances within a contract.
Sharing contracts	You can share counters and allowances across a sharing contract and its shared contracts (tightly coupled).
Shared allowances	Shared allowances can be shared between independent contracts using the same shared ID (loosely coupled).

The concept of sharing contracts and shared allowances are disjoint and cannot be used together; that is, a product that supports shared allowances cannot be used in sharing/shared contract.

Sharing Within a Contract

Subscription Contract

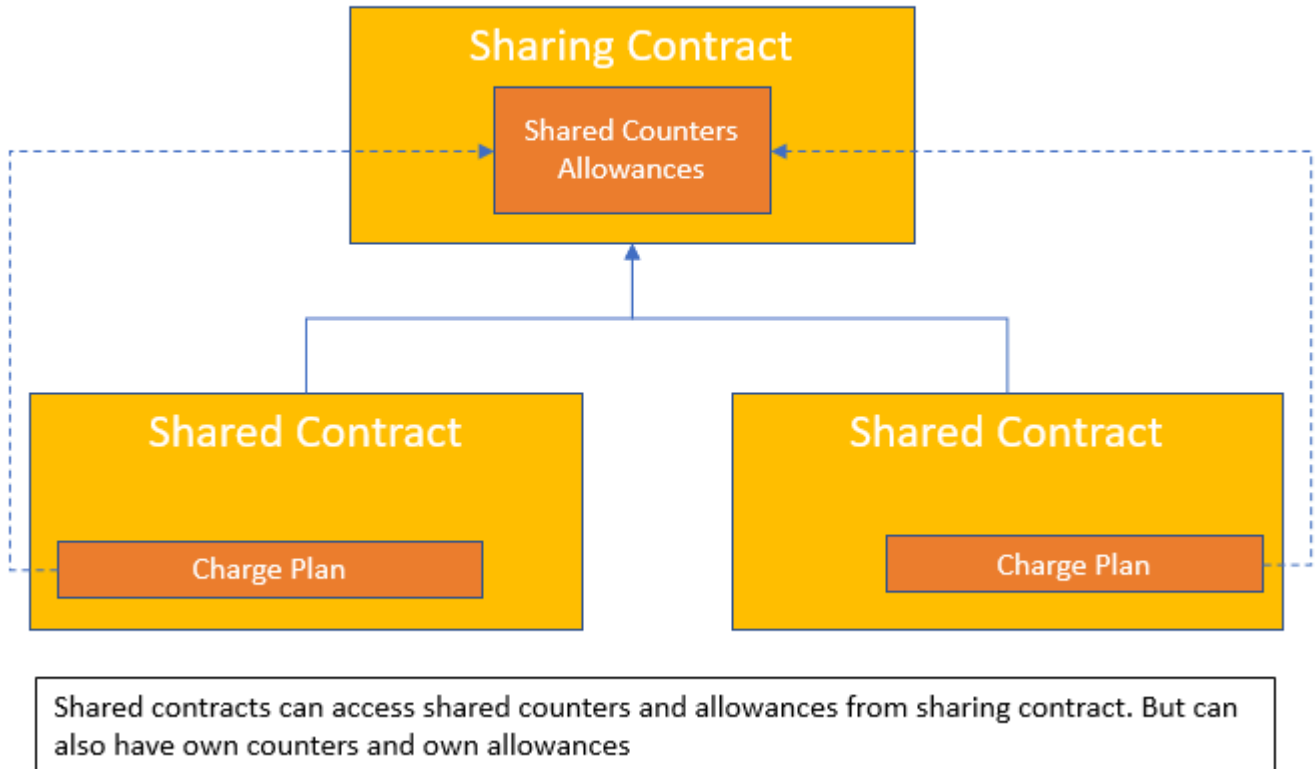


Charge plans of one subscription contract can access shared counters and allowances within the contract.

To share counters within a contract, you must define several charge plans for the same product that uses the same counter IDs and assign the same namespace to these counters.

Sharing Contracts

Sharing Contracts

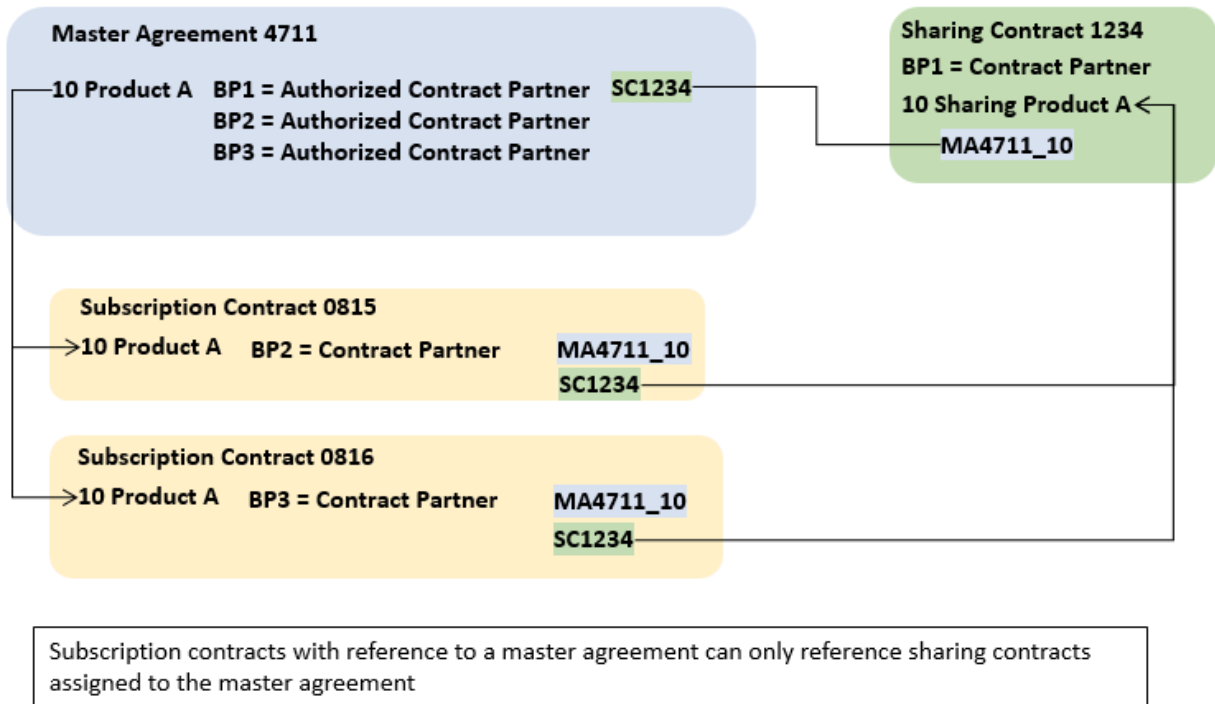


You can define sharing contracts with charge plans containing shared objects (such as counters and allowances). In addition, you create dependent (shared) contracts with reference to this sharing contract. These dependent contracts can then reuse the shared objects from the sharing contract. Sharing contracts and shared contracts do not need to share the contract partner since independent partners can be assigned to the contracts.

With sharing contracts, you can support B2C cases like family plans, where one family member owns the root contract (sharing contract) in which the shared counters and allowances are defined (for example, free downloads), and the dependent contracts (shared contracts) of the family members who can reuse the shared counters and allowances from its parent plan.

It is also possible to use sharing contracts together with master agreements.

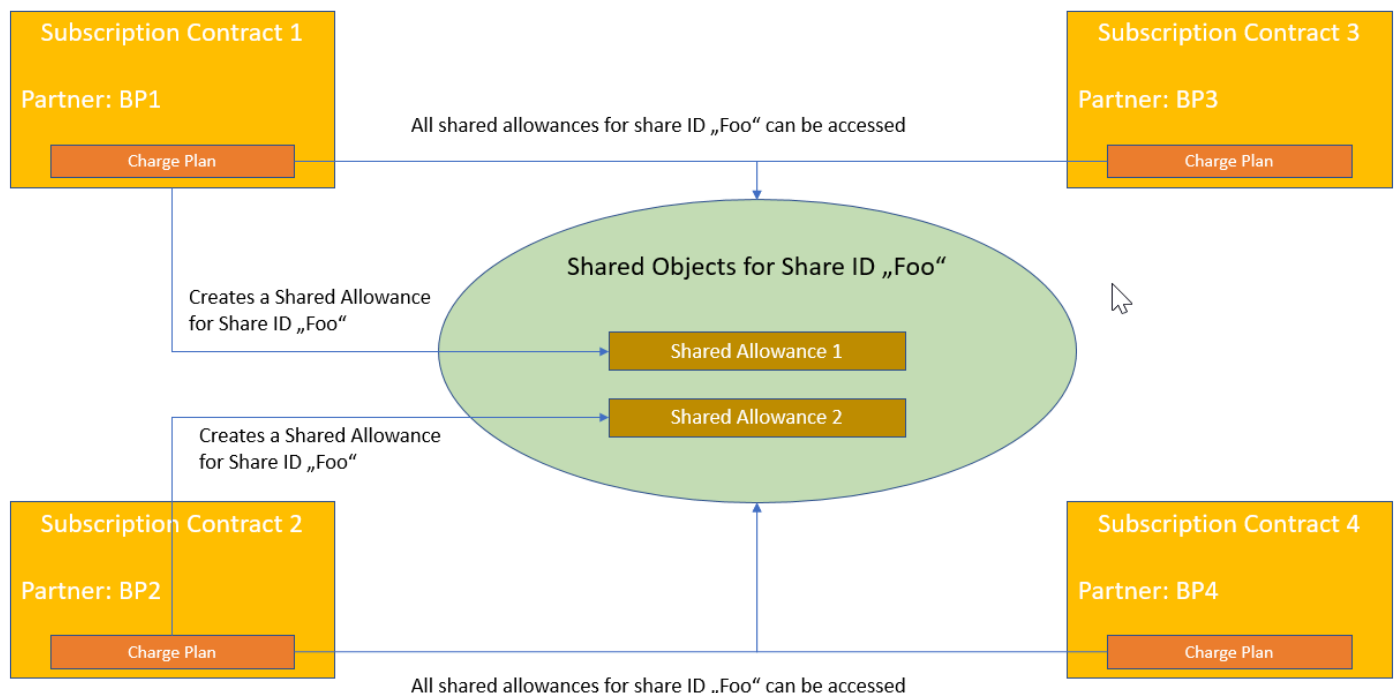
Master Agreement and Subscription Contracts for Sharing



Sharing contracts, in combination with master agreements, also serve the B2B scenario. A B2B customer agrees on a master agreement with a service provider. In this master agreement, they define a group of users that are allowed to request call-offs for a specific product for which a sharing contract is assigned. The charge plan assigned to this sharing contract defines counters and allowances. These counters can be used for all participating contracts collectively. All call-offs for this master agreement can join this sharing contract. The participating contracts will be linked to the sharing contract. Rating and charging processes in SAP CC will make use of the shared counters. Sharing contracts also support rerating processes in CI.

Shared Allowances

Shared Allowances



You can define charge plans using shared allowances. Contracts using shared allowances are much more loosely coupled than sharing contracts with the tight coupling of root and dependent contract. Shared allowances are identified during rating process via a share ID. The share ID is a combination of a share ID parameter defined in cross catalog mapping during product modeling, and a sharing group defined in the subscription contract. The sharing group is an ID which is defined via Fiori App "Manage Sharing Group". The Sharing Group must be assigned during subscription order creation or master agreement definition if shared allowances are in use. If contracts share the allowances across contracts, they must use the same sharing group.

Shared allowances can be used in B2B and B2C scenario. The contracts are not dependent on each other, each contract can be assigned to an individual subscriber account by using account split functionality. This is especially important when you work with many contracts assigned to one subscriber account (B2B).

Up to date shared allowances cannot be combined with rerating.

i Note

When you use sharing contracts take care of the number of contracts under one subscriber account in CC, as this concept cannot be combined together with account split functionality.

Sharing Contract

Counter Sharing is a concept for multiple customer contracts to share one or several counters.

Example

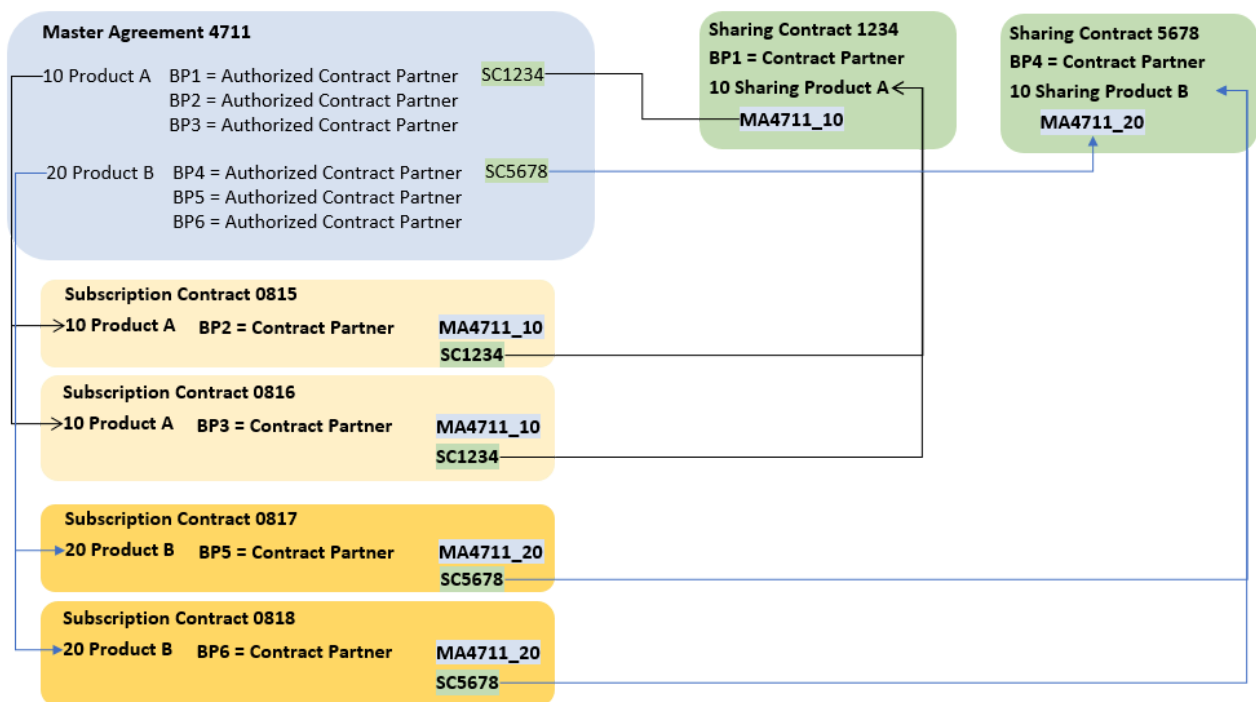
A corporate customer signs a master agreement with a communications service provider. The service provider offers mobile services to all units and employees of the customer. All fees and charges are paid by the customer's headquarters itself. The customer has two organizational units: sales and development. All service recipients of the sales organization get a monthly share of 1000 free minutes. As soon as the free minutes are consumed during a month, all additional calls will be charged according to the rate plan. The shared free minutes are consumed according to the **first come first served** principle. At the end of the month, the rest of free minutes expire.

Alternatively, all contracts of the service recipients of the sales department shall be paid by the service recipient itself.

In step one, the master agreement is created with two positions: one for the sales organization and one for the development organization. The master agreement contains the rate plan products, but not the sharing product.

The sharing product is created with a specific material type (delivered material type = SUBC). In the Cross-Catalog Mapping of both products defined with the same counter ID and same namespace. In SAP Convergent Charging (SAP CC), these counters are marked as to be shared between contracts: sharing contract and dependent contract. In a second step, the sharing contracts with references to items of master agreement items are created. In a third step, the subscription contract with the references to the sharing contracts is created.

Master Agreement and Subscription Contracts for Sharing



Master Agreement and Subscription Contracts for Counter Sharing

As an alternative of the upper use case, all contracts of the service recipients of the sales department shall be paid by the service recipient itself. You can also use sharing contracts and shared contracts without referencing a master agreement.

Display of Dependent Contracts

Dependent contracts of a sharing contract can be displayed in the UI role S4C_SOM_REP - Internal Sales Rep. (Subscr.) using the assignment block **Dependent Contracts**. This assignment block is visible only when a sharing contract is displayed.

Using Convergent Charging Tables

If your charge plan uses mapping or range tables and you redefine these tables in subscription contracts an agreement table in SAP CC will be created. You cannot use subscriber specific tables, only catalog or agreement tables with a reference to the business partner are supported.

Restrictions

In SAP Convergent Charging (SAP CC), counter sharing is only possible if all involved subscription contracts have the same subscriber account. As a result, the account split functionality is not possible. In other words, the concept of sharing contracts should not be used if there is a business partner with multiple contracts. However, the business partners in the subscription orders can be different. An alternative subscriber account (aSUAC) is used only when the contract is sent to SAP Convergent Charging. This is the same for the sharing contract and all its dependent contracts.

This subscriber account is a concatenation of a prefix **RC** and the sharing contract number. However, the aSUAC is not a real business partner and is used only in the SAP CC provider contract and acts as a subscriber account. The distribution of the aSUAC to SAP CC is handled by the FI-CA master data distribution, which sends the subscriber account and provider contracts to SAP CC. If you redefine a mapping or range table in a sharing contract or a shared contract a subscriber specific table is created. As the aSUAC is only known when the contract is distributed you need in your ODI schema the step STCO (Copy Subscriber Table for aSUAC) that the subscriber table is copied to the aSUAC. When you assign a CC table you can only use catalog tables or agreement tables created for the used business partner. a subscriber specific table cannot be assigned.

i Note

- You must create a separate item in the master agreement where the right business partners are assigned as service recipients.

It is possible to provide sharing for different products as long as the products have the right definition for counter sharing.

Products for Sharing Contracts

When creating shared counters, you must ensure that the product in the sharing contract and the product in the shared contract have mapping counters with identical IDs and namespaces in the charge plan.

You must define two products here: One product contains the definition of the sharing rate plan. The other product contains the definition of the rate plan product (both products have the rate plan type from a technical perspective).

In the sharing product, you can maintain which shared products can be used together with a specific sharing product. For more information about this, see **Shared Products**.

You must define the counters to be shared in Cross-Catalog Mapping for both products. In SAP Convergent Charging (SAP CC), these counters are defined as counters that can be shared between the sharing contract and the linked contract.

The master agreement contains the rate plan products. The sharing product is not part of the master agreement. The subscription contracts that are created in relation to a master agreement item contain the rate plan product. This subscription contract represents one specific line.

If counter sharing is to be performed for subscription contracts, a sharing contract must be created with the correct product to be shared. The sharing contract must be assigned to a master agreement item. This creates a relationship between the master agreement item and the sharing contract.

Product Type

We recommend creating a new product type such as SUBC (Subscription Sharing Product).

Assign the sharing product purpose to the new product type in Customizing for [Service](#) under ► [Master Data](#) ► [Settings for Subscription Master Data](#) ► [Subscription Product-Specific Data](#) ► [Assign Product Purpose to Product Types](#) ►.

Make a new entry, select the corresponding product type as a sharing product.

When you have created a new product type, assign the product roles to it. The product role is a subscription.

You make the relevant settings in Customizing for [Service](#) under ► [Master Data](#) ► [Settings for Subscription Master Data](#) ► [Subscription Product-Specific Data](#) ► [Assign Product Roles to Product Types](#) ►.

Make a new entry, select the corresponding product type, and assign the subscription product role to it.

In the sharing product, you can maintain which shared products can be used together with this sharing product. For more information about this, see **Shared Products**.

Counter Sharing Simulation

You can use the [Counter Sharing Simulation](#) button to display a counter sharing simulation for a selected cross-catalog mapping version for products. This view displays the following counters:

- Counters in the product you are currently maintaining, current version.
- Counters in the product you are currently maintaining, current version that are shared with other counters in this product.
- Counters that exist in a specific product version to be compared (only displayed once you enter a product and a Cross-Catalog Mapping version to be compared).
- Counters that are shared between current product, current version and the compared product, selected version.

Related Information

[Shared Products](#)

Transaction Types for Sharing Contracts

Use

Technically, the sharing contract is also a subscription contract with a different business transaction type. To create a sharing order or contract in Subscription Order Management, you can use the transaction types PRCO and PRCC. The business transaction type PRCO is used to create the sharing order. The business transaction type PRCC is used to create the sharing contract.

Prerequisites

If you want to use these transaction types, you have to make the relevant settings in Customizing of **Service** under ► **Transactions** ► **Settings for Subscription Transactions** ► **Define Settings for Transaction Types** ►.

Make a new entry for both business transaction types if they are not available in the table. Assign the document purpose type P.

To create sharing orders with reference to a master agreement, you must make the relevant settings in the Customizing for **Service** under ► **Transactions** ► **Settings for Master Agreement** ► **Define Target Document Type** ►.

Create a new entry for the transaction type used for the sharing order.

Master Agreement: Create/Assign Sharing Contract

Use

A sharing contract gives you the option of using the counter in a charge plan for all contracts collectively. The counter is found at sharing contract level and can be used from here by individual subscription contracts.

You can assign a sharing contract to several items in a master agreement or several different master agreements. Only the master agreement from which the sharing contract was created is relevant for billing.

When a sharing contract is created or assigned to a master agreement, the system only checks the product. All subsequent checks, such as the validity of the sharing contract, until the contract is released. You can use shared product to define restrictions.

You can also assign several sharing contracts to a master agreement item. During release in the master agreement, the system decides which sharing contract is to be used. During release, sharing contracts in the system are displayed in a separate assignment block. The following criteria must be fulfilled:

- The sharing contract must be active and error free.
- The sharing product must correspond to the product for the master agreement item.

You can select all authorized contract partners from the corresponding master agreement item.

A sharing contract is linked to the corresponding master agreement by the document flow. If a sharing order document is transferred to a sharing contract document, the system generates a reference to the sharing contract from the master agreement item. The system deletes any existing references between the sharing order document and master agreement item.

Scenario Without Reference to the Master Agreement

You can use this scenario if persons who are not employees at a company want to use the same counter in your subscription contracts.

Prerequisites

To create a shared counter, products in a sharing contract and subscription contract must have a counter with the same ID and namespace in the mapping for the charge plan. During release, you can only select transaction types that have the usage **P — Sharing Order** in Customizing.

Activities

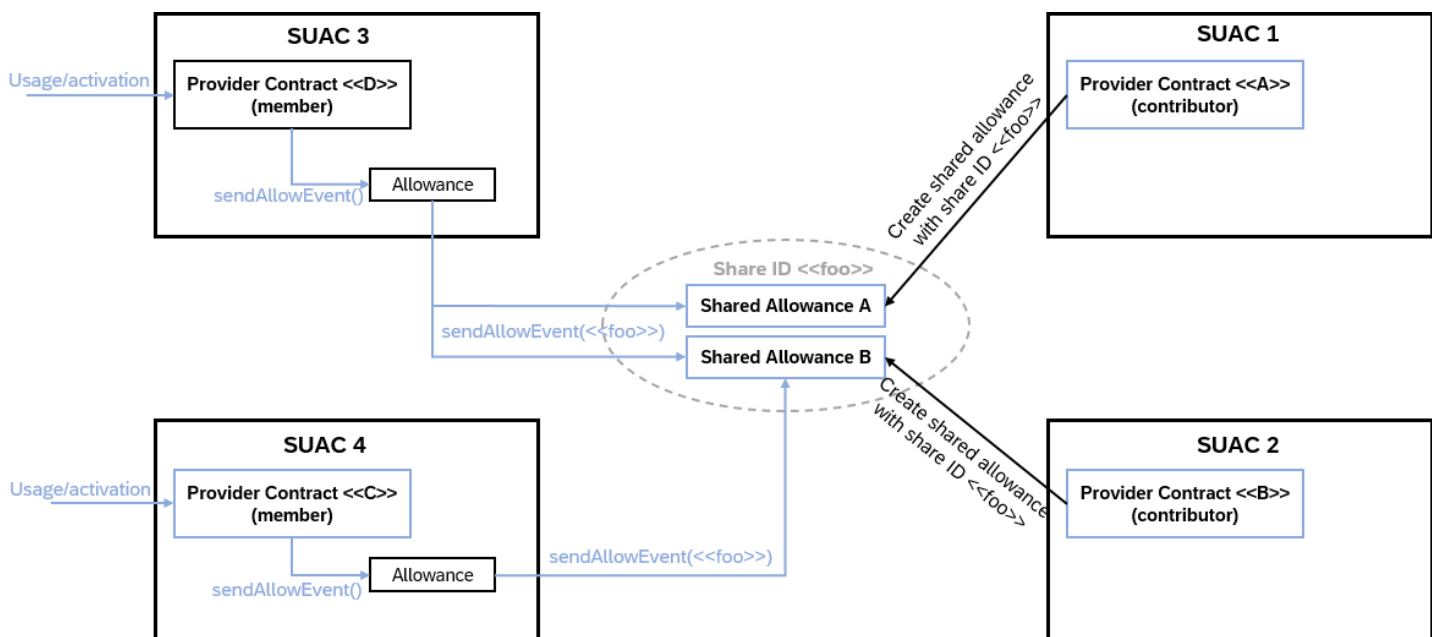
You create a sharing contract at item level in the master agreement. Choose **New** in the **Sharing Objects** assignment block.

Related Information

[Master Agreement](#)

Shared Allowances

Handling in SAP Convergent Charging (SAP CC)



For more information on how to model allowances in SAP CC, refer to [SAP Convergent Charging](#).

Settings in Subscription Product-Specific Data

If your charge plan defined in SAP CC uses shared allowances, it needs at least one parameter to define the share ID that is used to create and access a shared allowance. A sharing is only possible between contracts using the same share ID.

You must make the following settings for the Cross-Catalog Mapping version:

- Set indicator **With Shared Objects**

If this indicator is set, you must maintain a sharing group during order capturing. For more information see, [Manage Sharing Groups](#).

- Set the charge plan string parameter that is used for the share ID where **Value Used = Share ID**. When sending the contract to SAP CC, the parameter value is determined as a concatenation of the sharing group defined in the contract and the redefined value of this parameter.

To control creation of the shared allowance, you can assign an allowance definition group to your allowance table parameter.

Use in Subscription Orders and Master Agreements

When creating a master agreement or a subscription order with a product that uses shared allowances, you must maintain a sharing group ID in the subscription-specific data on item level. You can only maintain sharing groups which are defined in the Fiori app [Manage Sharing Groups](#).

If you create a call-off for a master agreement, the sharing group ID is copied to the subscription order item and cannot be changed. A sharing group ID cannot be changed by any contract change process.

Rerating of Shared Allowances

Rerating of shared allowances is supported with SAP Convergent Charging release 2025 FPS0. For more information about this, see [Charge Plan: Definition](#).

To enable rerating of shared allowances in subscription contracts, you must ensure the following:

- You are only allowed to add subscription contracts for the same business partner in a sharing group, that is, you must assign the payer in the sharing group. Sharing groups without a payer can only be used for shared allowances without rerating.
- Your cross-catalog mapping (CCM) version must use charge plans which support rerating of shared allowances.
- The sharing group must be unique within the subscription contract, that is, all the items (main and subitems) that are relevant for distribution to SAP CC must use the same sharing group.
- Only one active contributor contract can exist in a sharing group. You can create a new contributor contract in a sharing group only if the existing contributor contract operational status **Closed**. This means that in case of a contract transfer, the new subscription contract cannot be distributed to SAP CC until the previous contract is closed.
- You cannot change the sharing group or contract role (switching from contributor contract to member contract and the other way round) in a change order. This must be considered during a product change or the upgrade of a CCM version
- Only one payer can be used in a master agreement.
- Only one call-off document is possible in a master agreement for a contributor contract since only one contributor contract is allowed within a sharing group.

Constraints

Shared allowances have the following constraints:

- Distributed counters can only be initialized with positive values and allow only debit operations during rating.
- Too many distributed counters can have a negative impact on performance and memory consumption.
- In a subscription contract in Subscription Order Management, shared allowances are not displayed.

Related Information

[Sharing Contract](#)

[Allowances in Subscription Order Management](#)

[Manage Sharing Groups](#)

End-to-End Extensibility in Subscription Transactions

Adding Fields in Subscription Order, Subscription Contract, and Master Agreement

Use the [Custom Fields](#) app to add custom specific fields to one-order objects in SAP S/4HANA. The following object share the same business context:

- Service order, service contract, and service quotation
- Solution quotation
- Subscription order and subscription contract
- Subscription item and subscription contract item

This means that the custom fields you add will be available in all objects that share the same business context. In an app, a field will be visible only when you include it in the user interface. To add fields to master agreements, you use the [Agreement Header](#) and [Agreement Item](#) business contexts.

After adding the fields in the app, you must execute the post processing report CRMS4_EXTENSION_POSTPROCESSING for your business context in the backend system. After executing the report, confirm the following enhanced database tables (for example, in transaction SE11):

- CRMS4D_SERV_H (Subscription Order Header)
- CRMS4D_SERV_I (Subscription Item)
- CRMS4D_MAGR_H (Agreement Header)
- CRMS4D_MAGR_I (Agreement Item)

If you want to expose a new field on a Web Client UI, in the UI component Workbench (BSP_WD_CMPWB) you must use the following procedure:

1. Define the UI enhancement set for the Web UI enhancement and assign it to the appropriate client in the backend system.
2. Define UI enhancement sets in the BSPWDVC_CMP_EXT view cluster in transaction SM34.
3. Assign enhancement set to client in the BSPWDV_EHSET_ASG view in transaction SM30.
4. Enable configuration mode in WebClient UI via personalization. You have two options to display the field on the user interface:

- Visualizing the fields in the CUSTOMER_H or CUSTOMER_I assignment block.
- Add the fields to another view.

i Note

All custom fields added as described above to one order objects are only available either in the Customer_H or Customer_I BOL component or context nodes.

CUSTOMER_H and CUSTOMER_I BOL can be visualized in a separate assignment block which is already integrated in some of the UI components, such as the subscription order. However, they are not part of the default configuration. So, you must adapt the UI configuration as follows:

1. Launch the Web Client UI, select an appropriate business role and navigate to the UI you want to change, such as the subscription order.
2. On the overview page, choose **Configure Page** to open the overview page configuration.
3. Move BTCUSTOMER_H / I from **Available Assignments Blocks** to **Displayed Assignment Blocks** and save your changes.
4. On the overview page, choose **Show Configurable Areas** to enter the view configuration for BTCUSTOMER_H / I.

The custom extension fields used to enhance the subscription order/contract and CI/FI-CA processes through key user extensibility also transfer that field data from Subscription Order Management to CI/FI-CA automatically during contract distribution. You add these custom fields using the following procedure:

1. Add the business role SAP_BR_EXTENSIBILITY_SPEC and go to the **Custom Fields** app.
2. Enter the following field properties for the new custom field:
 - **Business Content:** CRMS4_SERV_I
 - **Label:** Item1
 - **Identifier:** You can modify the identifier proposed by the system as per your requirements
 - **Tooltip:** You can provide the required tooltip
 - **Type:** You can select the appropriate field type, for example, Text
3. In the **Business Scenario** tab enable **Create Provider Contract Item from Subscription Contract Item** to create the same label in CI provider contract.
4. After saving the KUT fields, publish and it will generate and will extend the Service and Provider contract tables.
5. Check the field with the data element ZZ1_ITEM1 in CRMS4S_SERV_I_INCL_EEW_PS include structure in CRMS4D_SERV_I and PRVCTRITM_INCL_EEW_PS include structure in DFKK_VT_I.

For extension of the fields in provider contract, go to the **Business Scenario** tab and enable **Create Provider Contract Item from Subscription Contract Item**. Go to **User Interfaces** and enable **Data Sources** to enable the field available on the provider contract SAP GUI screen.

Extending Search with Custom Fields App

You can extend the search for subscription orders, subscription contracts, and master agreements. Since the subscription contract search is an item level search, only the item level fields are available in the search parameter and search result structure of the subscription contract search. The fields that have been enhanced on header or item level can also be enabled as search

parameters and in the search result structure. To extend the search in the [Custom Fields](#) app, choose **UI and Reports**, and enable the usage of the fields. Publish the change.

You add the fields to Web Client UI using the following procedure:

1. In the Web Client UI, select an appropriate business role and navigate to the UI you would like to change, such as the subscription order search.
2. Choose **Show Configurable Areas** to enter the view configuration for the search parameter view.
3. To add the field to the screen, in the search criteria view configuration move the enhanced field from the **Available Search Criteria** to the **Selected Search Criteria**. Save your changes.
4. Repeat the procedure for the result list area.

SOAP API Extensibility Using Custom Fields App

The fields that you have added on a header or item level can also be enabled in the subscription order and subscription contract change SOAP API. To do so in the [Custom Fields](#) app, choose **SOAP API**, and enable the usage of the fields. Publish your changes.

Table Extensibility of Subscription Order and Subscription Contract

You can create custom tables for use in subscription orders and subscription contracts via the business role **AXT – Application Enhancement Tool** in the WebUI. To create custom tables, SAP provides the following enhancement objects:

- **PROVIDER_ORDER** (Subscription Order)
- **PROVIDER_CONTRACT** (Subscription Contract)

On the **Search Enhancements** screen, choose **New** and select the enhancement object to create the custom table. After you have added the table details and required fields in the respective assignment block, set the **GENERATE** indicator. Go back to the main screen and click **Save and Generate**. The status of the enhanced table then changes to **Active**.

To expose the custom tables on a Web Client UI, in the UI component Workbench (transaction BSP_WD_CMPWB), you must move the new tables from the **Available Assignments** block to the **Display Assignments** block.

Implement SAP Note [2963553](#)  to be able to copy enhanced table data in subscription order data to subscription contracts.

Data Management in Subscription Order Management

In accordance with the Data Protection and Privacy guidelines, Subscription Order Management delivers functions that enable archiving, deleting, and destruction of personal user data.

For more information, refer to the topics below:

- **Data Archiving in Subscription Order Management** under [Data Management in Service](#)
- [Blocking and Deleting Personal Data in Subscription Order Management](#)
- [Veto Check for HR Master Data in Subscription Order Management](#)
- [Data Destruction Objects in Subscription Order Management](#)

Integration

Subscription Order Management integrates with various external systems to provide extended functionalities.

Refer to the links below for more information about each integration.

- SAP CPQ for a front-end solution that enables easy creation of quotations based on specifically configured products. For more information, refer to [Integration with the Solution Quote Object in SAP S/4HANA: SAP Billing and Revenue Innovation Management](#).
- SAP Entitlement Management to trigger an EMS provisioning event that controls activation of a subscription contract based on the user's entitlement (for example, a subscription, license, or membership). For more information, refer to [Replicate Subscription Contract from SAP S/4HANA via Event Mesh](#).
- External consumers of your service using [APIs and Events for Subscription Order Management](#).
- [SAP Global Trade Services](#) for adherence to international trade compliance rules.

Integration with SAP Global Trade Services

SAP offers SAP Global Trade Services (SAP GTS) and SAP International Trade Management to streamline the process of trade compliance and enable adherence to international trade compliance rules. Subscription Order Management integrates with SAP Global Trade Services (SAP GTS) to facilitate multiple compliance checks such as Embargo Check, Sanctioned Party List Screening, and Product License (legal control) Check.

When you create a subscription order and either save or submit it, the following GTS relevant checks are implemented:

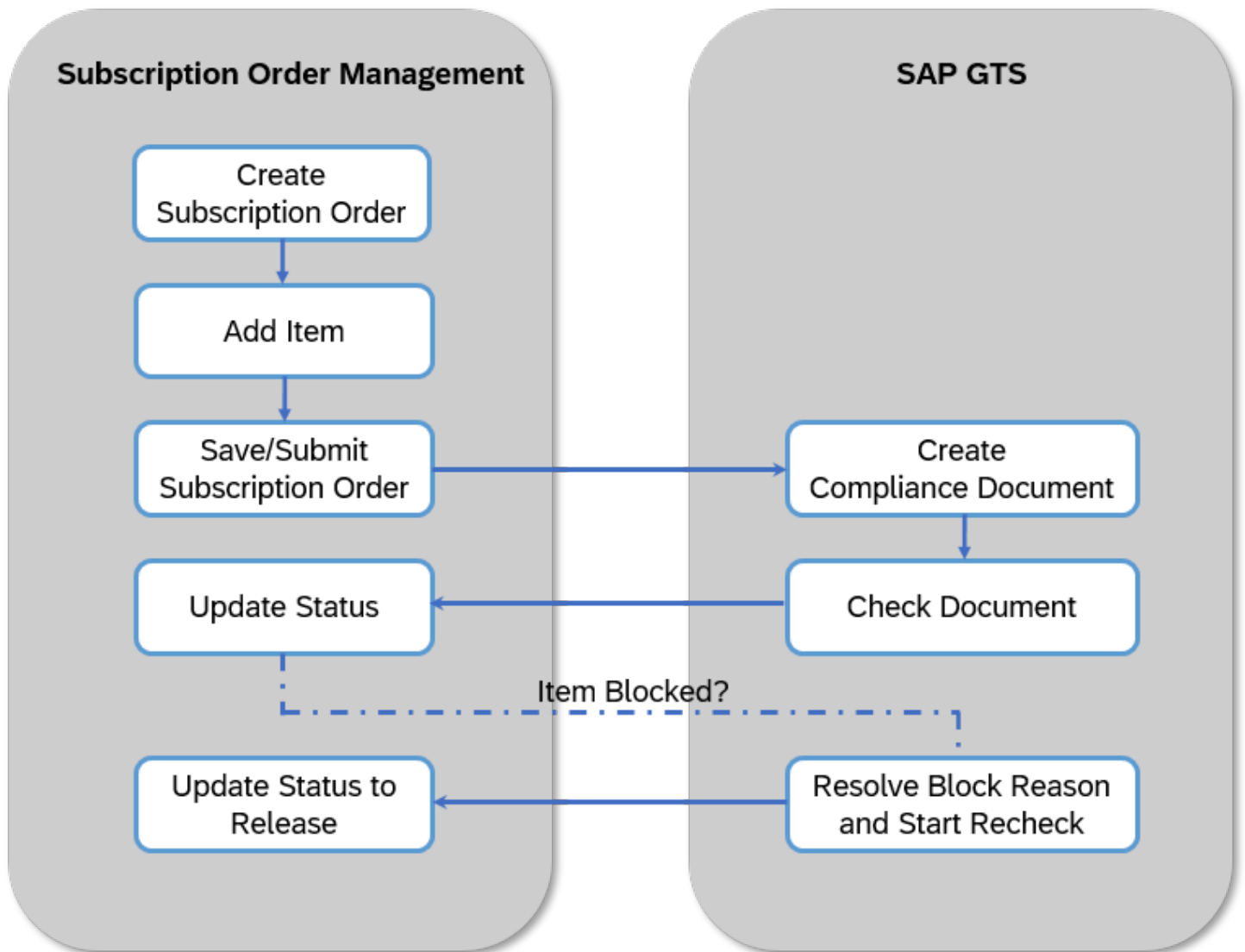
- **Embargo check:** The system checks for any existing trade restrictions placed on the import or export of any goods or services to a particular country or state. This check is implemented at the subscription order header and item level and sets the relevant status.
- **Sanctioned party list screening:** The system compares the business partner addresses with the addresses on the relevant sanctioned party lists. This check is implemented at the subscription order header and item level and sets the relevant status.
- **Legal control (import/export):** The system checks the products to see which ones might require a license and sets the relevant status.

For more information about the available statuses for the compliance checks, refer to **Compliance Check Statuses**.

You can use the individual statuses to search for and filter out the list of subscription orders with trade compliance issues for one or more items. For example, you might want to find all subscription orders in which at least one item is blocked by an embargo check. In addition, you can use the **GTS Status** field to search for those subscription orders where at least one of the three compliance checks is either blocked or is not relevant for the check.

Compliance Check Process

The following diagram illustrates a compliance check process for a subscription order:



Please note that image maps are not interactive in PDF outputs.

When you change the subscription order and save it, it's transferred to GTS again and a new check is performed.

For more information on compliance, refer to [Schedule Service Transaction Status Update - Global Trade Services](#).

Configuration

- Before integrating with SAP GTS, the business partner and product used in the subscription order must be replicated from SAP S/4HANA to SAP GTS.
- You must activate the BAdI implementation /SAPSL/PI_SRVA_TRANSFER (Implementation: GTS Transfer Service Document) in Customizing under **Integration with Other SAP Components** > **Integration with Governance, Risk and Compliance** > **SAP Global Trade Services** > **Control Data for Transfer to SAP Global Trade Services** > **Activate Transfer of Service Transactions (SRVA)**.
- To create a compliance document for a subscription order, you can do so in Customizing under **Integration with Other SAP Components** > **Integration with Governance, Risk and Compliance** > **SAP Global Trade Services** > **Control Data for Transfer to SAP Global Trade Services** > **Configure Control Settings for Document Transfer**. Using this Customizing activity, you can also configure the system to prevent subsequent processing of blocked subscription orders, for example, to prevent creation of a subscription contract if a subscription order is blocked.

Post Processing

In case there is an issue during transfer of document status from GTS to SAP S/4HANA, no error is raised but all the documents with a GTS failed status are saved in the /SAPSLL/SRVASTS table in SAP S/4HANA. In this case, you use the Fiori app [Schedule Service Transaction Status Update - Global Trade Services](#) to re-transfer the status of the GTS Trade Compliance check back to the SAP S/4HANA system. For more information about this, refer to [Schedule Service Transaction Status Update - Global Trade Services](#).

You can also update the status of the GTS Trade Compliance check via the report /SAPSLL/SRVA_DOC_UPD_REPROCESS. You can either schedule this report or execute it manually.

Related Information

[Integration with SAP Global Trade Services](#)

GTS Integration with Subscription Contracts

GTS integration with subscription contracts ensures that the subscription contract is aligned with international trade compliance rules by executing a GTS check when the contract is created or on a long running contract using a compliance document that will set the contract status to **Blocked**.

GTS block during contract creation

In this scenario, the distribution and activation of the subscription contract depends on whether a GTS block is detected during contract creation. If there is a GTS block, the contract is distributed and activated only after the GTS status is set to **Released**.

No GTS block on subscription order or subscription contract

In this scenario, a GTS check is executed on the subscription order. If no GTS block is detected, the order is submitted and a contract is created successfully.

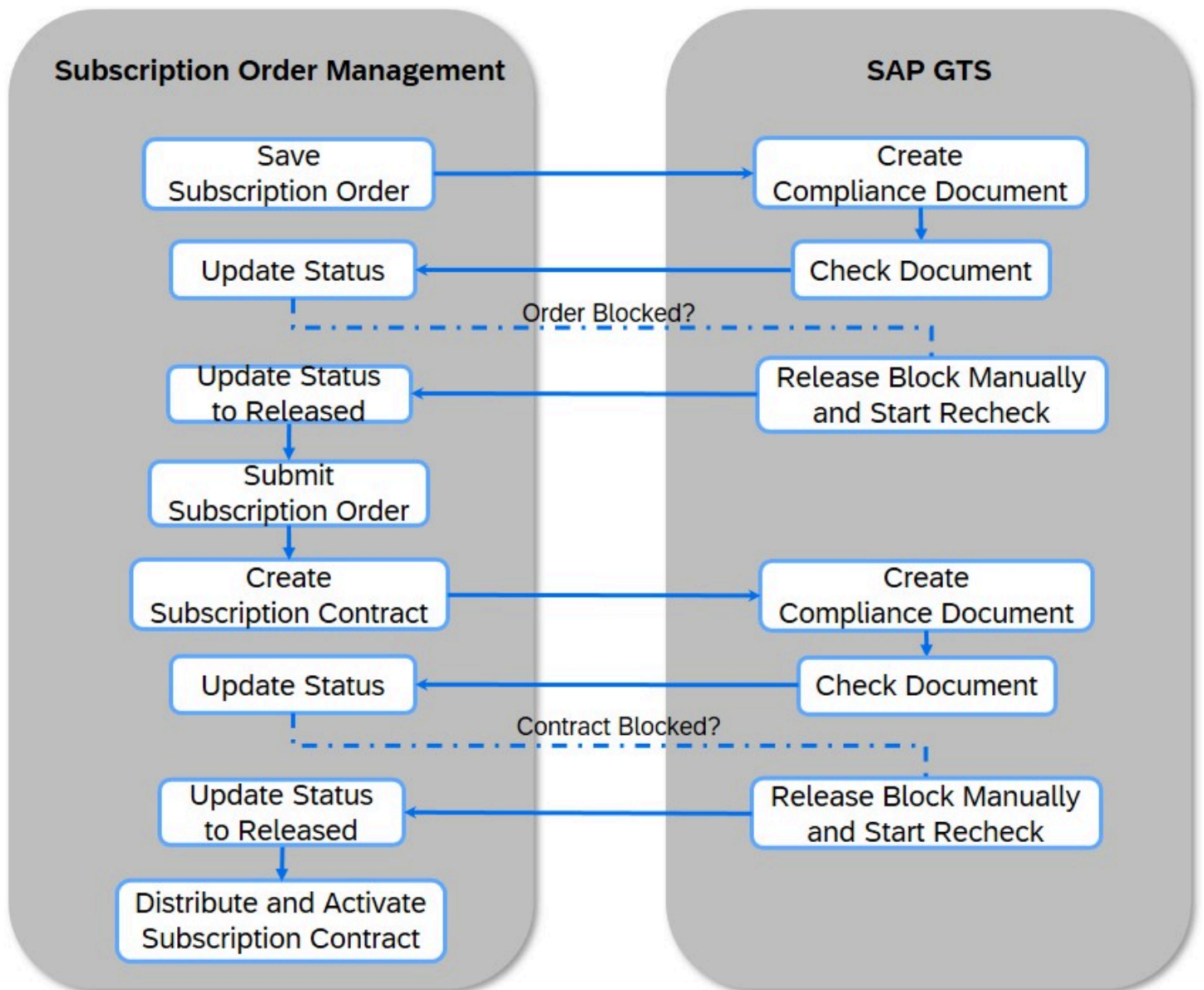
GTS block on subscription order is released manually and GTS block on subscription contract is released manually

In this scenario, a GTS check is executed on the subscription order. If a GTS block is detected, the order cannot be submitted and contract creation is blocked. The contract is created once the GTS block on the order is released manually. A GTS check is executed on the contract and if a GTS block is detected, the ODI step GTSB (**Stop distribution due to GTS block**) blocks the contract distribution and activation. Once the GTS block on the contract is released manually, the distribution proceeds to the ODI step PCEA (**Call FICA Message contract activation**) and the contract is created successfully.

i Note

The ODI step GTSB must be assigned as a prerequisite to the ODI step PCEA in Customizing under ► [Service](#) ► [Transactions](#) ► [Settings for Subscription Transactions](#) ► [Document Distribution](#) ► [Define Settings for Document Distribution](#) ► [Step Types](#) ►.

This process ensures that a compliance document is created in the GTS system on the contract object. If any GTS block is introduced during the lifecycle of the subscription contract, this compliance document serves as a trigger to update the relevant GTS status on the subscription contract.



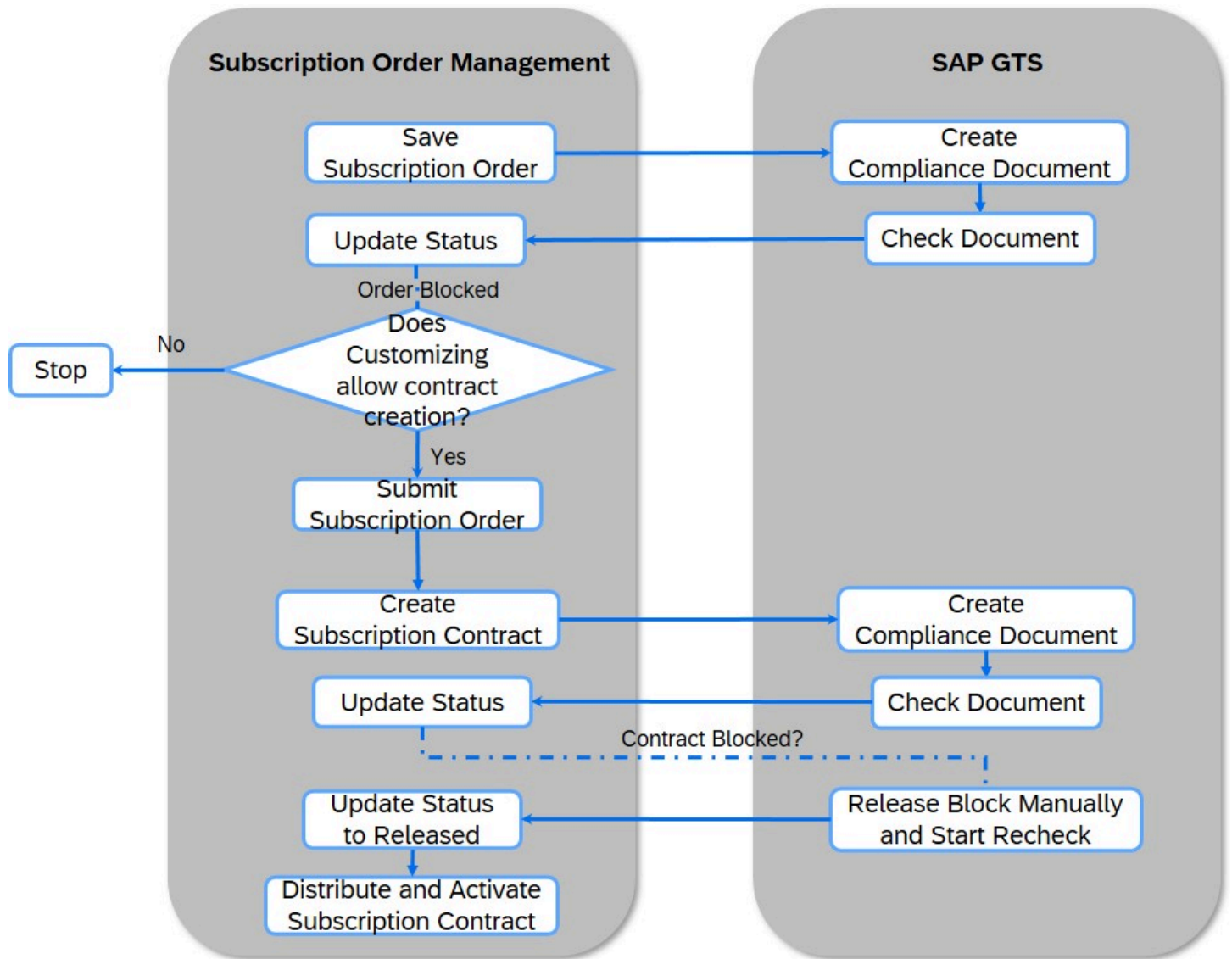
GTS Block on Subscription Order is Released Manually and GTS Block on Subscription Contract is Released Manually

Subscription Contract Created from Blocked Subscription Order and GTS Block on Subscription Contract is Released Manually

In this scenario, a GTS check is executed on the subscription order. If a GTS block is detected, the system considers the **Prevent Subscription Contract Creation for Blocked Subscription Orders** flag in Customizing under **Integration with Other SAP Components > Integration with Governance, Risk and Compliance > SAP Global Trade Services > Control Data for Transfer to SAP Global Trade Services > Configure Control Settings for Document Transfer**. If this flag is checked, the contract cannot be created and the process is terminated. However, if this flag is unchecked, the contract is created with a GTS block and the ODI step GTSB (**Stop distribution due to GTS block**) blocks the contract distribution and activation. Once the GTS block on the contract is released manually, the distribution proceeds to the ODI step PCEA (**Call FICA Message contract activation**) and the contract is created successfully.

i Note

The ODI step GTSB must be assigned as a prerequisite to the ODI step PCEA in Customizing under **Service > Transactions > Settings for Subscription Transactions > Document Distribution > Define Settings for Document Distribution > Step Types**.



Subscription Contract is Created from a Blocked Subscription Order, GTS Block on Subscription Contract is Released Manually

GTS block applied on an existing contract

In case of long running subscription contracts, it is possible that a GTS block is implemented **after** the contract is distributed and activated. In such a case, the GTS status on the contract is set from **Released** to **Blocked** and the list of available change processes are limited based on the **Allowed on Contract Blocked by GTS** flag in Customizing under **Service > Transactions > Settings for Subscription Transactions > Contract Changes > Settings for the Maintenance Framework (BTMF) > Define Change Process > Define Scenario Dep. Attributes**. If the flag is checked on a specific change process, then it is allowed on the blocked subscription contract. Additionally, you use the **Skip GTS Check for Change Order** flag in the same Customizing activity to control whether the GTS check is executed in a change order based on a blocked contract.

❖ Example

Assume a blocked subscription contract where only the **Cancel** and **Change Product** change processes are available. These are enabled by checking the **Allowed on Contract Blocked by GTS** flag.

If you choose to change the product on the subscription contract then you would expect a GTS check to be executed on the change order to ensure that there are no GTS blocks implemented on the new product. This is enabled by unchecking the **Skip GTS Check for Change Order** flag. If no GTS block is detected, then the change order can be submitted to update the subscription contract.

Now consider that you choose to cancel the blocked contract. In this case, a GTS check is typically not required on the change order and you can disable it by checking the **Skip GTS Check for Change Order** flag.

Integration

For information on how to setup the GTS integration, refer to the [Integration](#) section in the topic **Integration with SAP Global Trade Services**.

Related Information

[Integration with SAP Global Trade Services](#)

Compliance Check Statuses

For each item in a subscription order, SAP GTS executes a trade compliance check. The prerequisite is that the subscription order and the items are relevant for the check. Which items are relevant depends on the settings configured in Customizing for the GTS system. The status at the transaction header level is determined by the statuses of the individual items.

For each of the three compliance checks – embargo check, sanctioned party list screening, and legal control – the system sets one of the following statuses both on header and item level depending on the check result:

Compliance Check Status	Description
Not Relevant, No Check	The system hasn't performed a compliance check because the subscription order item is not relevant for the check.
Released	The system sets a subscription order or item to Released if no trade compliance issues are found.
Blocked	The system sets a subscription order item to Blocked if there's a trade compliance issue. The trade compliance specialist can resolve the block reason, for example, by maintaining the missing data such as an export license, and start the recheck of the subscription order.
Partially Blocked	This status is only relevant for the header level of a subscription order. The system sets it if at least one of the items of the subscription order is blocked. For more information about when this status is set, refer to the examples in the table below.
Pending – Not Yet Checked	The compliance check of an item is still pending.

Trade Compliance Status of Subscription Transactions

On the header level of a subscription order, the system sets a cumulative status depending on the statuses of the subscription order items. The following example explain how the status of a subscription order with three items is set:

Status of Item 1	Status of Item 2	Status of Item 3	Resulting Order Status	Comment
Not Relevant, No Check	Not Relevant, No Check	Not Relevant, No Check	Not Relevant, No Check	If all the three items are not relevant for a check, the status at the subscription order level is Not Relevant, No Check .
Released	Released	Released	Released	The subscription order status is set to Released

Status of Item 1	Status of Item 2	Status of Item 3	Resulting Order Status	Comment
				if there are no issues in any of the items.
Blocked	Blocked	Blocked	Blocked	The subscription order status is set to Blocked if all of its items are blocked.
Released	Released	Blocked	Partially Blocked	The subscription order is set to Partially Blocked if at least one of its items is blocked and at least one item is not blocked.
Released	Blocked	Not Relevant, No Check	Partially Blocked	
Released	Pending – Not Yet Checked	Blocked	Partially Blocked	
Released	Pending – Not Yet Checked	Released	Pending - Not Yet Checked	The subscription order is set to Pending – Not Yet Checked if the check is pending for one of its items and none of the items are blocked.

Integration with Joule

Subscription Order Management leverages the integration between SAP S/4HANA Cloud Private Edition and Joule to bring to you critical capabilities such as create subscription order and search and summarize subscription contract. For more information about these capabilities, refer to [Subscription Order Management](#).

i Note

You may require additional licenses to use Joule.

Related Information

[Joule](#)

Data Extraction CDS Views in Subscription Order Management

The following data extractions CDS views are available in Subscription Order Management:

- [Data Extract Subscription Order](#)
- [Data Extract Subscription Order Item](#)
- [Data Extract Master Agreement](#)
- [Data Extract Master Agreement Item](#)
- [Data Extract Subscription Contract](#)
- [Data Extract Subscription Contract Item](#)

Video Library for Subscription Order Management

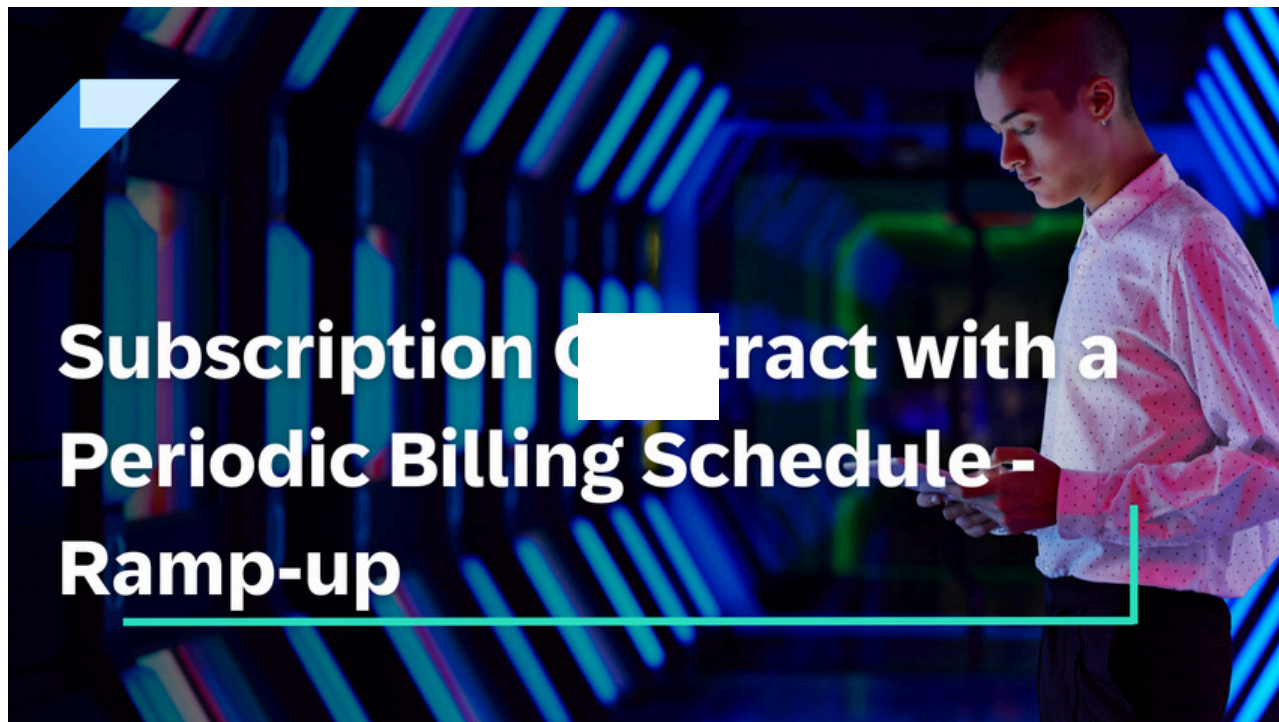
i Note

Captions are available for **multiple languages**. Use the **CC** button to see which languages are supported.

You can also use the **Search video** field to search for specific text in the captions.

Subscription Contracts with a Periodic Billing Schedule

Subscription Contracts with a Periodic Billing Schedule (Ramp-up Pricing Scenario)



[Open this video in a new window](#)

For more information, see [Billing Schedule for Periodic Subscriptions](#).

Subscription Contract with a Periodic Billing Schedule (Discount Scenario)

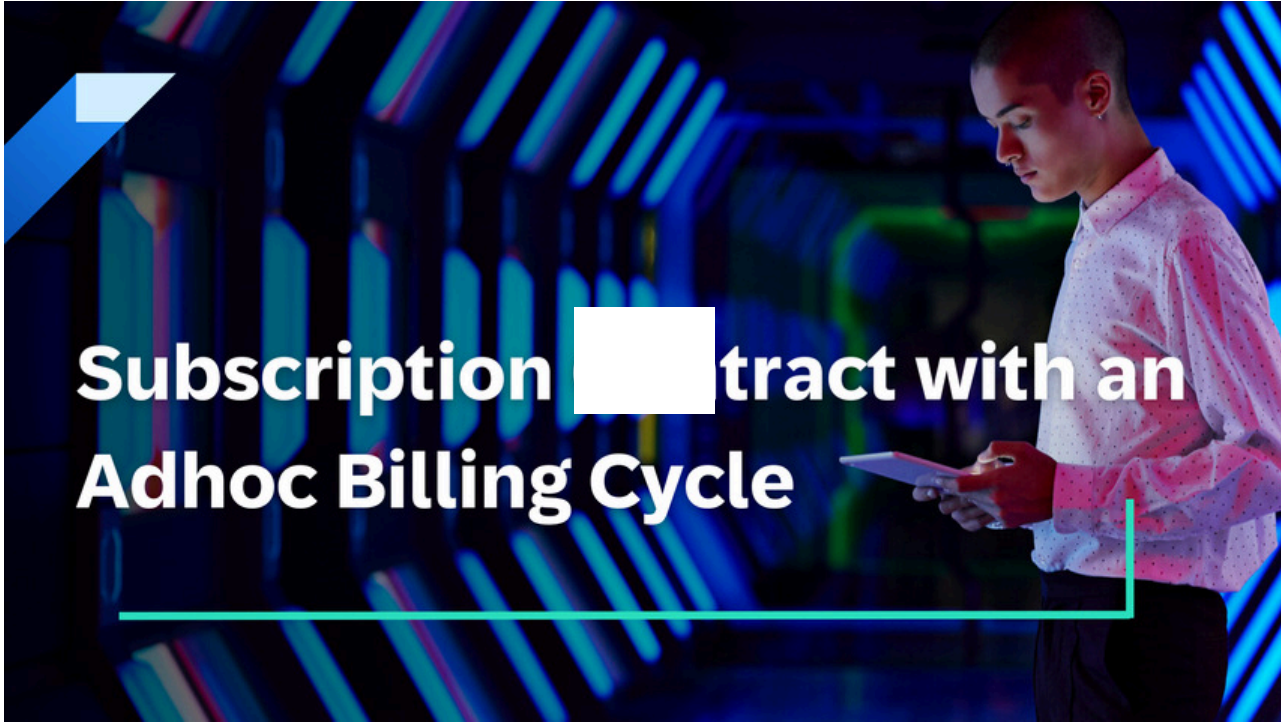


DISCOUNT SCENARIO

[Open this video in a new window](#)

For more information, see [Billing Schedule for Periodic Subscriptions](#).

Subscription Contract with an Adhoc Billing Schedule



[Open this video in a new window](#)

For more information, see [Adhoc Billing Schedule for Fixed Price Subscriptions](#).

Subscription with Recurring Delivery of Physical Products



[Open this video in a new window](#)

For more information, see [Subscription with Recurring Delivery of Physical Products](#).

Integration of Subscription Order Management with SAP Global Trade Services (SAP GTS)

Handling a GTS Block on a Long Running Subscription Contract



[Open this video in a new window](#)

For more information, see [Integration with SAP Global Trade Services](#).

Mass Processing

Repricing Multiple Subscription Contracts





[Open this video in a new window](#)

For more information, see [Mass Processing](#).